
the
semantic
turn
a new
foundation
for design
klaus krippendorff

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Dedication

To:

Horst Rittel

for being a dedicated and inspiring teacher and an enthusiastic explorer of complex design issues — greeting each new discovery with his mischievous smile — but above all, for entrusting me with my own ideas.

John Rheinfrank

for reintroducing me to design with cutting-edge projects — always playing with ideas and struggling for clarity while enabling his team members to make things happen.

Reinhart Butter

for practicing product semantics well before it had a name, teaching me what he learned with his students, and never wavering in seeing the virtue of the semantic turn. Without his longtime collaboration, encouragement, and practical projects, this book would have been unlikely.

Marge Thorell

for listening to the excitement and drudgery of writing this book and making her time with me generously available for it.

…philosophers have only interpreted the world…;
the point is to change it.

Karl Marx, 1845*

Laß us menschlich sein.—

Ludwig Wittgenstein, 1937**

…vocabularies — all vocabularies, even those which contain the words which we take most seriously, the ones most essential to our self-descriptions — are human creations, tools for the creation of such other artifacts as poems, utopian societies, scientific theories, and future generations. …changing the way we talk…(is) changing what we want to do and what we think we are.

Richard Rorty, 1988***

* The 11th Thesis on Feuerbach in *German Ideology,* Karl Marx (1845).
** Literally but not connotatively: "Let us be human." *Culture and Value*, Wittgenstein (1980:30).
*** *Contingency, Irony, and Solidarity,* Rorty (1989:53,20).

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Foreword

When Thomas Kuhn introduced the notion of a *Paradigm Shift* into the public debate, he was describing the periodic reconstruction of accepted ideas, roles, and procedures in scientific inquiry.* Most of what he said applies to ideas in design as well. But after the design profession embraced Sullivan's 1896 principle, *Form Follows Function*,** the celebration of mass producible forms by the Bauhaus in the 1920s; the affectation of "streamlining" adopted when design aligned itself with marketing in the 1930s; the influence of operations research in the late 1940s; the practice of minimalism by the Hochschule für Gestaltung (HfG) at Ulm in the 1950s and 1960s; the predominance of the systems approach in the 1970s; and the concern with concurrent engineering in the 1980s and 1990s are at best advances in design thinking that pale by comparison to the paradigm shift in design we are now witnessing.

What outsiders do not appreciate, perhaps, is the conflict and pain that theorists and practitioners experience during the transition from one paradigm to another. Reinhart Butter, Klaus Krippendorff, and I happened to be at the HfG Ulm during 1960 to 1961, when the prevailing functionalist paradigm came to be questioned and started to break down. The HfG Ulm was a school far ahead of its time. There, the protagonists of form-follows-function and minimalism, on the one hand, and the rather diverse advocates of gestalt psychology, operations research, systems theory, and sociology of technology, on the other, were engaged in continuous and bitter disputes that later fueled many debates in the larger design community. Although the dissolution of the prevailing paradigm seemed imminent then, a new synthesis was not yet in sight.

Significant ideas and inventions have their gestation periods. To take root, paradigm shifts require a new generation of proponents, a fertile cultural climate, but also significant technological advances. It is not surprising, therefore, that the term "Product Semantics" did not emerge until 1984.*** Some thought that semantic considerations would merely add to the design

* Thomas Kuhn (1962). *The Structure of Scientific Revolutions*. Chicago: Chicago University Press.

** Louis Henry Sullivan (1896). The Tall Office Building Artistically Considered. *Lippincott's Magazine*, March 1896. http://www.njit.edu/v2/Library/archlib/pub-domain/sullivan-1896-tall-bldg.html.

*** Klaus Krippendorff and Reinhart Butter, Eds. (1984). Product semantics; special issue of *Innovations* 3, 2.

discourse. However, they became not only the key to the design of human–machine interfaces, they are now the principal drivers of information technology. Product semantics, the way I have observed it taught at The Ohio State University (OSU), was only the first step toward that long-awaited synthesis.

The Semantic Turn reviews the history of semantic concerns in design, presents their philosophical roots, and lays out several compelling design methods that take seriously what Klaus Krippendorff considers axiomatic for human-centered design: that *humans do not respond to the physical properties of things* — to their form, structure and function — *but to their individual and cultural meanings*. This premise radically breaks with functionalist traditions in design. A new paradigm is recognized when its basis has become obviously true while rendering previous truths obsolete. I fully subscribe to the book's premise and its far-reaching implications.

The Semantic Turn boldly outlines a new science for design that offers designers a solid ground on which to state and validate their claims — not unlike the "hard" disciplines, but without the shortcomings of Herbert Simon's 1969 proposal.* I am particularly pleased about the inclusion of systematic design methods, as they are rooted in my own contributions. Klaus Krippendorff accomplishes this synthesis as a recognized designer and as a distinguished scholar of human communication and language use. Reinhart Butter contributes examples from his pioneering explorations at OSU.

The Semantic Turn is an indispensable guide for the future of the design profession and a must for students, teachers, practitioners, and employers of design. I applaud the work that went into this remarkable redesign of design.

Bruce Archer
London
May 15, 2005

* Herbert A. Simon (1969). *The Sciences of the Artificial*. Cambridge, MA and London: MIT Press.

Introduction and Overview

This book introduces a new way of conceptualizing design as a professional practice and as an activity that is constitutive of human beings generally. The etymology of "design" goes far back, of course, to the Latin *de + signare,* which means to mark out, set apart, give significance by assigning it to a use, a user, or an owner. Design has the same origin as "sign" and to "designate," calling attention to something other than its observer-independent existence: meaning. The sixteenth century English emphasized the purposiveness of design, and because design often involved drawing, or "marking out," the seventeenth century moved design closer to art. Based on these original meanings, one could say:

*Design is making sense of things.**

The phrase is conveniently ambiguous. It can be read as "design is a sense creating activity," which claims perception, experience, and perhaps appearance as its fundamental concern, and this interpretation is quite intentional. Or, it could mean that "the products of design are to be understandable to their users," and this interpretation is even more to the point of this book. But the phrase concerns artifacts, which are human creations, the tangible products of artisans, industry, and other social institutions. Making sense is the result of human activity as well, but not as tangible as artifacts often are. Acknowledging artifacts and sense to be inseparable from what humans do is to discourage the conventional distinction between subjective interpretations of artifacts and their objective nature, which reappears in several forms, for example, in the distinction between perception and what is perceived or between art and engineering. Since we cannot speak of things that make no sense, the commitment to avoid non-sense, including the objective–subjective distinction, will become clearer in the book.

It is important to respect etymology, which acknowledges that words, or artifacts for that matter — after all, words are artifacts as well — can change their meaning as they travel through different uses, different situations, different times, and different people's lives. In the course of their journeys, words designate and accomplish different things. The industrial

* This phrase is part of the title of Krippendorff (1969).

revolution, an era of historically unprecedented changes in mechanical technology, appropriated the word "design" for its emerging need to make mass production acceptable to many and expand the markets for its products. Designers became "industrial" designers. By agreeing to shape material products that could be produced in ideally large numbers, designers (e.g., industrial, furniture, graphic, and fashion designers) made themselves subservient to the needs of mass production and came to live in the shadow of industry. The contemporary meaning of the word "design," sometimes referred to as "applied art," still carries the stamp of the industrial revolution. This need not be so. In fact, this conception of design increasingly reveals itself as an anachronism, leading to the suggestion that design reposition itself in the fabric of contemporary society.

Having barely freed itself from its industrial origin, it is fair to say that design, at least the way it is presently taught at professional art institutes and universities and practiced in industry, has essentially exhausted its vocabulary of forms. The competing schools of thought that fueled design in this century — the arts and crafts movement of William Morris, the Bauhaus in Dessau, the Ulm School of Design,* Scandinavian design, and Memphis — have lost their distinctiveness and momentum and have largely disappeared. Most educational programs include a little bit of everything in what they offer their students. No compelling manifestos exist. Design journals, while proliferating in numbers, have become reproductive of sterile still photography. They advertise products, producers, or designers, and are clearly shrinking in social significance. Consumer research has positioned itself as a judge of designed products, encouraging informed consumption but consumption nevertheless, and marketing is using the word design in order to sell pricier brands.

Consensus on a coherent and overarching visual aesthetic has broken down, and any hope of recovering this lost unity is in vain. MTV, Punk, and the Internet actually thrive on violating previous visual conventions. Even within the design profession, the traditional distinctions, for instance, between industrial and graphic designers, have become largely obsolete. Technologies have taken over some of the skills that designers thought they owned — desktop publishing, for example — and have forced a redrawing of the boundaries between design skills and mere technological literacy.** The scientific disciplines on which designers could rely to rationalize their work, psychology of perception, ergonomics, and engineering science increasingly reveal their limitations and have become clouded in epistemological doubts. Functional, aesthetic, and market considerations that justified the products of design in the past have been replaced or overshadowed by other more social, political, and cultural concerns, such as ecological sustainability and cultural identity.

* Hochschule für Gestaltung (HfG) Ulm, 1953–1968.

** For an account of how desktop publishing, during just one decade, has made the language of typography a common one and thereby disconnected it from its profession, see Faber (1998).

Sensing the opportunities that new technologies seem to promise, new scholarly disciplines like artificial intelligence, communication science, plus various hybrid professions, cognitive engineering and design management, as well as such technical specialties as computer interface design, have emerged and are blazing trails into territories previously claimed by designers. In this information-rich, fast-changing, and increasingly individualistic culture, contemporary design discourse is no longer compelling. Thus, industrial design finds itself at a critical turning point.

Design's identity crisis, however, lies not in the postmodern uncertainties, in the cultural chaos some critics say currently prevails (and the above does not really dispel this impression) but in the futile effort to search for stabilities in the wrong places, on the surfaces of tangible artifacts that no longer fit traditional conceptions. While mass production became inscribed in the mass media of communication, fulfilling industry's dream of supplying universal markets with their goods and services, achieving nearly global coverage, products have become immaterial, informational, and entertaining. But as this is happening, heterarchical social forms, information networks, such as the Internet, are radically undermining traditional social hierarchies and the one-way communication of the mass media, encouraging a new kind of individual involvement in technology, enabling alternative conceptions of reality, and creating and reproducing diverse practices.

This is not to claim to know what the information society "really" *is* — of which most futurists speak with conviction and enthusiasm — but to conclude that designers have lost their professional prowess in contemporary society by busily creating marketable products with the criteria of a rapidly fading industrial era, by helplessly floating in a stream of technological changes, or running after those who claim to have mastered the emerging discourse and pose like futuristic intellectual fashion models.

This book recognizes that design can no longer afford to continue on its well-trodden path. Design has to shift gears from shaping the appearance of mechanical products that industry is equipped to manufacture to conceptualizing artifacts, material or social, that have a chance of meaning something to their users, that aid larger communities, and that support a society that is in the process of reconstructing itself in unprecedented ways and at record speeds. The emerging conceptions of what it means to be human and the role that technologies play in these conceptions, including contemporary preferences for a democratic way of living — at least in the West — offer designers a unique place to make contributions that are far more significant then they were under the aegis of industry. Designing artifacts to make sense, to have meanings and social significance, actually, going back to lost meanings in the original Latin word "design," entails a radical shift for the practice of design. It is a turn toward considerations of meaning — *a semantic turn*. It promises to revitalize design and give it an important place in a postindustrial society.

To this end, Chapter 1 reviews some of the dimensions of the changed realities that design is facing, beginning with product semantics, which can

now be realized as merely one aspect of a far more fundamental semantic turn. Chapter 2 describes several key concepts of human-centered design that give rise to four overlapping theories of what artifacts could mean.

Chapter 3 concerns a theory of meaning of artifacts in use; Chapter 4, of artifacts in language; Chapter 5, in the life of artifacts; and Chapter 6, in an ecology of artifacts — in networks of artifacts or technology for short. Chapter 7 draws upon these approaches to meaning and proposes a science for design, especially focusing on meaning-sensitive design methods. Chapter 8 makes clear how the semantic turn differs from the aims of other disciplines with an interest in design, and Chapter 9 finally offers some observations on how and to what extent the semantic turn is rooted in the intellectual climate at the Ulm School of Design before its untimely closing in 1968.

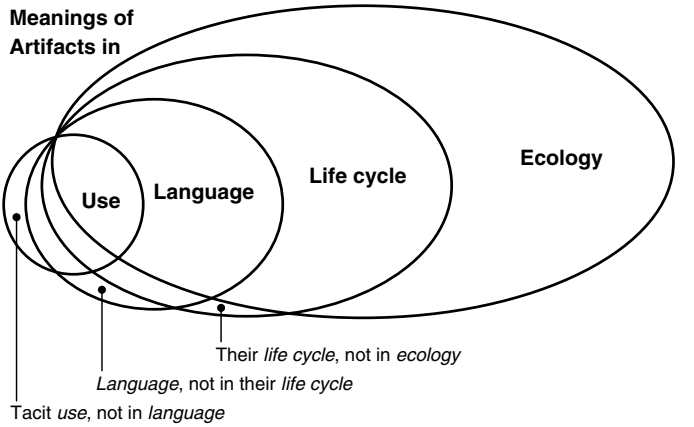


Figure 0.1 Relations between four theories of meanings of artifacts.

chapter one

History and aim

To provide a suitable context for this book, I start with a brief account of how the idea of product semantics developed, then review how design problems and the environments in which designers work have changed, and finally, I focus on what needs to be done.

1.1 Brief history of product semantics

Product semantics was a collaboration of several thoughtful designers and the starting point for the proposal to redesign design. The term first appeared in the Industrial Designers Society of America (IDSA)'s journal *Innovation* (Krippendorff and Butter, 1984). Reinhart Butter and I defined "product semantics" as both an inquiry into the symbolic qualities of things and as a design tool to improve these cultural qualities. Because this dual purpose conflicted with the epistemology of semiotics, we went back to the Greek word for the study of meanings, or semantics, and added product as our initial concern. This special issue on product semantics brought together the works of several designers and design researchers. They discussed industrial products, not as photogenic objects of exemplary aesthetic qualities, but regarding what they could say to their users, as communications, as having meanings. The contributors to this special issue sensed a new beginning for design, one that recognized a basic concern if not *the* basic concern of designers.

For me, however, the idea of a semantics for design goes back even further than this publishing collaboration, to two papers (Krippendorff, 1961a,b) written concurrently with my graduate thesis at the Ulm School of Design* (Krippendorff, 1961c). The idea of this thesis hibernated until the early 1980s when Reinhart Butter, with whom I had remained in contact regarding these ideas, thought it timely to join efforts and edit this special *Innovation* issue.

Immediately following the 1984 publication of *Innovation*, IDSA invited interested industrial designers to a summer workshop at the Cranbrook

* Hochschule für Gestaltung (HfG) Ulm, Germany, 1953–1968.

Academy of Art, conducted by Reinhart Butter, Uri Friedländer, Michael McCoy, John Rheinfrank, and myself. This workshop proved not only successful, but it attracted the attention of Robert Blaich, then Director of Corporate Design at Philips Corporation in Eindhoven, The Netherlands, where it was repeated, with only a minor change in participants. Philips' designers from all over the world attended. This repeat performance gave Philips Corporate Design not only several product ideas (one would become the by-now famous roller radio), but, more importantly, this engagement created a new momentum, a new conceptual orientation, a new methodological basis, and a new organizational identity.*

By 1987, these ideas found worldwide resonance. The Industrial Design Centre at the Indian Institute for Technology (IIT) in Bombay invited design practitioners and scholars to a major conference on product semantics called "Arthaya," an ancient Hindi word for meaning. Designers in multilanguage and multicultural India, with its visually rich mythologies, embraced the semantic turn with open arms for it promised to provide concepts, methods, and criteria that would not only serve industry's interests but moreover would respect diverse sociocultural traditions and support indigenous forms of development. The universalism prevailing in the industrialized West traditionally had denigrated cultural diversity as a sign of underdevelopment and deficient rationality. The new acceptance of diverse cultural meanings of industrial artifacts moved design into the center of cultural development projects.

In 1989, Reinhart Butter and I edited a double issue of *Design Issues* (Krippendorff and Butter, 1989), which has since become a standard reference in the product semantics literature.** Here, we agreed to define product semantics as both:

- *A systematic inquiry into how people attribute meanings to artifacts and interact with them accordingly*

and

- *A vocabulary and methodology for designing artifacts in view of the meanings they could acquire for their users and the communities of their stakeholders*.

Definitions focus attention on what matters. We wanted to avoid mentalistic conceptions of meanings and sought instead to embed meanings in how artifacts are actually used, in a meaning–action circularity. We soon realized that our concept of meaning could not be limited to what users do with artifacts but must also apply to what designers do with them, although designers have their own meaning–action circularities. Finally, we wanted product semantics to be not a mere scientific descriptive effort but one that provided designers with conceptual tools, a vocabulary for constructively intervening in processes of meaning making. The publication coincidentally

* Probably the most authoritative among several reports from this workshop and about its conceptual and managerial consequences is in Blaich (1989).

** Its key articles have been reproduced in Margolin and Buchanan (1995).

arrived just in time for the first European workshop on product semantics, organized by Helsinki's University of Art and Design, at that time called the University of Industrial Arts. Here, Hans-Jürgen Lannoch and Robert Blaich joined the original group. The reception of these ideas (Väkevä, 1990; Vihma 1990a,b) spawned further Helsinki workshops, conferences with an increasing number of international participants, and publications (Vihma, 1992; Tahkokallio and Vihma, 1995). This university has remained a leading player in these developments ever since.

In addition, workshops have been held in Colombia, Germany, Switzerland, Taiwan, Japan, Korea, and the United States. Semantics has found its way into the curricula of several design departments, notably at the Ohio State University in Columbus, Ohio, at the Cranbrook Academy of Art in Cranbrook, Michigan, and at the University of the Arts in Philadelphia, Pennsylvania. Courses taught at the University of the Arts produced a workbook (Krippendorff, 1993b) that has been reproduced and is currently being used in several other educational settings.

In 1996, the U.S. National Science Foundation (NSF) sponsored a major conference to see how designers could contribute to making information technologies more widely available. NSF was looking for consensus on a national agenda or at least policy recommendations for a national role for *Design in the Age of Information.* The report, by that name, (Krippendorff, 1997) outlined rising technological opportunities, new design principles, and educational challenges; it recommended several major research undertakings. The design principles for the new information technologies turned almost exclusively around issues of semantics: understanding, meaning, and interfaces that enable cooperation, honor diversity, even support creative conflicts. The educational recommendations focused on the demands of human-centered design curricula, and the research agenda identified significant knowledge gaps that research dollars might bridge. Semantics was recognized as a key to making the benefits of the ongoing information revolution more widely accessible.

In the summer of the same year, 1996, the Cooper-Hewitt National Design Museum in New York invited a small group of design historians, psychologists, communication scholars, architects, art editors, and museum curators to an interdisciplinary conference and workshop on "The Meaning of Things." This leading U.S. design museum realized meaning as a new unifying concept in design. Participants searched for common threads through such cultural artifacts as public spaces, industrial products, museums exhibitions, Holocaust memorials, computer interfaces, and folklore.

In the fall of 1998, Siemens AG and BMW sponsored a special symposium in München, Germany, entitled "Semantics in Design." It brought together 12 designers and experts from such fields as communication, consumer research, innovation research, and new media technologies to talk about their approaches to how artifacts acquire meaning. This workshop realized meaning as a central concern for design while acknowledging

several vocabularies, which I elaborate on later, to deal with its conceptual challenges.

Emotions have since become increasingly important aspects of semantics. The 1994 conference in Helsinki, already mentioned, addressed this idea in a workshop themed "Pleasure or Responsibility." Then in 2001, an International Conference on Affective Human Factors Design (Helander et al., 2001) brought human factors researchers together in Singapore. I delivered a keynote address on "Intrinsic Motivation and Human-Centered Design" (Krippendorff, 2004b). The question of how and which emotions are invoked while using artifacts naturally follows the question of what artifacts could mean. Many papers, therefore, explored issues of semantics, including a Japanese approach to it, called kensai engineering. The conference was repeated in 2003. Meanwhile, Donald Norman (2004) wrote a popular book entitled *Emotional Design*.

In addition to the works just described, there were parallel developments. One is an attempt within the Hochschule für Gestaltung Offenbach to develop what is called a *Theory of Product Language*. It has resulted in several unfortunately inaccessible monographs, among them one by Richard Fischer (1984) on "Sign-Functions" and one by Jochen Gros (1987) on "Symbol-Functions." Dagmar Steffen (2000) edited a comprehensive account of this theory. In Section 8.7, I comment on this approach. There also exists a proprietary in-house guide by Siemens (München, Germany) based on the metaphorical distinction between the form and the content of products (Käo and Lengert, 1984).

Semantics is attracting the attention of marketing too, pursuing the idea that meanings could "add value" to industrial products (Karmasin, 1993, 1994). There are at least six dissertations written on the subject.* Two other — until recently, unknown, — precursors of the semantic turn could be seen in works by Ellinger (1966), who developed the barely recognized idea that products have to convey the information needed to find their way to consumers, and by Abend (1973), who proposed to measure various product appearances. Perhaps also worth referring to was an effort in the 1960s, particularly in Italy, to address issues of meaning in architecture (Bunt and Jencks, 1980). Martin Krampen's (1979) work on meanings in the urban environment must be mentioned in this connection as well. Norman (1988, 1998), while framing the issues in psychological terms, had similar aims. A recent book on the economic perspective on design (Gerdum, 1999) moved meaning and product semantics into the center of explanations of a cultural dynamics of artifacts. Obviously, the semantic turn has many sprouts and is rooted in fertile ground.

Some of these developments may be traced to seeds planted in the 1960s at the Ulm School of Design about which I say more in Chapter 9. It surely is no accident that Butter, Fischer, Lannoch, Krampen, and I started our

* In design by Väkevä (1987) and Vihma (1990). In marketing by Reinmöller (1995). In architecture by Yagou (1999). In engineering science by Yammiyavar (2000). In anthropology and design by Diaz-Kommonen (2002).

careers there, and so did Sudhakar Badkarni who played a role in the Arthaya conference, and Shutaro Mukai (1991) who applies semiotic ideas to visual phenomena in his long-term effort to build a science for design at Musachino University in Japan.*

1.2 Trajectory of artificiality

As suggested in the foreword and overview, the industrial revolution managed to define design in relation to the mass production of material or, in the case of graphics, informative artifacts. In the belief that technological development would improve the quality of life for everyone, and committed to contribute aesthetically to material culture, designers worked without reflecting on their role in the larger context of expanding Western industrial ideals and replacing different cultural traditions elsewhere. There almost always was a schism between what designers believed they did and what they did in fact. The Bauhaus, for example, assertedly aiming at humanizing mass culture, managed to convert only very few of its designs into mass products. Instead, fueled by a fascination with geometrical forms and non-representational (abstract and experimental) art objects, it provided museums with its unique designs. The Ulm school, the Bauhaus successor, fully embraced mass production, even in its architecture department, and was successful in shaping post–World War II industrial culture. However, it did not realize that it thrived on an emerging postwar cultural elites' need to distinguish itself from the war generation with the help of new kinds of consumer products. When their demands came to be fulfilled, support for the school vanished and it closed. (This is not to suggest that this was the only reason for its demise, however.) Despite these vastly different efforts to situate design, virtually all of them subscribed to a functionalism, encapsulated in Louis Sullivan's (1896) dictum:

Form Follows Function

This dictum, elevated to a design principle, entails the belief that the form of tangible products would emerge naturally from a clear understanding of the function they are to serve. It does not question what they are to serve, where functions come from, and the legitimacy of those who define them for designers to start with. It signals designers' blind acceptance of the role they are assigned by society and by their industrial employers in particular. This dictum also reflects a hierarchical society in which specifications are written on the top and handed down or taken for granted, as if coming from an invisible authority. In a scathing critique, Jan Michl (1995) characterizes the notion of a function as a *carte blanche*, a metaphysical concept, that architects and designers could use freely to justify almost anything, their

* *The Idea and Formation of Design 6: 35 Years of the Science of Design* (at Tokyo's Musachino Art University) by Mukai and Kobayashi (2003).

designs as well as objects of nature (Krippendorff and Butter, 1993) — provided it stayed within well-established conceptual structures. The dictum is still widely believed and frequently recited. As of May 13, 2005, a Google search of "form follows function" yielded 122,000 hits. The semantic turn challenges designers' blind submission to a stable functionalist social order, which is anachronistic to the kind of society experienced today.

It is fair to say that the today's world is more complex, more immaterial, and more public than the world out of which this dictum grew. Design evolved beyond what the dictum could handle and designers are facing unprecedented challenges. Figure 1.1 shows a trajectory of such design problems. It starts with the design of material products and continues a progression of five major kinds of artifacts, each adding to what designers can do. This trajectory is not intended to describe irreversible steps but phases of extending design considerations to essentially new kinds of artifacts, each building upon and rearticulating the preceding kinds and adding new design criteria, thus generating a history in progress:

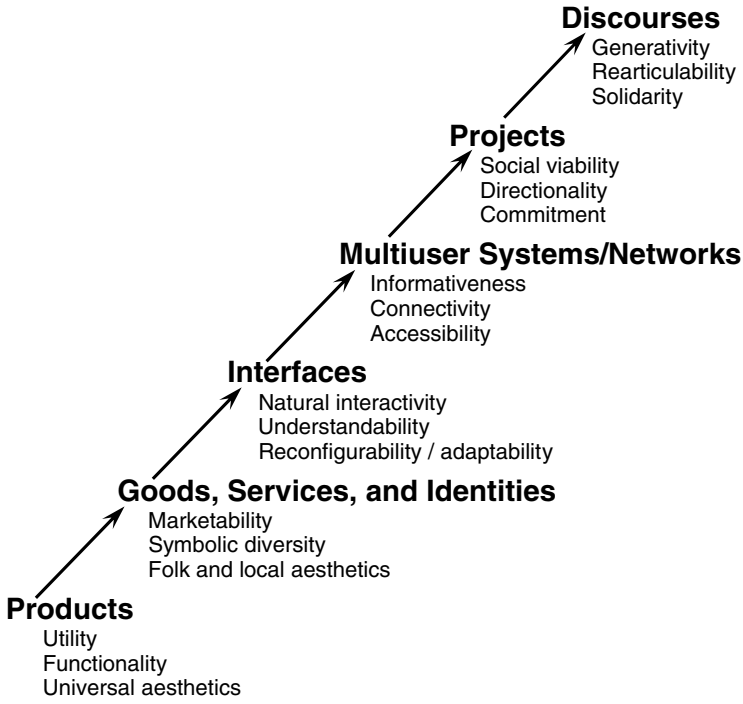


Figure 1.1 Trajectory of artificiality.

1.2.1 Products

Products, by dictionary definition, are what producers produce. The word refers to the end results of a manufacturing process. Designing products

means surrendering to manufacturers' criteria; for example, that they are producible at a *price* below what they can fetch on the market, and by the rational extension of this profit motive, that they have *utility* to their users as well. This producer–product–profit logic dominated decision-making during the industrial era (Simon, 1969/2001:25–49), an era of scarce material resources and hierarchical social structures, coupled with an unwavering belief in technological progress, not to mention frequent wars and increasing poverty for large sections of the population.

Designers could, of course, and good ones often did not stop at aesthetic considerations and improved on producers' product specifications, but because it was the producers who assumed the financial risk of failures, it ultimately was the producers' intentions that mattered. Misuses and misapplications were discouraged, which left producers blameless when people got in trouble because they used their products in unintended ways. To prevent unintended and incompetent uses, people could take courses, especially in the use of artifacts considered complex at that time, like typewriting or using washing machines, and it was not unusual for manufacturers to train and certify experts in what they manufactured. All of these measures demonstrate that producers were in charge of "correct" usage, and when designers talked about functional products, they had adopted producers' intentions as their design criteria.

With their affinity to art, product designers assumed responsibility largely for covering ugly engineered mechanisms with pleasing forms. "Form givers" is the literal translation of the German "Formgeber," used at least until the 1970s for the English "designer." Much of design discourse concerned forms — good forms, pleasing forms — governed by an ever-shifting *aesthetics*. It is precisely because designers had to justify the aesthetics of mass-produced products, products that would ideally be of use to everyone, not individual works of art, that the aesthetics of industrial products had to be a *universalist aesthetics*, one that was believed to be culture free and valid for everyone. The emergence of a universalist aesthetics led the industrialized West not only to deny its own cultural past, but also to render other, especially nonindustialized cultures, as primitive and in need of industrial development. Thus, in the industrial era, the idea of product design came to hide its subtext, the expansion of markets, the propagation of Western technological ideals to underdeveloped populations, and the refusal to take responsibilities for unintended consequences; the acceptance of this ideology was taken as a measure of modernity.

1.2.2 Goods, services, and identities

Goods, services, and identities are the focus of designers' attention since the 1940s, starting with styling as a new professional designation. Since that time, goods, services, and identities have become new kinds of artifacts. Goods are manufactured to be traded and sold, not merely used. Functions are secondary to their role in the marketplace and serve at best as sales

arguments. Services need to be designed to be recognizable and trustworthy so that customers return and develop loyalties to the service providers. Identities, whether they are encoded in logos, brand names, or corporate images, are intended to create various kinds of commitments. So, goods, services, and identities are products only in a metaphoric sense. Once artifacts become goods, they acquire exchange and use values. Once practices become services, they become institutionalized, for example, by using uniforms like the white coats of doctors and the handling of medical equipment that signify the nature of the services rendered and their professionalism. Associations of logos and products or services create brands, i.e., customer bonds or commitments. The qualities of goods, services, and identities are not of a tangible kind.

In designing goods, services, and various kinds of identities, designers are concerned with their *marketability,* that is, the ability to bring them to the attention of relevant populations, and with *diverse symbolic qualities* that encourage targeted consumer groups to acquire something, go somewhere, connect with a service, recognize a brand, or be committed to a seller, organization, or cultural practice. The design of goods, services, and identities ultimately directs the development of commercial, industrial, and corporate culture. Because these kinds of artifacts fundamentally are concerned with diverse people's perceptions — people that belong to different communities, have different habits, and pursue different goals in life — the design of goods, services, and identities can no longer rely on a universal aesthetic. Rather, it has to recognize a variety of less formalized and community-specific *folk and local aesthetics*. One valid finding of market research is that there are no universal aesthetic principles, only changing distributions of likes and dislikes, and reasons for them. It follows that the design criteria that are applicable here cannot be derived from those governing the design of products, as characterized above. Not that market advantages, brand recognition, and commitments to use services were unknown before the 1940s — even preindustrial craftsmen relied on their reputations and signed their products, but this was not their acknowledged concern and expertise. Products can become sales items and consumer goods at different periods during their life, at which point they are different kinds of artifacts.

1.2.3 Interfaces

Miniaturization, digitalization, and electronics have made contemporary technology nontrivial. Computers, airplanes, and atomic power plants are products for some, and goods, investments, and a matter of pride for others. But for most, their operation remains incomprehensible. Nevertheless, allowing their use by nonspecialists brought into being another kind of artifact, one that mediates *between* complex technological devices and their users: human–machine interfaces. Interfaces are most familiar in conjunction with computer use and often associated with what can be seen on a computer screen. Indeed, human–computer interfaces opened the possibility of

widespread use of computers and ushered in an information society. However, interfaces are everywhere, in the control room of atomic power plants, in the cockpits of cars, in the touch screen of Palm Pilots, even in the handling of ordinary tools, which can no longer be separated from their use but must be seen as enabling what humans do with them. The design of interfaces shifted designers' attention from a concern for the internal makeup and appearance of technology to what mediates between users and technology, the betweeness in which interfaces evolve.

The three most important features of interfaces are their interactivity, dynamics, and autonomy. *Interactivity* refers to the action–response sequences, command and execution cycles, or the give and take that is inherent to the human use of machines. *Dynamics* implicates time and refers to the human use of artifacts, which rarely ever returns to its point of departure. *Autonomy* emphasizes the containment of the process. Self-evidence and self-instruction suggest that everything that is needed to continue a "productive dance," is provided within the interface, ideally without instruction from its outside. We might add that interfaces extend human capabilities into the otherwise incomprehensible, and in the case of computer interfaces, they amplify the users' mind.

Interfaces constitute an entirely new kind of artifact, a human–technological symbiosis that cannot be attended to without reference to both. For designers, a key concern is that interfaces are *understandable*. Users' understanding need not be "correct" as intended by the producer, engineer, or designer of the technology. It needs to go only as far as needed for users to be able to interact with that technology as *naturally* and effortless as possible, without causing disruptions and reasons to fear failure. User-friendliness is a popular term that describes interactive understanding and seamless or uninterrupted use. Usability is another word for much the same interface quality.

An important feature of interfaces, largely absent in the products of the industrial revolution, is *configurability* or reconfigurability. This refers to the ability of users to interface with these machines so that they end up working the way users want them to. Computer programming provided an early model for configurability. A computer is a general-purpose machine, and programmers can program it to perform very many and totally different tasks. But programming requires mastery of a hard-to-learn expert language. In response, interface designers have become more concerned with *adaptability* or with interfaces that have the ability to adjust themselves during use. This is accomplished by a technology that learns user habits and anticipates their actions, thus bypassing the need for programming skills. The crown of such one-user-at-a-time interfaces is (the idea of) virtual reality.

1.2.4 Multiuser systems and networks

Multiuser systems and networks facilitate the coordination of many human activities across space and time, whether these are sign systems (e.g., for

guiding traffic, way-finding in architecture) information systems (e.g., scientific libraries, electronic data banks, accounting systems, ticketing and flight controls), communication networks (e.g., the telephone, the Internet, or USENET), or the somewhat archaic one-way mass media networks (e.g., newspaper publishing, radio, and television). Unlike in the design of interfaces, which emerge between single users and a machine, the design of multiuser systems must concern itself with the information that such systems can provide to their participants, which in turn depends on their literacy, their backgrounds, the diversity of their concerns, and even divergent personal aims. But *informaticity* is experienced only on the periphery of such systems. Most multiuser systems also network their participants and must make such connections easy to understand. Informaticity, *accessibility,* and *connectivity* also are conditions for communities to arise within such systems. The telephone network is a familiar and comparatively simple example. USENET is an electronic discussion system on the Internet. Within it, discussion groups have evolved and continue along diverse subject matter lines, ranging from serious science to politics to literature and to various hobbies. Participants not only read each other's contributions, they apply what they have learned in the system to their lives outside it. Designers can no longer determine how such a system will be used, but they will have to provide the facilities for its many users to organize themselves within and around it.

1.2.5 Projects

Projects are even more elusive kinds of artifacts. They typically arise around particular desires to change something, to develop a technology, for example, and leave something behind that is useful for others outside those directly involved. As artifacts, projects are realized in particular communicative practices among participants. Designing the technology for getting humans on the moon, launching a program of graduate education in design, getting a favorite candidate for a political office elected — all involve the coordination of many people. Managers also address such design problems, whether in the form of research and development (R&D) efforts, businesses, and political or advertising campaigns. They are design problems nevertheless, with designers shifting their emphasis, now to achieving cooperation among people without losing sight of what the project is about. A technological development project may involve designers, engineers, scientists, financial experts, sociologists, marketing researchers, and the like, and a project may have to satisfy the expectations of most if not all of the participants. Even the realization of mundane artifacts requires the cooperation of many, and this is elaborated in Chapter 5.

Designers can either face these challenges themselves or work for others who do. Projects always proceed in language, in narratives of what has to change, needs to be done, how, by whom, and at which time. They also have a purpose, a point, and an "objective," however vague this may be to start out with. Understanding this purpose must attract people, motivate them

to make commitments, coordinate their contributions, and *direct* their activities forward. Projects rarely can be designed single-mindedly by someone in charge, by a designer, for example, by issuing instructions of what should happen. Designers can launch projects but not entirely control them. In fact it is the very openness to details that energizes a project and motivates its contributors to excel. All that the designers of projects can do is to suggest a direction, generate open spaces for potential stakeholders to see possibilities for making their contributions, and to attract sufficient resources for the project to come to fruition. Projects are socially viable organizations, constituted in what people do, and last long enough to leave something behind: a particular design, a new technology, a self-sustaining institution. Projects are the models for participatory design.

1.2.6 Discourses

As a first approximation, discourses are organized ways of talking, writing, and acting accordingly. For a more formal definition see Section 1.4.1. There are many distinct discourses, professional, political, religious, and economic discourses. A discourse may be supported by a multiuser system — an information system, publications, and formal meetings — but does not depend on it. It may be enacted in a project, but does not require a common purpose. Discourses reside in communities of people who collaborate in enacting what constitutes their community, performing it, so to speak, and thereby creating everything that matters to the members of a community as members. Discourses direct the attention of community members, organize their actions, and construct the worlds they see, speak of, or write about.

We cannot be fully aware of our speaking, thinking, and doing. To the extent we do not, a community's discourse does some of the thinking for its members. Discourses entail a tension between being conservative, reusing established forms, and being creative in new ways. *Solidarity* means caring for one's community. It often means resisting change, complying with what community members are accustomed to, and preserving their habits, traditions, and conventions. *Generativity* means openness of discourse to new vocabularies and metaphors. It changes the organizational forms supported by the members of a community, improving the lives of some, perhaps at the expense of others. Designers, by their very professional commitments to each other, cannot escape and derive their identity from being part of a design discourse community.

Because language, being, and acting are so intricately connected, it seems difficult to redesign a discourse, at least not in the traditional sense, from outside of it. Think of the fate of the well-intended design for an international language, Esperanto, or how many years it took to weaken the discourse of obligations and duties that supported such hierarchical structures as monarchies, patriarchies, and dictatorships. However, it is not impossible to create and start using new metaphors, new vocabularies, and new ways of languaging, like poets and science fiction writers do, thus bringing forth new

ways of conceptualizing the world and encouraging new practices.* A requirement for introducing changes in a discourse is that the discourse remains *rearticulable*, that its users can understand, practice, and speak about these changes.

The semantic turn is a seed for design to redesign itself by means of its own discourse.

What does this trajectory — the progression of artifacts with different kinds of design criteria — tell us about design? Four observations follow.

- To meet contemporary challenges, design cannot possibly limit itself to industrial era conception of products, to products in the literal sense of being the terminal points of industrial production, of mass production in particular. When artifacts become goods, services, or identities, the criterion of producibility fades into their mobility, their ability to be traded, be remembered, produce loyalties, and create places to get something done. The 1940s styling responded to the need of manufacturers to not only supply functional products but also to compete. In adding market considerations, industrial designers ended up limiting themselves to the design of pleasing surfaces. This transition was the first step in the now ongoing dematerialization of artifacts, assigning the criteria of industrial production a subordinate place in what matters. Interfaces, multiuser systems, projects, and discourses, while undeniably based on technology, cannot be produced industrially.
- The artifacts along this trajectory increasingly concern fluid, indeterminable, and immaterial or virtual qualities. A well-designed industrial product can still be photographed and published. But a well-designed brand to a large extent resides in the memory of customers. It escapes the photograph of any logo, for example. One may videotape the interface between a pilot and the instruments in her cockpit, but not what happens on the Internet. Projects may be talked about and their reality is unquestioned, but this reality is partly cognitive, partly political, and certainly interactive. Their outcome is not entirely predictable by the project designer.
- In the course of this trajectory, artifacts become increasingly embedded in language. Negotiating the price of goods in the marketplace is still *about* the product. Its value is not intrinsic to it, however. In human–computer interfaces, texts, icons, and graphic arrangements are a transitory *part*. In multiuser systems, information *is* their currency, and in projects, the manner in which participants speak of what could or should be done *constitutes* a project. The trajectory shows the move from the production of functional mechanisms to the constructive use of language. In the course of this trajectory, the causal models of a uni-verse — a single version of what is — are replaced by linguistic models of how multi-verses come to be and are maintained.

* The most articulate proponent of this paradigm shift in philosophy is Rorty (1989).

- The trajectory shows the move from the belief in technological determinism — the conviction that technology develops autonomously and by its own logic — to the belief in the artificiality of the world, a world that is continuously constructed, re- and de-constructed, and contributors to that process are awarded but also held responsible for what they do. This is a move from the image of humans as having to adapt to technological progress and of designers as making adaptation less painful, to the image of humans as able to influence the direction of technological development and of designers as finding ways to support diverse practices of living, community, and the sense needed for individuals to feel at home. It is a move toward human-centeredness, the acknowledgement that meaning matters. This is the core of the semantic turn.

1.3 The changing environment of design

As may be imagined, the semantic turn that design is taking is correlated with several major intellectual, cultural, and philosophical shifts, some of which are woven into subsequent sections. It is also paved by radical changes in the social and technological environment in which design is now practiced.

1.3.1 Society

Living today, it is always hard to imagine what life was like generations earlier. The society that created the industrial revolution — then celebrated as an era of unprecedented technological progress and of unquestionable benefit to humankind — was far more oppressive and unfriendly to human concerns than imaginable. Factories attracted workers from agricultural communities into quickly overcrowded cities, and craftsmen lost their independence and pride in what they did. The social philosopher Karl Marx described the increasing economic inequality between "the owners of the means of production" and "the laborers" who had to sell their labor. Industry robbed the earth of irreplaceable resources and started to pollute the environment to an extent still felt today. The currency that regulated growth at that time was capital, which was seen as grounded in matter and energy, and perceived to be the prime scarce resource. Subsequently, the sociologist Max Weber observed the rise of large rational bureaucracies whose white-collar workers willingly performed machinelike functions. Hierarchy remained the dominant form of control. In social organizations — factories, businesses, and governments — hierarchy rested on contractual obligations. In sociology, logical hierarchies became naturalized and reinscribed into the fabric of everyday life. Theory and practice went hand in hand. Engineers knew what technological progress was, manufacturers knew what they could produce with profit, and the numerically increasing homogenized masses of people eagerly absorbed what was made available, ostensively for them, but in fact of benefit to the growth of manufacture. Photographs of smoking factories

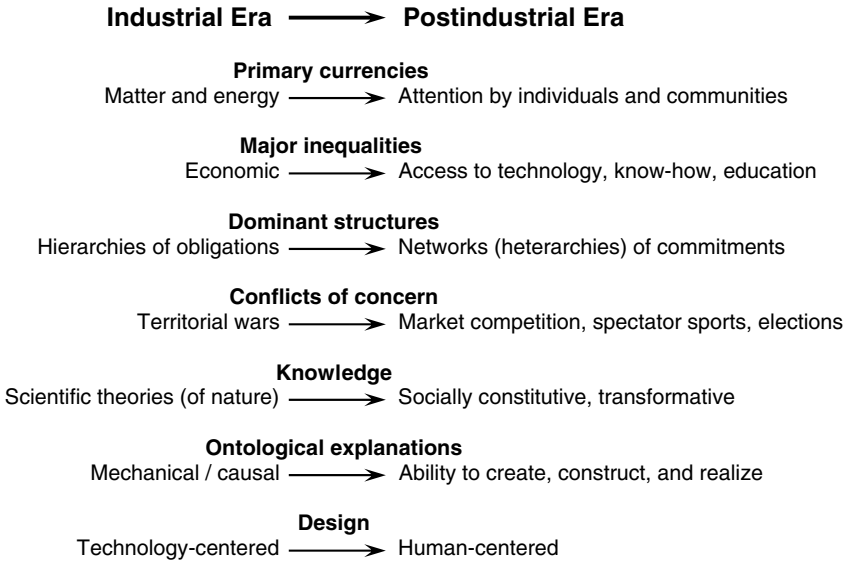


Figure 1.2 Shifts in societal dimensions.

and of the hard life in overcrowded cities visualize the quality of life at this time. Wars to expand markets and create colonies became increasingly gruesome by the mass production of ever more lethal weapon systems. The hierarchies in the military and of contemporary dictatorships probably are the last vestiges of the industrial era's dominant form of organization.

Design entered when this — by contemporary standards — oppressive rational consensus became weakened by competition for limited markets, which slowly opened choices. How far the postindustrial society has moved may be seen in Figure 1.2. Although energy problems have not disappeared since industrialization, the more important bottleneck now turns out to be individual time. Working for an income, conspicuous consumption, entertainment, surfing the Internet, and attending to family and friends are alternative ways to spend time. Attention is the new currency. Hierarchies, while not entirely absent, are now undermined by networks or heterarchies that are governed by individual commitments that participants make to each other horizontally. The telephone, interactive media, community rituals, and public events are social technologies that encourage participation more so than the top-down specification of functions.

The feeling of being able to accomplish almost anything as long as enough people are willing to put their minds and energy to it opens unprecedented opportunities for engineering (e.g., exploring space, developing computer hardware), for medicine (e.g., stem cell research, drugs), for science (e.g., mapping the human genome, experimenting with small particle accelerators), and above all for human-centered designers (with examples discussed in the previous section). The willingness to imagine worlds in which humans feel at home and initiate collective efforts to realize them for the

benefit of everyone who cares is a mark of a postindustrial society. It makes design the driving activity of the postindustrial society. As could be seen in the above trajectory of artificiality, design has shifted from being product- or production-centered to a human-centered effort.

1.3.2 Technology

The transition from an industrial to a postindustrial society was ushered in by new technologies that dramatically shifted emphasis from trivial mechanisms as prostheses of human action, to complex self-regulating systems as prostheses of human intelligence; from tools that had to be guided manually to systems that process information. The latter are more or less self-organizing, said to have an *intelligence* of their own, and their emergence marks the change from low to high technology. High technology is not be confused here with a style.

In the course of this transition, electronics has virtually destroyed the naturalness of form. One well established consequence of electronic technology is *miniaturization*. The bulky desk calculators of only 30 years ago have shrunk to wristwatch sizes and would become even smaller were it not for the need to operate them with a stylus and to read their displays. The bodies of high-tech artifacts are limited by what human users can do, monitor, and understand. What goes on inside such artifacts, besides the human sensory–motor coordination with them, is no longer relevant to their users. In fact, it can remain quite incomprehensible — as long as users can handle it adequately and are not be frightened by it. Consider what drivers actually know about the workings of their car, how little personal computer (PC) users know about the hardware architecture they are working with, or what the pilot of a jet fighter learns about why it flies. The forms of such devices are no longer derivable from how they are produced and what they do, but from their users' ability to conceptualize and handle them. Data gloves, wired for registering finger movements, and wearable computers as shown in Figure 1.3, exemplify the reduction of technology to a mere human interface. Virtual reality simulates worlds in which people can move more freely than in real life.

The fascination with high technology derives less from the arbitrariness of their forms and the artistic license it offers to designers, but more from new qualities of human involvement with them. Observe the visitors to electronic game parlors, their excitement, their submergence in the interactive worlds they control, backed up by a technology that offers rich imaginations. Competent players of electronic games can become one with these technologies, much like experienced racing car drivers, fighter pilots, multiuser domain (MUD) participants, and even sport sailors and skiers do. Artifacts that are intrinsically motivating exhibit variability, natural interactivity, multisensory involvement, continuous learning, a clear sense of place and direction, etc. (Krippendorff, 2004b). The new technologies could not have come so far without becoming increasingly human-centered.



(a) (b) (c)

Figure 1.3 Electronic artifacts become their interfaces: (a) Remote control by Clarion; (b) data glove by Fifth Dimension Technologies, 2005; (c) wrist computer by Lisa Krohn Design, 1987.

1.3.3 Manufacture

Manufacture is a necessary correlate of the technology it produces. It has been changing radically as well. Mentioning three interrelated developments may suffice. *Customization* started very simply when in the 1950s assembly lines began to accommodate variable color schemes and extras for cars. This idea has expanded to production lines that are easy to program and thus able to quickly respond to variable customer preferences. The example of the industrial manufacture of shoes to the exact measurements of customers is well known. It moves industry closer to craftsmanship, albeit of an impersonal kind. Systems by which customers design their own dresses on a manufacturer's Web site are already in place. The same is anticipated for the production of individualized furniture.

The role of the designer is affected by the closer coupling of the market and manufacture. If buyers can direct production, traditional product designers are no longer needed. Taking their place are designers of combinatorial systems of products and the multiuser interfaces needed to communicate with such systems. Informativeness, ease of use, and accessibility now become important, but so does the requirement that a great diversity of buyers, consumers, and users can understand the same system, each in their own way and finding the options they have in mind. Under these conditions, design becomes concerned with systems of variables, with the dimensions within which buyers think, and how these can be correlated with what manufacturers are capable of delivering.

What makes this coupling between market and manufacture possible is *programmable manufacture*, which has drastically lowered the minimum number of units of production that are economically feasible. In fact, it has become possible to produce unique products on an assembly line, which half a century ago would have contradicted the premises of industrial production.

There are book publishers who keep templates and produce just the number of books ordered the day before production, eliminating the cost of a warehouse. Programmable manufacture of smaller numbers of products has taken some pressure off users to conform to the most common denominator, enabled designers to concentrate on such variables as usability and desirability, and allowed communities to preserve their diversity.

To give users a voice probably was the least of manufacturers' concerns, but it allowed them to stay competitive in fast-changing markets. A second advantage of programmable manufacture is shortened time for adaptation. The 1980s crisis in the U.S. automobile industry was only partly attributable to better quality controls, worker morale, and hidden government subsidies abroad. The more important cause was that the U.S. automobile industry required far more time to react to changing markets and introduce new models than automobile manufacturers in Germany and Japan. Manufacturers who could turn out new models at a faster rate utilized a variety of computer aids but, more importantly, relied on *modular designs*, on systems of relatively standard or separately developable components that could be recombined into new products that differed mainly in their arrangement. Volkswagen's Rabbit was the first of several models that designers could develop from it, and in fact over time it evolved into various Golf versions. Honda's Civic started as a small sedan, but its modularity gave rise to a hatchback, a coupe, a sports car, a crossover sports utility vehicle, the CR-V, and a fuel-efficient hybrid, all on the Civic frame. Engineering such systems unwittingly follows folk conceptions of how a car could vary. Besides looking for safety, comfort, convenience, and economy, drivers inquire about a car's engine, its steering mechanism, brake system, stereo receiver, global positioning system, and other features. These are the very features that drivers can distinguish and talk about and that automobile engineers (therefore) develop separately, combining them into a manifold of vehicles.

1.3.4 Computer aided design (CAD)

Computer aided design (CAD) began quite innocently as computer-aided drafting but quickly developed into three-dimensional imaging, modeling, and simulations (Bhavani and John, 1997). CAD systems enable designers to create and explore far more alternatives than before and in shorter time, thus speeding up the design process and reducing the probability of future failures. CAD has also altered the design process qualitatively. The most important virtue is to obtain visualizations that can be shared by those who matter. It also allowed for the pretesting with clients and users, for example by "*walking*" *virtually* through complex products or buildings, to see how components interact and hold up in use, and to let potential users get a sense of what they may one day experience. *Rapid prototyping*, the ability to create three-dimensional models that potential users can actually handle, is a natural extension of CAD. It significantly shortens product development cycles.

The aforementioned workshop, Design in the Age of Information (Krippendorff, 1997), identified the need and possibility of creating visual data banks and making visual information generated by design research projects as easily accessible as electronic texts. Realizing that one of the most outstanding abilities of designers is to process information from diverse sources and disciplines, the merging of CAD and visual data banks transforms traditional designers, who processed information largely in their heads, into designers who can make use of information contributed by many and from anywhere.

Furthermore, it is increasingly realized that design involves coordinating diverse professionals playing different roles in the development of human-centered technology. *Cooperative software* has moved the development of artifacts from a sequential conception — one phase starting where the previous one finished, all subsumed to a common goal, and typically causing difficult negotiations across these experts — to a parallel process with opportunities of concurrent adjustment of the directions each participant is taking. Team members, not necessarily working at the same location, exchange information on what may affect each other's work and negotiate compromises before conflicts arise (Galuszka and Dykstra-Erickson, 1997). Naturally, it is the developers of large software systems that have taken the lead in developing computer aids for such cooperative developments. For designers, collaborative software has the potential to *involve diverse users in a design process*, representing them not merely as statistical data points, but in a variety of roles: as experts in a relevant area, as creative partners, as subjects, in short, as stakeholders who can shape the direction a development is taking by asserting their "stake" in its outcome. The lone ingenious designer, who could do everything by him- or herself without the need to listen to other voices, is rapidly becoming history.

1.3.5 Design management

Thirty years ago, the term "design management" was virtually unknown, and the idea of managing design was considered anathema to the creative impulse of designers. It now embraces a set of organizational skills for encouraging innovation and technological development, as well as for managing design departments in larger business organizations. A good part of the growth of this professional know-how is a result of the activities of the Design Management Institute (DMI) in Boston, Massachusetts, which publishes the *Design Management Journal.* This organization encourages design research and organizes conferences and workshops with the aim of helping design managers become effective leaders in their field and advocating design as a way to responsible economic growth. It recognized early that management is a design process in its own right, one that involves people in a purposive activity. Design management pursues strategic objectives much as design does but on a level higher than product development, in fact, and in terms of the above trajectory of artificiality, it considers design

as an organizational project to be pursued. This too has changed what design is or can do today.

1.3.6 Market research

Designing products entails a concern for production. Designing goods and services entails a concern for their sales. It renders "the market" as the ultimate arbiter of what is to be produced. Marketing — launching, promoting, and monitoring the sales of goods and services — entered design considerations since the 1950s and has grown since into a multibillion-dollar industry. Market research attempts to predict the behavior of markets and inform manufacturers about consumer needs and how well their own and their competitors' products sell.

For designers, market research data are of uncertain value. Early market research modeled itself on public opinion research, attempting to measure attitudes and opinions asserted by consumer groups with targeted socioeconomic characteristics. Because of the frequent disconnect between what people say they do and how they act in fact, between espoused theories and lived theories (Argyris and Schön, 1978), opinions turn out to be quite unreliable predictors of consumption. Then came consumer research. It seeks to account for consumer choices in terms of what motivated them. But choices can be observed only among available products and describe reasons for consumer preferences when people are aware of alternatives, the very alternatives that designers are to create. Thus, observation and questioning can say little about innovative designs, artifacts that are not yet available for comparison either as products, prototypes, models, or ideas. Moreover, seeking to capture large markets always favors the most common denominator and opposes the least expected: true innovations. By contrast, focus groups with small numbers of invited consumers who, while guided by a focus group leader, talk more informally about problems they encounter with the products of interest and in their lives, can generate unanticipated responses. However, focus group data suffer from the above-mentioned disconnect between asserting opinions and behavior (Zaltman, 2003), present severe sampling problems, and are, hence, not always reliable. Finally, even where designers are aware of and ignore the structural limitations of market research data, manufacturers rely on such data largely to reduce the risk of product failures. Market research therefore is inherently conservative and not supportive of innovative designs. The famous failure of Ford's Edsel, put on the market in 1957 and designed after the most thorough market research known at that time, demonstrates the points made above.

Yet marketing exists and designers learned to cope in this new environment. In an effort to transcend the narrow focus on sales, designers have assumed an advocacy role for the culture that their designs affect or for the individual users of their designs. The Ulm School of Design saw itself as opposing the dominant marketing considerations and as an advocate of

industrial culture. Recently, designers came to speak for individual users by adding user-friendliness and usability to their traditional ergonomic and aesthetic concerns. The notion of "ease of use" became institutionalized, however. The International Standards Organization (1998) defined "usability" as "the degree to which specific users can achieve specific goals in a particular environment with effectiveness, efficiency, and satisfaction." Unfortunately, this definition reintroduced the functionalism and goal rationality of the industrial era into the postindustrial discourse (Krippendorff, 2004b). Good intentions aside, usability research succeeded mainly in providing marketers with added sales arguments — who would not value something claimed to be user-friendly and usable? Undoubtedly, research on the cultural significance and use of artifacts is badly needed, perhaps not as it has emerged in market research. Market research has disappointed designers by not only constraining design innovations but also focusing on a very limited stage in the life of artifacts: sales.

1.3.7 Philosophy's linguistic turn

Philosophy's turn toward language, its *linguistic turn*,* and postmodernism — not to be confused with the architectural movement by that name, or the Memphis interpretation of postmodernism, for example — has systematically undermined the epistemological foundations of the Renaissance conception of science on which technology was constructed and industrial design grew to where it is today. With the emergence of technologies that extend, simulate, and replace faculties that were thought to be uniquely human — microscopes, telescopes, writing, the media, computation, information systems, and virtual reality, — and academic disciplines that address these innovations theoretically — cybernetics, artificial intelligence, and the cognitive sciences — one witnesses nothing more profound than the emergence of a new image of what human beings are, an image to which design will have to respond as well.

The most important part of this new image has become tied to the use of language, dialogue, and discourse. Language is a cultural artifact that enables humans to coordinate their conceptions, engage in joint action, and construct and reconstruct the realities they see. In the use of language, languaging, acting, and perceiving are inseparably tied to a constructive understanding. It is a truism that one cannot know what exists without conceptualizing it as such. Languaging is the primary source of conceptions. It also presupposes the bodily participation of human beings. Language is spoken, written, and communicated. Entering humans into this conception of reality entails this self-reference: *Humans are beings who language each other into being*. This self-reference is remarkable by implicating language as a condition for understanding oneself as a human being, understanding the understanding of other beings, and enacting this understanding in the face

* Rorty (1970) and Lafont (1999) to name but two specific conceptualizers of this paradigm shift in philosophy.

of one's understanding of others' understanding of reality. Reality as understood always is constructed by those who claim that understanding — claiming being a linguistic move. The linguistic turn in philosophy suggests that efforts to compare an objective nature with conceptions of it are the unfortunate entailments of a flawed vocabulary that is responsible for creating this and other epistemologically problematic constructions. It suggests that philosophers search for the linguistic basis of the problems that past philosophers have posed and pondered. These problems do not exist outside of language (Rorty, 1979). Closer to home, the linguistic turn also suggests that the known world is an artifact that someone, usually a community, constructs to be observable, experienced, talked of with others, and enacted. Both reality and language are recursively interwoven artifacts. They are not facts waiting to be discovered for what they are, they are brought forth by what people say and do (Rorty, 1970), and this includes the work of designers as well.

The Greeks, Plato in particular, thought that truths could be established in dialogue. For Plato, language was a container of propositions not connected to observations but to a world of ideal types behind mere appearances. The metaphysical problems that modern philosophers proposed to solve either put the human mind in the center of their theorizing (René Descartes), appealed to some kind of pure reason (Immanuel Kant), or made scientific observation their foundation (Bertrand Russell). None of these attempts to find a stable ground succeeded. Karl Marx turned the world of the idealists, as he called them, upside-down and considered "the material base" as the foundation of his philosophy. He was an important philosopher for design, asserting that "The philosophers have only interpreted the world in different ways; the point is to change it."* This indeed is the point of design. But his theory of society was deterministic, was not applicable to him as a privileged observer, and failed to account for how his own writing, his discourse, shaped the world he described but ultimately failed to create the reality it promised.

Ludwig Wittgenstein, more than any other twentieth century philosopher, pointed out that all of these systems have floundered on the rock of language. He pointed out the futility of trying to escape the pervasive fabric of language in search of a truth that lies outside it and started to conceive language as an interactive practice of people involving each other in "ways of life." Perhaps not so incidentally, Wittgenstein spent part of his life as a designer, and his approach to language reflects this albeit short occupation. Wittgenstein suggested that the meaning of words and utterances are not found in what they represent but in how they are used. Use can be recognized in what utterances accomplish in what he called "language games," conversations, dialogues, and discourses. Words do things,** create artifacts, and

* Thesis on Feuerbach in his *German Ideology,* Marx, (1845).

** See Wittgenstein (1953) as the originator of the connection between language and action, Whorf (1956) as an anthropological linguist who came to the same conclusion, Austin (1962) as the grandfather of speech act theory, and Searle (1969) as a codifier of the latter, unduly limiting the contentions of the originators.

change worlds. Postmodern philosophers (Rorty, 1989) are correct in asserting that the linguistic turn in philosophy brings to a close philosophers' long preoccupation with finding ultimate truths, master narratives, holistic world views, and efforts to construct a single "uni-verse," one version of THE world that fails to account for its observers or assigns them God-like privileges. All of these are but discursive constructions that reside in communication between people. They do not exist in the abstract.

Given the industrial origin of design, it is not surprising that its vocabulary is full of universalist conceptions. Already mentioned is the early commitment to a culture-independent aesthetic. But universalism also underlies many taken-for-granted notions such as of matter, form, gestalt, color, beauty, and the aforementioned idea of function. All of these industrial era conceptions purport to describe objective, that is, observer-, culture-, and language-independent, properties of objects. As such, these conceptions result from a careless use of language, allowing Cartesian notions to enter, and ending in ontological* claims. How this use of language can unwittingly draw one into such an ontology can be described in four steps:

1. **Being in dialogue**. All concerns are asserted in dialogue between real people. Consider a scientific experiment. Typically, it starts with a conversation between researchers and subjects regarding what they expect of each other. Suppose the experimenter uses color chips to elicit color terms. The procedure yields first person accounts like "I see a purple chip." The first person references may also be implicit when subjects merely name the color they see. The color terms thus generated are responses to questions concerning perceptions. They are embedded in dialogue.
2. **Entering the Cartesian consensus**. The belief in the knowability of an observer-independent universe of objects, together with the belief that competent use of language lies in the accurate description of its objects, justifies taking those accounts of perceptions on which subjects agree as descriptive of the objects in question, here the colors of the chips. Those accounts on which subjects do not agree are taken as the reports by unreliable or incompetent observers, who either did not understand the task (had inadequate language skills), or responded to conditions extraneous to the task at hand. It is these two beliefs, and only these, that lead one to interpret consensus as caused by something real and lack of consensus as subjective and not trustworthy.
3. **Ontologizing the consensus**. Consensus allows one to translate the first person accounts on which subjects concur into propositions about

* Ontology, by definition, is the science of being; more specifically, the construction of a world that is presumed to exist without its observers or constructors. By contrast epistemology is the science of knowing. An objectivist epistemology studies how the human mind comprehends or accurately represents ontology. A constructivist epistemology, by contrast, studies how humans or members of a community come to understand. Its criterion is not ontology but viability, the ability of knowers to successfully enact their understanding.

the colors of the chips: "I see …" becomes "It is …." While first person accounts resonate with experiences and are acceptable by dialogical partners, propositions are true or false. They no longer refer to the subjects' perception, to the experimenters' involvement, and render the linguistic nature of the color terms transparent — unnoticed and assumed shared by subjects, experimenters, and readers of the research findings.

4. **Denying steps (1), (2), and (3)** in favor of their result: an ontology that consists of objects and their properties, here, colors that now appear to exist independently of any one observer and without reference to the language that directed the process.

The transformation of dialogic accounts of human-specific phenomena into ontological propositions by way of Cartesian conceptions of reality underlies many of the common claims of objective facts. Color is just one phenomenon that shows, indeed, no simple correlation with physical measurements and cannot exist without human perception. Form, gestalt, beauty, weight, function, and especially meaning are human-centered, not natural qualities. The attempt to routinely or mindlessly objectify such human-centered conceptions is a fundamental epistemological mistake. The linguistic turn resists such objectifications. It is an attempt to locate the known worlds in the dialogues or language games in which they arise, including the abstract theories of physics and the problems philosophers have pondered.

The semantic turn needs to take the role of language in the use of technology seriously. It is the use of language that distinguishes forms, materials, functions, and problems, and directs designers' attention to what they are to do with them. It might appear obvious that artifacts do not exist without their makers, but colors are artifacts as well, being the product of the human perceptual apparatus. Without language one would have no inclination of what other fellow humans perceive. But language can also entrap, as seen in the above steps toward objectifying dialogical accounts. The linguistic turn in philosophy creates awareness of the use of language.

1.4 Redesigning design (discourse)

1.4.1 Discourse

In the above trajectory of artificiality, I already remarked on discourse as a design problem. Let me offer a definition of discourse, extending Wittgenstein's idea of language games in five mutually constitutive features of discourse,* and then give some examples.

1. **A discourse surfaces in a body of textual matter, in the artifacts it constructs and leaves behind** to be (re)examined, (re)searched, (re)articulated, and (re)produced with variations. Textual matter is the

* The following revisits a longer analysis of opportunities and pathologies of then current design discourse (Krippendorff, 1995).

literal heritage of a discourse. Its intertextualities, the connections created between these artifacts, are apparent in (a) relying on common vocabularies, (b) creating references across texts and building one artifact recognizably on the shoulders of others, (c) introducing reviews, histories, research reports that reassemble its textual matter or exhibit its artifacts from new perspectives. In a living discourse, textual matter (i.e., the artifacts it creates and attends to), (d) is always incomplete, ready to be elaborated and added to.

Dictionary definitions of discourse limit it to textual matter, and studies of literature conform to such definitions. The definition offered here extends text to all kinds of artifacts, the objects that a discourse constructs, and it adds the following four essential components. They characterize discourse as a social construction.

2. **A discourse is kept alive within a community of its practitioners**, in whose conversations (a) textual matter is continuously (re)read, (re)written, (re)produced, (re)worked, (re)searched, (re)articulated, (re)designed, and appraised for its exemplary nature or discarded. (b) In the process of elaborating its textual matter, a discourse community recognizes itself and its members. (c) Members of a discourse community understand the *meanings* that its textual matter and artifacts have for each other. (d) Members of a discourse community continually test each other's competencies and commitments to their community and offer incentives for belonging.
3. **A discourse institutes its recurrent practices**. Recurrent practices are habitually repeated or typical of the discourse to the point of being expected if not required. (a) They enable social institutions to thrive on their recurrences, regularizing individual practices, turning individual practices into legitimate procedures, including substituting technologies for them. (b) These institutions in turn assure that members conduct themselves as expected and control the selective distribution of knowledge of each other's contributions, methods, and responsibilities. (c) They also uphold discourse-defining axioms, exemplars, and paradigms, assuring that the textual matter and artifacts that a discourse generates is coherent, follows agreed upon guidelines, and moves in the 'right' direction.
4. **A discourse draws its own boundary**, distinguishing between what belongs and what does not. It does this regarding everything that matters to it: (a) its textual matter, the objects it constructs, and the problems it solves, (b) its criteria for individual membership in the discourse community, (c) exemplary practices regarding both the manner in which its textual matter is attended to and expanded, and the institutional practices consistent with it.
5. **A discourse justifies its identity to outsiders**, especially when its members come in contact or must work with nonmembers or members of other discourse communities. Here, a discourse's role within society

becomes an issue. Successful discourses manage to justify themselves in their (a) construction of reality (truths), (b) virtues (values), and (c) competencies (expertise). A discourse that fails in this respect may lose its respect, its members, what they are able to accomplish, and, hence, its viability.

In sum, discourses, so conceived, are not merely spoken and written, they are social systems with a life of their own. Discourses can be enormously productive of new artifacts, run out of steam, or vegetate by merely reproducing themselves. Their artifacts vary greatly, from abstract theories to medical practices to rather concrete material arrangements. Discourse communities can grow in size or shrink. When below a critical threshold, discourses die, leaving artifacts behind that other discourses may appropriate, as archeology does with artifacts from extinct cultures. Discourses can be more or less structured, asserting stronger or weaker identities, empowering effective institutions to regulate discourse practices or favoring individualism. For example, mathematics is a highly disciplined discourse; public discourse is not. Although discourses are organizationally autonomous, they respond to other discourses by redefining their identities and redrawing their boundaries. Within such boundaries, members of a discourse community know who they are and can feel to belong. The boundaries of discourses are more or less permeable, however. A weak boundary invites colonization by other discourses. For example, since the Renaissance, the religious discourse has given way to scientific discourse, not so much in the number of believers, but in how everyday life is talked about and organized. To the extent a discourse relies on other discourses — recruiting new members, obtaining resources to produce its artifacts, or revenues to support its practices — it needs to justify itself and validate its products in ways acceptable to the other discourse as well. The primary aim of a discourse is to stay viable.

1.4.2 Design

What is design that a discourse could sustain? For Herbert Simon, design is the improvement of what is. He suggests "everyone designs who devises courses of action aimed at changing existing situations into preferred ones" (Simon 1969/2001:111). This definition is a good start but needs to be elaborated. For once, there would be no point in designing "courses of actions" unless *design brings forth what would not come naturally.* This has profound implications for the difference between science and design. Also it is important to realize that the role of designers is to devise courses of action or specify artifacts, not actually realizing them: *design proposes realizable artifacts to others*. Acknowledging that artifacts are not necessarily good for everyone, and aid not just individuals but influence also how they live together, *design must support the lives of ideally large communities,* not or only minimally impairing the lives of others (Agre, 2000). Design cannot avoid ethical questions. And, finally, because improvements must be understandable and decidable by those affected, not imposed by lone designers or authorities who are not

acknowledged by the community in question, *artifacts must make sense to most, ideally to all of those who have a stake in them*. These qualifications lead to a human-centered concept of design that qualifies Simon's definition to the point of being a radical departure from it.

The difference between science and design needs to be addressed. Simon states it in simple terms. "The natural sciences are concerned with how things are, … design, on the other hand, is concerned with how things ought to be, with devising artifacts to attain goals" (Simon 1969/2001:114). The fundamental problem that designers are to solve, he suggests, is how to change an existing situation into a preferred one. Defining design as problem solving is common but commits designers to a technical rationality that is at home largely in engineering. It conveniently bypasses the qualifications suggested above. Simon's background in computer engineering and early cognitive science leads him to distinguish science and design in terms of the logic they respectively employ. In the natural sciences, he observes, the standard propositional and predicate calculi, which are able to state or deny factual truths, serve their researchers well. Designers, however, are concerned not with factual truths but with what should be. Replacing "is" with "should be" makes all the difference. "Should" statements are imperatives, and the logic underlying the design discourse, Simon suggests, is normative or deontic, not propositional. In conceiving design as progressing from analysis to synthesis to evaluation, Simon conceives synthesis as enumerating alternative solutions and evaluation as using optimizing techniques to identify the best or a satisfactory solution. This process manifests his technical rationality in action. It works where problems are clearly defined and the solution space is finite, which often is the case in engineering.

Simon's technical rationality extended simple engineering conceptions to larger design problems in defense systems, R&D efforts, National Aeronautics and Space Administration (NASA) projects, and operations research. But it started to falter when applied to city planning, the design of corporate strategies, and even of consumer goods. Not only does Simon's rationality assume consensus on what is to be accomplished, it also takes for granted that the outcome of the design process can be implemented by decree, similar to how the components of a mechanical system are installed. Technical rationality emerged as the mode of operation in the industrial era, is typical of how engineering problems are solved, and is still practiced in tight organizations — in the military, for example, or in bureaucracies, where users can be trained, correct use can be enforced, and dissenting voices are unheard of. Technical rationality is at home in coherent social hierarchies. But it fails when applied to problems that involve people as informed agents, in heterarchical forms of organizations like markets. In cities, for example, designs typically improve the lives of some at the expense of others; interest groups may be looking for particular outcomes and try to define the problem to their advantage; in the marketplace consumers make informed choices that cannot be understood in mechanistic terms. Under these conditions technical–rational problem solving breaks down, and design must proceed differently.

Horst Rittel (Rittel and Webber, 1984) made the shift from top-down technical problem solving to a conception of design that accommodates the participation of stakeholders. Working with planners of large social systems, he soon recognized that Simon-like problem solving could not provide guidance where intelligent humans — individuals, organizations, and communities — with interests in a design are involved. He called problems of a technical nature *tame* problems in contrast with what he identified as *wicked* problems, and he anticipated several of the issues that the semantic turn is now addressing. For example, he observed that in the social realm, problems are never solved (and thereafter forgotten); they more likely involve conflicts that may be *re*solved by consensus — only to resurface later as other kinds of conflicts, calling for further resolutions, and so on. In the social domain, stakeholders typically invest in the outcomes of design processes, in their own future. Coming to a consensus on what the wicked problem is, is the problem. His design conception moves argumentation into its center, making language and discourse the ultimate arbiter of what is desirable, achievable, and will be done.

Why is there a need to discuss such basic design conceptions? Simon writes: "In the past much, if not most, of what we knew about design and about the artificial sciences was intellectually soft, intuitive, informal, and cook-booky" (Simon, 1969/2001:112). He makes the academic culture in universities responsible for this malaise, which identifies respectability with the natural sciences and values propositional over deontic knowledge. Engineering and medicine, he suggests, have paved the way out of this confinement. Argyris et al. (1985) concur with this assessment but blame scientific discourse for this failure, which defines its boundary in terms of science vs. not-science, leaving the applied disciplines only the choice between accepting the role of an inferior subcategory of science, or casting themselves as outsiders to science: as artists or practitioners. Designers experience the same institutional dilemma.

What do mainstream natural scientists do? As researchers, they collect data by observation or measurement, and as theorists, they generalize them to similar data or situations not yet observed. Theories are propositions that can be true or false, being either supported or opposed by empirical evidence, and when statistics is involved, in degrees. For the intended comparison between what scientists and designers do, four points are noteworthy:

- Scientific research is essentially *re-search*, a repeated search for patterns within available data.
- Data always are of *past* happenings, whether they are found or generated for a purpose, for example, by counting a population or designing a controlled experiment. The patterns that data analysis is presumed to "find" have always existed *prior* to their analysis.
- Theories generalize what permeates the data — *common* properties, *stable* patterns, and *underlying* causalities. By definition, generalizations omit details that are irrelevant to the intended theory. The law of falling bodies,

for example, concerns theoretical bodies. Truly unique events, being not generalizable, are of little interest to natural scientists who would rather believe they do not exist. Predictions from the past to the future always presume that the theorized properties are stable and unchanging within theoretical limits.

- To preserve the idea of nature as an undisturbed object of study, scientific observers are not allowed to enter their domain of observation, are required to remain detached, spectators of happenings, and certainly must not affect the data they intend to analyze. This is to assure that scientific findings, theories, and laws are about an observer-uncontaminated nature.

Overtly, natural scientists justify their work as in pursuit of knowledge for its own sake. Covertly, scientific knowledge is limited not only to the kinds of knowledge that satisfy the four points above, it celebrates a representational kind of knowledge — know-what — which entails rarely questioned ontological commitments and is subject to the politics of available research funding, publication, and the institutions of science. Finally, by relying on measurement devices and computational techniques of research, the resulting knowledge, propositions, theories, and laws manifest the same kind of technical rationality observed in Simon's work.

Now, contrast the above with what designers do:

- Designers are motivated not by a quest for knowledge for its own sake but by:
  - **Challenges**, troublesome conditions, problems, or conflicts that have escaped (re)solution.
  - **Opportunities to change something for the better** — not recognized by others — to contribute to their own or other communities' lives.
  - **Possibilities to introduce variations** into the world that others may not dare to consider, creating something new and exciting — just as poets, painters, and composers do — aimlessly and for fun.
- Designers **consider possible futures**, worlds that can be imagined and could be created in real time. They are concerned less with what has happened, what already exists, or what can be predicted by extrapolations from the past than with what can be done. Designers' most outstanding ability is not being afraid to explore new ideas, to challenge theories that claim that something cannot be done, or to question what is commonly taken for granted. Thus conceived, the futures that designers envision are inherently unpredictable from laws of nature, though not necessarily contradicting them.
- To choose among them, designers **evaluate the desirability** of these futures. Desirable worlds must make sense and be of benefit to those who could realize these worlds and might come to live in them. Developing a consensus on the desirability of a possible future calls for deliberations

among its stakeholders, using a language that is capable of going beyond data or facts.

- Designers **search the present for variables**, things they are able to vary, move, influence, alter, combine, take apart, reassemble, or change. These variables define a space of possible actions, a design space as Phil Agre (2000) calls it. Designers need to know the efforts required to alter these variables and how likely they are in bringing forth desirable futures (and avoiding undesirable futures).
- Designers **create and work out realistic paths from the present toward desirable futures** and **propose them to those who can bring a design to fruition**. Successful designs depend on designers' ability to enroll stakeholders into their projects, even if these stakeholders pursue their own interests as well. The paths that designers invite stakeholders to take must be presented as realistic, affordable, of benefit to those whose effort is required, and, above all, open valuable opportunities to those affected by a design.

Obviously, the artifacts that designers create differ quite radically from those produced by scientists. Designers and scientists work within incommensurable paradigms and efforts to confound them or search for commonalities might not serve either profession. Consider the following differences:

The artifacts of science are generalizations, theories, or laws of nature, from which empirically testable propositions can be deduced. Generalizations populate scientific discourse about what exists, the objects and happenings of an unattended world, but designers cannot stop there. While the proposals for a design may originate in visions, they ultimately must become so specific as to be realizable in the form of working artifacts. After all, "art-i-facts" are facts that are made real in all essential details. Also, while the artifacts of science merely need to be verified for what they claim, the artifacts of design cannot be realized without the participation of stakeholders and ultimately their users who put them to their test. Scientific generalizations and detailed design specifications are located on opposite ends of a continuum.

Re-searching the past for generalizable patterns is the established method of scientific inquiry. Searching the present for available paths to desirable futures is the method of inquiry in design — but they obviously are of incompatible kinds. Scientific research, the theories it produces, is rooted in past observations and the predictions it suggests assume that the patterns found in the past will persist in the future. This kind of knowledge, celebrated since the Renaissance, is of little interest to designers who are intent to change the world in ways not predictable from natural laws. Scientific theories may warn designers about what cannot be altered or assure them of the stabilities not worth questioning: a designers' present point of departure, the means they may rely on, or the ground on which their plans can be expected to proceed into a desirable future.

The world seen through theories, accounts of patterns found in data, is essentially deterministic. The celebrated role of neutral or detached observers condemns scientists to these constructions. The use of mathematical theories, computational models, causal explanations, and verbal propositions about observed behaviors impose deterministic accounts. They leave no space for human agency. They cannot explain the practices of realizing alternative worlds, which would not come about naturally. And they have absolutely no place for meanings to which people respond. Good designers must distinguish between where causal determinism is justified, for example, in explaining mechanisms, and what they need to address: the meanings that artifacts have acquired in particular communities.

Predictive theories are validated by waiting for the anticipated evidence to come in their support. By contrast, designers' proposals about the future amount to constructive interventions. If a design would become real without intervention, it wouldn't be a design. Inasmuch as designers rely on others to realize what they are proposing, designers' ideas need to be narrated, visualized, communicated, and acted upon by those who can bring the intended artifacts into being. Like scientific theories, designers' narratives are linguistic artifacts. But, whereas theories are expected to satisfy truth conditions, the narratives, plans, or proposals of designers must enroll others to realize the artifacts they project. They cannot become "true" without human involvement, and this, not truth, is what any design needs to accomplish.

Since the Renaissance, the natural sciences have inquired into nature as if it existed independently of its observers, were unaffected by the language used to describe observations of it, and were unable to understand that it is being observed and how it is analyzed. By considering the systems they create to be functional or causal systems, engineering displays the same attitude to its objects' abilities. Mechanical systems are not assumed to understand anything either. But design cannot possibly thrive in this construction of reality. It concerns real people's actions vis-à-vis technology, people who understand their artifacts, create their worlds, have foresight, intelligence, and feelings, and can respond to being observed and communicate among each other and with designers. The image of humans provided by the natural sciences is impoverished relative to what designers need to develop.

What made the natural sciences such a powerful force in contemporary culture is its ability to state limits on what can be done. Information theory (Shannon and Weaver, 1949), for example, is a theory of the limits of transmission and coding. The three laws of thermodynamics state limits on the conversion of energy into its various forms. The laws of gravity define constraints on the movement of matter. These laws, but also theories and propositions, assume a timeless truth, and some such truths may well be so. To the extent that designers obey these limits, they are less likely to fail, but these designers also are relegated to merely fill the gaps left unattended by science. This is a rather undesirable world, and it is the uncritical acceptance

of this world that is severely limiting. Yet, one of the strengths of designers is to reconceptualize, recontextualize, and question what heretofore was assumed fixed and to thereby prove these limits to be malleable and artificial as, of course, they are.

Shifting gears, but only slightly, Simon is right in suggesting that design occurs in many professions: engineering, medicine, management, and education. Design is implicit in government, law, journalism, architecture, and library science. Terrence Love* has identified as many as 650 fields in which design is practiced. Schön (1983) and Argyris et al. (1985) see design as underlying all professions. Professionals model, compose, engineer, fabricate, program, construct, layout, draft, organize, direct, and institute practices not yet existing; and they tend to have well-established vocabularies to describe the changes they introduce in their worlds.

For Nelson and Stolterman (2002), "design is a natural human activity and everyone designs all the time." When someone plans a vacation trip, rearranges the furniture in the living room, treats a patient, reads or writes a letter, draws a cartoon, or tends to a garden, future possibilities are envisioned, evaluated, acted upon, and exhausted. *Everyday design is a way to realize not just artifacts but also their designers* — "realize" in the dual sense of materializing, making known, and rendering useful, but also taking responsibilities for it, as author, originator, artist, or designer. Not everyone who acts to make the world a better place calls him or herself a designer. Design as a professional practice differs from design in everyday life by relying on publicly acknowledged competencies, the use of methods, but above all on an organized way of languaging, a design discourse, that coordinates working in teams and with clients, justifies proposals for artifacts to their stakeholders, and distinguishes professional designers from those doing it largely for themselves.

Among the professional designers, there are those concerned with strictly technological artifacts, with artifacts that may well be designed without consideration of their users' conceptions, by engineers for example, and those concerned with human interactions generally and human interfaces with technological artifacts in particular. They are called here technology-centered and human-centered designers, respectively. Technology-centered design improves the world in designers' or their clients' terms. Making a machine cheaper to produce, more energy efficient, or more usable by more people may well be intended to and actually does benefit a community of users, but the measures of these benefits are the designers' choice, as discussed previously, imposed from above, by experts onto lay people. It thrives on hierarchical organizations of responsibilities and nourishes the kind of functionalist society that emerged during the industrial era. By contrast, human-centered design derives its criteria from a community of users in whose worlds designed artifacts may have to find a place together with their users, bystanders, instructors, and critics. Figure 1.4 depicts this distinction.

* Terrence Love, April 25, 2004, in the list PhD-Design@jiscmail.ac.uk

The arrow from human-centered professional design to design in everyday life will become clear in Section 2.5.

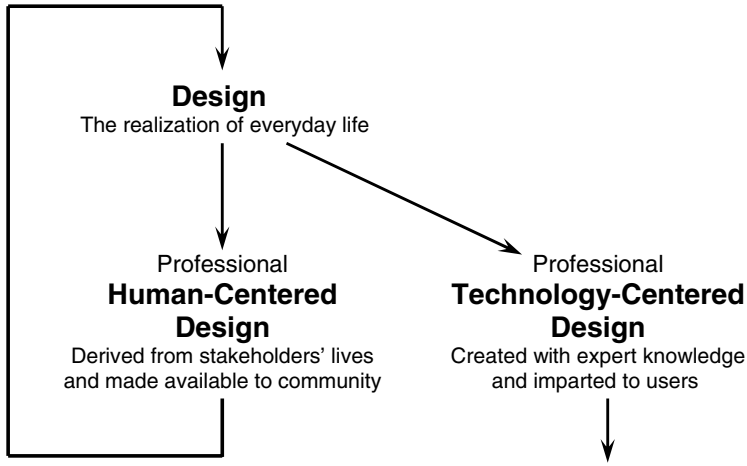


Figure 1.4 Distinctions within design practices.

1.4.3 Design discourse

In this section we discuss the state of design discourse, how supportive it is of design practices, and how well it stands up in challenges from other discourses. It relies on the five components of discourse defined in Section 1.4.1.

- **Its artifacts and textual matter.** Being on the surface of a discourse, its artifacts and textual matter are easy to identify. There exists a relatively large body of literature on design, books, exhibition catalogs, and numerous journals. Museums house outstanding examples of designed artifacts in special collections, variously organized by periods, styles, schools, product categories, cultures, or designers. The marketplace, albeit short-lived, is full of artifacts worthy of the design label. Designers have no problems pointing out and telling stories about designs, mundane ones, famous ones, and even their own. The artifacts and textual matter of a discourse can be photographed, read, reproduced, played with, and used, but they may not mean much to the uninitiated unless explained, which is what popular magazines have taken onto themselves.
- **Its Discourse Community.** When asked, designers know quite well how to distinguish fellow designers from nondesigners, who-is-who in their community. An important indication of membership in the community of designers is their ability to talk about design, designs, and designers; discuss design problems in a designerly way; and show that they have designed something worthy of the profession. In other words, membership in the design community is demonstrated by being familiar with

the current design discourse, with its latest topics and contributions to the textual matter of design. All of this sounds quite circular, and it is, because membership in the design community fundamentally involves experiences that are embodied, not necessarily written. The common assertion: "You have to be a designer to appreciate design" manifests the circularity in which the design community is constituted. The sense of belonging to a community of designers develops when entering a design school whose students learn to articulate design concepts, or hanging around long enough in a design office. In an Internet discussion group on design, Holt has this to say. To be "a designer is having a very special way of viewing the world, seeing possibilities — not just (having) certain skills. (It is:) … how you organize your day, design your kitchen, and prepare a meal. Questioning the way things are being done. A way of living, organizing and seeing — not something you … take off like a coat. … every profession requires certain skills, and a real professional questions, improves and refines his (or her) style of working over time … . This 'special way of viewing the world' is what I think makes the difference — and a designer."* The cultivated feeling of being special is a characteristic of community, of any community, but it says little about what holds the design discourse community together.

- **Its Institutionalization of Recurrent Practices.** Like any discourse, design discourse must reflect on its practices, recognize what is successful, and encourage the growth of institutions capable of preserving these practices. Programs in design education, university degrees, and requirements to teach design are well-established institutions. There are professional associations for designers to meet, show their work, learn from each other, and hone their vocabularies. Design offices keep books on graphical or ergonomic standards on their shelves. There are design journals — not necessarily read by everyone — that set agendas, define terms, publish outstanding examples of design, and promote their version of design discourse.

  By comparison to academic discourses, however, there are relatively few opportunities to earn Ph.D.s in design. Two recent conferences, in 1998 (Buchanan, 1999) and 2000 (Durling and Friedman, 2000), considered a doctorate in design, and a list for discussing Ph.D. studies on the Internet is active.** The lack of advanced degrees in design implies a lack of academic self-reflection. Unfortunately, most dissertations about design are written from outside the design community, by students seeking degrees in art history, education, cognitive science, or marketing. Also, most books on design are written by nondesigners. While designers might enjoy the attention by outsiders, the latter's narrow attention to consumer products, to artifacts that can be photo-

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graphed, interpreted, and made significant for the academic discipline granting the degree hardly helps the design community to understand itself. There are no widely accepted dictionaries of design, no standard texts. This is one serious weakness of contemporary design discourse.

Typically, professional inquiries into their own professional practices focus on finding generalizable methods, methods whose success recommends wider use, systematic treatment, and teaching as a methodology in its own right. Methodology entered the design discourse in the late 1950s at the Ulm School of Design where Bruce Archer (1984) introduced systematic design methods. Nigel Cross (2000) offers an overview of how the methods movement developed. The early momentum was fueled largely by the impressive rigor of the natural sciences. Scientific concepts were imported that, because of the above-mentioned incommensurabilities, not only failed to address what designers actually do but also encouraged terminologies that became straightjackets and diverted designers' attention from what really mattered. Simon's (1969/2001) *The Sciences of the Artificial*, considered groundbreaking in 1969, subsequently revealed its limitation to a technical rationality that grew out of engineering. In architecture, Christopher Alexander (1964, 1977, 1979) made the transition from a positivist to a human-centered approach to design methods. Schön's (1983) *Reflective Practitioner*, adopting a human-centered approach as well, analyzed professionals not as objective decision makers but as intelligent actors. He discovered that most professionals do not pursue worked-out plans, enumerate alternatives, and calculate utilities for each, but think in small incremental steps, and act and reflect on their actions recursively.

Argyris et al. (1985) followed these works with his ambitious project for a unified *Action Science*. While sympathetic to their efforts, one should keep in mind that all "sciences" provide institutionalized ways of knowing and typically grow reliable methods; the above approaches have yet to bear fruit.

To locate what this book is proposing, let me distinguish three concepts of science related to design, starting with Cross' two categories:

– "**Science of design**, … that body of work which attempts to improve our understanding of design through 'scientific' (i.e., systematic and reliable) methods of investigation" (Cross, 2000:96). Here, design is the object of research, undertaken by various academic disciplines, yielding knowledge *about* design, in the terminology with criteria from these disciplines, that is, from outside the design discourse.
– "**Design science**, … an explicitly organized, rational and wholly systematic approach to design; not just the utilization of scientific knowledge of artifacts, but design in some sense a scientific activity itself" (Cross, 2000:96).

– **Science for design**, a systematic collection of accounts of successful design practices, design methods, and their lessons, however abstract, codified, or theorized, whose communication and continuous evaluation within the design community amounts to the self-reflective reproduction of design practices. It also includes ways of consulting related knowledge bases in support of particular design decisions and project research. Its aim is to keep design discourse viable and productive.

Comparing these concepts, a "design science" manifests itself in a preference for systematic design procedures, cherishing the systematicity of practice. A "science *of* design" reflects various scholarly interests in design, their perspectives, vocabularies, and criteria of validity. For example, a history of design uses historical methods; a psychological study of designers' creativity may contribute to psychological theory; or a sociological inquiry into the design profession draws on sociological explanations. Designers might appreciate this attention and even learn something, but such inquiries contribute little to their design process. A "science *for* design," by contrast, does not surrender its criteria to other disciplines. It encourages designers to examine their own practices in their own terms and to disseminate proven design methods among designers, and embraces project research. Project research utilizes relevant knowledge from other disciplines but frames it in terms of given design objectives, for example, R&D findings, market research results, ergonomic guidelines, and theories from cognitive science. Contributions from other disciplines are often essential because artifacts may have dimensions for which experts exist and are immensely useful when forged into arguments for a particular design. One can summarize the aim of a science for design as a concerted effort to improve the language and practice of design, its capacity to generate new proposals, to justify them to those who matter, their stakeholders, and above all, to make the redesign of design discourse a routine obligation.

- **Its Boundary.** As already alluded to, the current community of designers does not do so well in maintaining the boundary of their discourse. While designers know quite well who and what belongs, there are competing definitions of design, competing interests in appropriating design for specialized interest, and a lack of appreciation of the role of language, especially professional language in the design process. Engineers consider themselves professional designers although their focus is limited to finding technical solutions to (often social) problems. Scientific researchers call themselves designers of experiments, surveys, and analyses of their data. Medical practitioners design treatment plans. Biologists develop genetically modified plants. Management consultants devise organizational interventions. I mentioned that design is also an essential ingredient of everyday life. While the word design is used widely, where is the boundary of design discourse and what happens there?

Studies *of* design attempt to redraw the boundaries for design, but from the disciplinary perspective of those studying design for their own reasons, for example, from the perspective of art history or psychology. This is possible largely because the boundary of design discourse is not too clear, uncertain. Someone who studies physics or medicine would find their boundaries in the textbooks of these disciplines.

Probably more problematic of the current boundary is its vulnerability to intrusions by competing discourses. Marketing, for example, having expanded the industrial era emphasis on products to goods, services, and identities, conceives of design as one of its departments, the sole purpose of which is to add value to products. Designers have resisted this definition, preferring a broader vocabulary that expresses human-centered, cultural, and ecological concerns. But the threat of a "hostile takeover" of the design discourse is always present. Marketing is not the only discourse that would welcome design as one of its subdisciplines. Donald Norman treats design as a psychologist, and recent writers on computer interfaces adopt cognitive science as a master narrative. These discourses can be described as colonizing the discourse of the design community. The process is not necessarily intentional. It is more likely due to designers' inability to recognize the importance of their own discourse, which is manifest by their constant search for fashionable concepts and methods from other disciplines and their willingness to put these in charge of what worked yesterday. Unfortunately, for most designers, the intellectual grass is always greener on the other side of the boundary and fertilizing one's own seems less fashionable than copying well-to-do neighbors.

- **Its Justifications.** Although natural scientists, and one might as well add engineers and medical professionals to their list, rarely acknowledge their languaging, their operating in discourses of "science," "engineering," or "medicine," respectively, the justifications of their artifacts have been enormously successful and of great benefit for the communities of practitioners of these disciplines. Their claims tend to be supported by test results, mathematical calculations, or clinical success stories. Textbooks on statistical tests are widely used and considered valid. But what is perhaps more important is that the criteria for accepting scientific theories in the natural sciences, designs in engineering, and the treatments of illnesses in medicine are well defined, widely accepted, and the public holds these discourses in high esteem. This was not always so, of course. Science's early struggles with religion were vicious on both sides and ended with religious discourse retreating into a domain that the scientific discourse did not care to enter. Contemporary struggles over the distribution of research funds are struggles between scientific, economic, and the political discourses over whose criteria should apply. Regrettably, design has not yet achieved a comparable degree of respectability that is

commensurate with its contributions to the well-being of communities and the workings of society.

Taking the previous discussion to heart, the aim of this book is to encourage a redesign of design by starting to talk differently about design, the world it can affect, what to do, and how to proceed. Based on what a strong design discourse would require (Krippendorff, 1995), I shall consider:

- **Redrawing the boundary of design** to include new responsibilities that are more productive than what design has been. To this end, Chapter 2 develops the basic concepts of human-centered design.
- **Developing several levels of approaches to meaning in design**, four in fact, using variations of the basic concepts. These are developed in Chapters 3 through 6.
- **Formulating reproducible design methods for human-centered design** from the stories of successful design practices and lessons of failures. Chapter 7 proposes a science for design. This is not to promote unnecessary rigor or methods for their own sake, rather to provide design discourse with compelling justifications for the designs it produces: tests for the claims that designers make about projected futures (truths), cost and benefit of their designs (virtues), and ways to demonstrate their expertise (competence) — all in the service of the new responsibilities of design and the viability of the design community.

chapter two

Basic concepts of human-centered design

The previous chapter described several dimensions of the semantic turn from technology-centered design to human-centered design:

- From the design of products to the design of artifacts that can play various social roles
- From a belief in technological progress to a concern for artifacts that are supportive of communities of users and are user-friendly for their individual members
- From universal and culture-free conceptions of design in a (single) uni-verse to an acknowledgment of the role of language in the construction of diverse community-specific worlds
- From imposing intended functions of products, even enforcing particular uses of technology through training or certification, to allowing people to use them in their own terms
- From designers as a lone genius or authority to designers who can work in teams, including with users, and are able to enroll the stakeholders of their designs in joint projects
- From an attention to objects, products, material artifacts (an ontology), to an awareness of the processes of constructing and reconstructing (ontogenesis or design) artificial worlds whose sole purpose is to make sense to us, remain useful, and enable us to feel at home with them

This chapter focuses on human-centeredness, where it comes from, who contributed to it, what its key conceptions are, and what it entails for what can be called a design culture. Human-centeredness must not be confused with humanism, the Cartesian fascination with the human spirit and a concern for human values in opposition to material functioning. Human-centered design avoids this dualism. It would also be a mistake to think of the semantic turn as a mere change in attitude from the Cartesian contrast between perfect objectivity and flawed subjectivity to the contrast between

dumb and severely limited technology and creative human intelligence, as Norman (1998:160) suggests in his slightly edited comparison in Figure 2.1.

Readers should not pick out a few concepts from the approach presented here, and operationalize them in positivist or behavioral terms. The semantic turn amounts to a shift in paradigm, a change in the foundation for design. It provides a network of interconnected concepts with quite radical consequences for practicing designers, not a set of isolated ideas. Foremost, the semantic turn recognizes the human involvement in the artifacts of design, acknowledging not only that designers are humans, communicate with others through and about the technology they develop, and participate in the social constitution of reality, but also that all those affected by technology bring their humanness to bear on what they do with it. Artifacts are prostheses of the human mind, being, and doing. Realizing the entailments of this shift is not a simple turn to take.

Technology-Centered View		Human-Centered View	
People	**Machines**	**People**	**Machines**
Vague	Precise	Creative	Unoriginal
Disorganized	Orderly	(accommodating)	Rigid
Distractible	(focused)	(context sensitive)	(context insensitive)
Emotional	(rational)	Resourceful	Unimaginative
Illogical	Logical	(many intelligences)	(fast but repetitious)

Figure 2.1 Attitudes about people and about machines that technology- and human-centeredness, respectively, entail.

2.1 Predecessors

Human-centeredness has a respectable intellectual history. It is carried on the shoulder of giants. The ancient Greek philosopher Protagoras is believed to have been the first to say that "Man is the measure of all things, of things that are that they are, and of things that are not that they are not" (Russell, 1959:47). He abandoned the idea of a representational truth and put human experiences in its place. The objectivity–subjectivity dualisms that Protagoras' statement undermined did not die, however. The early eighteenth century German natural scientist and humanist Johann Wolfgang von Goethe came to human-centeredness from an experiential perspective. His 1810 experiments with the color of shadows — actually continued from those first described in 1672 by Otto von Guericke — culminated in a color theory (Friedrich, undated) that opposed Isaac Newton's physical theory of the color spectrum. Goethe became convinced that sight, the human eye, was the most unfailing instrument human beings can rely on, quite unlike

Newton's effort to dismiss it as subjective and biased. Goethe fought against the then emerging objectivism that privileged physical measurements over human experiences; he considered Newton's theory a major epistemological mistake and a disservice to humankind. It was not that one theory provided better predictions than the other, but that Newton's theory predicted the responses of mechanical measurements, which have nothing to do with how people see. Goethe correctly recognized the fundamental epistemological mistake in Newton's and modern physicists' claim to be able to measure and theorize visible light and color by physical measuring devices, as if human perception had nothing to do with colors. Goethe considered light and color to be a human phenomenon, created by human perceptual organs and not reducible as such to physics.

Goethe's insights regarding color perception proved correct. Color turns out to be the result of the eye's physiology. It is literally made in the eye and has no existence outside of it. No correlations exist between measurable wavelengths of light and the experience of colors (Maturana and Varela, 1988:16–23). In contemporary times, Humberto Maturana and Francisco Varela have made a strong version of human-centeredness the cornerstone of their theoretical biology. They argue that no theory — not of the world (ontology), not of human perception (phenomenalism), and not of biology — should contradict what the human body is capable of doing, the body that observers require to observe and that theorists cannot escape from when articulating a theory and claiming its validity. Maturana and Varela (1988:141–176) take human cognition as an operationally closed system that develops its own internal correlations (constructs its own worlds), preserving the ability to live in the face of recurrent perturbations from an otherwise unknowable outside. This would imply that artifacts, like colors, may have external causes but are conceptualized, constructed, and experienced by our own nervous system — without knowable correspondences to what could exist without a human observer.

A separate path to human-centeredness begins with the eighteenth century Italian scholar Gambattista Vico,* a contemporary of Goethe, who took issue with René Descartes' belief that the mind was an organ whose function was to represent the world that existed outside of it as accurately as possible. Descartes, influential to this day, took the objective–subjective distinction as fundamental and believed nature was perfect, but the questioning mind was fallible in its task of mapping the world. His epistemological attitude was consistent with the technology-centered view shown in Figure 2.1. Vico, in contrast to Descartes, asked himself what humans could know (best) or may not be able to say anything about. He developed the insight that human knowing results from doing, from creating things, from constructing the world one lives in. Satisfying the demands of the church at that time, he argued that humans could not know how God made the world, only what

* *Vico's Science of Imagination*. In Verne (1981) the reader can find references to the original works in Latin and in Italian. A discussion of the impact of his work on contemporary thought can be found in Glasersfeld (1995).

they made in their own terms. Design is implicit in this assertion, and one could interpret Vico's *New Science* as providing a human-centered epistemology that is grounded in design activity, not in detached observation. One convincing proof for this principle is the absolute certainty of mathematics, which he correctly identified as a human invention, not a reflection of nature. Actually, Vico was not only a philosopher but also an effective university administrator and designer of legal structures. So he must have had a keen understanding of what it takes to design social realities, not just the material artifacts supportive of them. Equating truths with what had been done or could be made is also rooted in the etymology of the word "fact" from the Latin *factum*, "something done." "Artifacts" are facts that are created with craftsmanship or special talent. Their truths lay in their workings, not in what they are. Most recently, Ruth Hubbard, questioning biological facts, saw the need of reminding her readers of the obvious, "Every fact has a factor, a maker" (Hubbard, 1990:20). Vico would have liked this extension toward holding people accountable for the facts they make, conceptualize, arrange, manage, or destroy. Vico is considered the grandfather of constructivism (Glasersfeld, 1995). I am suggesting that construction and design are synonymous as far as what they describe, although the words are often applied to different empirical domains.

In the 1930s, the biologist Jacob von Uexküll (1934/1957) developed an ecological theory of meaningful world constructions by animals and humans. He considered the possible worlds of an animal as circumscribed by the sensory and motor organs that the animal has available. According to Uexküll, the world that actually exists for animals and humans to see, is created, comes about, is experienced, and is rendered meaningful, as a result of one's actions.* One could say that Uexküll answered Vico's question of what one can or cannot know by implying that people may talk of what they do not know, but live in and see only the worlds they have constructed to make sense of what their sensory–motor organs are (capable of) doing. In contemporary terms one might say that all living beings coordinate their sensory and motor organs, and develop meaningful connections between what they do and how their doing affects what they sense. The world of ticks, bats, and humans differ, first, in the sensory and motor organs available to them, second, in their respective ability to coordinate the two, and third in the complexity of their constructions of reality. So, bats and humans differ not only in their sensitivities (to radar signals versus visible light) and actions open to them (flying, holding on to edges as compared to the innumerable activities that humans can perform), but also in the richness of correlating sensations with the actions that affect them. People can drive cars because they are capable of meaningfully connecting what they see to what they do or wish to accomplish. Uexküll's principles apply not only to species-specific world constructions but also to intraspecies differences, to people who differ

* This gives the word "actually" the meaning of being actionable, a reality that is experienced in action.

in culture, language, occupation, history, and desires, thereby making different sense of their environment and treating it as different worlds.

From experiments in visual perception during World War II, James J. Gibson (1979) developed an ecological theory of perception, quite compatible with Vico's and Uexküll's conceptions. Gibson recognized the need for a human-centered language that would neither physicalize human perception, as criticized by Goethe, nor psychologize the terrestrial environment. It is important to realize that his experiments were aimed at improving the human use of technological systems. The two conceptual keys to his theory are affordance and direct perception. Affordance is the perception of one's ability to do something with what is sensed. So, Gibson describes features of the human world as graspable, walk-on-able, sit-on-able, step-on-able, movable, distinguishable, or edible. An affordance is a reciprocal relationship between an observing actor and features of the environment. Affordances describe what is meaningful to the user of artifacts, not *something* independent of human involvement or described by a detached observer. Regarding direct perception, Gibson noticed that the affordances of ordinary, everyday, or familiar features are understood directly, effortlessly. Children learn to see affordances, what they can do and what things in the world can do for them. For most adults, going up steps, driving a car, writing a letter, and eating at a table are routine and unproblematic activities, which means that their perception and action are in such a perfect fit with the environment that there is no need to reflect or decide on what is seen or what comes next. This fit shows people as knowing what needs to be done without reflection and doing what is called for almost automatically. Direct perception is essentially what product semantics called self-evident, intuitively obvious, and natural. Gibson clearly realized the importance of human-centered conceptions for his work, and he and his followers continue to contribute empirical studies of this ecological fit of perception and action. Only recently have the human-centered conceptions of Gibson entered the discourse of interface designers, traditionally dominated by the technology-centered conceptions of ergonomics (Flach et al., 1995).

None of these "giants" speak about language, however. Gibson struggled to find or create words that were appropriate to his human-centered conceptions, and described this difficulty, but he did not reflect upon the role of language in obtaining information from his experimental subjects, much less upon the relationship between language, perception, and action, which is Benjamin Lee Whorf's (1956) contribution to human-centered conceptions. Before Whorf became a famous twentieth century American linguist, he worked as an accident investigator for an insurance company where he experienced the very phenomena that he later studied and wrote about. In one reported incident, workers had dropped a cigarette butt into an empty gasoline drum, and they were disbelieving that it could explode. The word "empty" signaled a vacancy, a void, and a container ready to be refilled with all kinds of things. This conception seemed all the more obvious as there was no visible evidence that these drums contained anything. The fact that

gases replace the objects that one removes from a container is not part of the conceptual system associated with "empty," even less that residual gases could form an explosive mixture with the air that entered. He conjectured and later substantiated that the language used to describe a situation directs individual perception to available options, to what can be done. The fundamental insight of Whorf and Edward Sapir, his later teacher, was that the vocabulary and grammar of a language has much to do with how speakers of that language think of their world and act in it.

The connection between language use and perception has been substantiated in numerous studies. For example, cross-cultural experiments in color perception by Berlin and Kay (1969) demonstrated a clear correlation between the color terms available in a language and the colors that its speakers could recall and conceptually handle (not to be confused with the ability to detect color differences). Grammatical constructions of noun phrases, consisting of adjectives and nouns, which are typical of standard European languages, lead to the perception that objects have properties. Or, because the linguistic distinction between male and female is an either or proposition, leaving no acceptable middle ground, the present culture makes it difficult for people to cope with transvestites or androgynous individuals. The ambiguity of what lies between the well-defined extremes can lead to violence against those that do not fit the convenient linguistic categories of male and femaleness. Whorf's work has been narrowly interpreted as the Sapir–Whorf hypothesis, the hypothesis that language determines thought, and it is this narrow hypothesis that has been criticized by positivists. Whorf and Sapir's point is that language, thought, and culture (behavior, including the use of artifacts) grow up in close correlation with each other and that human perception is not a mapping of objects in a culture-free environment into abstract concepts, but that conceptions of the world have much to do with how one speaks of that world. This basic finding would explain how people living in a relatively homogenous culture and speaking a common language could easily come to live in the illusion that they perceive things the way they are and judge members of other cultures as deviant and inferior in this respect.

Of all of these theorists, Ludwig Wittgenstein has been most influential for me. His work initiated the linguistic turn in philosophy, a turn from abstract philosophical problems, such as the logic of reality, truths, and the nature of mind, to what humans do while speaking a language (including making such abstractions). Wittgenstein had at least three careers. The first culminated in his dissertation in which he carried a picture theory of meaning from its first proposition, "The world is the totality of facts, not of things" to its logical conclusion, "Whereof we cannot speak, thereof we must remain silent" (Wittgenstein, 1921: 31, 189). In the second career, he taught elementary school in rural Austria but also worked as an architect. In his third career, he taught at Cambridge University and wrote his *Philosophical Investigations* (Wittgenstein, 1953). In the latter work, he developed a theory of meaning that rendered much of philosophy up to this point, including his

earlier writing, literally meaningless. He discovered what seems now quite obvious, that language is always spoken in real social situations, by real people, in conjunction with actions and in the presence of artifacts. He abandoned his philosophical heritage in favor of investigations of how language is actually used in socially coordinated practices, which he called language games. Using language, he argues, is a human activity, and its effective meaning must be sought in the activity that it accomplishes. The meaning of a word, he wrote, is its use in language. Language does not represent, it does something, and the criterion of its use is not truth but its appropriateness, what makes it the right thing to say in a given circumstance, and what makes it meaningful to the participants — as judged by the participants in the language game. My conception of discourse in Section 1.4 elaborates Wittgenstein's language games into the social domain.

So, how do these conceptions connect?

Goethe, Gibson, Wittgenstein, Maturana, and Varela, clearly are all concerned with embodiment, the acknowledgment that human activity cannot be separated from the body of real actors, users, even the designers of their worlds. Physiology, sensory–motor coordination, and background experiences are omnipresent in what people do, albeit taken for granted, perhaps mutually understood, and therefore often escape propositional logic and rational arguments. Abstracting rules from what is being said is not what is being said. This should resonate with the concerns of designers for details and uniqueness, not for generalizations. Artifacts do not work in theory nor can they mean anything to anyone in the abstract. They may be more or less tangible constructions that nevertheless support or object to enacted conceptions. In studies of language, this amounts to a shift from studying written words to studying utterances, from studying grammatical rules to how speakers and listeners make things happen in cooperation with and for each other.

Wittgenstein started seeing meanings neither as representations, nor as currencies of exchange, but in what is brought forth in language. In one of his examples for a language game, he describes the utterances that construction workers use in the process of building a house. Such interweaving of language and action spawned speech act theory (Austin, 1962; Searle, 1969), which is the key to understanding interactivity. In Wittgenstein's rejection of representational theories of meaning and truths, one may recognize Vico's equation of knowing and doing or constructing, as well as Maturana and Varela's notion of language as the coordination of coordination of actions.

Wittgenstein also acknowledged that meanings arise within boundaries, in the particular situations in which language is used, in the language games that are being played, or in "ways of life," with which he equates specific uses of language. The boundary that a professional discourse maintains is one example. Refusing to generalize meanings beyond self-drawn boundaries goes well with Uexküll's meaningful worlds (minus the role of language) and with Whorf's explorations of cultural differences based on different language structures (minus the interactions implied in language

games). The significance of the role of meaningful boundaries extends to Maturana and Varela's biological notion of autopoiesis into the social domain as self-organizing networks of meanings. According to Wittgenstein, meaning is neither fixed nor enforceable from outside a conversation or determinable (often not even observable) without participating in a language game. I prefer the metaphor of co-ordination to that of a game because it does not have the unintended entailment of winning or losing, nor of the need for rules that players must accept before playing. Written with the "-", co-ordination suggests that the responsibility for what happens is distributed over those within a boundary.

The vain attempts of natural scientists to deny or bypass the embodied and language-based nature of their own conceptions often ends up celebrating technology-centered constructions. This is what Goethe's polemics against Newton was about, and what fueled Uexküll to argue against "physiologists" who attended to the mechanical functions of organs at the expense of the biology of living organisms — "bio-logy," after all, is the study of the living, the bios, of organism, not of how one can distinguish conceptually convenient parts within a body. The tension between technology- and human-centeredness also underlies Bateson's (1972) distinction between behavior that is energized by physical causes and behavior that is triggered by information, as he defined it. For example, when a stone is kicked, its trajectory is determined by the physics of the kick. A dog, however, responds to being kicked by what it means to being so treated by the kicker (1972:460). For living beings, meaning (Bateson would say information) is neither an entity nor a causal agent. In the 1930s, George Herbert Mead developed a wholly new sociological approach, called symbolic interactionism (Blumer, 1972, 1990), on the premise that meaning is central to human behavior and negotiated in social interaction. Although Mead and his students said nothing of design and little about linguistic structures, the social nature of meaning is what design semantics will have to embrace in greater detail than that provided by either of these scholars.

Along with Mead's reasoning, linguistic meaning is never that of only one individual. There is no private language, as Wittgenstein insists, suggesting that the meaning of words lies in the history of their acquisition, for the individual as well as a culture, thus, meaning is always social and the result of conversations, collaborations, and histories of coordinations among people. The same is true for artifacts. All artifacts have experiential histories, which are woven into social or cultural histories, always involving many people and their use of linguistic categories and artifacts. Also, new artifacts always emerge from or develop from familiar ones. Artifacts are language in interaction.

The foregoing should not be construed as a complete family history of human-centered contributors. It would take another book. It had to be highly selective. But it deliberately excludes well-known names for reasons that they take opposing epistemological stances, for example cognitive theorists who hope to explain thinking processes in computer terms, semioticians who

believe in the primacy of the distinction between signs and what they represent or signify, or engineers who start with the reality of mechanisms and let ergonomists worry about efficiency of use and designers about covering ugly and unsafe mechanisms with attractive forms.

I contend that the fundamental changes toward human-centeredness, reviewed above, fuel an equally fundamental shift in design: the semantic turn toward meaning.

2.2 The axiomaticity of meaning

Section 1.4 asserted, citing others, that there are many professions that engage in design activity without calling themselves designers — engineers, managers, medical professionals, politicians, even architects — not to forget ordinary people who organize their future with creativity and purpose. Designers know quite well who is and who is not a designer although their criteria are rarely clear. Holding a design degree does not necessarily assure self-reflexivity, and being employed as a designer may not suffice either. Lack of clarity in this regard supports the earlier observation that the boundary of design discourse is weak, uncertain, and difficult to draw when several other discourses are trying to usurp the field of design for their own purposes. Most outsiders see design as an applied art, as having to do with aesthetics, unlike a solid profession unto itself, with technical knowledge, skills, and responsibilities to rely on. Insiders to design, by contrast, talk of innovative ideas, coordinating the concerns of many disciplines, being advocates for users, and trying to balance social, political, cultural, and ecological considerations. The underlying thread in designers' self-perception is a concern for what people do with artifacts. It differs from the concerns of other professionals by addressing issues of human interfaces with technology, technology understood quite broadly as what humans create — artifacts.

Section 1.2 presents a trajectory of artifacts that demonstrates increasing human involvement. Human-centeredness, therefore, is a way to distinguish design from other creative and purposive activities. Since the advent of product semantics, it has become increasingly clear that human-centeredness defines what designers call design and offers the clarity that was missing from the design discourse of the past. In this discourse, meaning occupies a privileged position. The early work on product semantics showed that meaning matters more than function, leading to the axiom that:

> *Humans do not see and act on the physical qualities of things, but on what they mean to them.*

This statement is axiomatic to a human-centered design discourse in terms of which designers can conceptualize their aims, organize their work, and make compelling arguments for their designs. This axiom also suggests a productive distinction between design and what other disciplines teach and do. In the past, designers have moved between engineering, art, market

research, process planning, visual persuasion, and consumer advocacy. They had to know a little bit of everything, without being respected as authorities in any one of these endeavors. Designers' successes were hard to explain. Hence, designers often ended up rhetorically disadvantaged and overwhelmed by criteria not of their choosing. *Designers' extraordinary sensitivity to what artifacts mean to others, users, bystanders, critics, if not for whole cultures, has always been an important but rarely explicitly acknowledged competence*. Putting meanings into the center of design considerations will give designers a unique focus and an expertise that other disciplines do not address. Moreover, the apparent irrefutability of this axiom gives design professionals a solid rhetorical ground from which to justify their work.

Although the primacy of meaning is not widely recognized, it has always played a central role in design. Obviously, the drivers of a Lamborghini spend awesome amounts of money and endure considerable inconveniences because owning and driving such a car means something special to them and its use in public communicates this to others as well. True, a car needs to be driven, which entails that a network of engineering functions works reliably and responds to the driver's command. But for drivers of Lamborghinis, the functionality of its components is taken for granted and hence subordinate to how they see themselves driving it and how they see themselves being seen with it. Aesthetic theory hardly explains the extraordinary value put on such cars. That design has everything to do with what a Lamborghini means for their drivers and for the public is unquestioned. Why traditional design discourse rarely addresses issues of meaning and takes rhetorical advantage of the remarkably simple truth of the above axiom is less comprehensible.

The axiom on the centrality of meaning applies to nontechnical artifacts as well. For example, people need to ingest food, of course, but dine in fine restaurants, enjoy culinary delights for their gustatory and visual pleasures, and the reasons for likes and dislikes show up in the way they speak of food. Armed with the vocabulary of French cuisine of a certain area, people enjoy one kind of food, and when counting calories, they are drawn to another kind of food. One's ideas of the chemistry of food may explain some choices of what, how, and how much one eats, but in fact, these ideas hardly affect digestion, which goes on independently of one's understanding. Thus, people eat what they believe is culturally acceptable, is tasty, and makes them feel satisfied afterward, what it means to them. What "really" goes on inside the human digestive system is literally outside of one's reach. Whatever one learns about the mechanism of digestion, food pyramids, calories, fats, and vitamins, right or wrong, it is entirely conceptual and resides in the realm of linguistic meanings, not in what is actually happening inside the human body.

Clothing is not much different from food. To wear clothes, a person does not need to know how fabric is woven, the chemistry of dyeing, and the plight of the underpaid Indonesian workers who sew them. What matters is whether one feels good in them or not, and this feeling depends, at least

in part, on expectations of what other people might say about how one looks in them.

For a similarly mundane example, consider ladies' high-heeled shoes. They are said to look elegant, make a woman's legs look longer, and the wearer several inches taller, all of which are semantic attributes that the wearing of such shoes encourages. The fact that they are ergonomically unsuitable for walking even normal distances and in the long run ruinous to their wearer's feet, apparently does not put a stop to their mass production or discourage widespread use. When a woman comes home from a formal occasion, the first thing she changes are the uncomfortable shoes she wore essentially to define herself for others.

Clearly, meaning matters more than function. While cars, food, and fashion are foremost social or cultural artifacts, meaning need not be entirely conventional. It governs the use of technology, especially where its artifacts escape their users' understanding. Indeed, if the use of computers, electronic appliances, the telephone, the Internet, airplanes, atomic power plants, even the government were limited to those who knew how they worked, very few if any would be able to use them. The point is that the understanding needed to use something reasonably complex, computers, for example, is entirely different from the knowledge needed to design, produce, fix, sell, or engage in the politics surrounding it. One cannot ignore that designers, engineers, business people, politicians, cultural critics, and users, all live in different worlds, act according to the different conceptions they bring to what they encounter, and create different meanings for what seems from any one perspective to be the same thing. Although physicists claim extraordinary privileges in describing the way the world objectively IS, there rarely ever exists one correct description of anything. Not only are physicists known to have changed their mind about what the universe is, there is no basis to say that any one world construction — the users', psychologists', designers', engineers', physicists', economists', or environmentalists' — is superior to all others. Choices of worlds always are a question of feeling comfortable in them and being able to do what one has set out to do. The semantic turn offers designers a world of their own.

Is it possible to observe the real behind what it means? This is of course what many philosophers have asked themselves since Plato conceptualized ideal forms behind visible appearances. The question has survived when natural scientists ask themselves what it is that they actually measure. But all answers to these related questions are given in language. Facts do not speak for themselves, and I suggest that such questions are the linguistic artifacts of grammar. Propositions of the form "A has the meaning B" presuppose "A" while claiming "B." In such propositions, it is assumed that objects are in the possession of an entity called meaning. The dialogical origin of such ontological propositions is discussed in Section 1.3.7. It is not surprising, therefore, that the users of such propositions are mislead to a reality with discoverable meanings instead of realizing that it is their conception that makes it so. Every conception is someone's conception, every

artifact makes sense to their users, but different senses to different people. The engineers' functions are meaningful to engineers, but these functions are not the only truth, and not necessarily shared by nonengineers. In the world of physics, functions do not exist. And while the profit from sales is real to the salesperson, it may not be to the environmentalist who sees profit motives as ruinous of the environment. There is no escape from the axiom. It states an undeniable truth that is so strong that one might as well embrace it fully for the strengths it provides for design discourse. Without realizing its strength, designers are doomed to chase the ghosts of other discourses.

2.3 Sense, meaning, and context

Symbolic interactionists, such as Mead and Blumer (1990, 2000), would probably embrace the axiom presented above. But because sociologists focus on explanations, not design, their stake in the concept differs from that of designers. For human-centered designers, this axiom defines the boundary of design discourse, including design methodology and designers' professional competencies, and it is, hence, of considerable rhetorical significance. When adopting meaning as central to design, designers must be far more specific than merely stating the axiom. The following attempts to define meaning and context by way of sense.

2.3.1 Sense

Sense is the feeling of being in contact with the world without reflection, interpretation, or explanation. It involves all senses: seeing, hearing, touching, tasting, smelling, and even the kinesthetic sense. Were it not for Charles S. Peirce's (1931) insistence on a representational ontology of signs, one could have equated sense with his "firstness, … the *quality* of what we are immediately conscious (aware) of."* Uexküll (1934/1957) described what is here called sense as ecological fitness, an environment's recurrent support of what an organism does. Gibson (1979) called it direct perception. Sense is neither identical to nor a coded version of the stimuli that enter a receptor organ, the retina, for example. It has already been mentioned that colors are not a property of the stimuli that trigger them, but what the human eye creates. Sense is the background against which one notices what is unusual, unexpected, or different. Sense is the tacit, taken for granted, and largely unconscious monitoring of what is. Most examples of sense concern the familiar, common, and unproblematic surroundings that allow one to focus attention on something else, like sitting on a sofa while reading the newspaper, or riding a bicycle while attending to the traffic. Sense is also informed by dispositions, needs, and expectations, including emotions, all of which have to do with the human body.

* Paragraph 1.343 in Peirce (1931). The parentheses are mine, which is to say that I would consider the feeling of being self-evidently in touch with the world as awareness, not consciousness.

The normal or desirable state of sense is the feeling of being comfortable within one's world, at home, safe, and assured that one is on track, in tune or in sync with what is going on. As such, a particular sense defies explanation. It is a dimensionless feeling with numerous deviations, such as feeling discomfort or pain, being frightened in unfamiliar situations, losing control, or being uncertain about what to do. From the perspective of an outside observer, the sense of comfort may well be recognized in the recurrence of interactions, in a stable or persistent correlation between sensory attention and motor actions. But such explanations do not enter the sense of being in the world. Sense simply is. Here are a few characteristics of sense:

- Sense is always *someone's* sense. It is an embodied phenomenon. No other person and no physical instrument can substitute for or replicate anyone's sense, and the sense that something makes is not observable by anyone else.
- Sense is *patterned*. It usually is grounded in a great number of conceptual details without awareness of these details. For instance, in entering a familiar room one typically faces many things without consciously distinguishing among them, yet if there is something out of order or missing, one may feel something between puzzlement and discomfort before noticing what is different and in need of explaining or correcting. An outside observer may well attribute someone's sense to a history of being in, living with, using, and perhaps even easily talking of one's environment and consider sense as a matter of memory-based "re"-cognition, but sense does not contain this history and does not mean access to this memory.
- Sense takes place *in the present*, neither in the past nor in the future. Sense always "is." It is being in touch. Whatever happened before and whatever can be expected to happen afterward is not the always-current sense. Sense is not a snapshot. One may have a sense of direction while being in motion.
- Sense is *indistinguishable from its cause*; and *from one's expectations*. The chair as seen is the chair that is and why it is seen at a particular moment does not enter the sense of it. People are constitutively unable to see beyond or behind their present senses. That things-as-such are inaccessible was already known by the skeptics. Immanuel Kant made this insight the cornerstone of his epistemology and constructivists developed it further. As Maturana insists, in the present there are no illusions, no mistakes.
- Finally, sense is *never in doubt*. Sense may result from a chosen action but offers no choice. It may differ from one person to another, but this difference cannot be seen. It may later turn out to be an illusion, but when sensed it is not. Truth and falsity do not apply to sense. Sense self-evidently is, and it is therefore always trusted.

To speak of interfaces as self-evident, natural, and intuitively usable is to speak of sense. To consider sense as an embodied phenomenon prevents

one from confusing two incommensurate phenomenal domains: the *description* by an observer of what someone else attends or responds to, and one's own *sense of the observed other* to which one may respond. Even theorists construct their world starting from their own senses, not from the senses of observed others. Understanding is always someone's understanding. The notion of sense acknowledges one's being familiar, in touch with and monitoring something that is fundamentally inaccessible otherwise.

2.3.2 *Meaning*

On the most basic level, meaning restores perceived differences between what is sensed and what seems to be happening. It is a way to remain in touch with a world that had become uncertain or in doubt. More complex meanings could be likened to explanations for how a sense is embedded in the context of other senses, its role in such a context, or why something said could be said better, shorter, or more understandably in the same context. In language, meaning is the answer to the question "what does this mean?" It is asked when one's seemingly familiar sense has become somehow insufficient or "questionable," and its answer points to what else needs to be considered in its place.

Below, five complementary manifestations of meaning are distinguished: in perception, in reading, in language, in conversation with others, and as re-presentation, followed by the next section, in the context of something else.

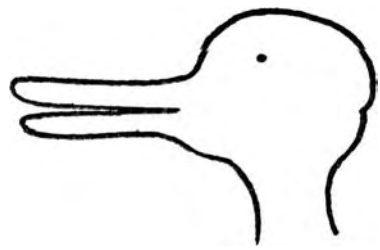


Figure 2.2 Duck–rabbit: alternative meanings.

- **In Perception.** In perception, meaning arises in the awareness of the *possibility of different ways of seeing*. The simplest examples of "seeing" something "*as*" something else are so-called flip-figures, like the drawing already discussed by Wittgenstein (1953:194), shown here as Figure 2.2.* It can be seen in three ways. Without much attention, it is a line drawing that takes up space on a page. With attention paid to its shape, it can be seen *as* the head of a rabbit or *as* the head of a duck. Casual attention does not distinguish between alternative ways of seeing. It is a meaningless figure, the head of a rabbit or the head of a duck. Naturally, and

* Wittgenstein's (1953:154) adaptation from Jastrow (1900).

without reflection, one may go on from there to other things, never realizing these alternatives. Awareness that the drawing can be seen in either way, entails being cognizant of the difference between the figure and its two ways of seeing it, which leads one to recognize the drawing as a flip-figure with two meanings. As a matter of cognitive economy, casual attention typically restricts perception to a single sense.

Similarly, one can sense a chair but can also see something "as a chair." "Seeing something as…" entails the distinction between something sensed and how this something is seen as, and introduces into perception a difference, not otherwise noticed, between sense and meaning. Seeing something as a chair renders that something open to the possibility of other meanings among which chairness, the perceived ability to use it to sit on, is but one alternative. Thus, in perception, artifacts, their sense, mean all the possibilities and constraints that come to mind.

As previously stated, sense cannot be reduced to the physics of its triggers. Nor can meaning be separated from one's bodily involvement with a perceived artifact. In fact, perception can be thought of as making one's bodily coordination with features of its environment meaningful, manageable, and usable. Gibson (1979) used this insight as the basis for his human-centered so-called ecological theory of perception. His affordances are meanings that suggest the human ability to act so as to change an existing sense to a preferred one. For example, in noting the grasp-ability of a handle, the perceiver anticipates a sequence of senses, of which the current one is a part, potentially leading to grasping and moving it until a desirable sense is achieved. When seeing the step-on-ability of stairs, the bend-ability of a wire, the sit-on-ability of a chair, the eat-ability of food, and the drive-ability of a car, sequences of senses are invoked by what is present to the senses. The meaning of a car surely includes the ability to drive it, to change one's sense of where one is to the sense of being where one wants to be. However, a typical car has other affordances as well: showing it off to neighbors, using it for a private conversation, transporting a payload, repairing it, etc. Thus, experientially, *artifacts mean their affordances, the set of their imaginable uses*. Recalling the history of becoming acquainted with a particular artifact, being able to distinguish it from others, knowing its composition, predicting its motion, imagining who else would use it and how, fearing what would happen should one fail to handle it right — all of these meanings constitute what an artifact "is."

- **In Reading.** In reading, meaning arises in attending to the composition of text. Reading and writing are complementary operations. One cannot write without being able to read. Written language offers nearly endless possibilities for grouping characters into words and combining them into larger expressions. But because any text must make sense first to the author and then to readers, word choices are constrained by what is sensible and meaningful to the author and to the author's idea of who

the readers are. One can distinguish between syntactic constraints, a grammar, for example, semantic constraints (which words go with which other words), and situational constraints (what assertions are appropriate in the presence of particular others who hold us accountable for what we say). But these are analytical distinctions, not used by readers. Readers may be quite aware of sensing strings of characters, but see right through them to what they mean to them. Reading typically is routine, fast, and largely unreflected, so much so that readers cannot spell out the rules of grammar they are using to break the character strings into units and the assumptions they have internalized about how the text they read came to be, why the text was composed the way it was, what its writer had to have in mind to write and read it the way it now appears. The answer to the question of what a text means is mostly given in language, perhaps in the form of another text, and sometimes by some kind of demonstration that those interested in that answer need to read as well. Thus, much as for artifacts, the meaning of text is certainly affected by what is sensed, but the sense of it merely restricts the great number of possible interpretations. There are no causal relationships that could determine the meaning of a text, but there are cultural constraints on alternative readings. Without reference to social habits, history, and language use, the relation between what a text means to an author and what it means to readers remains incomprehensible, just as there is no necessary relationship between what an artifact means to its designer and what it means to its user. The truth of a text is always secondary to its reading, just as the use of an artifact is always secondary to its meanings.

- **In Language.** What artifacts mean is also informed by what is being said of it, of its dangers, of the difficulties in handling it, of its costs and benefits, and of the approval or disapproval expected from others who witness their use. The conceptions provided by language are an indispensable part of how artifacts are perceived, conceptualized, and talked about. Already mentioned was Whorf's (1956) example of containers — because they were described as "empty," they were therefore considered void of anything, including the invisible gasses that turned out to be explosive. Narratives place artifacts into grammatical constructions that provide not only the linguistic context of the noun object but also define the dimensions in which the reader is likely to view the mentioned artifact. Advertisers do their best to associate qualities with products on the market that highlight qualities that could enhance sales. Efficiency, user-friendliness, lightweight, valuable, and attractive are verbal attributes of mentioned artifacts that may well be measurable but are not considered important unless talked about in that way. How powerful a car really is, how much gas it consumes, how it compares with those made by competitors is first heard in language; it may never be experienced as such, but most certainly influences the perception of a

car, as well as buying, driving, and enjoying it. Language has a way of highlighting the aspects of artifacts that are important, worth talking about, and seeing as such. In language, *artifacts mean the possible grammatical and semantic positions their names can justifiably occupy in speech and written text*. Assertions are justifiable when they can withstand challenges to the contrary.

- **In Conversation with Others.** Questions of meaning can also arise when we become aware that others seem to see things differently, when others use words or handle artifacts in ways we would not, when others account for their world in terms different from our own. Experiencing such discrepancies challenges the obviousness of our own perceptions, and accepting the possibility of versions other than our own calls for explanations of these apparent differences. Because we cannot observe others' senses, have no access to the meanings they construct, and no way of knowing why they see the world so differently, conversationally insensitive explanations dismiss the perceptions of others' as "obviously" flawed. But considering that most people can give good reasons for what they do, the most practical way to render these discrepancies meaningful is to ask open-minded questions and to seriously consider their answers. Questions concerning meanings generate stories in which artifacts play the roles that their tellers give them, consistent with the experiences they have with them. While stories can never capture all the meanings that informants could bring into a narrative, especially their feelings and tacit understandings, conversations provide a window into the understanding that others have and that designers need to understand. The key to this understanding is unprejudicial listening, avoiding our own categories, and being careful in rearticulating these stories in our own terms. This is what ethnographic research aims to do. In conversations, *artifacts mean the possible responses to questions concerning the meaning of an artifact, everything that can be said about it with mutual respect.* Thus meaning is not negotiated but equals *all the roles that an artifact can play in a conversation.*
- **As Re-presentation.** Photographic images are said to represent what was in front of a camera's lens. Indeed the optics of a camera performs a mapping function. Even for viewers who were not at the scene, a photograph has meaning. They cannot judge the mapping function, however. They have nothing to compare the image with but with what they can imagine. Photographs are sensed, no doubt, but this is not the point of taking them. Photographs are best understood as making present something that occurred elsewhere and at another time. Photographs and other images can also be sensed repeatedly, make something present again and again — perhaps each time a little differently — but present nevertheless. Thus, *images mean what they can re-present to their viewers,*

make present again. The semiotician Peirce* talks of icons, a category of signs based on figurative resemblances. But it is not the resemblance of a sign, the image, and what the image signifies, reality, but the viewers' ability to use their senses to vividly imagine something else, its meaning.

Re-presentations are far more common than intended representations, such as photographs, drawings, or witness accounts. For example, when seeing a book, one sensed that book only from a particular point of view, with a particular perspective. It is one sense among many possible ones whose joint meaning is book. One never senses a computer, always only one view of it. Re-cognizing what it does, the many ways it can be of service, is its meaning, and this is what human-centered design is to assure.

The five applications of the concept of meaning have their human-centeredness in common. While the meanings derived from reading and re-presentation may not have obvious implications for actions, the meanings of technical artifacts almost always concern possible uses.

- Meaning is *a structured space, a network of expected senses, a set of possibilities* that enables handling things, other people, even oneself. They guide actions much as a map shows all the possible paths from where one stands. The capability of creating possibilities for action distinguishes human agency from mechanisms. Pushing a bell button causes a doorbell to ring. The doorbell has no choice. But how one responds to the ringing has a great deal to do with the options one can imagine.
- Meanings are always *someone's construction,* just as sense is always someone's sense, and, hence, meanings are always *embodied in their beholder.* Not even communication enables the sharing of meanings. What communication may achieve is a conceptual coordination, a con-sensual demonstration, and the conversational affirmation of the sense of being understood.
- Meanings *emerge in the use of language* but especially involving human *interactions with artifacts*. Meanings are neither intrinsic to the physical or material qualities of things, nor can they be located within the human mind. Just as the meaning of a text emerges in the process of reading, the meaning of an artifact emerges when interfacing with it (and through it with others). Humans constitutively participate in this process, in the part that entails conceptual *openness*.
- Meanings are *not fixed*. Human participation in interfaces with artifacts is characterized by *conceptual openness*. Meanings are *constructed* from previous experiences, *expanded* on them, and *drift,* much like imagination does. What constrains that flexibility is the viability of human involvement and its economy.
- Meanings are *invoked by sense,* and sense is always part of what it invokes. Thus, current sense is a metonym of what it means, especially

* Paragraph 1.343 in Peirce (1931).

of what one can do in its presence. For Gibson,* meanings are perceived affordances. **

Umberto Eco (1976) once defined "Semiotics is in principle the discipline studying everything which can be used in order to lie." This definition implies intentionality, which is not required in the definition of meaning adopted here. But the two definitions agree on the importance of alternatives. One cannot lie without having the option to say something untrue. There is no meaning without awareness that something could be seen (read, interpreted, or used) differently. With the noting of alternatives comes the possibility of contradictions. Figure 2.3 depicts two examples concerning artifacts. The fur-covered cup, saucer, and spoon by Meret Oppenheim makes what obviously appears to be a cup virtually unusable as such by a surface that is repulsive to take to one's lips. Man Ray renders a clothes iron dysfunctional by gluing the heads of 13 nails to its most important identifying feature, the highly polished metal surface for ironing clothes. Nothing could be more damaging to fabric, which an iron is to care for, than bringing it in contact with a series of sharp nails.



(a) (b)

Figure 2.3 Semantic contradictions. (a) Fur-covered cup, spoon and saucer by Meret Oppenheim (1936); (b) Iron by Man Ray (1921).

* The theory of affordances. Pages 127–135 in Gibson (1979).

** The conception of meaning advocated here clearly goes against semiotic notions that look for meanings in what signs stand for or represent, in the contents of messages (as distinct from their forms) or in the properties of sign-vehicles. In an interesting discussion of meanings *in* architecture, the semiotician Umberto Eco (1980:58), after exploring numerous approaches to their conceptualization, ends up pleading for leaving buildings open to variable interpretations *by* their occupants. His observation may be taken as an endorsement of human-centered explanations, of the need of artifacts to afford many meanings, and even of the conception of meaning as a space of possibilities — which is the starting point of the semantic turn's conceptualizations.

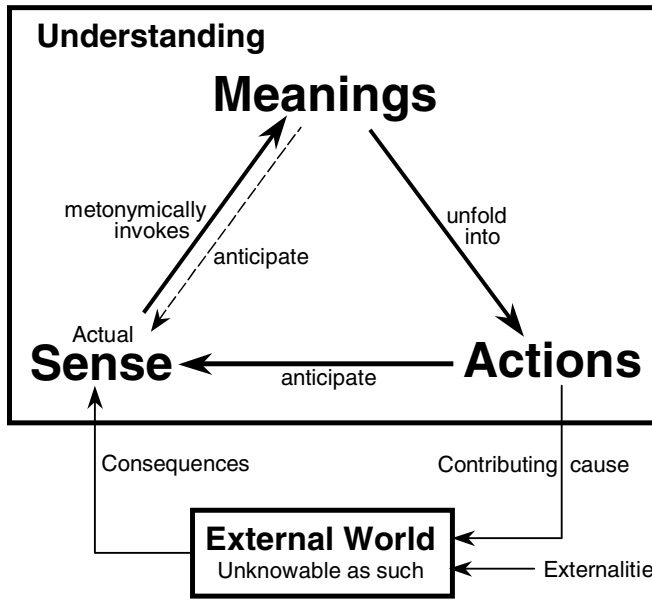


Figure 2.4 Sense, meaning, and action.

Some of these points are summarized in Figure 2.4. Here, sense, always actual and unquestionable, is depicted as metonymically invoking meanings — metonymically because the present sense is always part of what it invokes: the meanings of what one senses. Multiple meanings entail choices among actions that are geared to handle something, to steer it in the right direction, or to reconfigure it so as to achieve desirable senses. Meanings unfolded into actions also produce expectations of what is to be sensed, which are either supported by what is sensed as a consequence of actions or frustrated by it. The world outside, the artifact, imagined to be controllable, but only in part, is not knowable as such — except through sensing the consequences of one's own actions and revising its meanings when failing.

Meanings do not merely occupy the central focus of human-centered design, they offer designers a new kind of "becausality," which is captured by the simple insight that

One always acts according to the meaning of whatever one faces.

This statement does not parallel causality in physics. It always concerns sets of possibilities and presupposes human agency. Just as a question delineates possible answers without being predictive of one, so do meanings indicate a range of imaginable uses and exclude unimaginable or inappropriate ones.

2.3.3 Context

Just as figures are recognizable only against a ground, so do possibilities imply constraints. The above considerations of meaning must be contextualized. Not all meanings are recognizable all the time, not all perceived affordances are in fact afforded when acted upon, and not all imaginations lead to practical designs. Meanings and contexts are twins, but they behave quite differently. What does a context do?

- **Contexts Limit the Number of Meanings.** Typically, meanings are polysemous on their own. They occur in multitudes. There is a logical temptation to assign one meaning to one thing, but this is a fiction, sometimes useful to assume, but a fiction nevertheless. Without a context, a dictionary usually lists several meanings of a word. Competent users of a dictionary tend to know which meaning applies in their situation because they have a sense of the context in which the unfamiliar word occurred. Artifacts too mean different things in different situations and to different people. A traffic sign on a street may direct traffic, inform or warn a driver. On the wall of a student loft, that physically same sign may serve as a decoration or perhaps as a sign of defiance, certainly not seen as directing traffic. A hammer in the hand of a craftsman is a constructive tool. Observed next to a dead person it may be the murder weapon. An African mask in the museum means all that someone can explain, none resembling what it meant to the performer of a ritual dance. Recontextualizing changes meanings.

  Designers can benefit greatly from knowing and playing with contextual constraints. Some contexts are compelling, seemingly without discernible reasons; others must be enforced. The idea that artifacts have an intended meaning from which users should not deviate is common to institutional arrangements: traffic signs, uniforms, national flags, seating arrangements in court. In the industrial era, producers were inclined to educate users to handle their products as intended. In a postindustrial era, products are increasingly expected to be self-evident in particular contexts. One is inclined to say all meanings result from seeing something in the context of something else. Consider Figure 2.5. It shows five characters. Cover the top and bottom character and the center one reads as "B." Cover the left and right character and the center one reads as "13." Evidently, it is difficult to escape the influence of the adjacent characters. Here, seeing "as" is compellingly context dependent. Well-known principles of gestalt perception may explain this particular example. The dependency of meaning on the context of what is sensed is far more general than commonly assumed. Some perceptual theorists virtually *equate* context with meaning. Here, however, one can conclude that *artifacts mean what their contexts permit.*

- **Human Agency Is Manifest in Social Contexts of Use**. Because meanings and assigned functions are not physically measurable, yet guide



Figure 2.5 Alternative meanings in alternative contexts.

human interactions, how could they be amenable to observations (as distinct from individual involvement with artifacts)? The short answer is that one cannot observe meanings, only their effects on behavior. For example, short of being able to ask questions, when one observes someone using a chair not just for sitting, but also as a depository for books, as a valet for clothes, as a stepladder to reach high up, as a stand to entangle wool for knitting, as a privileged family member's seat, or as something with exchange value, one can presume that, whatever that chair means for its user, its meanings permit these uses. Or, if someone is observed to put a fallen chair back into the upright position, restore its broken leg, or clean it, that chair is not treated as a physical object but shows evidence of meanings that the user is intending to preserve. Observations of this kind are evidence of human agency whereas routine behavior may not be. Artifacts can have meaning in many more ways than are observable, not only because some meanings do not make a noticeable difference in behavior, emotional ones, for example, but also because not all meanings are enacted in the particular context in which their use is observed. Thus, observing the effects of held conceptions says little about the nature of the conceptions actually held.

Much can be learned from observing or videotaping and analyzing the uses to which people put their artifacts in the environments in which they work. Careful observations, especially of failures or awkward practices, often are good starting points for design. However, the observation of such happenings is always limited by an artifact's contexts of use — by the material support on which it depends and without which it could not function the way it does, just as cars need a system of roads and pencils need paper and a flat surface for writing — but also because such uses may be governed by rules and expectations of the social situation defined by its participants. With an interest in inferring meanings, the challenge for observers is to separate the effects of meanings from automatic, habitual, or mechanical behavior. The latter tend to be recurring in temporarily unalterable sequences. Meanings, by contrast, require users' awareness of alternative ways and their ability to go different routes, succeed

against odds, or correct possible mistakes to a degree better than chance. Thus, from the perspective of outside observers — and videotaping the use of artifacts favors this perspective — *the effects of artifacts' meanings can be observed in the diversity of contexts of their use* provided that their uses could not be explained by physics, habits, or chance.

Observers are often tempted to assume that what they see, others see as well. The assumption of this equality — privileging the outside observer's meanings at the expense of the meanings that guide an observed user interface — accounts for the failure of functionalism in design and behaviorism in social contexts; this assumption is now adopted in the cognitive sciences. In everyday life, people not only observe each other's uses of things, they also judge or comment on deviations from expected uses, reward novel uses that are seen to benefit one's community, or criticize uses that could endanger others. Comments of this kind carry the use of artifacts into the domain of language and demonstrate social and interactive meanings at work. They also assume and enact human agency in that people hold each other accountable for what they do. Putting observational accounts in the context of the social processes that generate linguistic explanations, criticisms, instruction, and advice, therefore enriches observational records with meanings otherwise missing. In combinations of observational and verbal accounts, then, *the meanings of artifacts become manifest in the set of all social contexts in which artifacts can be used without inviting disapproval or sanctions*.

- **Mutual Contextualization**. Just as figures and grounds are often interchangeable in that one provides the context for the other, this also applies to the meanings of artifacts. The meanings of an artifact's parts depend on the meaning of their arrangement, just as the meaning of its arrangement depends on that of its parts. There is no simple entry into this circle of mutual dependencies. Understanding a complex artifact is like reading a text, though artifacts can be touched and played with, not just seen. Written characters are grouped into words, words are arranged into sentences, sentences form paragraphs, and so on. To understand, the meaning of a sentence, for example, implicates the well-known hermeneutic circle. It may start with some initial set of assumed word meanings. One must then hypothesize a grammatical structure to make sense of these words. Assuming a meaningful structure for the whole usually calls for a revision of what the words mean in that context, leading to revising word meanings in terms of sentential meanings and sentential meanings in terms of word meanings — until an understanding emerges in which all apparent incompatibilities are resolved to the reader's satisfaction. The same is true for figuring out the meaning of an artifact as a function of the relationships among parts, mutually contextualized by their arrangement, and as a function of how the whole is related to other artifacts and users' intentions. Figure 2.6 depicts this process in its generality.

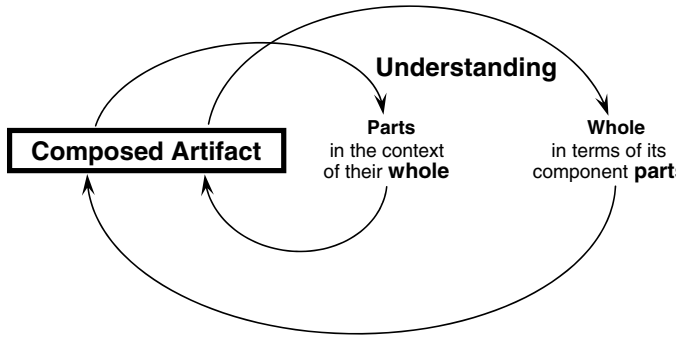


Figure 2.6 Hermeneutic circle: creating coherent meanings.

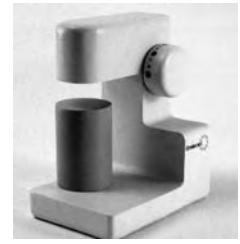
A hermeneutic circle explains how people render multicomponent artifacts meaningful, be they texts or material objects. Figure 2.7 exemplifies mutual contextuality with the help of an exercise. Reinhart Butter gave his OSU design students a set of basic and inherently meaningless geometrical forms, and asked them to combine these forms and only these forms to make different electronic products of recognizable kinds. The exercise was to show how parts receive their meanings in the context of each other and that the meaning of the whole resulted from their arrangement. The exercise unequivocally demonstrated that a great number of rather different artifacts could be constructed from these basic building blocks. In the context of each other, these geometric shapes assumed different meanings.



(a)



(b)



(c)



(d)

Figure 2.7 Meanings in mutual contexts. (a) The given set of shapes, meaningless as such; (b) L to R: ironing appliance (half-scale), juicer, can opener; (c) kitchen appliance; (d) telescope.

This section argued for meaning as the defining concern for designers and presented its general idea: built on sense, rooted in the recognition of alternative ways of seeing on the one hand and acting on the other, and dependent on the context of the artifacts' use. Chapters 3 through 6 are more specific and develop four ways for artifacts to have meanings that are central to design. Specifically, Chapter 3 concerns meanings of *artifacts in use* and presents a theory of human interfaces, typically between a user and an artifact. Chapter 4 addresses meanings of *artifacts in language* and develops a theory of the many roles that artifacts come to play in human communication. Chapter 5 focuses on the meanings in *the life cycles of artifacts* and suggests a theory of how artifacts come to be. Since designers are constitutively involved in the genesis of artifacts, it could be considered an outline of a theory of professional design. Chapter 6 attends to the meanings of artifacts in an *ecology of artifacts* and proposes a theory of technology, of how species of artifacts are brought into interaction with one another. It can also be considered a social theory of design culture.

The following sections attend to three other concepts that are central to the semantic turn, the concept of human beings in human-centered design, the concept of knowledge, and the concept of the culture that nourishes design.

2.4 Stakeholders in design

Designers are surrounded by intelligent professionals who have an interest in the outcome of a design process: clients, engineers, CEOs, financiers, sales people, and the members of institutions that provide data in preparations for a design or do research after prototypes are available. Design literature says little about them while much is written and argued about users. In fact, most of the professionals just mentioned talk about THE user, as if they were familiar with THE typical individual, as if he or she existed. User-friendliness is an important sales argument. A quality called "usability" has recently entered design discourse, and the phrase "user-centered design" is to suggest that designers need to take care of users. Being human-centered, the semantic turn clearly embraces user concerns but is uneasy about references to the stereotypical user for several reasons:

- THE user is a myth, or at best a statistical artifact. For example, ergonomic charts feature average human body dimensions, separate ones for males and females. They do not say how few people if any actually are average in all dimensions. Statistically, there are about as many males of the average women size as there are women of the average male size.* People who conform to all statistical attributes are rare and may not even exist in real life; the proverbial family with 2.3 children is an example.

* This is true under the assumption that males and females have about the same standard deviation from their respective means.

- THE user is an invention of the industrial era and reflects the conveniently restricted responsibility of producers to deliver functional products to a majority of targeted "end users." The technical rationality of this era blinds against unintended uses and users, and takes intermediate users for granted. Before a product reaches its intended user group, it passes through the hands of many who use it for a variety of reasons: to solve an engineering problem, to keep jobs in a factory, to profit from increased sales, or to supply supporting gadgets. After its intended use, it may become of interest to repair shops, benefit recycling companies, and become an ecological nightmare for communities that live near dumps. All of these people and many more could be said to "use" industrial products, albeit in ways that are at variance with the simple end-user or consumer concept. Limiting design concerns to this fictional end-user at the expense of all the other people who are touched by a design presents a serious problem.
- Some designers consider themselves advocates of THE user. This may have been an admirable ethical stance during the industrial era, during which time users had no voice. However, in view of the emergence of highly intelligent users, effective consumer advocacy groups, strong environmental protection agencies, and consumer research, diverse voices do exist that will have to be listened to. Claiming to be an advocate of users (plural) without their consent not only stereotypes a whole population of people, it also smacks of paternalism — as if users had no mind of their own and needed designers to understand what is good for them
- The statistical definitions, mentioned in the first bullet, agrees with the above in denying users a voice, but moreover, it cannot cope with users' agency, leaving no space for users to define themselves and act in their own interests

Fundamentally, the semantic turn requires respect for the concepts, values, and goals of those affected by the technology being designed. The axiom on meaning entails as much. Respect is granted by attentive listening and acknowledging what people say — not necessarily complying with what they want, but giving fair considerations to their views and interests. For designers, listening may take several routes from engaging in ethnographic research to inviting those interested and able to participate into a development team. Participatory design starts with overcoming the stereotype of the voiceless user.

Product semantics began replacing the concept of an average individual, THE user, by *networks of stakeholders*. By definition, stakeholders

- Claim their stake (interest) in a technological development (a design or its consequences)
- Are experts in their own worlds and usually very knowledgeable about the stake they claim in that development
- Are willing to act in support or opposition of that development

- Are willing to mobilize the resources they command: information, expertise, money, time, connections to members of their communities, and the powers of the institutional roles they occupy.

In product semantics, the acknowledgment of stakeholders was prompted by the obvious recognition that those who buy a technology may not be the ones who use it; that there are far more and highly differentiated kinds of people involved in bringing a design to fruition; that stakeholders tend to be not only well-informed but also creative in their own terms; and far from the predictable and simple minded caricature that the concept of THE user makes them out to be.

By turning away from the concept of the serviceable stereotype of THE end-user to that of a network of stakeholders or community amounts to a shift in the conception of design from a technical or rational problem-solving activity (Simon, 1969/2001) to a social process that relies on stakeholders with different and potentially conflicting interests. Rittel's distinction between tame and wicked problems (Rittel and Webber, 1984), reviewed in Section 1.4, is a clear statement of this difference. Designers need to enroll stakeholders in their projects, convert opponents into supporters, negotiate diverging perspectives, utilize diverse expert knowledges that are found elsewhere, and rely on stakeholders to move the development of an artifact forward, or they simply fail.

Stakeholder conceptions are hardly unfamiliar to designers. Designers always had to deal with sophisticated clients and spent much of their time on presentations of their ideas rather than creating them. Designers know well that they almost always face supporters and opponents, sometimes behind the scenes, and mostly unidentified. Instead of complaining about something from which there is no escape, the politics of stakeholders needs to be acknowledged and worked with. The polyphony of voices of all those involved needs to be embraced and listened to. To find ways to work with the stakeholders of the technology under consideration is to succeed in the postindustrial era where authority and knowledge are no longer centralized and institutionalized, but distributed over many active stakeholder communities. Seeing users, bystanders, collaborators, and opponents as stakeholders grants them the respect they deserve.

2.5 Second-order understanding

As elaborated in Section 1.3, meaning is always someone's meaning. Human-centeredness must therefore acknowledge that the meanings of interest to designers are embodied in individuals, who, as members of communities, coordinate their understanding by interacting with one another. As such, individuals cannot escape speaking from a position — as a designer, researcher, investor, sales person, or user — and designers need to understand the positions from which their stakeholders understand their world. The following shows that this kind of understanding is incommensurate

with the paradigm of Renaissance science, which fueled the industrialized West and is now being challenged. The semantic turn presents an exciting alternative to the traditional form of knowing.

To start the argument, it is not hard to see that professional designers of technology ought to have an understanding that *differs nontrivially* from that of the users of this technology. For one, designers are educated professionals and occupy a position from which they can act in ways that users might not. During the industrial era this difference was viewed largely as a gap in sensitivity and knowledge between expert designers and the masses of ignorant users. However, the general increase in technological sophistication and the acknowledgment that artifacts could mean different things to different people in different circumstances, and in different communities, renders the idea of such a gap no longer tenable. Today, we need to realize that all stakeholders, by the very fact of their living and asserting themselves, are experts in their own lives, and this applies to designers and users as well. *Designers and their stakeholders merely understand differently,* and the issue is how human-centered designers can bring these different kinds of understandings into a design.

When artifacts are designed to make sense to others, two intertwined understandings are necessarily involved: (1) designers' understanding of the artifact being proposed and (2) designers' understanding of different users' understandings of that artifact. While our own understanding of our world usually is perfectly clear, when observing artifacts in the hands of others, it is often surprising to see how different these others' understanding can be. Such differences speak against the most widespread assumption that all rational individuals must see the world in the same way or be wrong. When designers are asked to check their conceptions of what they are designing against how users see a design, designers usually are either stunned in disbelief (how could they not comprehend the obvious) or are profoundly humbled (how could my assumptions be so wrong). It is when experiencing that our own understanding of something differs from that of others and willing to explain and learn from these differences while respecting those involved, that questions of interindividual meanings arise.

We must realize that the understanding of someone else's understanding of something is qualitatively different from the understanding of that something. Understanding someone else's understanding is an *understanding of understanding*, an understanding that recursively embeds another person's understanding in one's own, even if, and particularly when, these understandings disagree, contradict one another, or are thought by one to be wrong or appallingly unethical. This recursive understanding of understanding is a *second-order understanding*. Inasmuch as human-centered design is fundamentally design for others, it must be grounded in second-order understanding.

Figure 2.8 locates a semantics for artifacts in a designer's second-order understanding. There is the world that a designer understands first hand, and there are the many worlds that stakeholders understand. A designer's second-order understanding makes room for other stakeholders' potentially

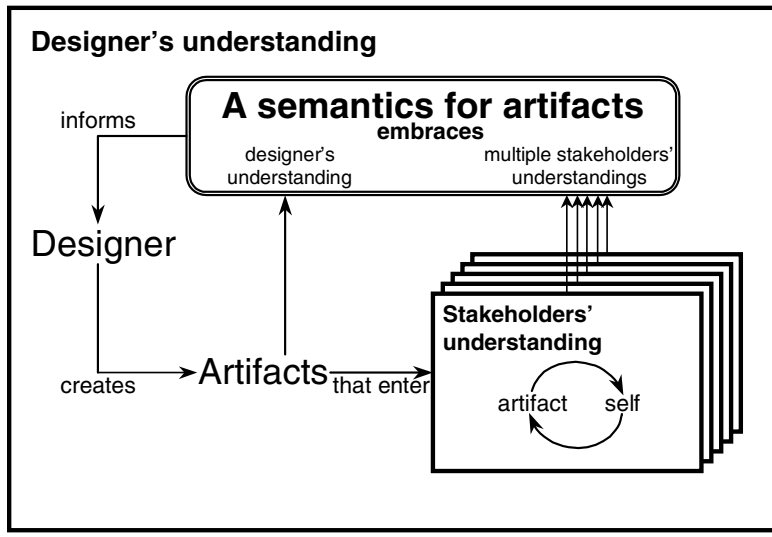


Figure 2.8 Design semantics as second-order understanding.

rather different worlds, worlds that differ not merely in clarity or in the perspective from which different stakeholders are seeing, but foremost in their construction, as may become evident in the use of incommensurable vocabularies, logics, values, and goals. Each is positioned in his or her own understanding; the designer in the designer's world, the client in the client's world, the ecologist in the ecologist's world, and the user in the user's world. Design semantics crosses these worlds by relating the artifacts as seen by designers to the same artifacts as interacted with and described by their stakeholders.

How can such crossing of worlds be accomplished? Actually, by asking questions about the meaning and use of con-sensual objects — con-sensual being jointly sensed as in the right image in Figure 2.9. Such questions can open a window into the conceptions held by others. Answers to questions like "What does this A mean to you?" "Why do you use this A the way you



First-order understanding

God's eye view renders
the observed as mechanism
without capacity for understanding

Second-order understanding

Each assumes the capacity for understanding artifacts,
for understanding other's understanding of artifacts
and ways to interactively confirm these understandings

Figure 2.9 Observer with first- and with second-order understanding.

do?" and "What did you think when using this A?" relate three things: First, there is the questioner's object A. A is what it is, but only for that questioner. Second, the "this" refers to con-sensuality, the questioner's assumption that both see and talk are about the same con-sensual object. Third, the reply "This A means B to me" enables the questioner to compare his or her A with the asserted B and construct an explanation, its meaning. Figure 2.8 shows how design semantics weaves the two worlds yielding A and B, respectively, into the designer's second-order understanding.

In other words, Figure 2.8 depicts how the stakeholders' worlds appear in the conversations with a designer's world. This may appear obvious and simple but has profound implications for how designers must now proceed and how design discourse needs to develop without falling into the trap of conventional thinking. Second-order understanding is important enough to contrast it with the kind of understanding that human-centeredness must turn away from, conventional first-order understanding. First-order understanding, the understanding of something that cannot understand, is manifest in representational knowledge, which is what the natural and behavioral sciences provide. These sciences require that their practitioners make three metaphysical commitments, which are incompatible with the semantic turn in design:

- The commitment to relying on *causal explanations*. Causal explanations are grounded in the belief that physically measurable forces and events determine what happens in the world. Causality is incompatible with human agency, however. If humans had no choice, taking responsibility for their actions would be meaningless. Unless freedom of choice is an illusion, as some natural scientists claim it to be, design activities that change the world in ways that cannot come about naturally defy the determinism of causal explanations. Commitments to causal explanations limit us to first-order understanding and prevent the recognition of human agency, design, and meaning (as elaborated in Section 2.3).
- The commitment to *objectivity*, defined as the quality of propositions, theories, and laws to describe nature without the interest and involvement of the theorists as human observers. The idea of an observer-independent world is evident in the concerted efforts on the part of scientists to prevent or reduce what is called observer "biases," and replace "unreliable" or "subjective" observers by mechanical measuring instruments. That this phenomenon is seen as a threat to objectivity attests to the difficulty, if not impossibility, of scientists being able to observe from no particular position, and to understand without eyes, a brain, a body, and language. Moreover, natural science discourse is constructing its object, nature, as unable to understand and respond to how it is studied, who studies it, and why. Thus, the commitment to objectivity not only prevents scientific observers from interacting with their object of observation, it declares this a meaningless undertaking. While objectivity is not a logical requirement of first-order understanding, it treats people as

subjects, dismisses manifest variations in their perception as subjective, and has no place for second-order understanding (of how different people could conceive con-sensual objects differently).

- The commitment to *consistency*. Consistency is a property of systems of explanations. Striving for consistent laws and theories of nature leads to the conception of a uni-verse, which is inherently regular, rule governed, and ultimately mechanical or computational in nature. Alternative versions, multi-verses, cannot coexist and are dismissed as subjective, illusionary, or mistaken. First-order understanding cannot consider incommensurable world constructions as equally valid, which is where second-order understanding begins.

These are metaphysical commitments, beliefs, or convictions as there can be no evidence for any of them. When taken seriously and acted upon, the first-order understanding they provide rules out as implausible, design that transforms the world in ways not predictable from natural laws and must dismiss alternative versions of con-sensual objects, which is the basis of intersubjective meanings. First-order understanding is just not powerful enough to support human-centered design.

Having shown from where the semantic turn must depart, I now examine where second-order understanding leads.

- Second-order understanding treats humans not as mechanisms but as *knowledgeable agents.* By definition, knowledgeable agents, stakeholders, and intelligent users are held accountable for their actions. While not every observable human behavior is conscious and intentional, a conceptual framework that allows the attribution of agency is superior to one that precludes this possibility.
- Second-order understanding grants knowledgeable agents their own understanding, the ability to apply their own logic to their own actions, and construct their own worlds. Granting this possibility to others is far from subscribing to radical relativism. One's world constructions are not arbitrary. They must prove viable in guiding one's interfaces with artifacts and communication through them with members of one's community. Being alive demonstrates this viability at least regarding constructions of reality that are actually applied. While it is not impossible that different users' understandings turn out to be alike, the same, or generalizable within a particular community, second-order understanding cannot presume commonalities and single truths. It embeds different understandings in one's own understanding.
- Since the meanings held by others cannot be observed, they may be inferred from the interactions they inform and from how they surface in conversations, that is, second-order understanding is *dialogical* in nature. By contrast, first-order understanding is *monological* in the double sense of (1) privileging only one logic, the logic of the observer, and (2) communicating in one direction, from the vocal expert to the voiceless

layperson, and from the master designer to the design-ignorant user. Dialogue opens spaces for different understandings to be coordinated and to coexist as such, the designers' included. Referring to Figure 1.4, the dialogical nature is depicted by the arrow from professional design to design in the realization of everyday life, present for human-centered design but absent for technology-centered design.

- Second-order understanding recognizes meanings as polyphonous, enabling *a range of options* for human actions. While meanings are not causal — as reduced to in the natural sciences — or computational — as theorized in the cognitive sciences — or representational — as conceived of in semiotics — respecting someone's meanings of artifacts nevertheless informs us of what their users consider possible and appropriate within particular situations.
- Understanding is always embodied in or enacted by people. Those who inquire into others' understanding, ethnographers, for example, must *include themselves in the conversations* in which second-order understanding is to surface. Detached observers, researchers taking a God's eye view (Putnam, 1981) of the world and of other fellow beings, can speak only about their own conceptions. They are constitutively prevented from appreciating very different understandings in action and are likely to dismiss others' deviating from their own understanding as subjective, biased, or unreliable.

While second-order understanding (of stakeholders' understanding of artifacts) may include first-order understanding (of artifacts by their stakeholders), first-order understanding does not have the logical power to speak of second-order understanding. This is why the semantic turn cannot rely on semiotics or cognitive theories, which are objectivist and individualist, respectively, and remain caught in first-order understanding. Figure 2.9 attempts a graphical comparison of the two kinds of understanding.

Second-order understanding does not come easily. It calls for a conceptual gestalt switch that involves seeing meanings as recursively woven into interaction with others. Without second-order understanding, the new information technologies, which thrive on the multiple perspectives that their users bring to them and to each other, remains incomprehensible. The reason that technology is so conceptually challenging is that most theories of it are stuck in first-order understanding. Second-order understanding is a prerequisite for design in the postindustrial age. Without embracing the concept of second-order understanding, readers may find what follows incomprehensible.

2.6 Ethics in a design culture

Design is more common than recognized. For instance, birds are architects of their own nests, leaf-cutting ants practice agriculture, crows use tools, and chimpanzees form coalitions against rivals. These "innovations" have developed incrementally and over considerable periods of time, so slowly that

they appear like inherited traits, but they are not. The communication among bees is culturally determined, not biological. Primitive tools have been found with humanoid skeletal remains dating to about 100,000 years ago. About 50,000 years ago, artifacts began to become more complex, cave paintings emerged, settlements developed, and long-distance trade started to spread various practices. This sudden change has been correlated with the appearance of language (Wade, 2003). Technology and language are both combinatorial systems — using a limited number of elements to create new combinations. It stands to reason that early humans regulated their affairs by bringing these systems together. So, design has been with us for a long time, and the use of language is a constitutive part of it.

It already has been suggested that design is an everyday activity. Planning a meal, planting a garden, decorating one's home, writing poetry, and even voting for a political candidate fit Simon's (1969/2001) definition of design as an effort to change existing conditions into preferred ones. But why do people not recognize these ordinary activities as design? The reason is historical. As elaborated above, design — industrial design, design as a profession — emerged in response to industry's need to expand its markets. It usurped what the crafts did since the Middle Ages and took charge of voiceless users and consumers. In the functionalist society of the industrial era, craftsmen ended up on assembly lines and ordinary people eagerly surrendered their design abilities to institutions that were thought to know better and provided them with work and essential products. But competition in the market place and a certain amount of material abundance changed all that. It created previously unavailable choices and began to undermine the flagrant inequities of the industrial era.

The changes in design thinking due to the current rise of information technology are far more radical than the ongoing demands made by markets. Computers basically are general-purpose machines that do nothing without being programmed. Not only can ordinary computer users install various operating systems, acquire specialized software, and link different communication services, but what is far more important is that this technology enables its users to customize their artifacts, to redesign them in order to make sense of them and feel comfortable within the world of which they are a part. In this respect, a computer is a very new kind of artifact, one that offers choices, not merely between available products but in its functioning. Communication technology does much the same. It provides channel capacity with few restrictions of what to say over long distances and to more people than can be reached naturally. Artifacts designed to offer choices were unthinkable during the industrial period. Information technology not only values everyday design, it depends on it. To be fair, historically, systems of components that users could use variously have been known for some time. For example, musical instruments enable infinitely many compositions to be played, constrained only by what artists can do and connoisseurs are willing to listen to. Furniture is such a system as well, leaving ordinary users plenty of room for self-expression. Not all such systems are literally designed;

language, for example, largely evolved and continues to evolve in use, and so does music. What these systems have in common is their designability. They exist not so much on account of professional designers, but on their ability to amplify design in everyday life.

The suggestion here is that information technology and human-centeredness are in the process of ushering into existence a historically unprecedented design culture. To appreciate this new culture, compare science with design. The objects of scientific discourse are generalizations of past observations — finding theories and laws of nature whose truths are established without reference to scientific observers. Design, by contrast, searches for present openings to change something so it is better, to find what can be moved and consider moving it, ideally for a large community of stakeholders, and unquestionably including designers. If one were to add the hours that a population of people spends on improving its members' lives, planning, reconfiguring, communicating, or designing something — professionally, in everyday creative activities of making something about what is available, including building, writing, and performing — and compare this with the hours that people spend in testing the truths of propositions — from scientific theories and journalistic accounts to legal proceedings — the judgment is clear: Contemporary culture is dominated by design activity, not by science. This is evident in the increasing willingness of people to assert themselves and act on their own behalf; to think and write about the future; to experiment with newness; to create art, entertainment, and technical gadgets; to grow businesses; to become stakeholders in public projects; to enroll others into their projects; or to run for political office. While scientific activities are highly valued, rewarded, and heavily relied upon, as they should be, the scientific culture of the Renaissance has been surpassed by a culture that has moved design activity into its center, *a design culture* for short. Figure 2.10 compares the dominant features of both.

Functionalist Society	Design Culture
Technology-centered	Human-centered
Hierarchical knowledge structures	Stakeholder networks, advocacy, and markets
Rationally derived and assigned functions	Interactively negotiated and supported meanings
First-order understanding	Second-order understanding
Technology serves to predict and control	Technology facilitates design in everyday life
Finding technological solutions to social problems	Proposing desirable futures and paths to them
Re-searching past records for patterns	Creating and exploring required variability
Know-what was, how something worked	Know-how to transform possibilities into realities

Figure 2.10 Ingredients of emerging design culture.

Nelson and Stolterman (2002), who develop a slightly different view of design, nevertheless come to a similar conclusion about the need to acknowledge design culture in contemporary society. They discuss design as based on four pillars: the real, service (to the community), systems or design thinking, and holistic considerations. Revisiting Figure 1.4, we can say that human-centered designers, working from within a community, do assume the responsibility of serving the larger whole and implicitly create a design culture that embraces everyday design activities. Technology-centered designers, arising out of the common ground of design in the realization of everyday life, but working from outside a community of users and according to specifications that limit their concerns to what is demanded of them, implicitly encourage a functionalist society. This society is marked by hierarchies of authorities communicating specifications from larger systems to their subsystem, and ultimately to individual users of technology.

In view of the role of professional designers in a design culture, saying that the current societal shift is toward an information society turns out to be a misconception. The reference to information keeps one foot in the industrial era, in a culture that valued scientific knowledge, factual information "about" things, over other ways of knowing. It legitimized "know-what" through formal education, publication, and science, while regarding practical "know-how," emotional involvement, the conceptions negotiated in public conversations, and the design of technology as an inferior kind of knowing. The artifacts that fascinate now are precisely those that enable us to alter our own world and the technology that attracts more and more users, the Internet, for example, answers questions concerning "where to" get something, "how to" do something, and "who can" help, rather than searching for evidence for particular propositions. The shift is not from scarcity to an abundance of information, but from the know-what exists and is (or was) true provided by science to the know-how to undo, rebuild, change, and construct, which could become true, in other words, design.

Probably the most important reason for the increasingly prominent design culture is that design activity is *intrinsically motivating*. Making something or having a hand in creating something worthwhile is always inherently satisfying. This is why poets write, even without rewards, why designers design, and why people paint their living space, buy their own furniture and arrange it to their liking. Design is a way to understand things, to make them meaningful, to feel at home with them, and to make them part of one's life. In that process, people realize who they are to themselves and in view of others, of the members of their community. This is true not just for professional designers. It occurs in everyday life. Alternatively, when people are prevented from deciding who they are or how to live their life, they are essentially reduced to appendixes of systems, machine parts, robots, or prisoners — anonymous and interchangeable. Fundamentally, humans resist being imposed upon and always seize on opportunities to realize

themselves in their own terms and with the artifacts they can handle. This points to a basic proposition in human-centered design:

Design constitutes being human.

Some designers fear that this proposition, together with the prospect of an emerging design culture, threatens their livelihood. These worries are justified only for industrial era conceptions of design, such as Simon's (1969/2001), and the failure to realize that the trajectory of artifacts, described in Section 1.2, has moved on from products to other things worthy of attention. The advent of desktop publishing computer programs in the 1980s exemplifies this threat. Suddenly, an ordinary computer user could play with fonts and use different layouts to design stationery, pamphlets, and books in shorter time than it would take to explain the task to a typographer. But desktop publishing could threaten only those who tried to hold on to computerizable skills, suddenly finding themselves exposed and empty handed. This does not need to be so, however, provided professional designers are willing to apply their own competencies to themselves, redesign themselves, change their self-conception, and move to where others cannot go. The ability to redesign themselves as a profession distinguishes between human-centered designers in the postindustrial culture and designers of a bygone era. Following are three imperatives for professional designers in the emerging design culture. The first is ethical, the second practical, and the third political.

- *Design artifacts that increase their stakeholders' ability to design on their own*. This is what poets do most naturally: inventing novel uses of language that in due time enriches the lives of their readers or listeners. By analogy, designers can do no better than making compelling proposals that could not have occurred without them, yet speak to their stakeholders' lives. Once accepted and realized, designers must move on. As an ability that constitutes being human, design cannot be "disciplined" for long or usurped by persistent interest. It needs to increase options.
- *Invite the stakeholders in a design to participate in the process*. Since design may lead a development but cannot be realized without others, and since it is impossible for anyone to know the concerns of those who matter, the stakeholders in a design, it is imperative for professional designers to take part in the deliberations among the stakeholders of their designs. Participation makes the resources of the stakeholder community available; renders the stakes that people have in a design addressable; enhances the acceptability of design conceptions and criteria, and increases the likelihood of realizing a design. Given that design is intrinsically motivating for everyone, the more stakeholders have a hand in a design, the more likely will it come to be.

- *Assure that the discourse of the design community grows and remains reliable.* Being able to draw on histories of successful designs, apply reliable design methods, and utilize established techniques for validating meaning claims presupposes a collective effort to contribute to this discourse, evaluate it in practice, and disseminate it to the members of the design community. Educational institutions have much to contribute to keep this discourse viable and up-to-date. It is the public acceptance of this discourse that awards designers the ability to author compelling proposals, backed by defensible demonstrations of the virtues of their designs.

As design becomes a dominant feature of postindustrial culture, designers need not fear losing their jobs. By committing themselves to a human-centered design discourse, one that fully embraces the new participatory design culture, designers can distinguish themselves publicly by being able to propose artifacts that nobody else could imagine, that are substantiated better than anyone else can, and that mobilize the stakeholder support required to bring a design to fruition.

chapter three

Meaning of artifacts in use

A theory of meaning for artifacts in use addresses how individual users understand their artifacts and interact with them in their own terms and for their own reasons. It is tempting to regard such a theory as a psychological or cognitive approach to the meaning of everyday things, as suggested by Norman (1988). However, because use essentially involves interactions, such a theory must embrace the nature of artifacts as well. It must follow Wittgenstein's suggestion to locate the meaning of artifacts (words) in their use (Wittgenstein, 1953:11–14, 421, 432, 454, 489), not as referring to other things. Such a nonrepresentational theory of meaning naturally embraces Gibson's ecological theory of perception, which concerns the fit between human activities and how the material world affords them, but it must go beyond asserting this fit by addressing user-directed interactivity. It must recognize the dynamic nature of use. The social context of use is described in Chapter 4.

Any theory necessarily reflects the understanding by outside observers, by theorists who stand apart from what they theorize, in the case of designers, with the intent of improving or designing the artifacts that would support desirable user–machine interactions. Granting users the ability to understand in their own terms, Section 2.5 defines the understanding of users' understanding as second-order understanding and distinguishes it from the ordinary understanding of tangible artifacts, artifacts that may be understood but cannot understand. Designers are concerned with proposals, plans, drawings, models, prototypes, and arguments for how to realize them, which are artifacts that are quite different from what their stakeholders or users live with: products, objects of exchange, services, appliances, consumable goods, gifts, markers of identity, and means to accomplish ends. Thus, a theory of meaning for artifacts in use by others has to be embedded in second-order understanding.

Artifacts are not entirely stable entities. Their meanings change with use. For example, most uses start being guided by users' preconceptions. As users become competent, they change their understanding without a necessary end in sight. People learn all the time, which means changing one's understanding. Also, artifacts tend to undergo several transformations in meaning,

without necessarily changing their material composition. A valuable dinner plate stacked in a cupboard is different from one on the dining room table or one displayed on the wall. A designers' theory of what artifacts mean in use must embrace not only changes due to learning but also these different kinds of uses. Designers are not the only conceptualizers. Their version of reality is not more legitimate than the realities of divergent others.

This chapter outlines the key vocabulary of this theory, much like a tool kit, to be used for analyzing, conceptualizing, and ultimately designing artifacts for use by others who interact with their world according to their own concepts. The semantic turn demands of designers that they shift their attention from the design of material artifacts to the design of artifacts whose meanings enable desirable interfaces to arise. The material artifacts that designers can control through design, planning, and compelling arguments are mere supports of how people use them. Section 3.1 presents interfaces as a new kind of artifact. Section 3.2 considers their disruptions as threats to the usability of artifacts, which is what designers need to avoid. These sections are followed by an account of three levels of uses that designers need to enable: recognition (3.3), exploration (3.4), and reliance (3.5). Each develops concepts and outlines tools for designers to enable use on these three levels. Section 3.6 concludes with several design principles for use.

3.1 Interfaces

Interfaces are a new kind of artifact. They can be observed everywhere but have become objects of design only since their more obvious use in personal computers. Typically, personal computers are extremely complex electronic devices whose operational architecture escapes any single individual's understanding, especially the users'. One purpose of interfaces is to render usable that which can no longer be understood in functional details. In the 1970s, when computers were still mainframe machines, only a few experts could handle them. The shift to an information society was ushered in largely by the design of interfaces that distinguished between user-relevant and user-irrelevant features of computers and could present the former in easily recognizable and actionable terms. But this is not their only purpose.

Today, the interface has become the prototypical artifact of the postindustrial era. The semantic turn generalizes this concept to all human interactions with technology. The face of an analog clock that hides the mechanism for turning its hands is an interface — not a particularly interactive one, but one that allows users to coordinate their own activities with other users of watches. Handling the remote controls of a VCR, sitting comfortably on a chair, cutting fabric with a pair of scissors, and walking up a flight of stairs are all interfaces in the sense that they cannot be explained by human cognition, nor by the physical properties of the artifacts handled in the process, but by the interactions between these two rather different ways of being. An important property of interfaces is that they weave human sensory–motor coordination and an artifact's reactivity into a dynamic whole

that human participants in the interface can understand and feel comfortable being in. The human body is as much part of an interface as the artifact interacted with.

Figure 3.1 shows a range of interfaces: From left to right, interfacing with a traditional wall phone, holding on to an adjustable walker, turning a crank, using a remote control, pushing buttons, and obtaining feedback from a screen, cutting paper with scissors, etc. Even such mundane activities as sitting on a chair can be described as an interface. Lisa Krohn's truffet is not just a stool. It responds to one's shifts in weight and its layered wood and foam rings afford easy carrying from one place to another. All of these interfaces consist of interactions, very different ones to be sure, not only as in the more frequently mentioned human interfaces with computers. Human knowledge of technology arises in interfaces with that technology, not independent of it. Users may be shielded from unnecessary complications, whether this pertains to the workings of a gambling machine, the wiring of a lighting control panel, or the mechanics of an airplane, all of which are experienced as interfaces. Where design is concerned with use, interfaces are the issue.

Because of the newness of this artifact, several concepts of interface have evolved, most of them not yet quite adequate. Largely informed by the tradition of graphic design is the equation of interfaces with systems of recognizable icons on the surface of a computer screen, readable and attractively composed. This concept ignores the two central features of interfaces, interactivity, users responding to and changing what they see, and dynamics, the particular coupling of humans and machines that evolves and creates movements in space and in time. There is the German word *Schnittstelle* that emphasizes the place where the world of humans can be separated (by a cut = Schnitt) from the world of machines. This concept fails to recognize the between-ness, already indicated in the word "*inter*-face." In an interface, user concepts penetrate the domain of a machine to the extent that the machine affects them and relevant parts of the structure of the machine enter human cognition as needed to inform actions. Computer programmers tend to favor another interface concept. They often distinguish the computational engine, which does the actual work of the computer, from the interface, which is that part of the machine that is programmed to represent that work on the monitor. This concept renders the design of computer interfaces as an engineering or coding problem, which falls short of acknowledging the part that human cognition plays in the process. The word "interface" is also used for the protocol that governs the transmission of data between two machines. When applied to human–machine interactions, this would amount to defining a code that users would be asked to learn, thus trivializing the role of human intelligence and demanding that humans conform to computational terms. It reduces users to mechanisms of sorts.

Here, *interfaces are prolonged and ideally intrinsically motivating interactions* between human actors and their artifacts. The interactions of which an interface consists arise spontaneously in a wide range of practical situations,

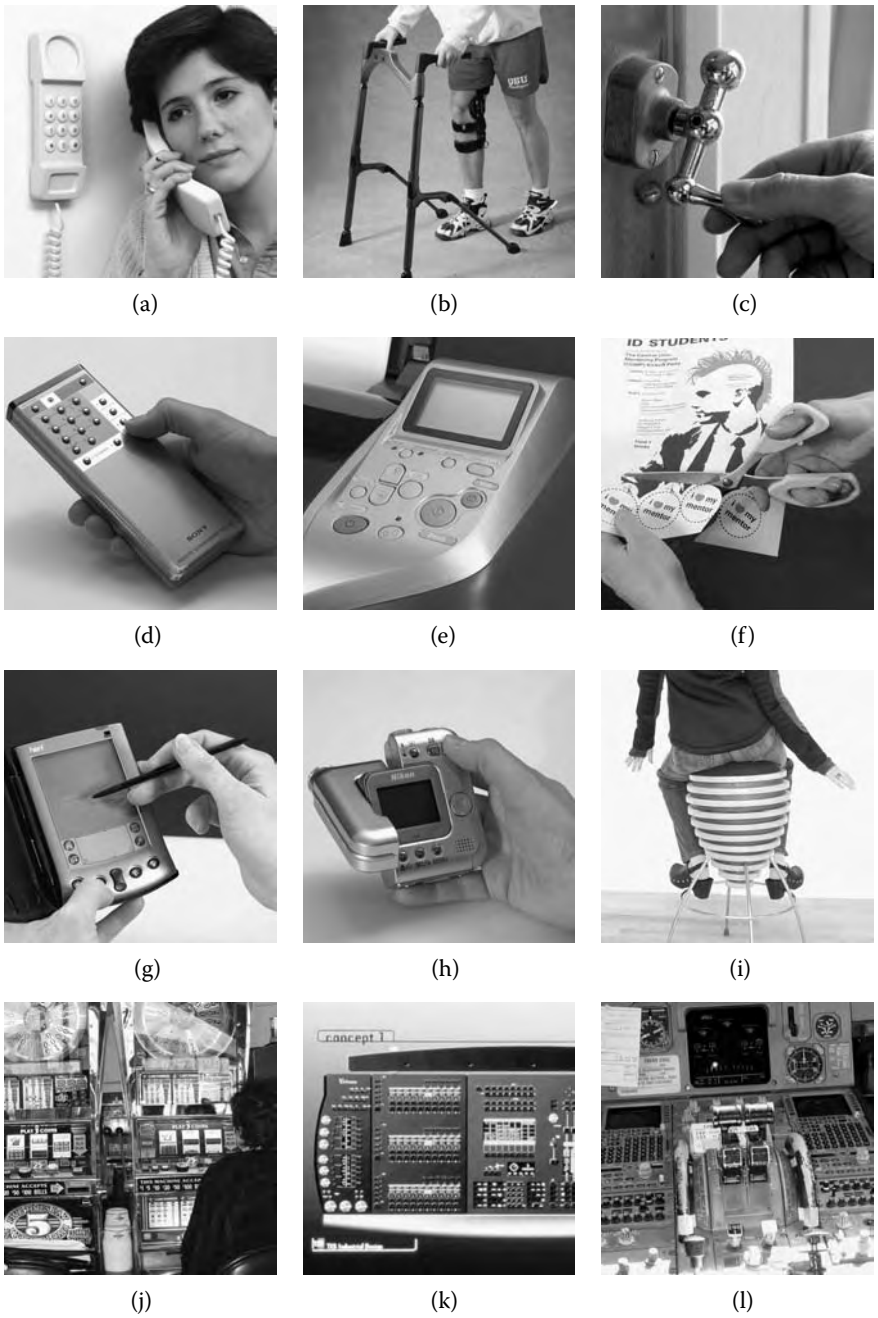


Figure 3.1 Diverse interfaces. (a) Telephone; (b) walker; (c) crank; (d) remote control; (e) printer controls; (f) scissors; (g) touch screen; (h) digital camera; (i) tuffet by Lisa Krohn; (j) one-arm bandit slot machine; (k) stage lighting panel by Tim Terleski; (l) airplane cockpit.

from handling simple tools, as in Figure 3.1, to participating in complex social systems, computer interfaces being somewhere in between. Interfaces have symbiotic qualities. Human use of technology always extends some human abilities, often by accepting some less important constraints: human memory by reading and writing, preventing the participation of smell, taste, and sound; human decision making by delegating routines to computational algorithms, restricting knowledge to what is computable; and extending the reach of the human voice by the use of the telephone, preventing visual, tactile, and olfactory forms of human communication. Whereas some futurists have written about the symbiosis in terms of cyborgs, wholes composed of living and mechanical parts, humans with artificial organs, for example, the preceding definition of interface emphasizes a fundamentally asymmetrical relationship between human agency, the ability to understand and act purposively, and material artifacts that, while made by humans, are fundamentally different from them. To understand how these two worlds interact, we must keep their different modes of operation separate, least of all because humans are not designable and machines cannot recognize meanings. For these very reasons, interfaces evolve in somewhat unanticipated ways, but it is the human user who can change the course of their evolution, a machine merely enables or constrains.

How interfaces happen can be described from within a user's understanding, from the sensory-motor experiences of a participant, or from the position of an observer, including that of the artifact's designer, who may be intent on improving an interface by redesigning the artifact (not the user).

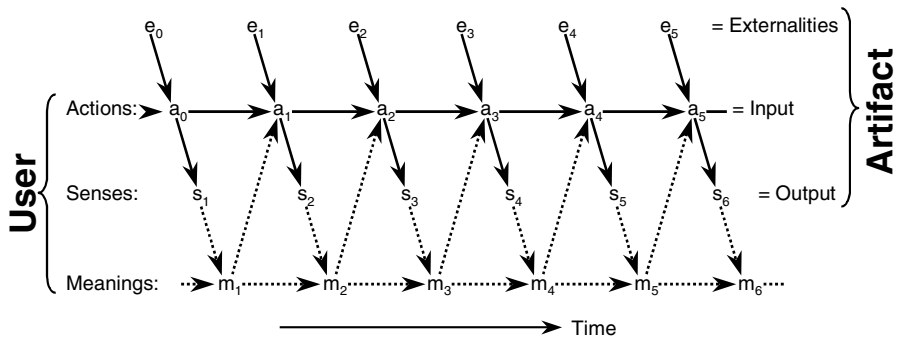


Figure 3.2 Interaction protocol of an interface.

From outside an interface, an observer will describe an interface as a sequence of human actions, followed by the artifact's reaction, as depicted in Figure 3.2. In that figure, arrows indicate dependencies over different times t, indicated as subscripts. Solid arrows depict the determinisms of the artifact, and dotted lines indicate what the human user of the artifact embodies. Technological artifacts are causal mechanisms. Their internal makeup accounts for their responses to (a) external conditions e_t at various times t,

not under control of or unknown to the user, so-called externalities, (b) the history of inputs received in the form of human actions, $a_0 \rightarrow a_1 \rightarrow a_2 \rightarrow \ldots \rightarrow a_t$, and (c) the current user's action a_t, and (d) displaying certain but not necessarily all internal states $s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow \ldots \rightarrow s_{t+1}$, including the immediate response s_{t+1} to a_t. Machines can be constructed to conform to a collection of triplets: the machine's present state s_t at time t, the input a_t at time t, and the machine's next state, s_{t+1} after one unit of time. Triplets of this kind describe the consequences of pushing a bell button as well as what pointing and clicking with a mouse on a computer screen commands the computer to do. Some consequences of these actions, s, are displayed for users to see; others remain inaccessible or internal to the machine.

As previously mentioned, the dotted arrows in Figure 3.2 depict the part of the interface for which the human user is responsible. Human participation complements that of a machine, but it cannot be explained causally. Humans do not act on what an artifact physically is or displays but on how they sense it, what it means to them, and what they wish to accomplish. Figure 3.2 shows the intersection between these two domains of existence, an artifact's behavior and its user's understanding. To understand an interface, we must keep separate the two domains of being. In the natural sciences, including ergonomics, the preferred construction of the human user is as another mechanism, unreliable perhaps and perceptually biased, as suggested in Figure 2.1. Cognitive science, for example, prides itself on developing computational models of users, which are complex but essentially causal mechanisms. While humans can opt to follow strict routines, they are not destined to do so and can achieve mechanistic behavior by surrendering their agency to an authority or by submitting themselves to rigorous training. Understanding the human part of interfaces nonmechanistically is the challenge of second-order understanding, one that human-centered designers always face.

From within an interface, those involved find themselves in a particular sensory–motor coordination, acting either to sustain an intrinsically motivating sequence of senses, one that is enjoyable or fun to experience; and/or bringing forth an extrinsically motivated sense, a goal or objective. For example, when a car is driven, the interface amounts to a person literally sitting in the driver's seat and using his eyes, ears, hands, and feet to preserve the correlation between his conceptions of the car, the road, and the destination. In that process, the driver places the choice of an action a_t — moving the steering wheel, pressing or releasing a pedal, shifting gears — in the context of where he started, m_0, and where he wants to be, m_{final}. The interaction $s_1 \rightarrow a_1 \rightarrow s_2 \rightarrow a_2 \rightarrow s_3 \rightarrow a_3 \rightarrow \ldots \rightarrow s_{\text{final}}$, where in the world of the user, s_t is the sense made of the output of the artifact, and a_t is the user's acting on that sense, serving as input to the artifact — is guided by meanings $m_1 \rightarrow m_2 \rightarrow m_3 \rightarrow \ldots \rightarrow m_{\text{final}}$ or the user's conceptual models comprising these meanings. Along the way, s_t might be interpreted as an obstacle to be circumvented, as a traffic sign to be obeyed, or as signaling an opportunity to pass another vehicle. Such interpretations occur in a user's understanding,

not in the mechanism of the artifact. The meaning of driving comes from putting our sense–action–sense sequence in the context of our world construction, the past, the current, and the desirable. This amounts to knowing what the various controls do, knowing the range of actions available when sensing s_t, and knowing how a_t will change the current s_t to a desirable next sense s_{t+1}. In driving a car, just as in interfacing with any artifact, the user works from one sense to another. Monitoring the sequence of senses is all that a user can do. In a good interface, we always know where we are, s_t; and what we need to do, a_t; the consequences of our action, s_{t+1}, are available without delay. In a good interface, we also always know where an action, a_t, leads to, s_{t+1}, and whether we are on the way to where we want to be, m_{final}.

In writing a book, for instance, writers can observe themselves composing a text, letter by letter, highlighting, cutting, and pasting portions of that text and using various tools that support writing: a thesaurus, a spelling checker, different typographies. By pointing, clicking, and using a computer keyboard, authors shape their text into what they hope others will understand. In that process, writers are their text's first readers, and they are likely to read it from different perspectives, observing its length, editing with the tracking tool, printing a hard copy of the document, and so on. Thus, here too, authors steer from sense to sense, guided by the meanings they invoke for them, continually revising their assessments of where they are and where they want to be, in the case of writing, in view of what imagined readers are expected to understand.

The interface with scissors is far simpler than that of writing. Unlike in writing, which is always to be read by someone, cutting a piece of paper rarely involves second-order understanding. Scissors do not require much thought to understand how they work. They allow one to put fingers in convenient openings, and opening and closing the blades in succession comes naturally. Because the blades extend beyond their handles, one is able to observe the effects of the cutting action and to ensure that the emerging cut follows a prescribed or imagined line. Here too, interactions with scissors and paper are guided by continuously monitoring the unfolding sequence of senses from a beginning to an end. The use of scissors mediates between the paper and one's own criteria for what is to be accomplished.

As shown in Figure 3.2 and in the previous examples, we are only partly in charge of the senses that are brought forth. A driver may not react as quickly as possible; a computer may run out of memory; and scissors may not be strong enough to do the job. In interfaces, constraints always exist to be reckoned with. In use, at any present moment, *the meaning of an artifact is the range of imaginable actions and senses* that users can expect, both as a result of their action and in the absence of actions. In the context of use, meanings are not referential, connotative, or associative, but situational, locating possible actions and expected senses in processes of interaction, showing the user of artifacts how the process could continue. This meaning is part and parcel of a concept — correct or incorrect — of the artifact that one is facing. We can liken the user's world to a conceptual map — without

insisting on its accuracy or representativeness relative to what is. Revisiting Figure 2.4, every sense of where we presently are acquires its meaning by finding a location on that map, suggesting where we are, what we can and cannot do, how far we have traveled, and what paths are available for us to continue on our way. Senses are always metonyms by being in a part–whole relationship to their meaning (on a conceptual map).

While in the context of use, meanings are the imagined possibilities for acting and sensing, what explains the choices among these actions, how an interface actually unfolds, has much to do with human participants' understanding, habits, and motivations: why they are where they are. When users come to their wit's end, when they can no longer imagine what to do, when they see no way to get out of the situation they have gotten themselves into, that situation of total uncertainty and inability to continue has lost its meaning.

To generalize the foregoing:

Humans always act so as to preserve
the meaningfulness of their interfaces.

This proposition could be regarded as a *theorem* in the context of use of the axiom on meaning presented in Section 2.2. Accordingly, the users of artifacts do their best to preserve the meaningfulness of their interfaces with their artifacts, whatever direction this may take them. Designers must do their best to support the continued meaningfulness of these interfaces materially and in ways that come most naturally to the users.

3.2 Disruptions and usability

Because meanings always extend beyond actual senses, they may lead to unfulfilled expectations. A computer freezes unexpectedly. A wrong turn is taken at an intersection. We cut our finger. A usually reliable tool breaks unexpectedly. Martin Heidegger (Dreyfus, 1992:77ff.) calls such a condition "breakdown." However, what breaks down is neither the artifact (although it is indeed possible that it falls apart unpredictably) nor the users' concepts or memories (although they too could be inappropriate, drift, and lose touch with "reality"). What breaks down is the meaningfulness of the interface of which we are a part. This happens when meanings and actions lead us to expect something that is contradicted by what is sensed, when expected and actual senses deviate significantly. Ordinary language has many expressions to distinguish among such disruptions. For example, and without an intended ordering:

- **Glitches**: Execution errors due to drifting into a routine when interactions require attention
- **Inaccuracies**: Straying from a norm beyond what is tolerable
- **Mistakes**: Taking something for what it was not, finding oneself to have made a wrong turn or not followed known procedures

- **Misuses**: Applying something inappropriately or in unintended ways
- **Distractions**: Attending to something irrelevant at the expense of what really mattered
- **Dilemmas**: Being unable to choose between two equally unsatisfactory alternatives
- **Entrapments**: Being stuck in a place without exit, lacking the imagination to proceed, going around a loop without progress

Figure 2.4 can be elaborated upon to include the behavior of a machine, its inputs, internal states and outputs, and the occurrence of disruptions, the mismatch of experienced and expected senses. In Figure 3.3, meaning is invoked metonymically by what is sensed, which always is part of a user's world construction. Meaning translates into appropriate actions and predicts expected senses. In continually meaningful interfaces, actions bring forth what is continuously confirmed and realized (considered real). When experienced and expected senses do not match, disruptions occur.

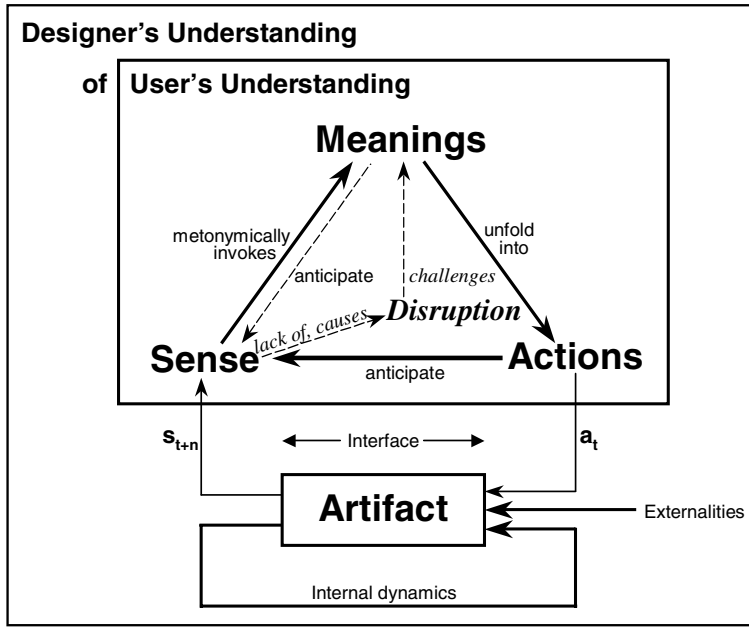


Figure 3.3 Designers' understanding of a user interface.

To recover from such losses of meaning, we need to pause and reconstruct the concept of the artifact or object we are facing. Humans have no access to what "really" is. They are destined to rely on their understanding, their constructions of reality that available senses are expected to confirm. Occasions of disruption are the only moments in which reality asserts itself. However, these moments do not reveal why and how we were wrong, only that our conceptions were inadequate and always only in respect of our actions. Constructions never represent an outside reality; they merely

respond to the history of disruptions, to momentary failures to preserve the meaningfulness of an interface.

There are two kinds of explanations for the occurrence of disruptions: explanations that fault human conceptions, including the actions that follow on these conceptions, and those that fault the artifact for not performing according to the way the user thinks it should. The preceding list of disruptions is of the former kind. From a user's perspective, explanations that pertain to the user's conduct encourage learning. Explanations that hold the artifacts responsible for not conforming to the user's conception include malfunction, equipment failure, material defects, wear and tear, lack of integrity, design flaws, ambiguity, misleadingness, and bad workmanship — most of which blame other people — to which one may add mishaps, accidents, and contingencies (entropy) that blame nature. Blaming external causes signals unwillingness to learn but is of interest to designers because it is where learning is refused, made difficult, or plainly impossible, for example, when mistakes are fatal to the user, that designers may have to find constraints that prevent users from getting into trouble.

Back to disruptions, here are three points and a recommendation:

- Users cannot deliberately cause disruptions in their interfaces. Mistakes are never intentional. This insight goes back to an old Aristotelian axiom, the principle of noncontradiction. When applied to human epistemology, not to ontology, it asserts that humans cannot contradict themselves.
- The causes of disruptions are always found after the fact; they are explanations, reconstructions, interpretations, not observations, and in the case of human-centered explanations, these explanations concern options overlooked, senses misinterpreted, or actions carelessly executed.
- Recurrent disruptions have a good chance of disappearing. Users learn from past mistakes after exploring their reasons.
- All artifacts should be designed so that
  - Disruptions debilitate users only minimally and pose no serious threats.
  - Disruptions are "undoable" or correctable with suitable help: flagging errors before they can cause serious problems is one solution. Restoring a state to what it was before the disruption became evident is another.
  - Disruptions become welcome opportunities for learning, for acquiring mastery, for example, by enabling users to retrace their steps in order to discover unused alternatives. Disruptions should not demonstrate user incompetence.

User-friendliness, or ease of use, is indicated by the absence of disruptions. To prevent disruptions, designers need to anticipate the meanings that users bring to their interfaces (the range of imaginable consequences) and to ensure that an artifact can actually support at least the actions that these meanings imply. When something looks flexible, it should indeed be flexible.

When something looks like a (virtual) button on a computer screen, it should be (virtually) capable of being pushed (i.e., clickable) and have the anticipated effect. Meanings that imply actions with unanticipated consequences are invitations to possible disruptions. The use of full glass doors is a good example. The entrance to the building of the Annenberg School for Communication (where I teach) once had double glass doors, both opening to one side only. They had identical handles inside and outside and no clues for whether the doors opened by pulling or pushing. Visitors were left to try one action, and if this failed to open the door, try another action, and they were mistaken about 50% of the time. Worse, sometimes, during cold weather, one of the two doors was kept locked without any indication which. This reduced the chance of entering the building without disruption to 25%. It is indeed frustrating to try to enter the building by pushing one door, then pulling it, and, without succeeding, reasoning that that door was locked, pushing the other door, only to find out that it had to be pulled open. Disruptions like these are frustrating, though not life threatening. The current glass doors have horizontal handlebars for pulling and vertical ones for pushing. Frequent visitors may learn the difference. However, in the absence of established conventions for the use of horizontal versus vertical handle bars, and since both can be used for pulling and for pushing, about 50% of the newcomers trying to enter the building still experience disruptions. The hardware of user-friendly doors should not promise what cannot be done with these doors.

This kind of disruption is frustrating but not too costly. When there is a chance of losing all the files on a computer without warning or remedy, when drivers can be mislead to the wrong side of a highway, or when missiles can be launched by mistake, then designers have to take serious precautions. They must discourage serious disruptions. Appearances must be unambiguous. Actions must be afforded, their consequences must be foreseeable, and actions with drastic consequences must not be easily available. Probably one of the most important accomplishments of interface design is to make disruptions less dangerous or costly to the user and more easily correctable.

Psychologically, disruptions are discouraging. The inability to correct them causes feelings of being stuck, disorientated, powerless, and incompetent. People's fear of technology is the fear of being seen as failures, of getting into situations without meaningful escapes. The concern for safety is a concern for seriously debilitating disruptions. User-friendliness is not merely an issue of the number of errors made per unit of time. It is rooted in the confidence of being able to handle disruptions. Design can support that confidence.

So far, the meaningfulness of interfaces with artifacts has been highlighted. The point was made repeatedly that humans cannot respond to physical stimuli but to what such stimuli mean to them and that such meanings play important roles in how the interfaces between humans and their artifacts, their constructed worlds, unfold. The temporality of disruptions or losses of meaning points to different modes of attending to artifacts.

According to Heidegger, disruptions, or "breakdowns" as he calls them, signal a change in concerns from being tacitly involved with the world to engaging it as "equipment." For him, we are always involved with the world, mostly without paying much attention to that involvement, if any. The human body is in constant contact with the external world — its boundary has motor and sensory organs and a breathing skin — but it is also in constant motion, not just at its boundary but also its interiors (i.e., the nervous, immune, pulmonary, and digestive systems). In everyday life, a person notices very little of his or her own bodily activity. Heidegger calls this mode of attending to the exterior world, to its artifacts or natural objects, "ready-to-hand." Disruptions render the "equipment" of the world "present-at-hand." In the latter mode of attending to artifacts, we seek to understand how something present is constructed, why it works the way it does, where it leads, and, after a disruption, what went wrong, and how things could have been otherwise. It is only through the experience of disruptions that we become attentive to artifacts as "equipment," as something "in-order-to." When relying on a mouse to move the cursor on the computer screen, we are not attending to the mouse but to what it affects. Sure, the mouse is controlled by hand and finger movements, and it is necessary to handle it appropriately, indeed, but when everything goes well, this activity is transparent, tacit, taken for granted. In Heidegger's terms, the mouse is then used "ready-to-hand." When the cursor does not seem to move the way we want it to, for instance, when the mouse happens to reach the edge of the mouse pad, then the focus of attention shifts to the mouse, which then becomes "present-at-hand." Once it is understood what went wrong, and appropriate corrective actions have been taken, we can once again rely on the mouse, relegating it to the background of what is to be done on the screen. This unproblematic state of involvement can be called "*reliance*," and the attention to the makeup and affordances of artifacts — Heidegger's "present-at-hand," — can be called "*exploration*." In these terms, when writing an e-mail, we rely on our computer, the telephone network, switching stations, telecommunication satellites, the Internet, and much more, without fully understanding the equipment that supports the electronic communication, while attending to what we wish to say. When the e-mail bounces back, it is then that we are forced to explore why this is happening until the problem is solved. Thereafter, we can resume attention to what we want to say and to whom.

There is a third mode of attending to artifacts, mostly prior to their use, which entails knowing of them. This is technology, literally, the logic of techne, of the artificial world: what is available, what can be made and how, what it can do, and how it can be accessed and in which terms. Knowledge of technology does not pertain to a particular artifact but to systems of realizing them: their taxonomy, principles of construction, production, and functions. As a discipline, technology is taught in engineering.

For ordinary users of artifacts, however, this third mode of attention is conceptual. It amounts to understanding the artifacts that we are surrounded with, where they are located, how to access and make use of them. Language, taxonomy, and recall are crucial for people to see the equipment nature of what is available. For ordinary users, technology is neither abstract nor full of details. It is knowing our way around the furniture of the world. For example, computer users tend to know about different operating systems although they may be familiar with only one. They know of available software and their reason for having chosen the ones they actually load on their system. They also know people that can fix their problems and give advice when disruptions are debilitating. This knowledge defines a community of users. Another example is dentists who must be familiar with many more tools then they have in their office, of which they use only a fraction for treating any one patient. A well-equipped medical office should be able to handle not just the most typical cases, but also highly unusual ones, and good dentists have equipment for every foreseeable eventuality, know what each is for, and where it is to be found. What dentists cannot do they delegate to dental labs, who surround themselves with a different technology and know how to handle each of their artifacts. Attention to technology is largely manifest in the ability to search for and recognize artifacts when needed. Hence, this mode of attending to artifacts is called, "*recognition*," which includes distinguishing them by function, name, appropriate use, places where they are kept, and features that identify what they are.

Thus, artifacts must be designed to afford users' three modes of attention:

- **Recognition**: Correctly identifying what something is, what it can be used for.
- **Exploration**: Figuring out how to face something, how it works, what to do to achieve particular effects.
- **Reliance**: Handling something so naturally that attention can be on the sensed consequences of its use.

These modes of attention are shown in Figure 3.4. The figure is intended to suggest that users' attention to the artifacts in their environment starts by *recognizing* something that they are either looking for or come across, recognizing opportunities not initially envisioned. Recognition involves determining the kind or category of artifact we are facing. The ability to recognize crucially depends on previous experiences with similar artifacts.* Recognition could also mean reconceptualizing something that by exploration turns out not to be what we originally thought it was. Figure 3.4 labels as "acquisition" the path from recognition to exploration. Artifacts may be acquired by purchase, or more likely by taking it from the place where it waited to be noticed and putting it where it could be used. *Exploring* includes orienting oneself to the artifact's face, determining what it could do if one were to interface with it, and how it might be held or handled, its affordances,

* Recognition always is re-cognition, cognizing again.

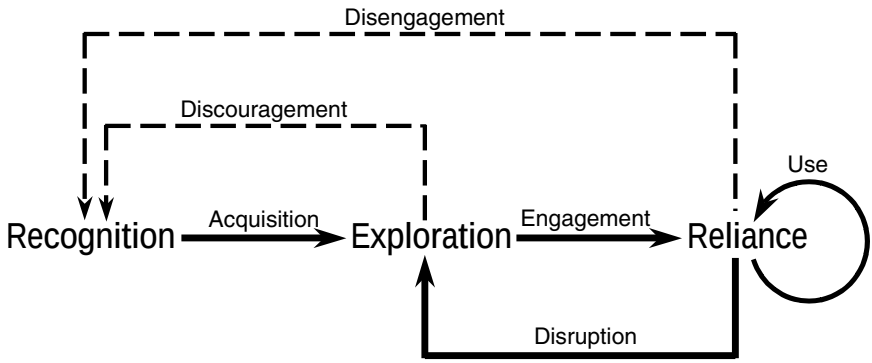


Figure 3.4 Transitions between three modes of attention.

what it does, its states, and what the instructions for the users say about it. Exploration generates anticipations, whether guided by a purpose or being fun. Obviously, designers are advised to enable both: informative interfaces and the pleasure of exploring them. As users build their confidence to handle the artifact in question, they come to increasingly *rely* on it, and attend to what the artifact enables them to do, what really matters. Disruptions bring the users' attention back to explorations, to figuring out what went wrong, and how the interface "really" works. Eventually, the artifact is returned to its holding location, retired or discarded, and the experience of interfacing with it leaves residues in the users' mind, available for future recognitions and uses. This is the use cycle of almost any artifact.

In Figure 3.4 "use" appears as a recurrent reliance and attendance on the consequences of that use. From a human-centered perspective, usability, the ability to use an artifact, essentially means continuing reliance without disruptions, however small. With the concept of intrinsic motivation in mind, Mihaly Csikszentmihalyi (1997) calls being in perfect reliance "flow."* Artifacts that are difficult to use are those whose interfaces require frequent attention, forcing users to go from reliance to exploration and back, upsetting the rhythm of the interface. By contrast, the International Organization for Standardization (ISO) defines usability as "the extent to which a product can be used by specific users to achieve specified goals with effectiveness, efficiency and satisfaction in a specific context of use."** Effectiveness is defined as "the accuracy and completeness with which users achieve specified goals," efficiency as "resources expended" in relation to that accuracy and completeness, and satisfaction as "freedom from discomfort, and positive attitudes to the use of the product" (ISO, 1998). In this definition, the reference to specific users is noteworthy. Being forced to cycle between reliance and exploration is both inefficient and frustrating in users' terms. However, coming from the ergonomic tradition, ISO does not appreciate human-centeredness and

* See also Krippendorff (2004b) on intrinsic motivation.

** ISO 9421-11 (1998) Ergonomic requirements for office work with visual display terminals (VDTs) — part 11: Guidance on usability.

makes (external) specified goals the key to formal usability measures. This excludes intrinsic motivations or relegates them to a "positive attitude." Others, Jacob Nielson, for example, add to ISO's usability criteria the attribute "learnability," or the ability of users to "rapidly start getting some work done" (Neilson, 1993), which concerns the path of engagement, moving from exploring an artifact to relying on it.

This section distinguished three modes of attention. In the ergonomic tradition, which is behaviorist in orientation and institution-dependent, not human-centered, usability tends to address only one, reliance, and deals with it in functional terms (goals). However, to achieve meaningful interfaces, artifacts have to be designed to support all three modes of attention: recognition, exploration, and reliance.

3.3 Recognition

Recognition is cognizing again, identifying something by its kind (name) and in view of the use to which it could be put. Recognition is motivated by a need or interest and is based on possessing a conceptual system of categories that reflects a history of involvement with things similar to those recognized. Recognition is mentioned in Aristotle's *Poetics* as "anagnorisis," which, when applied to persons, is said to employ tokens, artistic contrivances, memory, reasoning (including false inferences), and lastly, the events themselves.* Aristotle describes anagnorisis as a change from ignorance to knowledge. Because cognizing something again always presupposes some prior knowledge of the world, recognition is not as radical a change in knowledge as envisioned by Aristotle. It largely concerns the location and category of artifacts, not how they work, which is part of exploration. In situations that do not provide users the time or opportunities to acquire new concepts, designing artifacts that are easily recognizable depends on expected users' past experiences, common sense, and prevailing conventions.

3.3.1 Categories

The positivist or technology-centered concept of categories is based on the classification of objective properties, which partitions the world into distinct sets of objects with shared qualities. Unions and intersections of such sets give rise to conceptual hierarchies with the most common property on top and the most unique properties below. Linneus' system of classification in biology is the best-known example to which this concept leads. Biologists identify features of the animal or plant in hand and with the help of the system find its proper name and what is known about it. The belief that the human brain recognizes by performing similar tests has entered several fields: cognitive science, for one. Such an approach to recognizing artifacts is far removed from human reality, however.

* Aristotle's *Poetics* Chapter X [English translation by Butcher (1951)].

Inquiries into how humans form categories, starting with Eleanor Rosch (1978, 1981), have challenged the logical positivist concept. Now, categorization research is indispensable in any experiential approach to semantics (Lakoff, 1987), which designers need to embrace. Its basic insight is that people recognize objects by their typicality, how closely they resemble the ideal type of a category, often called prototype.* Accordingly, artifacts do not form homogeneous classes; their membership in categories is graded. Near the ideal type, artifacts are unambiguously recognizable. The less typical they are, the more uncertain is their classification into categories, and the more likely they invite competing interpretations. Thus, from a human-centered perspective, membership in a category is not an either/or proposition, but one of degrees. The ideal type defines the center of its category. Boundaries around them are fuzzy, and recognition is a function of the typicality of an artifact.

Operationally, typicality is the answer to the question of how typical something is, for example, in pair comparison: "Which of the two cameras is the more typical camera." Such questions tend to be easily answered, largely because they are the subject of natural conversations about what things are. Answers concerning the typicality of a great number of objects of a particular kind reveal some kind of map of the degree of membership of these objects in a category, its ideal type being the center. Among birds, for example, ordinary people recognize long-necked birds, big-beaked birds, big birds and tiny ones, and birds of different colors. Penguins are so atypical that many people have difficulties recognizing them as birds. Assertions about the distinctive features of objects are always relative to the usual, the typical, the ideal type of the category of the objects we are trying to recognize. Regarding birds, in the United States, the robin probably comes close to the ideal type bird. Figure 3.5 reproduces the image of a bird used in

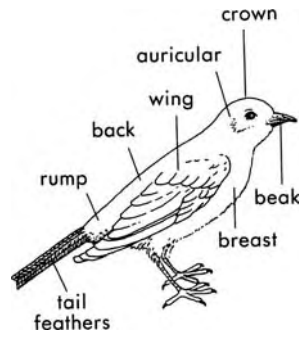


Figure 3.5 Ideal type bird found in a dictionary.

* I chose "ideal type" in preference to "prototype," which is commonly used in the literature on categories (Rosch, 1978) but would be confusing in the literature on design where prototypes designate the first model of a product, usually for testing.

The American Heritage Dictionary (1992:190) to define its parts. Naturally it comes close to the robin. However, even the robin has some features that make it a robin and hence different from the ideal type of the category bird. In reality, ideal types do not exist.

The *ideal type* of a category is a cognitive construction from which all accidental and irrelevant features are absent. Ideal types are simple, skeletal, present the "deep structure" or "gist"* of something — its "Wesen" (Gros, 1984) or essence. Ideal types could also be characterized as having no redundancy, or materiality. Human memory seems to store ideal types from which we notice deviations. This is evident when someone describes a bird (the ideal type) as being white, having a long neck, and standing up straight. Any assertion of this kind spells out the dimensions along which that particular bird deviates from the ideal. When people search for something, a tool, a container, a bus stop, or a bank, they search with the ideal type in mind and look for a close match, becoming aware of how it deviates, and then arriving at a recognized category.

Exploring ideal types is a common design effort (Athavankar, 1989). Drawing an approximation to an ideal coffee cup calls for omitting its ornaments, smoothing irrelevant curvatures, and deleting references to its materiality. However, the very fact that it is a drawing with pencil lines, or any one material realization suggests that artifacts can always only approximate their ideal type. Figure 3.6 shows the results of computer morphing, a process of averaging several shapes, here of coffee cups, to find what they all have in common.

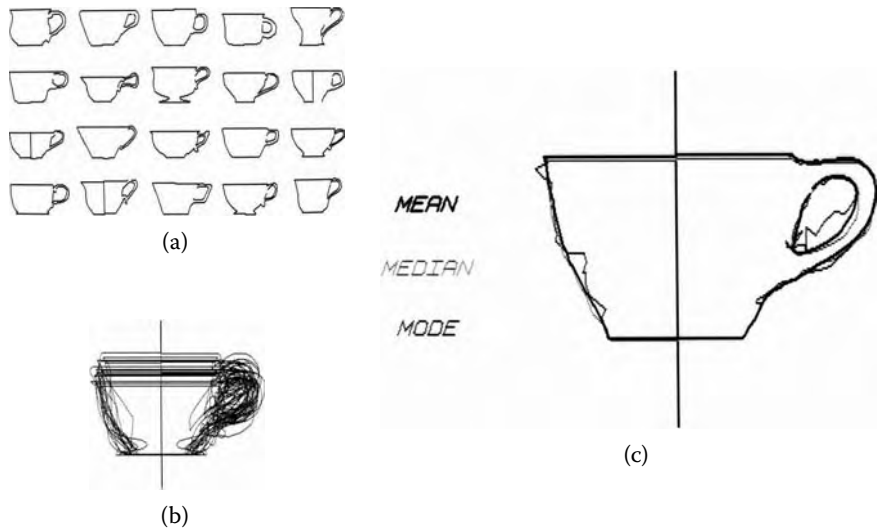


Figure 3.6 Approximating the ideal type coffee cup by morphing. (a) Various cups; (b) superimposition; (c) three kinds of averages.

* John Rheinfrank, personal communication, 1986, and from several of his presentations.

S. Balaram (1989) describes a fascinating political example. Mahatma Gandhi symbolized his effort to unify India under British rule by using implements that were common throughout that subcontinent — the spinning wheel, the Gandhi cap, the mud hut, and the staff, among others — from which all local identities in design were removed. He thus surrounded himself with artifacts that were close to their ideal type, offending no particular group, and making it possible that the masses of ordinary Indians could transcend local differences and identify with his movement.

Research has shown that artifacts that are seen as more typical of their category are also easier to recognize and more reliably identifiable and in less time than those less typical of their category. This finding is quite general and applies to colors, words, signs, and just about any phenomenon. It makes the case for designers who redesign a familiar artifact to be cognizant of the effects of deviating from an established form. Public buildings, the layout of fast food restaurants, fire extinguishers, traffic signs, money, automobile interfaces — artifacts that require fast recognition of what they are by many people and without much time for learning — need to have more typical forms than private artifacts and those used by a few experts who may be trained or can be instructed to recognize unusual forms. Even in situations where designers are free to choose their forms, they will always have to balance recognizability against the benefits of innovation. Ideal types are culture specific and, within any one culture, may shift as new variations of artifacts appear in use. Figure 3.7 shows three still cameras. The one on the left, an ingenious design, basically a lens to look through and focus, deviates quite radically from the ideal type camera, which is close to the single lens reflex camera in the center of Figure 3.7. The camera on the left is in fact often confused with a video recorder. On the right, one finds a digital camera for which an ideal type has not yet been formed. The history of photography is a succession of ideal types, starting with a wooden box with a small lens in front. Recent technology is in the process of no longer supporting a single ideal type camera by adding a camera as a mere feature to other ideal types, of cell phones, for example.



(a) (b)

Figure 3.7 Cameras with different typicalities. (a) Canon Fotura; (b) single lens reflex camera by Pentax and digital camera by Nikon.

Ideal types are not restricted to the visual, however. Movie directors are very much concerned with the typicality of characters. Moviegoers recognize a typical mother less by relying on her face than on how selflessly she cares for her child. In Carl G. Jung's psychology, the term "archetype" corresponds closely to the ideal type. It describes individuals' guiding concepts, cultural in origin and of deep psychological significance. A "stereotype," by contrast, is an oversimplified characterization of the members of classes, ethnic or religious groups being prominent examples. They are likely sources of misidentifications. Because social stereotypes often are self-serving, they are easily assumed but rarely actually tested. When actually traveling to, say, Sweden, a stereotype of what Swedes are like or Scandinavian design consists of is likely to be challenged. If the ideal type represents the essence of a category, then stereotypes create a farce of it.

Deviations from ideal types may be of two kinds:

- **Dimensions** *are variables within which artifacts are constituted*. They are always there. For example, all physical objects have a weight, occupy space, and occur in time. They may vary along these (and several other) dimensions without ceasing to be physical objects. A car may transport one or more people, may move at various speeds, is more or less safe, and occupies space wherever it is. Without any of these (and several other) dimensions, it would not be recognizable as a car.
- **Features**, by contrast, *are dispensable additions that do not alter the identity of an artifact*. Whether or not a telephone has a redial button, caller identification, blocks unwanted callers, or transmits images, it still is a telephone.

In designing recognizable artifacts, an exploration of their ideal types and an understanding of the cognitive correlates of deviating from it are essential.

3.3.2 Visual metaphors

In language, metaphors probably are the most powerful tropes for creating new realities. They are the tools of poets, inventors, and politicians. Lakoff and Johnson say: "The essence of metaphor is understanding and experiencing one kind of thing in terms of another" (Lakoff and Johnson, 1980:5). For designers, visual metaphors are important in making new technology comprehensible. In the context of use, *metaphors enable the recognition of artifacts in terms of the dimensions and features of other and more familiar artifacts.*

For example, personal computers (PCs) were born in the 1970s, when computers were largely mainframe, housed in cabinets behind climate controlled glass enclosures, and accessible only to designated operators. Designers combined two well-known ideal types, the typewriter and the television screen. Neither technology had much in common with computation. However, as metaphors, the typewriter plus TV screen enabled first designers but then more importantly, users, to draw upon experiences from these

two well-understood empirical domains. The typewriter was known to produce text on paper; the television screen reproduced images electronically. Putting the TV screen on top of a typewriter suggested a new meaning, a paperless typewriter, one in which a text can be created character by character and scrolled up a screen, in a manner similar to the way that paper came out of a typewriter. Indeed, this was all that the first PCs did. The combination of two well-known ideal types of technology made this new device recognizable to potential users and simultaneously dispelled their somewhat mysterious nature, which had prevented untrained people from seeing virtues in it for personal use.

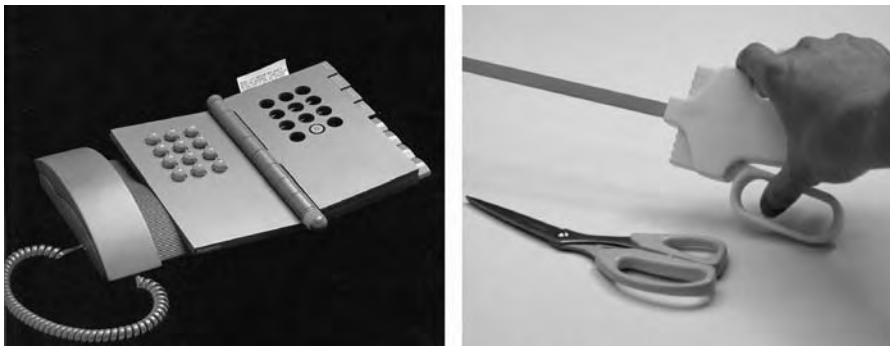
Once introduced, visual metaphors tend to drive technology further. In the case of PCs, because TV screens reproduced colors, it took very little time before "clever" designers thought that PCs should use color as well. TV screens also reproduced images, and it was therefore natural for PC developers to expand the use of computers to the manipulation of images also. Today, and particularly with the simultaneous invention of visual interfaces for pointing, clicking, and moving icons with a mouse, PCs are increasingly a visual medium, tied into already existing telephone networks, the Internet, and the entertainment industry, continually expanding their use.

The birth of the PC shows how, in the absence of familiar forms, metaphors can render novel technologies recognizable. But its subsequent developments also demonstrate how metaphors, which have the ability to change perception, can also dissolve themselves with frequent use into new ideal types. The fate of most metaphors is to end in what poets call dead metaphors and graphic artists call clichés. The terminal point of a metaphor is not so undesirable, however, because it eases recognition and speeds up categorization. But it also leaves something else behind, a network of similarities and differences. The shape of current PCs, while their ideal type is far from settled, still carries its developmental history into the present. They still have a keyboard, screen, and box containing the hardware. Current users are more familiar with PCs and laptops than with typewriters, hence, the metaphor that drew on familiar experiences with other technologies eroded. PCs now have become independently recognizable artifacts with undeniable connections to the past.

Visual metaphors facilitate recognition, usually quite unconsciously. They tend to emerge when we are faced with an artifact, image, graphical display, or sculpture that is ambiguous for what it is, does, or stands for, but we believe there must be more to it than what it seems. The *feeling of incomplete or inadequate understanding* is indicative of the absence of a readily available ideal type for the artifact in question. Such a condition may arise accidentally or by design. If by intention, designers would have to deliberately deviate from ideal types that could be misleading and need to anticipate how the following three-step process unfolds:

1. Initially, there seems to be no ideal type available for the artifact in question. Despite the *typicality of its component parts*, its composition does not seem to make sense in *the given context.* But the ideal types of its parts bring several heretofore separate empirical domains to one's attention.
2. Then, the dimensions and features of the component parts are projected onto each other and into the given context, selectively opening or ruling out possibilities for meaningful interconnections. Moving from parts to context and back amounts to *a hermeneutic circle* (see Figure 2.6) in which hypotheses about the artifact's composition are recursively entertained and tested against the possible meanings of its component parts and what the given context could afford. This process goes on until a hypothetical pattern emerges from any empirical domain and establishes itself as a meaningful way of seeing the artifact's initially unfamiliar composition: a metaphor.
3. Finally, this *metaphor's entailments are tested while interfacing with the artifact* in hand. If repeatedly successful in the context of a variety of similar artifacts, an *ideal type eventually emerges,* creating a category of the artifact(s) in use.

Key to most visual metaphors is the resemblance of one or more parts of an artifact with objects in a familiar empirical domain, the source domain of the metaphor, from which desirable meanings are imported. Lisa Krohn's widely published Phonebook in Figure 3.8 (Aldersey-Williams et al., 1990:47) draws on the pages of books that are turned in the process of reading; here, however, it changes how a telephone functions. The buttons remain physically the same but perform different operations depending on which leaf is open.



(a) (b)

Figure 3.8 Metaphors for self-evidence and reliable handling. (a) Phonebook by Lisa Krohn Design, 1987; (b) surgical instrument by Reinhart Butter, 1982.

Reinhart Butter's design for a ligating device exemplifies draws on another well-known artifact: scissors. Figure 3.8 shows a soft foam model of the instrument. Ligating means crimping blood vessels to stop the flow of blood during a surgical procedure. Devices to crimp staples are well known in office environments and in construction, but here the crimping works differently. It uses tiny V-shaped clips that need to be closed at the tip of the device. Pressing the finger rings together results in a smaller but correlated movement that closes a clip. Ligating demands precision handling and attention to what is done with the clips, not to how the device has to be handled, hence its use had to be obvious. Butter provided the affordance of entering the fingers of a hand into the two openings, regardless of handedness. Doing so sets in motion the metaphorical process from experiences with scissors to performing the desirable hand movement. He also placed part of the mechanism into the surgeon's palm for additional control (not provided by scissors). It is the perceived resemblance with scissors, however, that transfers experiences with them into the reliable handling of the surgical device — self-evidently, as it turned out.

Another example is the well-known desktop metaphor that the Xerox Corporation employs in the design of its large copiers. Copiers accept paper documents and produce duplicates by processes that users do not need to fully understand. The earliest copiers produced multiple copies from a single original. When Xerox copiers were able to copy larger piles of originals and could produce different kinds of copies, they became more like printing plants. They consisted of complex hardware but also created possible confusions about where different kinds of paper (originals, supplies, and copies) would have to be inserted or removed. In search of a good metaphor of how to handle this rather complex machinery, designers knew that users of copiers live in a paper world of desks, piles of paper kept in trays or drawers, and their work flows from in-trays to out-trays. They reasoned that users of copiers would understand a copier best if the knowledge they required for handling them derived from paper handling routines they were familiar with.

Designers arranged the mechanism of the copier below a surface of what resembled a table in size and appearance, on which office workers customarily worked with paper, here, however, elevated to the level at which they could do this standing. So the copier could be seen as a desktop. Office workers also use trays for storing papers, face up to see what is in a pile. To be consistent with the desktop metaphor, Xerox designers provided a tray-like bin, with a paper-size opening that invited people to enter originals, naturally many and face up. Putting originals into a tray rather than into a slot also conveyed a sense of knowing what is happening to them. For copies to exit, the designers observed the usual flow of work, from one side to another, providing an opening on the opposite side of the tray for the originals. And to avoid confusion between where the originals were to go and the copies were to exit, the designers made it difficult to feed originals into the exit. Typical desks also have drawers. Drawers usually are for keeping

things. Guided by the desktop metaphor, the copier featured drawers to keep the paper supply, distinguished from the body of the machine by color and handles to say that the drawers could be opened by pulling.

Mixing metaphors is a frequent source of confusion. Creating resemblances on many levels strengthens understanding. Metaphorical coherence made the use of this revolutionary copier easily recognizable. Since Xerox copiers dominated the market in the 1980s, technology has become smaller and massive copying has become more common. The technology of contemporary copiers has shrunk somewhat, leaving the desktop metaphor behind but keeping most of its original features.



(a) (b)

Figure 3.9 Metonym and metaphor in emergency telephone designs. (a) Metonym with signal qualities; (b) metaphor of safety.

The use of visual metaphors, as illustrated above, informs the design process as much as it enables users' understanding. Shaping a handheld computer to be recognizable as a tablet to write on draws on the familiarity of writing. Making a television screen flat enough to be hung on a wall draws on the familiarity of hanging a picture that provides a window into past or distant worlds. These and many technological developments are driven by metaphors. Their technology usually follows.

Not all metaphors transfer practical knowledge from elsewhere to the present. They can also transfer feelings. Butter's design students at The Ohio State University demonstrated this possibility in the design of a public emergency telephone. Figure 3.9 shows two proposals. On the left, one finds a telephone receiver, probably the most outstanding metonym of the telephone. Its super size superbly signals the location of the telephone. However, it was found not to serve the psychological needs of a caller in distress. On the right side (the final proposal and an early stage), the designer used a

metaphor that utilized a shield for the whole human body, which draws the user into a semienclosure, providing the needed sense of protection, security, and privacy.

Figure 3.10 shows three designs for elevator interiors. Butter asked his students to explore metaphors that could reduce some of the well-known anxieties associated with being confined to a small space. In the elevator design for office buildings, on the left, abstract landscapes and sky lighting are used to create a sense of space and openness. For a hospital, in the center, recessed areas are provided for people to step into and feel safe when having to share the elevator with medical equipment, hospital beds, and wheelchairs. The elevator for public housing, on the right, draws on the metaphor of a cave. The user enters an inviting elevator space that is already "prevandalized," intended to discourage willful damage.

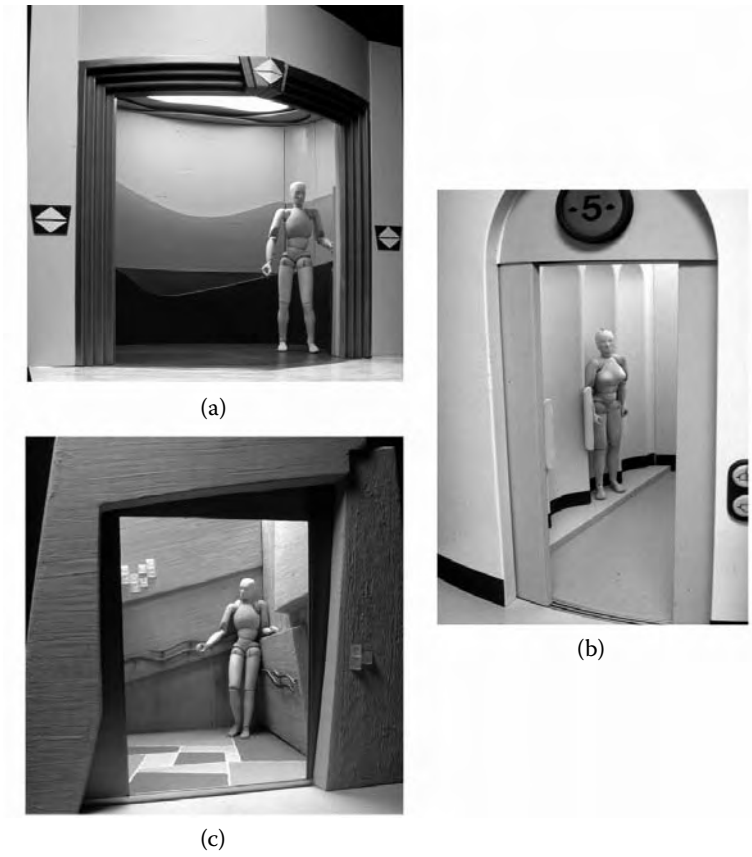


Figure 3.10 Metaphors in elevator designs. (a) For an office building; (b) for a hospital; (c) for a public housing project.

An important class of metaphors that needs to be mentioned here is the orientational metaphors (Lakoff and Johnson, 1980: 14–24): front and back, up and down, and right and wrong direction. Most artifacts are designed

with a typical orientation in mind and their ideal types tend to be tied to what faces their users. Buildings have a facade. Cabinet doors have a face, and TV sets have a front and a back, just as do books, credit cards, and waiting lines. The source domain of orientational metaphors is human communication. We face our partner in conversation, and human faces are what matters most. If a photographer is asked to take a picture of a TV set, even of a friend, it would be odd to be given one taken from their back. It is no accident that merchandise in stores is displayed the way a user would face them. In electronics stores TV screens look at customers, and in bookstores, the front cover of a book is what matters. The human face is the most expressive and variable part of the human body. The metaphorical face of an artifact is what contains its variables, controls, and displays, which is what its user typically interacts with. Recognition of artifacts is often limited to what typically faces their users.

Flatware may not have a front and a back, but it is natural to place forks, knifes, and spoons with their handles toward the user and usual to eat with the dangerous blade pointing away from us. By contrast, public sculptures may need to be recognizable and attractive from all sides. In politics a round nonorientational table is said to be more conducive for negotiations to succeed than a long table with preferred seats on either end. When orientational metaphors do exist, designers are inclined to spend more effort on the "faces" of artifacts, neglecting what is on the other sides. Observe the back of TV, what is underneath a car, behind a computer, or under a rug. Lannoch and Lannoch (1989) explored the use of such metaphors in the semantic organization of living spaces.

Deep-rooted cultural correlations exist between up and good, and down and bad (Lakoff and Johnson, 1980:22–26); between front and important, and back and ignorable; and high and superior and low and inferior. While there is no need to design artifacts that way, supporting this way of recognizing artifacts has the advantage of importing very basic and widely familiar everyday experiences from the use of all kinds of artifacts, especially those from interpersonal communication. The very notion of "interface" presupposes experiences of two humans facing each other, suggesting a concern for what happens between them.

The use of metaphors should not be confused with the idea of forcing a design into alien clothing, such as making telephones appear like the duck decoys used by hunters or Mickey Mouse for children. Playfulness aside, metaphors can aid but also mislead recognition. When designing multiuser systems, for example, or artifacts with rare occasions of use, such as fire extinguishers in public buildings, it is important for designers to be conservative and rely on the most common and widely understood shapes, unambiguous ideal-types, or dead metaphors, as discussed above.

Nor should visual metaphors be confused with verbal analogies. For example, after the computer mouse — the device for pointing, clicking, and dragging objects on a computer screen — was well in use, a designer had the "great idea" of proposing a new shape that reproduced the shape of a

real mouse, literalizing the word mouse. This added nothing to user understanding and likely distracted from its efficient use.

Similarly, metaphors should not be confused with symbolic meanings, i.e., arbitrary signs of something else, as illustrated by a contemporary chair named, "Louis XX." Its designer, Philippe Starck, describes it as "a chair for the countless uncrowned 'kings' of the 20th century" (Steffen, 1997:23). Whether ordinary people would see the chair that way is doubtful. But if there indeed were discernible statistical associations with nobility, it would be an empirical question of how these associations translate into the use of this chair. What the designer considers symbolic of nobility, even if agreed to by users, may not add anything to how people can and will use it.

Finally, visual metaphors are not found in observer-independent forms. They account for a process of how users make sense of unfamiliar artifacts in terms of familiar ones. Inasmuch as design is primarily concerned with newness, visual metaphors can render this newness understandable. Designers may well convince themselves that they have "found" a metaphor, but it may mean nothing to users unless it brings familiarity to the design. One may encourage interpreting an artifact metaphorically by avoiding undesirable ideal types and, instead, suggesting resemblances between the parts of an initially ambiguous artifact and familiar artifacts from a domain whose meanings are to be brought to bear on it. Much of design is metaphorical in this sense. Self-evidence, the seemingly effortless recognition of what something is, often amounts to nothing more than a dead metaphor.

3.3.3 Attractiveness

Making artifacts appear attractive, outstanding, and desirable was the nearly exclusive preoccupation of design in the industrial era. This era produced much writing on aesthetics, style, composition, pleasing geometries, and, recently, on so-called pleasurable products (Tahkokallio and Vihma, 1995; Green and Jordan, 2002). The view espoused here, however, is a human-centered one, which emphasizes not particular qualities of forms as being attractive on their own, but the conditions that attract potential users to particular artifacts and not to others — before acquisition, exploration, and eventual use.

Attractiveness is **a relative quality**. It exists only in contrast to what does not attract attention. When all objects and their background are of the same color, none is outstanding. When all are stored in uniform gray boxes, one would need labels to distinguish among them. What differs from other artifacts is more likely to be noticed than what blends or is camouflaged. This is not to say that artifacts should be attention grabbing, make themselves known by noise, bright colors, unusual shapes, or moving parts. When not needed, artifacts should remain background and not distract from current activity. Package designers prefer their products to attract buyers. Surgeons may want their tools to be there only when needed.

Utterly repulsive and potentially dangerous artifacts may be as noticeable as beautiful and useful ones, but since attention selects among many alternatives, the more beautiful ones are more likely to be looked at, acquired, explored, relied upon, and used. In making artifacts attractive, designers who make *aesthetics* their primary objective walk on thin ice, however. Aesthetics is a philosophical discipline, providing theories of what counts as beautiful. In fact, beauty always lies in the eyes of observers, their community. Theories are notoriously blind to changes. Attractiveness varies over time. A car that was fashionable 20 years ago may look utterly unappealing today. Although much has been written about what appeals — soon outdated, as one's perception shifts unnoticeably — here are a few general contrasts that designers may want to consider to enhance the attractiveness of their designs in the context of what else is available.

- **Newness:** This is probably the most important attractor. Newness contrasts with what is old, used, or worn out, but can lose its attractiveness when the artifact is so unusual as to prevent being recognizable. A related attractor is **uniqueness**. It is especially attractive for people concerned with their own shifting identity. Fashion is a good example of building newness on the familiar.
- **In the right place:** Much of attractiveness is tied to things being where they are expected to be. Not finding something is frustrating, whether because it is not clearly distinguishable from its surroundings, hides behind other things, or is where least expected. Being in the wrong place, not finding the appropriate button to push, can be annoying if not confusing and ultimately distracting from the hoped-for attention.
- **Simplicity:** Often, simplicity is appealing, but only in contrast with confusing complexity, on the one hand, and boringness, on the other. In designs for computer interfaces — computers often are mystifying to begin with — simplicity can promise understandability, ease of handling, and naturalness of use, but it must also fascinate.
- **Unity:** Unity comes in many guises: a common theme, parts that seem to belong together, transitions between unequal shapes, a frame, "container," or form.* Unity highlights artifacts that are seen as having it and hides those that do not: dissimilar components, scattered arrangements, lack of commonalities, or mixed visual metaphors.
- **Regularity:** The appearance of orderliness in the arrangement of component parts; repetition of the same or similar parts; a natural progression from left to right, top to bottom (note its analogy to reading — at least in much of the West); continuity, which contrasts with disorderly arrangements, seeming arbitrariness, and abrupt breaks in contours.

* Interestingly, the morphology of "con-tainer" points to the sharedness of the vessel, which is shipped from one place to another where content can be entered or retrieved. It serves here as a metaphor for the appearance/essence distinction.

- **Symmetry:** A kind of regularity. It is due to recognizable transformations of forms, for example, when the left side mirrors the right, as in the leaves on trees, the human body, and in the design of cars (except for the controls and the driver's seat). Symmetry attracts by its redundancy, which simplifies attention. *A*symmetry is more difficult to understand or focus on. Yet, deliberately deviating from an expected symmetry can attract as well.
- **Balance:** A somewhat doubtful verbal metaphor from weighing unequal objects, used to describe the perception of a composition as being harmonious, that the parts of an artifact, their proportions, sizes, shapes, and colors are in tune with each other. The contention is that balanced compositions attract by being comfortable to look at while imbalanced ones attract by being visually disturbing.
- **In grid/against grid**. Grids in typography, interface design, and architecture are a form of regularity, the imposition of a framework, an organization, or an arrangement. Artifacts that use a grid become attractive at points where their grid is disrupted, at their edges, or set in contrast with other grids based on a different geometry or none. Placing something diagonal, for example, attracts more attention than when inside a grid.
- **Intentionality.** Designers assume many freedoms to deviate from conventions, expectations, current senses of beauty, and social norms, but may not want to do so without conveying the sense that such deviations are deliberate, serve a purpose. Here, "intention" does not denote a mental state, but what users consider as having been intended. The perception of intentionality is common. Even children who find their toy to have a button "know" its intention and are attracted to pushing it. For any discontinuity, an opening, a gap, parts differentiated by form or color, or an unusual arrangement, users are always inclined to "find its reason" or, failing to do so, consider what they see as badly conceived, poorly put together, or arbitrary, in sum, unattractive. Being able to see something as well designed, skillfully assembled, or ingeniously conceived — not contrived, artificial, or meaninglessly embellished — is an inherently attractive quality. It was the pride of medieval craftsmanship and what designers have always pursued, but it must be made recognizable to serve as attractor.

3.4 Explorations

Exploration precedes use. Users of artifacts have to know what they might get into, how their interfacing with the artifact could evolve without losing its meaning. It probably is reminiscent of Cartesianism that designers tend to focus their attention more on what an artifact is, what it is for, and how it functions, than on how it can be explored, handled, and sensed, and with human-centeredness in mind, more on the ultimate reliance than on how its use is acquired during successful explorations. There are crucial differences between the three kinds of involvement. Recognition is dominated by sense,

exploration by cognition and action, by trial and error, for example, while reliance unquestionably depends on users' bodily involvement. According to Figure 3.4, exploration has two entry points: acquisition and disruption. Acquisition denotes any transition between recognizing what something is and figuring out how to proceed, be this the act of moving a kitchen appliance from a shelf to where it is needed; buying a new computer and putting it in place; or deciding to enter a training course for becoming an airplane pilot. Disruption, as already described, amounts to a temporary refocusing from what one intends to do with an artifact to why it didn't quite work out as intended. In either case, exploration provides users with know-how, with an understanding of how an artifact could be handled, with what to do and not to do with it. It allows for the confidence in the possibility of engagement or reengagement.

3.4.1 User conceptual models

It has been customary to assume that recognition relies on so-called user conceptual models or UCMs, which explain experiences that users bring to artifacts they intend to use. *A UCM is a network of operational concepts assumed to tell its beholder how an artifact could work, when to do what with it, and what to expect consequent to any actions taken*.* Just as we rely on the ideal types of familiar artifacts to recognize what a particular artifact is, so we rely on a system of equally simplified and skeletal concepts to ascertain what the parts of an artifact could mean in use. It is important to note that UCMs are designers' constructions of what users are capable of doing and learning. They manifest a designer's second-order understanding. It is also important to acknowledge that whenever an artifact is to be designed for use by many, such models cannot be one cognitive model of an idealized user. They need to represent the full diversity found in a community of users. Finally, UCMs can represent only what users are able and willing to assert and what observers can construct from user accounts augmented by conclusions from observations. Humans are not conscious of everything they do. The cognition of other individuals can be observed only in terms of their consequences. Usually, people think differently from how they think they think.

Figure 3.3 depicts a designer's understanding of a human interface. This includes the designer's understanding of a user's understanding, his or her construction of a UCM, which is intended to account for how senses invoke meanings, how meanings unfold into actions and entail expected senses. This may or may not be confirmed by the actual senses brought forth by the user's action. Three examples of UCMs follow.

First, maps: At any one moment in time, the sense we make of our surroundings locates us not in "reality" but on a map. We rely on mental maps to find our way on familiar ground, in our neighborhood, or at home, say when the light is out. When driving in an unfamiliar city, we may consult

* Lakoff (1987) calls them idealized cognitive models or ICMs.

a street map to find our way. Without some kind of map, we could not possibly know where we are, what we have to do to get from here to a distant destination. Nor would we know what to expect along the way. A road map may transform the church that we see into a landmark, confirming our way. Correlating the increasing or decreasing street numbers with the numbered lines on a map informs us about whether we are driving in the right direction. A map shows opportunities for short cuts, distinguishes major and minor thoroughfares, or indicates crossing county lines. These landmarks, shortcuts, county lines, and destinations have no reality without a map. From a human perspective, the fact that maps depict fewer details than what our location can reveal, a fact that semioticians are so fond of theorizing, is less important than that it renders our senses meaningful and enables us to act in anticipation of what will happen beyond what we can see at any one moment. A UCM enables designers to anticipate what an artifact could mean to its user. Thus, UCMs are the key to understanding how users interact with an artifact. They do not describe an artifact's nature but inform designers of the user's possible actions, which is what designers are concerned with and may observe indeed.

Second, electricity: Most people, including electricians, have ideas of how electricity works. There are several folk models of electricity, typically informed by metaphors of more familiar systems, plumbing for example. The metaphor of electricity as *flowing* through electric wires is common. Wires come in different sizes; the larger they are, the more electricity can flow through them, as is true of water pipes. Charged electricity flows through these pipes to where it is needed, and used-up electricity comes back to be recycled. Another folk model conceives of electricity as contained in tiny particles, called electrons, which run through wires and carry energy from where it is plentiful to where it is wanted. A third and more sophisticated concept is to think of electricity as analogous to a stone dropping into a pond and creating radiating waves that fade with distance. At the load wire, the electrons oscillate very powerfully; at the groundside, their movement is diminished, driving something along the way. Installation of electric wires at home can be accomplished by all three of these UCMs. Accuracy is clearly not the issue here, although it is doubtful that any of these UCMs would enable an electronic engineer to design a radio.

Third, home heat control: Willet Kempton (1987) studied how different people interacted with their thermostats at home. He charted the changes in temperature setting and interviewed users about what they were doing when changing the temperature setting and why. He found people to use two different conceptual models of how their home heat control functioned: as a valve and as a feedback system. People who controlled the temperature with the UCM of a valve changed the setting each time they felt either too cold or too hot, which occurred frequently, and this conception accounted for radical temperature fluctuations in the home, as recorded over time. Those who had the idea that they could set the temperature, and thereby command the heating system to work toward it, ended up changing the

thermostat far less often and enjoyed more even temperatures. These two conceptual models resulted in demonstrably different behaviors. Luckily for these users, the engineers' design for the thermostat accommodates both UCMs, probably without intention or awareness of the difference. The use of neither UCM resulted in disruptions — except that couples with different UCMs, unknown to either partner, often find themselves in endless struggles over the best temperature setting. To support meaningful interfaces between artifacts and their users, who can be expected to bring different conceptual models to the scene, artifacts need to be designed to accommodate ideally all those available among potential users, or face discouraging disruptions.

The UCMs that Kempton found were derived from extensive interviews with users. They explained how users handled thermostats, and this was evident in the recorded room temperature fluctuations. His was an analysis of verbal descriptions, and the two models were in fact verbal metaphors for what these users knew. These metaphors provided a window to the UCMs in use. UCMs for more complex situations may be constructed from protocols obtained by asking users to verbalize what they are doing while doing it, step-by-step. Setting a temperature is a comparatively simple action.

UCMs may be constructed from the questions that users ask when exploring an artifact and the instructions given to them by experts. They may also be built by observing how users handle a class of artifacts that are similar to the one being designed. There are even computer techniques available to build UCMs from observed behavior. For example, in the construction of automatic pilots of very large ships, there are computer programs that can learn from how the ship's captain responds to noted deviation from a course and thus acquire knowledge of the behavioral properties of the ship as understood by the captain. Similar techniques are used to program production machines that learn to mimic the movements of expert workers, say on an assembly line, and thus develop and apply the UCMs of users.

Thus, UCMs are heuristic constructions that operationalize the triangle of sense, meaning, and actions, as shown in Figure 2.4, and help designers explain how a community of potential users could interface with their design. Scientific approaches to constructing UCMs — ethnographies that attempt to describe existing concepts within a community and computer models of cognition that mimic the behavior of experts — seek to describe what is. However, Heidegger's observation that disruptions cause us to pause and think of an alternative approach, one that bypasses the disruption, suggests three things.

- UCMs are constructions of the reality as enacted by users, not representations of an outside world. They enable designers to explain what people can do and how they may act to bring forth desired senses. Like road maps, UCMs do not predict anything; they are neither accurate nor inaccurate representations of what people do, but what they can do, the

alternatives they have available in a situation. However, they do prove themselves by being more or less viable in use.

- UCMs resemble toolboxes, more so than a causal system of reactions or specific instructions, computer programs, for example. When disruptions are experienced, alternative concepts are created and tried out. Brains are far more variable than computational models of cognition can represent. Ordinary users have the capacity to handle many different and especially novel artifacts by applying concepts that worked in one empirical domain to another.
- UCMs are subject to learning. If one thinks of UCMs as maps, one has to qualify that they are constantly improved, revised, redrawn, or replaced by better ones. The designers' construction of UCMs must account for their modifiability, their ability to grow in complexity, and their increasing efficiency in use.

In sum, UCMs constitute an inventory of the meanings that a community of users have available, can easily acquire, and are likely to enact, given the possibilities that artifacts suggest.

3.4.2 Constraints

The range of possible uses of artifacts is usually far larger than anticipated by its designers. Milk crates are intended to transport milk to grocery stores, but designers can hardly prevent unintended uses: as bookshelves, playthings for children, bins to store tools, dividing walls, stepladders, or bicycle baskets. For a homeless person, a milk crate can hold priceless possessions. Tied to a pole with its bottom removed, a crate is a basketball basket. In the hands of an angry person, it can become a weapon. For a grocer, who paid for them, these uses are illegal, but what can he do? The range of the imaginable uses of an artifact is the range of its meanings. Milk crates are simple enough for users to understand any constraints on their meanings, which are a function of size, shape, capacity, sturdiness, stackability, and so forth. When the meanings of an artifact are relatively harmless, usability does not need to be constrained. Otherwise, designers may have to think of ways to prevent dangerous uses or make them difficult. It is possible, of course, to write on the milk crates that illegal uses can be prosecuted, issue warnings of certain dangers, or post prohibitions, all of which need to be read and understood, and may be ignored when inconvenient. This section concerns practical constraints on possible meanings that designers can control — like producing milk crates with big holes that cannot hold items smaller that milk containers. We recognize several kinds of constraints.

- **Natural laws:** Constraints that affect designers and users alike: the thermodynamic law concerning the conservation of energy, Einstein's mass–energy equation, the Shannon's laws on the maximum capacity of a communication channel (Shannon and Weaver, 1949). They limit the

heights of buildings, the acceleration of cars, and the amount of information that a CD can store. Designers may not be able to control these laws but may have to use them skillfully.

- **Physical constraints:** Nonnegotiable restrictions like the fence on a bridge that prevents suicides; the walls of buildings that force their users to come and go through doors; the safety shields that prevent factory workers from operating a machine when their hands are in danger; the device on large filing cabinets that allows only one drawer to be opened at a time, preventing it from toppling; electric fuses that prevent wiring from being overloaded and producing electric fires; the printer shutting down when it runs out of paper; the keypad that accepts only one character at a time; the toggle switch that is either on or off; the microwave oven that switches off as soon as a user opens it; and more. These constraints are effective whether a user understands their reason or not, preventing the user from getting into trouble. For these reasons, Norman calls them "forcing functions" (Norman, 1988:137ff).
- **Constraints that discriminate among users**. There is a large class of constraints that are intended to affect different users differently. Child-proof medicine bottles keep children from ingesting potentially harmful medications. Placing electrical controls deliberately above children's reach to limit access to adults. Hospital patients may be prevented from getting to medical supplies or playing with equipment because they do not have the know-how of trained personnel. But the most widespread way to discriminate among users is by selectively issuing keys. An ignition key is only given to the owner of a car so that children and thieves are unable to drive it. Keys for doors, safes, spare tires, medicine cabinets, jewelry boxes, or bicycle locks are found everywhere in the world, and the possession of a key distinguishes among legitimate and illegitimate users. A more recent kind of "key" is the personal identification number (PIN) that is needed to access bank accounts, information networks, security areas, and combination locks. Because keys can be stolen and the use of PINs can be observed and copied, efforts are under way to tie the legitimate use of certain artifacts to unique user identities. This reasoning underlies efforts to replace the safety locks on handguns by electronic hand recognition devices and for developing retina identification of individuals in high security environments. All constraints that discriminate among users are defined by known user characteristics, be they
  - Unequally distributed physical abilities
  - Expert knowledge
  - Possession of keys, and the knowledge of codes or passwords by legitimate users
  - Physiologically unique user attributes that are not reproducible by unauthorized individuals

- **Overrideable constraints:** These constraints make users aware of dangerous or irreversible consequences of their actions and enable them to lift these constraints for their own good reasons. They shift attention from how an artifact is used to what is to be accomplished with it, from reliance to exploration, and they acknowledge that many errors are the result of being carried away by routines. For example, to prevent computer files from being accidentally deleted, software writers warn users of the consequences of their action and ask whether the user really wants to go through with it. In banks, safe deposit boxes need to be opened by two keys, the customer's key and the banker's key, which assures that there is an identity check of the customer; neither can empty the box without the other. In the U.S. defense system, to prevent the mistaken firing of missiles against assumed enemies in case of a nuclear attack, the system does not allow a single commanding officer to initiate such an action. Should one officer notice an attack, carefully worked out procedures require deliberation and concurrence before such an irreversible action can be taken.
- **Unnecessary constraints.** Many constraints that users experience result less from a deliberate effort to prevent disruptions or potentially devastating situations than from technical constraints or from not-too-well thought out impositions. One example of a technical constraint is the aforementioned pair of glass doors that open to one side only. The choice of another kind of hinge could prevent half of the disruptions mentioned earlier. Another example is the use of 3.5-inch floppy disks for PCs. These disks are almost square. There are eight possible ways to insert it into its receptacle. In four cases, users experience a physical constraint: the disk is too big. In three cases, the disk does not go completely into the slot, leaving the last option to be the correct one. Although one may notice a tiny arrow on the disk and after a while learn how to insert such a disk, a different form might have communicated this better. The use of credit cards in money access machines presents a similar situation. Because a credit card is not square, the small slot to receive it means that there are only four ways of inserting it. On these machines, manufacturers tend to offer images that suggest how the card is to be inserted either by showing where the magnetic stripe is to be or from which side the face of the credit card is to be read. The machine detects rather quickly when the card is wrongly inserted. But instead of devising a test to reject three out of the four possible ways a credit card can be inserted wrongly, one could also enable reading the magnetic stripe in either of the four ways without requiring instructions that can be misread. Suitable instructions surely can reduce technical constraints from becoming frustrating; semantically obvious solutions are user-friendlier; avoiding them altogether might be a design challenge but well worth the effort. Engineers learn to be creative within technical constraints; users may have to learn them the hard way,

so designers of user-friendly interfaces are advised to find ways to prevent possible meanings from being so limited.

An example of a well intended but in the end unworkable constraint was a U.S. law requiring the use of seat belts in automobiles before starting the car. It prevented the car from being driven unless all occupants buckled in. Because seat belts save lives, this made sense. Whether a seat was occupied and its buckle locked could be easily detected. Buzzers warned when the conditions of driving were not met. This physical constraint was so disliked, however, that new car owners asked their mechanics to disable it and eventually succeeded in having the law changed. Norman suggested several reasons and a conclusion: People did not like being forced to wear seat belts and preferred to take the risk of getting hurt in a car accident to its inconvenience. The forcing function could not distinguish legitimate situations for not wearing seat belts from illegitimate ones; for example, by not being able to distinguish between whether a person or a package occupied a seat. The mechanism was not reliable, buzzing and stopping the car inexplicably. People kept the seatbelt permanently buckled under the seat, making it difficult for passengers to actually wear them. Norman's conclusion: "It isn't easy to force unwanted behavior upon people. And if you are going to use a forcing function, make sure it works, is reliable, and distinguishes between legitimate and illegitimate ones" (Norman, 1988:134–135).

3.4.3 Affordances

The word "affordance" was coined by James J. Gibson (1979:127–135) as part of a theory of perception, known as ecological theory. This theory had its origin in experiments conducted during World War II in which Gibson inquired into what aircraft pilots needed to see when trying to land their aircraft, especially under less than optimal conditions. Gibson observed that pilots did not literally see the physical features of a terrain before deciding whether to land their plane. Experienced pilots saw the "landability" of their aircraft and did so rather directly, without abstract logical processes. Physical features certainly contribute to but cannot cause human perception. It turns out that there is little correlation between the two, a fact that Maturana had observed regarding the perception of colors as well. The lack of this correlation also underlies the axiom of meaning (Section 2.2). Gibson described perception as a fit between human physiology, the disposition to act, and what the environment could support. Generalizing his theory to all animals, Gibson defines: "(t)he *affordances* of the environment are what it *offers* the animal, what it *provides* or *furnishes*, either for good or ill… . ("Affordance") refers to both the environment and the animal in a way that no existing term does. It implies the complementarity of the animal and the environment… (Physical) properties … (can be) measured with the scales and standard units used in physics. As an affordance of support for a species or animal, however,

they have to be measured relative to the animal. They are unique to the animal" (Gibson, 1979:127).

The lesson of Gibson's work is that one does not perceive objects but usability: the sit-ability of a chair, the carry-ability of a box, the walk-up-ability of stairs, the movability of an object, the open-ability of a door, the edibility of food, the possibility of being harmed by a knife, even the solvability of a mathematical equation. Perception, he points out, has much to do with the makeup of the human body. For example, a 5-inch cube, which is a geometric description of the form of an object, is graspable, whereas a 10-inch cube is not. To be graspable, an object has to have two opposing surfaces separated by a distance less than the span of the hand, but that distance should not be smaller than what two fingers can pick up. In other words, human perception is intrinsically tied to what humans can do with their body.

It follows that everyday objects are perceived and conceptualized in terms of affordances: cups by our ability to hold and drink from them, knobs by our ability to turn or push them, switches by our ability to move them from one position to another, and ballpoint pens by our ability to hold and write with them. Gibson called his approach ecological because he realized that what ecologists call a niche and define as the environmental conditions that support a species' life is nothing but a system of affordances, not to be confused with a phenomenal or subjective world. To be perceivable as useful, everyday objects have to be designed in terms of their users' abilities, i.e., in human-centered terms (for examples, see Figures 3.12).

Focusing on affordances was Gibson's way of overcoming the Cartesian objective–subjective duality, which is still hounding popular concepts of perception as the subjective representation of objective facts. This duality is also built into the idea of semiotics as the discipline concerned with the relationships between physically manifest sign-vehicles and what they convey. Although the term "human-centeredness" did not exist at Gibson's time, his theory of perception is a human-centered one. It neither physicalizes perception nor psychologizes the world outside.

A careful reading of Gibson's work suggests that his *affordances, the perception of possible uses, the awareness of usabilities, equal the meanings* of artifacts in use. Perceptions are always anticipatory of the consequences of one's actions. Actions that succeed in bringing about the anticipated senses are *ipso facto* afforded by the environment. Thus, an affordance is a unit of perceptual fit between the circularity of present sense–meaning–actions–anticipated sense and a world outside, which is unknowable, except in that fit (see Figure 3.11). The world outside seems to be remarkably pliable for numerous sensory–motor coordinations to acquire such a fit and become reliable without causing disruptions. Walking, for example, creates a path and a well-trodden path makes walking easier. The shape of a baseball bat is such that even someone who has not heard of baseball would be able to hold it on the 'correct' side and use it for swinging, banging, or similar actions that engage both hands. A pair of scissors has two holes in which

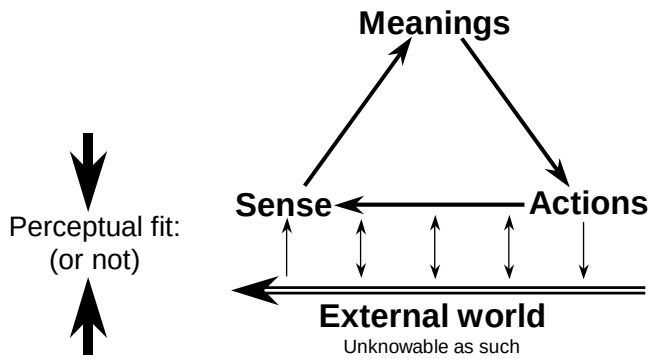


Figure 3.11 Affordance: the unit of direct perception.

the thumb and one or two opposing fingers of the hand fit so naturally that its use becomes obvious. A knob that can be grasped by two opposing fingers, such as a radio dial, affords twisting; one that can be held with a whole hand, such as a doorknob, affords turning; one that extends only slightly above one surface, such as a doorbell button, affords pushing. The perception of such uses is direct, not because affordances exist without a perceiver, as Gibson insisted they do not, but because the sensory–motor coordinations in question are afforded by nature or numerous artifacts made that way. Affordances signal the habits of being-in-the-world.

What Gibson studied was the ecological fit between the behavior of organisms and its material support, in terms used here, the reliable components of interfaces. The directness of perception, he always insisted on, suggests that to perceive an affordance is not necessarily classifying or naming it. Designers need to be aware of the affordances that are common to a community of users; acknowledge how their senses, meanings, and actions routinely follow from each other; and find ways to materially support the affordances that keep an interface meaningful. With reference to affordances, one can distinguish between the following:

- **Direct perception of usability:** In Gibson's sense, direct perception is the perception of reliable, and therefore taken for granted, meanings of environmental features. It is anticipatory of what can be done at present.
- **Enacted affordances:** The unproblematic enactment of meanings, as in Figure 3.3. By Gibson's definition, an affordance is the routine support that an environmental feature provides for a particular behavior. Perceived affordances need to be distinguished from enacted affordances. The former are anticipatory, the latter are their present manifestations. The enactment of perceived affordances may fail to be supported by the environment, cause disruptions, challenge their meanings, and give rise to learning (see Figure 3.3). Successfully enacted affordances strengthen the taken-for-granted nature of direct perception.

- **Constructed affordances:** While affordances are perceived, in Gibson's terms directly, and acted upon relatively automatically, users may develop more abstract conceptual models (UCMs) of their artifacts and their worlds. The UCMs of complex artifacts, of a computer for example, undoubtedly are constructed on numerous affordances, but involve hypotheses about how it works and embrace convenient metaphors that can no longer be perceived directly and whole. Nevertheless, such models may be enacted and relied upon much as perceived affordances are. By involving higher-order concepts, inferences, reasons, a logic, and language, however, the enactment of such models is more vulnerable to disruptions on a level above Gibson's affordances: keying in instructions, moving objects, and clicking on buttons.

The semantic turn takes affordances as the most reliable building blocks of interfaces. Like common words, affordances are relatively standard in use, hence the ability of perceiving them directly, effortlessly, and without thinking. Everyday language provides a rich vocabulary for describing affordances. With its help, people explore unfamiliar artifacts and construct their own UCMs and that of other stakeholders. Much like complex linguistic expressions, the meanings of complex artifacts are no longer as obvious as the simple affordances that Gibson's definition addresses. Human-centered designers may draw on this common vocabulary in developing artifacts that are more likely to be understood than those developed in technical terms.

3.4.4 Metonyms

Metonyms are parts taken to stand for the whole to which they belong. In language, examples are numerous. The "White House" is taken as standing for the U.S. presidency; the "crown" for a king or a monarchy. When a moderator acknowledges "the redhead" in the audience, it is the person that is meant, not the red hair he or she has. A "string" quartet is played by four musicians whose instruments require strings to generate the sound. "Wall Street had a good year," asserts something about people owning U.S. stocks and bonds that happen to be traded on Wall Street. Social stereotypes are metonyms that take typical features to define a class, often simplistic and self-serving, for example, when talking of the attributes of Frenchmen, politicians, vegetarians, or bachelors, but also of the user. Metonyms are cognitive shortcuts, justified or not. Lakoff defines the conceptual model of metonyms:

- There is a concept A to be understood.
- There is a conceptual structure containing both A and another concept B.
- B is either part of A or closely associated with it in that conceptual structure. Typically, B uniquely determines A within that conceptual structure.

- Compared to A, B is either easier to understand, easier to remember, easier to recognize, or more immediately useful for the given purpose in the given context.
- B is used to stand, metonymically, for A (Lakoff, 1978:77–90).

In cognition, the ideal type B stands for the category A. In language, most metonyms concern features of persons or places associated with larger entities as previously exemplified. In exploring an artifact, metonyms are the outstanding parts or key features of that artifact, provided they lead users to its affordances, or enable them to figure out its makeup, function, or the support they can expect from it. In short, a metonym informs users about something on account of being a part of it, about the features of an artifact not in view, about the possible contexts of its use.

Metonymy underlies the relationship between sense and meaning, as depicted in Figure 3.3. Perceiving more than meets the eye results from "sense metonymically invoking meanings." If an artifact equals everything one knows or can imagine of it, then, surely, any of its present views, current states, and particular uses is always part of what is known of the artifact. For example, the view of a rock metonymically leads to an expectation of how it looks from the unobserved side, how cold or heavy it is. Similar part–whole inferences are involved when driving a car. The steering wheel and foot pedals and the landscape passing by are only a small part of what happens for a car to work the way it does. And only after locating ourselves on a roadmap do visible things become landmarks en route to where we want to travel. With a map, the driver's position and landmarks noted along the way become metonyms of the journey.

The part that is chosen to be a metonym of its whole is not arbitrary. Such a part must be in some sense outstanding, easily recognizable, and play a unique role in the whole. Sense is always outstanding by being vividly present, whereas meaning is virtual, a residue of past experiences, the product of one's imagination, and often schematic. A steering wheel is a unique part of the ideal type car. Even children playing at driving a car tend to rely on a steering wheel, even if only imaginary, as seen in the movement of their hands. A steering wheel would be a good metonym for driving, a violin a good metonym for a classical orchestra, bread a good metonym for a baker's shop, a file folder a good metonym for organizing documents in a computer.

Metonyms provide the basis for a human-centered *theory of signs*. Traffic signs, for example, might employ pictograms of the road, a car, bicycle, or pedestrian, but they do not represent anything beyond the part–whole relationship. They are not subject to truth or accuracy conditions. When drivers encounter a traffic sign, they may take it as a warning or instruction for what they can or must not do. Traffic signs are learned, and the learning experience is the whole in which these signs occur. There are various ways to acquire their meanings: by studying traffic rulebooks, by imitating other drivers, or

by learning the lesson from being given a traffic ticket, all of which include such signs as visually distinguishable parts. Generalizing from this example, all signs mean the roles they play within a larger system — traffic signs within the system of traffic regulations; pushbuttons on the side of a door within the assumed connections between activating a sound and someone opening the door; keyholes within the UCM of opening a container or securing it against illegitimate use, and telephone receivers as the most important part of the (traditional) telephone.

The "yes, I do," uttered at the appropriate moment within a marriage ceremony, plays a key role in moving a marriage forward and establishing a relationship between two members of a community. Signs are metonyms, not by representing something, as semiotics insists, but by being seen as part of a larger complex of experiences.

Designers can support the exploration of artifacts by using common metonyms as metaphors. For example, Apple computers started to use the image of a garbage can for users to drag unwanted files to it for deletion. In everyday life, garbage cans are containers to dispose of waste materials. The icon of a garbage can on a computer screen draws on this well-established metonymy, but metaphorically: While a computer file cannot be disposed of as garbage is — files exist as addressable states of a computer's memory whereas garbage exists as transportable matter — the icon of the garbage can encourages users to take what they know about garbage disposal into the domain of file management. This metaphoric use of metonyms can be described by this conceptual model:

- There is a concept A (disposing something) to be understood in the conceptual domain of a present artifact.
- Concept A also occurs in the familiar artifact A° (a municipal waste disposal system), which resides in a conceptual domain other than the artifact's.
- B° (a garbage can) is an outstanding and essential part of A°, and B° is typically taken as a metonym for A°.
- B (the icon) depicts B°.
- B suggests A on account of B depicting B°, B° being a metonym of A°, and A° embodying A.

Computer interfaces heavily rely on this conceptual model: the image of a magnifying glass on a clickable button that can enlarge an image or text, the paragraph sign "¶" for showing hidden characters, or the likeness of a floppy disk to initiate the saving of a file. A floppy disk, for instance, still is the most obvious metonym for the larger mechanism by which data are stored in and retrieved from a medium. The icon of a floppy disk and the document to be saved appear in the same conceptual domain, on the computer screen, which differs from the conceptual domain of the mechanism that actually saves files. Because the operation of "storing," A is difficult to

indicate, the icon of a floppy disk, B, suggests "storing," A, by way of depicting B°, the metonym of the storage mechanism, A°. Knowledge of A° is transferred to A. Semiotics has developed a vocabulary for describing how B indicates, depicts, or symbolizes B°. For B to be taken as a sign for users to act, indicality, iconicity, and symbolic representation is only a part of the conceptual model involved here.

Metaphors and metonyms probably are the most important conceptual models for understanding what artifacts come to mean to their users. Human-centered designers need not merely acknowledge the cognitive dynamics that these two tropes describe; they are needed to enable users to explore their artifact naturally. Metaphors especially are indispensable in facilitating interfaces with novel or conceptually difficult artifacts. Metonyms, by contrast, increase the efficiency of exploration. Parts and wholes belong to the same conceptual domain; they rarely create newness.

3.4.5 Informatives

Informatives report on what is happening elsewhere, review the history of an interaction, or lead one to expect what an action brings forth. The term is intended to parallel speech act theory in language. Searle, for example, distinguished between assertives, declaratives, commissives, directives, and expressives based on what a speaker intends a linguistic utterance to do and how a listener is expected to respond (Searle, 1969). Superficially, informatives resemble the traditional notion of signs as "standing for something other than themselves." Their differences are crucial, however. An informative informs users who are capable and willing to take it as such. From a human-centered perspective and regarding the use of artifacts, meanings reside in the relationships between users and artifacts. "Stand for" or "reference" relationships have little place in a theory of meaning of artifacts in use unless users realize them and act on them accordingly. The colors of traffic lights, so frequently discussed as an example of a sign, do not stand for physical or mental objects, as alleged (e.g., Fisher, 1984:15). Rather, they tell drivers whether crossing an intersection is legal or not, with violations understood to be punishable. The second point of difference is that informatives are embedded in the dynamics of human interfaces with artifacts. Referring to Figure 3.2 and Figure 3.4, what makes something an informative is how it affects the continuation of an interaction, how it moves an interface forward. Three broad classes of informatives can be distinguished, those that redirect users' attention to what matters; those that inform users of where they are, where they came from, and how far they have to go; and those that inform users of what can be done, their possible paths, opportunities, and dangers. The following is a tentative list of the major informatives available to users for exploring what they are doing with an artifact and serving designers as a kind of checklist for considering whether their designs provide enough user support.

- **Signals** *call attention, first to themselves, but then, perhaps, to other things as well*. By this definition, all signals must succeed in competing for sensory attention with the other features of an artifact. They accomplish this by psychophysical contrasts, for example, a bright flashing light in the context of motionless illumination, the pulsating cursor in the midst of stationary text, a sound that cannot be ignored, the vibration of a cell phone, or a visual display that stands out against its background. Art and advertising often use psychologically disruptive contrasts as signals. The most obvious signals are warnings, which are needed at points when users are likely to overlook possible dangers, false turns, depleted resources, and errors with irreversible consequences (e.g., permanently deleting computer files). Warnings that are printed on an artifact, for example, are often ineffective because their permanence invites relegating them into the background. Choosing signals, their placement, and the kind of contrast they exploit amounts to finding a delicate balance between the qualities of features that the user can put into the background when not needed yet that can make themselves known when attention is required, anticipated or unanticipated. Signals may become a distraction when calling attention to themselves when not needed, useless (easily ignored) when unreliable, and a nuisance when persisting beyond redirecting the user's attention, for example when it cannot be turned off.

Informatives concerning where users are, where they came from, and what they have done are suggested in the following list.

- **State indicators**, which *show users what their artifact is presently doing, which mode of operation it is in.* For example, the tone following the dialing of a telephone number indicates the state of the addressee: ringing, busy, out of order, or not in service; a red sign near the handle of a door to a public toilet indicates "occupied." It has become customary for the top line of text on a computer screen to tell what software is being used and which file is open. There are three kinds of state indicators.
  - *Direct state indicators* are identical with the states that a user needs to know. In using simple tools, scissors, a hammer, or a pencil, the states of interest are whether the scissors are open or closed and in which direction they are pointing, where the hammer is relative to the nail that one intends to hit, or where on a piece of paper the pen is making its mark.
  - *Mediated state indicators* become important when the complexity of artifacts prevents users from having direct access to its operation. A fuel gauge, the temperature of the car engine, and the buzzer of a remote door opener mediate states of interest by a causal mechanism, making the states known to the user so that action can be taken if necessary. Direct indicators are more reliable than mediated ones. The position of a switch may tell whether a machine is on or off, but the humming of its motor is a more reliable indicator.

  - *Calculated indicators* abstract multiple states of an artifact and present them in terms that users can understand. The speedometer of a car is an almost too simple example. Speed is distance traveled divided by time. The mechanism of the speedometer "calculates" this proportion from two points on these two variables and presents it to drivers, largely because they can think in terms of speed and know the legal limits in these terms. Wattmeters, Geiger counters, wind chill factors, and the Dow Jones Industrial Averages are state indicators that transform something measurable, not necessarily visible as such, into numerical indices that are observable and comprehensible. Good economic indicators, for example, give people a sense of an economy's health.
  - Delays in the indication of states make the navigation of an interface difficult. In steering a large ship, for example, inertia intervenes between the position of the rudder and the intended change in direction. Compensating for these delays is not easy. This may call for a computed index that is more directly related to what the helmsman has to do. The hourglass icon on the computer screen fills delays, telling users to be patient and that something is in the works. This can be frustrating, however, because there is no indication of how long the wait could be. A computer could be stuck there. A more preferable indication is to turn state indicators into progress reports, replacing the hourglass icon by something that indicates the expected wait or spells out what the computer is momentarily doing.

- **Progress reports** locate the present along a continuum between starting an effort and finishing it, or show a user how much was accomplished up to now. Many elementary engagements in everyday life generate records of what has been accomplished and where one finds oneself in it. A penned letter is the written record of how one's coordination of thoughts and hand movements unfolded. The page numbers of a book tells the reader how much has been read. A house, built brick by brick, reports on how far the builder went. The horizontal bar graph on a computer screen, showing visually and/or numerically the percent of completion of a task, for example, downloading a computer program, amounts to a progress report. Although not all uses of artifacts are as linear and finite as writing a text, building something, or downloading a file, progress reports respond to the users' need to stay in touch with their own history of involvement. In a project of providing electronic support for the handling of insurance cases, we defined milestones that had to be completed along the way.
- **Confirmings** acknowledge to a user that their messages have been received and their actions accepted without yet showing their intended consequences. Confirmings make feedback about actions instantaneous. Their importance in everyday life is rarely recognized and mostly taken for granted, especially when handling simple tools. In cutting paper with

a pair of scissors, the user hears a supportive sound. In writing with a pen, the marks left behind on the paper immediately confirm the movements of the writer's hand. In hammering a nail, the carpenter receives feedback from the sound of hitting it, having a lower frequency when it stands out and a higher one as its head approaches the surface. Imagine a pen whose ink becomes visible a minute after it is applied. It would make writing nearly impossible. Experiments with human subjects who heard what they were saying but after a delay of few seconds prevented these subjects from formulating coherent thoughts and led them into an unintelligible speech discontinuity. Designing a noiseless keyboard seems a reasonable aim, presents no technical problem, and has in fact been done — only to discover that typists failed to get a sense of whether they were actually typing, whether the computer had accepted what they wanted to enter. Lack of such confirmations usually is extremely disorienting. In the design of complex artifacts, where actions and visible consequences are delayed or mediated by mechanisms that are not transparent to their users, designers may have to make special efforts to confirm the users' actions. For example, to confirm that clicking on an area of a computer screen was effective, interface designers have come to use the image of a push button whose change of shadows can be used to confirm that a click was taken. Such an image works as a visual metaphor whose changes are confirmative of the clicking action.

Informatives that offer guidance about the variables of an artifact and their consequences are as follows:

- **Affordings** *indicate the state of readiness of the artifact or parts thereof to be acted upon.* Section 3.4.3 defined an affordance as the perception of usability, which in Gibson's terms is direct and unreflected. Affordings include direct perceptions, such as of bendability, twistability, pushability, movability, graspability, steerability, heaviness, and dangerousness — Gibson's affordances — but also what users can derive by logical reasoning from clues, icons, shapes, and patterns. For the buttons on the toolbars of computer interfaces, for example, it might be obvious that they are clickable, but to figure out what they do may require users to engage in more elaborate cognitive processes. Knobs, levers, handles, steering wheels, and the metonyms on a computer screen become affordings when they can be seen as pushable, movable, turnable, and clickable, respectively — as exemplified by a collection in Figure 3.12. What contributes to the perception of affordings is their handiness — being of the right size for a hand to grasp, of the right height to hold onto, in the right place to push, and so on. Affordings are context sensitive. They become foreground when certain actions need to be taken and recede into the background when disruptions demand attention elsewhere.

  Cataloging affordings has been an approach taken by the Hochschule für Gestaltung Offenbach's project of a "Produktsprache," especially for

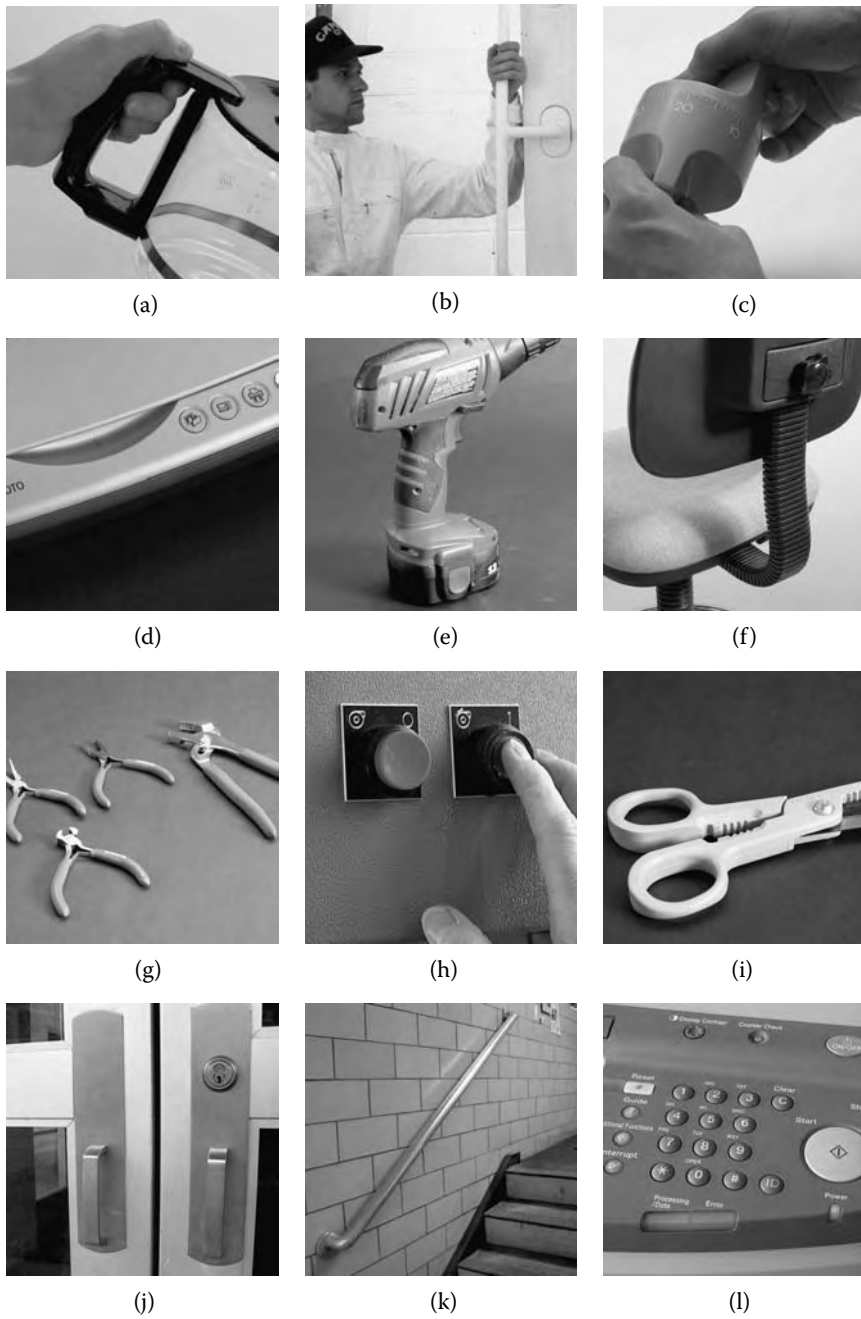


Figure 3.12 A sample of affordances. (a) Handleable; (b) support providable; (c) twistable; (d) openable; (e) graspable; (f) bendable and adjustable; (g) squeezable; (h) pushable; (i) finger insertable; (j) pull-openable; (k) hold-on-able; (l) key-in-able.

so-called indicator functions or "Anzeichenfunktionen" (Fischer, 1984). Working within a functionalist framework, Richard Fischer gives numerous examples of affordings of orientation, stability, handle-ability, movability, and precision.



(a) (b) (c)

Figure 3.13 Discontinuities that suggest differences in use. (a) Pavoni expresso machine; (b) Braun juicer; (c) Fiskars shears.

- **Discontinuities** *in parts, features, or arrangements suggest differences in their meanings*. The wood handles of a pan do serve as insulators against the heat needed in frying. But making these two parts visibly different serves as an informative for using them differently. Figure 3.13 gives three examples of using contrasts to suggest different uses. The LaPavoni espresso machine consists largely of chromed metal components, which can become quite hot in use. Its designers used contrasting black handles and knobs that users may touch safely while making coffee. On the Braun juicer, the overall circular body is interrupted by a concave indentation, suggesting a purpose, here where a glass for the juice could be placed. Additionally, since the opening from which the juice exits is not easily discernable, the designers marked its place by a red dot. As such, it says nothing. But on its otherwise white and uniform body, the dot directs attention to the metaphorical "point" of a juicer. The shears by Fiskars exemplify the use of color, here orange, for parts that the user may have to pay attention to, in this case to lock the tool for safekeeping. Another example is discussed in Figure 9.5. There is a general principle underlying these examples:

  Users are inclined to perceive discontinuities in the appearance of parts or features of artifacts as meaningful — especially in the absence of compelling technological reasons.

  Besides discontinuities in material, forms, and colors, there are other kinds. Unequal spacing of control knobs usually means that those closer to each other have something in common whereas those further apart are treated as unrelated. Larger controls are often seen as more important than smaller ones. An unexpected change in the sound of a car engine

usually indicates something to be attended to or worried about. The change in the cursor on a computer screen informs of something happening. Designers need to be aware of the human inclination to make sense of any discontinuity, encourage appropriate concepts for their interpretations, and avoid possibly misleading ones. Aesthetic considerations and graphic embellishments, which often guide a designer's choice of colors, patterns, and arrangements, should be secondary to keeping artifacts explorable and interfaces meaningful.

- **Correlates** bear physical or conceptual correlations between the movement, arrangement, or location of the controls of an artifact and the movement, arrangement, or location they affect. Norman calls correlates "mappings" and illustrates the lack of one by the widely recognized problem of knowing which knob controls which burner of a typical kitchen stove. Each burner is controlled by one knob. Whether for lack of imagination or space, the controls tend to be arranged in a row, bearing no obvious geometrical correlation with the typical two-by-two arrangement of the burners. The lack of correlation can cause dangerous confusions. Verbal designations on the knobs, for example, "left-front," … "right-rear," or icons rarely eliminate such control errors (Norman, 1988:75–79). Another albeit minor example involves translations of analog to digital time. Late visitors to the building where I teach are asked to sign a log when they enter and leave. While they do, they stand in front of an analog clock. It is amazing to observe how often people talk to themselves while trying to figure out what numbers to write down for the time or ask bystanders to confirm what they think the time is. After years of nudging a digital clock was installed and nobody had difficulties copying the numbers.

  Correlates may be positive or negative. When a control and what it affects move in the same direction, mimic each other's spatial arrangement, or are proportional to each other, the correlation is positive. For example, arranging the floor buttons in an elevator so that those for getting to higher floors are above those for reaching lower floors is natural, obvious, and a positive correlation as such. When the movements or placements of controls are oppositely related, there may well be a one-to-one mapping but the correlation is negative — much like correlation coefficients in statistics. For example, some cars that are put in reverse gear by shifting forward, which amounts to a negative correlation, while others are put in reverse by shifting backward, which establishes a positive correlation. There seems to be a general principle:

  When facing the controls of artifacts: users expect direct, simple, and positive correlations between what an artifact affords them to do and the effects their actions control. Indirect and complex correlations may be learned but take time to acquire and counterintuitive correlations are especially prone to cause disruptions.

Users follow their expectations, and even when told that correlations are negative or counterintuitive, they easily fall back on what is customary and natural, especially at moments of inattention. When controls are not obvious, with time, users learn to handle their artifacts. Drivers learn to drive their own cars, of course, but for new car owners and those who have to drive many different kinds, garage attendants for instance, negative correlations can create costly accidents.

The absence of a mechanically obvious relationship between users' actions and their consequences enters the domain of cultural conventions. Such conventions result from consistent use of the same correlations across numerous technologies. For example, turning buttons is common, but their effects are not. There is no necessary physical connection between the direction in which to turn a shower knob and the increase or decrease of the water temperature. Because the feedback on the effect of turning the knob is often delayed, travelers across cultural boundaries often find themselves chased out of a shower by extreme temperatures. In the domain of electronics, there is a growing convention for clockwise turns of buttons to increase something. The center of Figure 3.14 shows all three buttons of a radio that increase volume, frequency, and operation (on-off), by clockwise turns, respectively. Designers are advised to not only rely on such conventions, but also reinforce them by applying conventional icons, directional arrows, wedges, and codes using color, size, or arrangement, and relying on descriptions and warnings only as a last resort.



(a) (b) (c)

Figure 3.14 Correlates and a confusion of correlation. (a) Car seat controls mimic seat; (b) radio buttons clockwise increase something; (c) confusing elevator instructions.

Some of these population stereotypes are linguistically supported. For example, in most Western countries, to unlock a right-hinged door, you turn the key to the right (clockwise) and to unlock a left hinged door, you turn the key to the left (counterclockwise). This convention is an old one and derived from the mechanisms of early locks. European car doors work that way, but American car doors are inconsistent in this regard, and Honda's open the opposite way. Today's locking mechanism can go either way, and users tend not to understand them anyhow, which

suggests that mechanisms need no longer dictate the correlations that users need to learn. Figure 3.14 shows three examples. On the left are the controls of a car seat. These are on the side of the seat and felt not seen. Their shape mimics the two parts of a seat and their up, down, forward, and backward movements are positively correlated with the movements they control. On the right, one can see the positive correlation of elevator buttons and floors, which is contradicted, however, by instructions that follow top-down writing conventions, exhibiting a negative correlation to the floors.

When correlations are absent or not obvious, designers may have to rely on icons or suggest metaphors. In the United States, the toggle switches for home lighting point to the ceiling when the lights are on and to the floor when they are off. Pointing to the source makes sense. In the United Kingdom, these positions are reversed but made equally plausible: Switching the light on means bringing the light down from the ceiling and switching the light off means putting it back up to the ceiling. Exciting correlates between hand movements and data input to computers (see the data glove in Figure 1.3) and their use in virtual reality applications are being explored. The space for new controls with positive correlations is still wide open, but the lesson is always simple: The more the users' actions are positively correlated with or map proportionally onto what an artifact is to do, the easier is it to handle. The more designers rely on culturally established correlates, the shorter the learning curve and the fewer disruptions are experienced.

- **Maps of possibilities** *present the space within which users can navigate their interface with an artifact*. Users' conceptual models were characterized as cognitive maps for possible actions. They give their beholder a sense of where they are in their world. Where established models are not detailed enough or do not quite fit the artifact in question, the designer may have to provide the user within an interface, information about what can be expected, the actions that will become available, and what each of these will bring forth. Road maps are the most familiar examples of networks of possibilities. They do not predict what anyone will do with their help but show the possible paths their users can take from any one point to the next. The decision tree of the possible moves of a game, of tic-tac-toe for a simple example (Norman, 1988:120), is a map of possibilities and so is the diagram of options for telephone banking, and the graphical representation of work to be done and by whom in the planning room of a political candidate's election campaign. The outline of an article to be written defines a space that a writer intends to fill. An organizational chart defines which decisions are delegated to which departments or individuals. At a restaurant, a menu describes which foods are available for selection. In computer interfaces, menus serve as a metaphor for maps of possibilities that can guide the user into a desired future.

The reality of future possibilities can overwhelm users by its multitude and complexity. Maps are manageable simplifications. There are a few noteworthy guidelines that designers may consider. One is to use abstract graphical representations that leave out many details in favor of the overview they can provide. Patterns are easier to discern when seen than when described. A second kind of representation is centered on the present, showing current choices and as many future ones as the user can comprehend. The roadmap provided by a global positioning system to the driver of a car shows the car's current position in its center and the choices available at that location. Checklists of what to do are common in everyday life and useful in interfaces where there are several prerequisites for the available choices. A third representation is goal-anchored. In chess, endgames are important, and as the saying goes, there are many roads to Rome. With a clear goal in mind, presently available alternatives may be less decisive for reaching a designation. The challenge for designers is to provide a practical combination of these three, carefully considering the missing information that users can observe along the way while staying on course. Maps of possibilities fulfill the desire of users to make sense of their choices in view of where they want to be and what they wish to accomplish on the way.

- **Error messages** explain what can be done when an action does not accomplish what the user had intended. To clarify: (1) The word "error" tends to wrong an actor, here the user of an artifact. However, if a design can affect the rate at which errors occur and the severity of their consequences, then designers cannot entirely deny responsibility for users' errors. They are actors as well. (2) Since, by the above definition, errors are tied to users' intentions, and designers cannot possibly know all the intentions that users could have adopted for their actions, error messages can be provided only for paths taken frequently and leading to situations generally known to be undesirable, if not harmful, or bringing the artifact to a halt. Errors are cultural constructions, not entirely of mental or mechanical origin. (3) Even simple mechanical tools can stop working, fall apart, even kill someone, not necessarily because of reckless use or misconceptions. Things can just happen. Errors matter only to the extent their consequences are serious to those who care, cause unanticipated efforts, or inflict harm. Errors vary in their degree of attributed seriousness. The following describes errors in the order of increasing seriousness. The point of error messages is not to prevent errors but to reduce the seriousness of their consequences.
  - *Immediately correctable slips:* For example, typos, which are recognizable by the user and undoable. Designers may support recognizing such errors, for example, by underlining illegitimate words and providing spell checkers, or flagging slight deviations from a charted course of action and aiding their correction by undoing the

consequences of the last action. These errors can slow down an interface, but flagging can prevent irreversible harm.

- *Actions with irreversible consequences that are often taken by mistake:* For example, deleting a file or overwriting a document. Since only users can know whether a chosen action is intended; to prevent errors that are known to be typical, designers may install an overridable constraint (see Section 3.4.2) by reminding users about the consequences of their action and asking them to confirm that it was indeed intended. A confirmation overrides the constraint.
- *Actions that lead to a temporary dead end:* Such actions are often called "illegitimate" because the artifact is not designed to be used that way and the user is somehow expected to know that. Most such errors are design or programming errors: semantically suggesting and physically allowing actions that the machine cannot afford. Appropriate error messages should identify why the action or sequence of actions lead to the dead end — allowing users to learn what not to do next time — and give clear instructions for how users can get out of that situation.
- *Irrecoverable breakdowns:* Designers may not always be able to provide advance warnings or error messages that give reasons for a breakdown. Error messages require a minimally functioning system. A device that runs on batteries cannot indicate the loss of its battery — only its weakness. Here, the challenge is to design artifacts that fail safely. Building a weak part into a mechanism that breaks before the more important ones are affected is a traditional safeguard, for example, fuses that protect an electric circuit against overload. Routinely backing up computer files to protect losses due to a virus or unexpected power failure is another solution. Error messages may have to respond to but should not be disabled by the failure they are to indicate or help to correct. When a computer cannot connect to the Internet, Internet advice is meaningless. Error messages should be written in the users' language, avoid jargon, provide step-by-step instructions, and consider the likelihood that users will read them when desperate and without tolerance for blame or irrelevant matter.
- *Errors with potentially catastrophic consequences for users* call on designers to provide protective devices against harmful equipment failures, shields that prevent hands from being caught in a machine, fire extinguishers, airbags in cars, lifeboats on ships, etc. Because such errors should be rare, protective devices should be triggered automatically where possible or designed to be widely recognizable and usable with few if any instructions. Here, obvious meanings, ideal type forms, and widely held visual stereotypes are preferable to aesthetic elaborations.

- **Instructions** *guide users along a preferred path through the network of possibilities that an artifact affords*. Traditional artifacts, such as paper clips, screwdrivers, and binoculars, were either obvious or in the case of calculators, fax machines, and cars, came with instruction manuals. Generally, users do not read instruction manuals unless problems arise. A good deal of product semantics aims at the self-evident use of artifacts, relying on images or icons instead of words, thus minimizing the need for written instructions. Most contemporary electronic artifacts can contain three kinds of users' instructions and make them available when need arises.
  - *Enhancements:* To improve the efficiency of interfaces, to discover alternative uses and shortcuts
  - *Help:* To resolve apparent uncertainties of how to proceed
  - *Corrections:* To recover from apparent errors, especially after reaching dead ends

  These three needs call for different instructions. When errors are identifiable in terms of clearly circumscribed conditions and the users desire to repair possible damages or regain the ability to control the artifact, error messages are relatively easy to formulate: situation-specific step-by-step guides to get back to normal operation. The instruction for resolving users' uncertainties are more difficult to write because designers may know where the users are when asking for help, but not where they wish to go. Here, instructions need to be situation specific as well, but must be searchable in terms of what users want to accomplish. A map of possibilities, if available, can visually aid the search for help. Instructions to aid the exploration of the interface with an artifact can be more open. Good indices to available instructions are important. Answers to freely formulated questions should be provided in the context of what users already know, provide alternatives to the scenarios previously followed, suggest shortcuts, and encourage users to develop new conceptions in place of those that got them into trouble.

  Obviously, instructions should guide the user through a system of interaction. Nielson (1993:142–144) spells out four principles. Instructions should (1) rely on the vocabulary of common users and avoid obscure codes; (2) be specific by referring to what can be seen and done, for example, instead of saying that a document cannot be opened, it should say which document cannot be opened and which of its properties prevent it from being opened; (3) offer constructive step-by-step help, spell out the sequence of actions users can take, what they are expected to see, on the way to recovery; and (4) be polite and not intimidate the user. Users rarely feel good when they have to consult help files. There is no need to add blame to injury by calling an action an "error" — users always act in good faith — or "illegitimate" when users have no reason to know what the designer had intended. Nielson adds two very sensible

recommendations to the above. First, images usually are clearer than words. They can reproduce features that users need to observe and avoid the ambiguity of translating written instructions into images. Second, users should be able to apply the instructions while reading them.

3.4.6 Semantic layers

As previously mentioned, the industrial era demanded of its designers that they cover the ugly mechanisms of everyday consumer products with beautiful forms. It assigned designers the role of applied artists, experts on aesthetic matters with a sense for the economic reality of industry. Aesthetics has to do with form, not with meaning, and balancing aesthetic concerns with the necessity of function left meaning out of the picture. Covering functional mechanisms with attractive forms had the effect of separating two spheres of artifacts, the sphere of the user, consumer, and connoisseur of forms and the sphere of the expert technician who could build and repair what was beneath their surfaces. From a human-centered perspective, the form–function differentiation is mistaken. The distinction is not between the aesthetic and technical, nor between subjectivity and objectivity, but between two legitimate realms of meanings. Engineers are human as much as users are, and they do recognize, explore, and work with artifacts as their users do, albeit differently. Automotive designers may not see a car the way its driver does; they ask different questions and handle it in ways ordinary users may not, but neither perspective deserves special privileges. The semantic turn reinterprets a cover as embodying a distinction between two systems of meanings, applicable on different occasions, evoking unlike interactions, and often supporting the actions of different kinds of people. It suggests the possibility that meanings are layered.

A Xerox copier, for example, can be seen to have at least three layers of meaning. The first is for regular users to make copies — loading documents, choosing among options, and removing the desired copies. The second layer becomes available when something goes wrong. At this point, users who open the doors of the machine find themselves in a different world, where they can follow the well-marked paper trail through the copier and are led by instructions to the probable cause of the disruption. By reference to specially colored and numbered handles, levers, and knobs, users can rectify the failure and go back to making copies — the reason they came to the copier to begin with. The third layer is reserved for repairpersons, who, with the help of specialized tools, are able to enter a world that is shielded from the users' view and reserved for the technicians. These three worlds consist of three different systems of meanings, much like different languages, requiring different kinds of competencies to interface with the artifact. Layers of meaning are not necessarily related to different users, as suggested in Section 3.4.2. Most users of copiers have no problem with shifting mental gears, entering the world behind the doors of the copier, fixing paper jams, and



(a)



(b)

Figure 3.15 Semantically layered interfaces. (a) Canon copier on the level of copying; (b) on the level of correcting paper jams.

returning to the world of operating the copier. The first two images in Figure 3.15 show the two layers of a Canon copier.

The idea of layers of meanings did not emerge accidentally, however. It started in an environment in which products and services were sold separately, creating the distinction between users and technical experts. Early copy machines jammed far too often, necessitating long waits for service. Layering meanings enabled users to clear simple paper jams, and reduced repair costs, delays, and above all, frustrations. Layers of meanings are more common than it seems. A woman's purse has a public layer for everyone to see, and a private layer for the owner's personal belongings. A camera consists of one layer when taking pictures and another when inserting film. A fax machine has a place for blank paper and a place for the fax to exit, but when it comes to replacing ink cartridges it becomes a very different artifact that some may be reluctant to handle.



(c)



(d)

Figure 3.15 (Continued) (c) automobile on the level of driving; (d) on the level of servicing.

Designing cars that reveal a powerful and attractively designed engine under the hood, well before the car mechanic can explore any malfunction, has become of recent interest to automobile manufacturers. The second two images in Figure 3.15 show two semantic layers of an automobile. After opening the hood for whatever reason, a driver does not need to face a bewildering mess of wires, tubes, containers, belts, and dangerous looking, hot, and incomprehensible machine parts. Recent design efforts have focused on enabling drivers to better understand the routine maintenance work they can perform on their own. One motivation of such a design effort may be to impress car owners with the power of their car's engine, conveying the sense of owning "a tiger under the hood." While this aim may have little effect on usability, it demonstrates a shift in attention toward a semantic layer between

what a car owner can service and what an expert may have to take care of. Applying outstanding colors to those parts that users ought to watch and may need to handle is a further step in the direction of layering meanings.

Computer interfaces have, of course, capitalized on the idea of semantic layers. At least since the beginning of graphical user interfaces by Apple Computers, it is now common to open a window with its specialized tools and affordances, do some work there, then open a second window with different abilities, a third, a fourth, and go back and forth as needed. Users can do different things in different kinds of windows and freely travel though layers of meaning — without losing sight of where they came from and what they are dealing with. An artifact that supports multiple layers of meanings does not merely afford to be looked at from different perspectives, it becomes something different in each semantic layer. The greatest virtue of providing users with different layers of meaning is that they can become experts in the layers they feel comfortable with. They may enjoy one layer without the need to know all others. Users who fear to travel into unknown worlds do not need to learn the languages that define these other worlds. Semantic layers provide choices among kinds of interfaces, kinds of worlds.

All artifacts have different meanings at different times and for different users. The distinction made above between recognizing, exploring, and relying on artifacts already defines three kinds of uses. It would make sense, therefore, to design artifacts that analytically distinguish between different worlds, any number of worlds, not just for different users but also for different purposes and times. Whether there have to be marked boundaries that users need to cross by opening doors, removing covers, or clicking on new windows is a question of organizing different systems of meanings.

3.5 *Reliance*

When exploring artifacts, we are concerned not with what they are but how they work and may be handled. When relying on artifacts, we no longer ask "how-to" questions but are concerned instead with what they do to our world, what is happening while the artifact is in use. Under these conditions, artifacts become background, like breathing, wearing shoes, and walking, which can be taken for granted in view of the activity that really matters. For example, after a person has learned to ride a bicycle, after acquiring balance and no longer being afraid of falling, a rider does not need to remind herself to pedal. She can attend to riding, the traffic, the wind, the road, and, beyond that, what she will do when she has arrived at where the bicycle enables her to go. In fact, while riding it, she is amazingly unaware of what she has to do to actually stay on the bicycle. The automatic, reliable, and routine sensory motor coordinations enacted in the body take care of that. This is what relying on an artifact means.

All concepts of user-friendliness and of usability take reliance as the point of reference, whether ergonomists seek to measure it in terms of the rate and severity of disruptions, or whether ethnographers listen to what users say they are doing. To achieve reliance is also the primary aim of human-centered design, making interfaces meaningful to start out with but then able to disappear in favor of other concerns.

Surprisingly, designers have rarely been explicitly concerned with reliance. There is no good theory about reliance. Two reasons are likely to account for this lack. First and foremost, the long tradition of product-centered design aims at artifacts that are aesthetically pleasing, are functional (work as intended and are comprehended as such), and have utility (economic benefits) to all stakeholders. The idea that artifacts should be designed to become invisible, even only in use, is an anathema to what industrial designers worked so hard for and sought to achieve. Industrial design thrived on being noticed, not on creating artifacts that dropped from public interest. Second, it is just difficult to explain states of reliance. What can be said about it has less to do with the formal aesthetic qualities, functionalities, marketability, and economics of artifacts, which are rational concerns of extrinsically motivated users, than with the fit between actual users and their environment. In use, artifacts justify themselves by disappearing from their users' concerns. The concern for these artifacts is replaced by concerns for being and acting as humans in a technologically supported world. Such concerns go beyond the traditional designers' emphasis on material artifacts and their rational, that is, extrinsically motivated use. The following details several concepts that aim to overcome the tension between detached observation by designers or theorists, and involved participation in reliable interfaces by users.

3.5.1 Scenarios

Scenarios describe human–machine interaction sequences of what happened or what should happen in the life of an interface (Nielsen, 1993:99–101). There are various ways to obtain scenarios for a particular artifact. One is by video recording how a particular user interacts with an artifact. Such video recordings are not always easily interpretable, but they may be enriched by eliciting comments from subjects, either by asking them to narrate, while interacting with the artifact, what they are thinking and doing and why; or by inviting them to comment on their interactions when subsequently watching themselves on the video screen. The latter is more informative because most actions take place faster than the user can verbalize what is going on, at least not in needed details. Users may not see everything worth knowing while being interactively engaged. Such scenarios give designers a dual record of what happened and what the users saw themselves doing or responding to. In the tradition of ethnography, usability evaluations of existing products and prototypes make use of this approach.

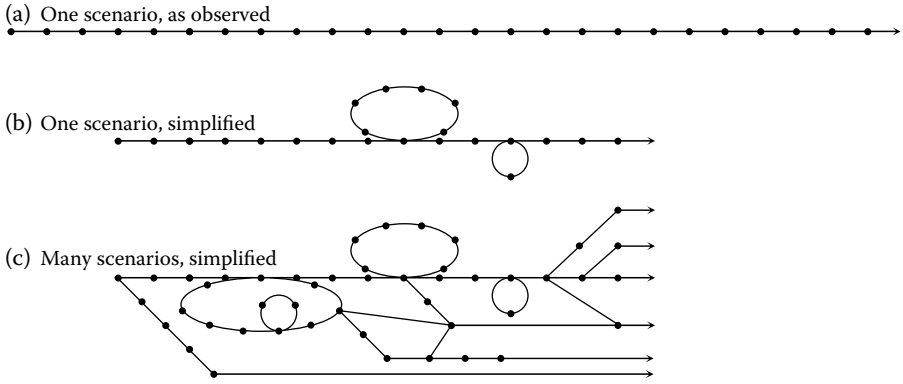


Figure 3.16 Analysis of several scenarios into maps of possibilities.

A second kind of scenario, which may well be combined with the first, is to hypothesize how users might or should be using an artifact and create a step-by-step account of what they would see and how they would have to respond to complete a task. For example, in designing an e-mail system, a designer may want to spell out the steps that potential users will use to access their e-mail, open it, answer and/or dispose of it, and make use of available clerical resources. They then might explore several ways to accomplish the same task, the icons that users are familiar with or could learn to steer through the options that the artifact needs to afford. Such hypothetical scenarios may serve designers as a means of communication with their clients to verify whether their development is in agreement with the clients' intentions; as a prototype, in various degrees of realism, including on paper (Snyder, 2003), to test users' ability to learn what is required to gain reliance or to discover what else the artifact needs to be regarded as useful and user-friendly.

In its most elementary form, a scenario records a single sequence of interactions as in Figure 3.2. Because reasons for choices cannot be observed, it is essential to record users' verbalizations of why they acted as they did, as suggested in the previous paragraph. For designers, the challenge is to analyze a collection of scenarios obtained from a representative sample of users and to construct a compound map of possible scenarios, similar to the maps of possibilities described in Section 3.4.5. The maps can serve as specifications for the intended artifacts to support them materially. Figure 3.16a depicts a single scenario, a temporally ordered sequence of interactions involving but one user. Although, literally, each situation in which people respond to what they see is unique — recall Heraclites' saying "you cannot step into the same river twice" — by ignoring features that are irrelevant for a design, one is likely to observe recurrences of the same actions at different times. Constructing now a simplified diagram, Figure 3.16b, in which each action of the original scenario occurs ideally only once, reveals loops, which are recurrent interaction sequences that repeat a process until the user is

ready to go on. For example, taking care of e-mail involves repeatedly going through the sequence: selecting an e-mail from a list, opening it, discarding, filing, or responding to it, then going back to the list. Most user-friendly scenarios contain many such recurrences, whether working down the list of e-mails, riding a bicycle from here to there by repeated pedaling, or cutting a piece of paper with scissors by alternating the opening and closing of its blades. Putting a large collection of interaction sequences that have been obtained from one or typically many users of the same artifact into a single diagram, Figure 3.16c, can reveal decision points, alternative paths to the same destination, and shortcuts that experienced users may have found.

Invariably, the most natural and, hence, reliable scenarios are those that consist of recurrent interaction routines, utilize familiar principles or common metaphors to conceptualize them, may be executed without much thinking (Krug, 2000), and can be learned easily. Obviously, detailed maps of possible scenarios are more informative about the usability of an artifact than descriptions of its function from the perspective of a nonuser. Scenarios must also acknowledge different kinds of users. Because observing thousands of potential users' scenarios can be costly, designers have studied typical users selected from each targeted user community.

A typical user is a myth, of course. But when designing new artifacts for usability and for not yet existing users, designers have resorted to constructing just such myths, so-called personas (Cooper, 1999). A persona is not a psychological concept. Rather, it consists of a set of scenarios, habits, or practices of living that an imagined archetypical user is expected to routinely engage in or is willing to acquire in the face of a novel or redesigned artifact. The idea is to construct several personas, and use them in deliberations among designers on how each would use the artifact, in the hope that the projected communities of users would behave just like their archetypical user, the persona. Examples of personas include: a young executive in a start-up company; an IT professional in an academic setting; a mother with two children working from home; a fashion-conscious upper-middle class urbanite; or a homebound invalid with an advanced degree. The ideal description of a persona includes all the characteristics — social, economic, cultural, lifestyle, but above all, available technical know-how and preferred learning styles — that could have an impact on the usability of the artifact and are sufficiently representative in the population of possible users. To participate in design decisions, a persona's description needs to imply behaviors that designers can utilize in conceptualizing and testing the artifact in question. While personas are specific users, the danger of designers arguing for them lies in the well-known fact that personas are designers' constructions. They represent designers' expectations, and often are constructed to be favorably inclined to a design. Ultimately, actual designs have to work with real people that need to be able to rely on a multitude of rather detailed scenarios. While personas offer a way to introduce users' personality characteristics into the design process, they should be constructed to be critical if not opposed to what designers wish to accomplish. Only when a design

can be shown to work for the most resistant users can designers be somewhat assured that they are on the right track. The conceptual benefit of the use of personas is that they replace aesthetic considerations with usability criteria.

3.5.2 *Intrinsic motivation*

Clearly, people surround themselves with artifacts they understand, are competent in using, and feel comfortable living with. Such are the artifacts that can be relied on when needed. However, this quality is not sufficient to explain why people engage more with some artifacts than others. What explains this selectivity is motivation. Motivations are justifications for actions, not to be confused with drives, values, or mental states, which invoke causal explanations devoid of human agency. Motivations are offered when asked why one was moved to act. For the use of artifacts, two kinds of motivation are distinguishable; both build on the reliance on artifacts in use.

- **Extrinsic motivations** *justify actions by reference to goals to be reached, tasks to be completed, and rewards expected upon successful completion*. Extrinsic motivations are rational, and their conception is indebted to the enlightenment. Surprisingly, Heidegger always saw technology (his ready-to-hand reliance as described here) as instrumental, as motivated by a concern for something else (Dreyfus, 1992). Short of providing maps to users for locating themselves as to where they are, where they come from, and where they could go, designers can do little in support of extrinsic motivations because they refer to states of affairs outside of their control.

There is more to keeping the interface with artifacts meaningful. People drive expensive cars, although cheaper ones would serve the function of transportation as well. People are willing to live with dysfunctional furniture as long as they like its looks, and what suits one is not necessarily acceptable to others with the same physiology. These are facts of life that cannot be explained instrumentally, rationally, and by reference to measurable criteria of success; hence, the need for the second kind of motivation:

- **Intrinsic motivations** *justify actions in their own terms. They refer to the sheer pleasure of being involved in a process* that Csikszentmihalyi called *flow* (Csikszentmihalyi, 1997), regardless of extrinsic conditions. Intrinsic motivations may be encouraged by interfaces that users consider inviting, exciting, and that draw them in. Intrinsic and extrinsic motivations, while clearly distinguishable, may well co-occur. In fact, they often collaborate in producing an effect. Intrinsic motivations account for what makes an interface exciting and refer to emotional states that come with involvement, not merely observation, which are always present and are not expected to be experienced in the future.

Intrinsic motivation enhances usability immeasurably and because designers can support it, it is an important aim of the semantic turn. To

highlight the difference between extrinsic and intrinsic motivation, I provide three examples and a list of conditions favorable for intrinsic motivation to arise.

In the mid-1980s, when PCs came to be purchased in noticeable numbers and entered workplaces in the United States, the Wharton School of Business and Finance, University of Pennsylvania, conducted several statistically sophisticated cost–benefit analyses of PC use.* Mainframe computers had already demonstrated their worth for hierarchical organizations with central accounting and scheduling problems. PCs, however, were different. They did not speed up typing, but wrestled editing, formatting, desktop publishing, and data manipulation out of the hands of experts and enlarged the scope of office workers. They brought previously centralized intelligence into everyone's reach. Subtracting the costs of purchasing and maintaining PCs from the benefits they created for the company yielded a surprising conclusion: *There was no justification to use PCs at all*. Little did the researchers appreciate that cost–benefit analysis is measuring extrinsic motivation.

History, however, proved cost–benefit considerations are not the only drivers of technology. Humans have wholeheartedly embraced information technology as part of contemporary culture. Where PCs entered the workplace, the hierarchical structures that provided tight and efficient controls became weakened. Where it allowed electronic games, it became exciting. Connected to the Internet, it changed the way we see the world. Its myriad of meanings pervades every sphere of everyday life. Ergonomics and functionalism, economic theories that are built on extrinsic motivations, can explain only a small part of these developments. Intrinsic motivation is what brings life to the use of technology.

Consider, secondly, downhill skiing. Standing in long lines at ski lifts that bring people, like cattle, up a mountain from which they have to come down on their own to where they had started certainly does not make much sense, especially in winter when it is cold. There is no apparent goal, no financial reward, and no rational reason to do this. Additionally, there is a high risk of accidents. Skiers fall, break their bones, get killed. Why on earth would anyone want to risk their lives like this? The short answer is that motivation is intrinsic to skiing, not justifiable by reference to delayed rewards or the achievement of objectively measurable goals.

Or, thirdly, when entering a gaming parlor as a naive observer, one might be struck by the buzzing, humming, and ringing sounds emitted by machines that take money from their players in return for letting them control certain images on a televisionlike game screen for a very short while. Yes, some of these machines announce winners and losers or give performance scores by which people can judge their own successes — but even losing does not seem to matter that much. Participation has no measurable consequences. Yet people are deeply involved in the process. Only an outsider would ask, "Why?" Players consider this question already answered

* James C. Emery, 1992, personal communication.

by the very fact of being there, by their ability to get lost in a game that these machines interactively support.

Those who are involved in such uses of artifacts (e.g., PCs, skis, and game machines), might find outsiders' why questions irrelevant, but after engaging with such artifacts are eager to talk with peers, relating their experiences in the glowing terms of intrinsic motivation. Adjectives like "exciting," "thrilling," "elating," "exhilarating," "blissful," "effortless" are typical. Feeling "fearless," "certain," "unencumbered," and "free" goes with it. Such extremely positive emotions go beyond judging their artifacts as "easy to use" and their involvement "effortless," or "natural." Experiences like these become "memorable," "unforgettable," "extraordinary," "profound," and motivate further involvements of this kind, with only minimal concerns for the costs and benefits of use.

Further primarily intrinsically motivated activities include: dancing, playing basketball, flying a glider, climbing Mount Everest, composing music, reading a novel, riding a motorcycle (Pirsig, 1999), or simply doing enjoyable work. Csikszentmihalyi (1990) interviewed numerous people who reported what he called "optimal experiences." Such activities vary widely, as the above examples suggest, but the experiences they described are virtually alike. Malone (1980) and Malone and Lepper (1987) studied intrinsic motivations in the use of toys and involvement in computer games. Alison Andrews (1996) interviewed designers of advanced computer games and produced a list of 18 intrinsically motivating features. The following summarizes their account of what intrinsically motivated activity consists of (Krippendorff, 2004b):

- Intrinsic motivations arise in meaningful human–machine **interfaces.** As described in Section 3.1, an interface is dynamic and relational, and consists of interactions. It is a process. Intrinsic motivations, therefore, cannot be attributed to the properties of an artifact, physical or aesthetic; be explained as a mental phenomenon (e.g., intentions); or be measured with mechanical instruments, typically favored in ergonomics.
- They presuppose **reliance,** the unproblematic interactive fit between human behavior and its technological support that allows users of an artifact to concentrate on what matters beyond the handling of an artifact. For users to be able to rely on an artifact, its responsiveness must be predictable. Disruptions upset such reliance and make intrinsic motivation unachievable.
- They demand a substantial amount of **user autonomy.** Users must be able to set their own goals and define their own criteria of success, have a sense of making their own decisions, and proceed in their own terms — at least where it matters. Most technologies simultaneously restrict and enable. Writing restricts communication to what others can read, preventing voices from being transmitted, and limiting visual experiences to still images, but it enables preserving messages in time and carrying them

over great distances. User autonomy refers to the area that artifacts enable. Noticing the constraints of a medium, having to follow precise instructions, or working toward goals set elsewhere to please someone else, for example, turns out to distract from intrinsic motivation.

- They involve **multiple sensory–motor coordinations**, ideally of the whole body. In skiing, virtually every part of the body participates in influencing speed and direction. Notwithstanding the Western cultural bias for the visual, there is the contact with the ground, the sound of the ski gliding through the snow, the wind blowing in the skier's face. Similarly, using the keyboard of a computer is not just pressing keys. What happens as a consequence of its use is or should be seen as well as heard and felt. In addition to the five external senses, there are two internal ones. The kinesthetic sense of where and what one's motor organs are doing, and the e-motions, the "ease" and "dis-ease" of otherwise unrecognized bodily processes. Distinctions among e-motions are called feelings (Krippendorff, 2004b:54). Limiting the participation of the available sensory modalities impoverishes an interface and is likely to reduce the chance of intrinsic motivation.
- They call for **continuous learning**. Csikszentmihalyi (1990) describes this feature in terms of the delicate balance between using exceptional skills and managing challenges. Skills are rooted in a history of doing something repeatedly and well. Indeed, skiing at a level at which it becomes enjoyable calls for considerable skill. But continually using an acquired skill becomes boring; for an activity to remain exciting, challenges are required. When skiing, these challenges come from the terrain, mediated by the skis. When playing computer games, these challenges may have to be programmed into the game. For an interface to remain intrinsically motivating, challenges must not be so large as to disrupt the process of engagement, but large enough to maintain a sense of creative mastery. Mary Catherine Bateson (2001) speaks of the human right of continuous learning.
- They grow on the **confidence** of being able to handle what might actually happen. Some designers have argued that intrinsic motivation comes from users being in control (Andrews, 1996). But perfect control can never be achieved. Intrinsically motivating activities provide a sense of not losing control, an absence of the fear of getting into trouble.
- They cannot be observed but **must be practiced**. The interactions in which intrinsic motivations arise may be video recorded, their disruptions counted, and their history described. However, observations cannot provide access to users' bodily involvement. What may be observed from the outside is what users take for granted, reliance, the background of their interfaces. What really matters escapes observation from the outside, from another body. Observers may conclude that intrinsically motivating activities are self-pacing. But for those involved there is no time. E-motions are felt in the present. Recalling or recording them is something altogether

different. Witness how we lose our sense of time, fail to keep track of other obligations when involved in something exciting. Feelings simply are.

- They are grounded in a clear **sense of place and direction**. While painting, for example, artists are extremely alert to where they are and seem to "know" what they are doing, without necessarily being able to explain it. Painters may have visions of the final product, but these visions often interfere with the excitement of painting. A painting is not finished until the artist knows it is — observe famous painters' frequent revisions. In downhill skiing, every slope ends somewhere, of course. But what matters at any one moment is the sense of getting somewhere. Intrinsically motivated activities can be said to be purposive without a goal.
- They are conceptually **self-sealing**. Intrinsically motivating activities typically require extreme concentration to the point that nothing else is noticed; for example, time, already mentioned, but also other people's concerns, potential dangers, and the original objective that may have initiated the activity. Action and the awareness of the action merge into one, into Csikszentmihalyi's (1990) "flow," or into a state of being absorbed without regard for anything but the continuation of the interface.
- They stay within an **uncontaminated reality**. When absorbed in an intrinsically motivating activity, there is no place for alternative realities, no place for playing a role, faking something, or lying; it means simply being what one is. This feature of intrinsically motivating activity has been described as a loss of self-consciousness. However, no consciousness is lost; its facades merely disappear, pretend selves vanish, and a self emerges that is liberated from all the usual constraints, and this self is very real during the involvement.

These features of intrinsic motivation make clear why what seems to drive postindustrial artifacts remains hidden to technology-centered or objective measurements and defies causal or rational models of human behavior. They also challenge ergonomic approaches to usability. This is not to say that all artifacts need to be designed to be fun and exciting, or that extrinsic motivations should be abandoned in favor of intrinsic ones. There are things that simply need to be done and for which users require artifacts that they can rely on or take for granted to attend to other important things. Where appropriate, usability or user-friendliness is significantly enhanced where intrinsic motivations are enabled.

3.6 Principles for designing usability

Principles for the design of artifacts that are meaningful in use are emerging slowly (Dykstra-Erickson, 1997). Several minor recommendations have already been woven into the previous discussion of concepts of the semantic turn. The following sections summarize a few.

3.6.1 Human-centeredness

As this book suggests, human centeredness entails conceptualizing technology in human terms. Its currency is meaning. In the special case of user-centered design, it must be recognized that communities of users determine the meanings that their artifacts have for them, not outsiders, designers for example. Meaning presupposes understanding and use. Because all understanding involves metaphors — understanding one thing in terms of familiar other things — designers should rely on metaphors from positive human experiences. Humans thrive when they can realize themselves on their own terms, unfolding autonomously. Thus, where designers have options, artifacts should be designed to be open to multiple uses, enhance community and dialogue, survive in the marketplace of ideas, and encourage heterarchical networks: cooperative, democratic, diversity sustaining, and nonhierarchical (Krippendorff, 1997:30). Metaphors derived from playful activities, from enjoying each other, or collaborating in the social construction of realities, are preferable to metaphors from technology.

3.6.2 Meaningful interfaces

On the level of use, the aim of design is to encourage interfaces between humans and users that move effortlessly from *recognition* to *exploration* to *reliance* and to stay there until the user is satisfied. Each phase amounts to a different kind of user involvement. Artifacts must be designed for seamless transitions from one phase to another and support each to their fullest. Aesthetics in recognition is a minor phase in the use of artifacts. Design for usability amounts to encouraging sensory–motor coordinations that lead users toward reliance without disruptions. Usability may be measured by the periods between disruptions and by the time required to correct them. But to assure usability, designers should concern themselves with ways of discouraging meanings that could distract users from achieving reliance on the artifact under consideration.

3.6.3 Second-order understanding

Designing for others draws on knowing the conceptual models and familiar scenarios of users (i.e., it requires entering users' understanding in designers' understanding, even if they are incompatible). Designers cannot assume that their own perceptions, their own world constructions, are the same or similar to those of the potential users of their artifacts. Design should aim at providing material support for existing and easily acquirable conceptual models, scenarios, and interpretive schemes. Insisting on being able to inscribe intended functions into artifacts is a designer's myth. Artifacts can constrain some uses, but design must create spaces for users, not force compliance with technology.*

* This recalls von Foerster's (1981:308) ethical imperative: "Act always so as to increase the number of choices" — I would add "for others."

3.6.4 Affordances

Affordances are the basic units of meaning in the use of artifacts. They are perceptions of what can be done with an artifact, its parts, and its controls (affordings). Affordances entail a range of meanings that can be acted upon with the expectation of bringing forth particular senses. Affordances are irreducible cultural habits and difficult to modify. Designers should make them easily recognizable and bring forth the consequences that users expect, ideally without unnecessary constraints.

3.6.5 Constraints

Constraints restrict the actions that an artifact can accept. Constraints can be frustrating when pointless, when what gets in the way of natural use are avoidable limitations due to designers' inattention, producers' convenience, or an ill-chosen technology. Constraints that prevent users from selecting their own goals and measures of success, and force compliance with a particular way of proceeding prevent users' intrinsic motivation (Andrews, 1996). Designers should avoid meaningless limitations and resort to constraints primarily to prevent illegitimate uses or protect users from being harmed. Generally, semantic ways of directing the attention and interactions of users are preferable to physical constraints.

3.6.6 Feedback

Feedback on any user action should be as immediate and as direct as possible, starting with confirmations that users' actions are accepted by the machine, and executed, with timely information about the intended and unintended consequences arising from these actions, up to indicating the readiness to accept further actions and directions.

3.6.7 Coherence

Comprehension is enhanced by the use of visual metaphors, metonyms, and icons that support each other. Humans are inclined to trust direct experiences more than written user instructions that, by merely informing about conceivable experiences, could be misleading. Multiple metaphors with conflicting entailments present problems, as do icons that stem from incompatible systems or instructions that fail to establish relations to what can be seen or needs to be observed. Designers are asked to avoid mixing metaphors and using similarities, proximities, and correlations that could lead to contradictory meanings. Defining semantic layers is one way to separate incompatible systems of metaphors, metonyms, informatives, and instructions, creating several coherent worlds between which the user can switch, interfacing differently in each.

3.6.8 Learnability

Humans learn all the time. Meanings change in use. Design merely intervenes in that process. Meanings, user conceptual models that organize them, and scenarios that roll them out in time are always somewhat in flux. Learning takes place always and everywhere, albeit in different measures. Some interfaces change faster than others; some resist newness and some are eagerly abandoned when new opportunities present themselves. A scientific survey of the competencies of a community of users is always already outdated when completed. It follows that artifacts should be designed for learnability: Users should be able to enter the interface with an artifact on any skill level and advance their competencies at their own pace. Users should also be permitted to introduce variations on successful scenarios, finding shortcuts, economizing efforts, and having fun. Users should be enabled to apply their own learning styles. Some learn by reading instructions, some by playing with untried alternatives, some by modeling what successful others do, and some prefer to be walked through by a mentor or expert. (Also see Section 3.6.11, Robustness.) Artifacts must acknowledge what users know today but be designed for learnability.

3.6.9 Multisensory redundancy

Most artifacts appear as multisensory phenomena. They can be touched, seen, heard, smelled, even tasted. Humans differ in sensory modal preferences. Some are oblivious to smell, some are hard of hearing, not to speak of possessing severe sensory disabilities. Humans compensate for deficiencies or weaknesses in one sensory modality by relying on another, without necessarily being aware of it. About 10% of all people are color-blind. Most do not know it because they have found alternative ways of coping with what would otherwise escape them. For example, color-blind drivers can read traffic lights by their relative position (top, middle, and bottom) and may well speak of their colors (red, yellow, and green) as if they could recognize them as such. We may never know what users actually relied on in a particular situation. Multisensory redundancy means providing the same affordance by alternative senses. Redundancy is not wasted effort when the information provided is mutually supportive.

Our culturally dominant sensory mode is the visual. Notice the comparative poverty of vocabularies for the senses of sound, touch, and smell — taste being reserved largely for food and kinesthetic senses and feelings being notoriously ambiguous. Designers demonstrate their preference for the visual by publishing their designs in colorful magazines, which cannot convey information on other modalities. The design for nonvisual senses is a neglected area (Ginnow-Merkert, 1997). Worse, by video taping interfaces, designers may never know whether a user relied on touch or on sound and whether observed users see what observing designers consider obvious. The design of interfaces with artifacts must assure that all senses are accounted

for and well coordinated, and that each sense is able to serve as an alternative source for the same meanings. Users who can shift sensory reliance from one mode to another without loss of meaning are not as easily distracted and can engage in parallel activities. Lately, car manufacturers have been designing the smells of car interiors and the sounds of closing doors. While these efforts may have little to do with usability, they attest to the increasing attention to nonvisual sensory experiences. The point made here is that designers' attention to all senses not only provides richer experiences with artifacts, it can also enable their users to operate in their most preferred sensory modalities.

3.6.10 Variability — diversity

The variability of artifacts should match the diversity of their users. During the industrial era, designers believed that optimizing the efficiency of use by criteria they or other authorities had stipulated was universally desirable. Their products aimed at an enlightened majority of consumers, in the conviction that the remaining population could be taught or would comply, hardly realizing that this benefited mass production more so than individual users. Much of the foregoing speaks against this practice. The new information technologies have encouraged interfaces with artifacts that enable users not merely to interact with them but also to connect and communicate with other users, who may live different lives and have different meanings for the artifacts they surround themselves with, yet use the same kinds of artifacts. The challenge for today's designers is to develop artifacts that are flexible, interactively modifiable, provide for multithreading, task migration across different systems, and customization or reconfigurability. Ideally, artifacts must have enough variability to enable a multitude of users to construct their own worlds in them. In other words, *the variability of artifacts should match the diversity in the population of users.**

3.6.11 Robustness

Section 3.2 suggested that the concept of an error entails finding someone to blame. This semantic entailment of the word error is not altered with Norman's generalization: "To err is human" (Norman, 1988:105–140). Is it the user's fault to have assumed that an artifact could do something that its designer did not think to allow? Humans cannot deliberately err. They may be momentarily inattentive but always act as best as they can in the circumstances in which they find themselves. It is a challenge for designers to anticipate what the users of an artifact might do and provide the necessary clues to prevent interfaces from being disrupted, or worse, causing harm. Artifacts should not be able to cause catastrophes, should have protective devices if they could harm somebody, fail safely should they break down, provide help when reaching a dead end, give warnings when an irreversible

* This is another version of Ashby's (1956:206-213) "Law of Requisite Variety."

action could be a mistake, and render disruptions easily reversible. An ideally robust artifact is one in which most foreseeable mishaps are prevented; when something unintended does slip through, it provides an opportunity for learning something new.

3.6.12 Delegation of design

Users' ability to create meanings for their surroundings and act on them is not radically different from designers' ability to develop a design and encourage its implementation by others. Users and designers may differ in intent, in their reliance on first- versus second-order understanding, and in whether they work through an institution or populate the market. Practices of living with artifacts and practices of designing artifacts are creative endeavors and entail agency. Contemporary information technology blurs the distinction between professional designers and intelligent users by enabling, if not demanding, that users design their own world with or within their artifacts. The advent of desktop publishing was a shock to graphic designers who thought typography was an exclusive professional skill. Computer hackers often know more about certain areas of computers than the designers who conceive them. As a creative activity, design is proliferating into many disciplines, but, most important, customizing, programming, and constructing information artifacts has become a fact of everyday life. In view of this technology, and according to what has been suggested above, professional designers can no longer maintain the exclusive privilege of designing artifacts for others. Design competence needs to be given away, delegated to users, and encouraged everywhere. Artifacts should be complex enough and designed to embrace enough uncommitted variety that users are enabled to design their own realities with them. Delegating design decisions to users is the *sine qua non* of human-centered design. Concerning the use of artifacts, designers cannot do better than to stay ahead of everyone else, at the frontiers of interface development, while teaching their craft to everyone willing to listen.

chapter four

Meaning of artifacts in language

The use of artifacts, their involvement in our interfaces with technology, is only a part of their life, an essential part but often only a small one. Most artifacts occur in language before they are put to use and often continue in language well after they fall out of favor. One may hear about new artifacts from experts or trusted peers, or learn about them through publications, advertisements, or consumer research. Adoption is simple and automatic. Acquisition tends to follow information about what artifacts do and whether they are worth their price. People feel compelled to justify their uses of artifacts to friends or superiors, or teach what they know to those who might benefit from them as well. Technical instructions make artifacts usable, and engineering texts analyze their features for further development. Artifacts that are no longer used enter museums and are kept alive in history books, literature, and entertainment — how else would one know of steam engines, Benjamin Franklin's use of kites to draw electricity, medieval swords, and primitive arrows. Without verbal explanations we would not know what they were.

Artifacts that are being talked about enter the language of particular communities, and become social or cultural artifacts, often well before they are actually used by individuals. The meanings that artifacts acquire in use are largely framed in language, and enacting these meanings occupies only a small part of their life. Thus, an independent theory of meaning of use, even the idea of individual users as distinct from nonusers, is a conceptually convenient fiction. Technology after all is a "-logy," and as such constructed and conveyed. This chapter looks at how language is implicated in the conception, application, and disposition of artifacts, and in understanding technology more generally.

To start the exploration, consider that most uses of artifacts are accompanied by what Bruce Archer called bystanders* who pass judgment, offer advice, or talk about uses and users with others. The sheer awareness of

* Bruce Archer observed this while teaching at the Ulm School of Design between 1960 and 1962.

being observed, the belief in being judged, or the need or desire to demonstrate one's competencies to others transform individual human–machine interfaces into public interfaces. Seeking approval, or fearing what others say about what we do with artifacts implicates language, conversations among stakeholders with an interest in the human use of artifacts.

When noting in Section 1.3 that advances in marketing and consumer research have altered the reality of design, it is important that we realize that such research depends entirely on verbal communication between researchers and potential consumers. Interviewers, for instance, ask questions and record answers. What cannot be expressed in words cannot be asked. Focus groups are managed conversations about consumer products and their users, from which verbal attributes are extracted and possible markets are explored. The definition of meaning in Section 2.3 and of disruptions in Section 3.2 in effect distinguishes among explanations that users tend to give for either of these. User conceptual models, UCMs, and scenarios (in Sections 3.4.1 and 3.5.1, respectively) are constructed from users' narrative accounts or from observations that are enriched by what users say they try to accomplish while acting. Warnings and error messages (Section 3.4.5) may use icons and images but can rarely bypass verbal formulations, whether users are asked to read them or whether artifacts "talk" to them. And motivation, both extrinsic and intrinsic (as distinguished in Section 3.5.2), can be revealed only in justifications for getting involved. Without verbal reports, designers would have little or no access to users' understandings. Language is ubiquitous. While not all meanings in use are verbally expressible, see Figure 0.1 — hence the previous chapter — those we talk about undoubtedly are.

In the process of designing, the role of language is easily overlooked. Designers cannot possibly do their work without getting some idea of what their clients want — design briefs, for example. Designers deliberate among themselves about what their clients expect, agree on the approach to be taken, share relevant knowledge, develop conceptions appropriate for the situation, and end up presenting their proposals to those who matter. While a picture is said to convey more than a thousand words, pictures never tell the entire story. They cannot say "no"; they cannot give reasons; they cannot plan; and they cannot provide feedback on how they are being understood by others. In fact, good designers often talk more than they believe they do, not just in presentations to clients but also among themselves, evaluating published ideas, weighing their options, soliciting opinions, and judging solutions. In fact,

The fate of all artifacts is decided in language.

A theory of the meaning of artifacts in language concerns how artifacts come to live in the narratives that their stakeholders tell each other and enact, in effect making the artifacts available for use or preventing it. Given that language directs the concerns of at least two people, but more typically that

of a community, a theory of the meaning of artifacts in language is an inherently social theory. It addresses the second-order understanding that individuals have of each other and because this understanding concerns artifacts as well, it must also embrace a theory of use.

Furthermore, whether a theory is spoken or stated in writing, it occurs in language as well. Therefore, a theory of the meanings of artifacts in language has to be self-reflexive. It has to treat itself as an artifact that is continually reconstructed and refined in use. Language is always part of a world that is in the process of being made, and so one must not be tempted to treat language as an immutable given, as a fixed tool to describe the way the world is. Because designers attempt to change the world, they must use language creatively as well.

On this level of theory, the notion of an interface is expanded from single users to third parties, to communities of stakeholders, whether they are would be users, judges, commentators, people with special interest, or marginally interested spectators. Stakeholders identify themselves by claiming to have a stake in a particular class of artifacts or technology. Figure 4.1 suggests that artifacts are not merely used but more importantly enter processes of human communication among stakeholders, including users. In language, artifacts are conceptualized, constructed, and communicated; their meanings are negotiated and their fate is determined. Such processes can no longer be described or measured in cognitive, ergonomic, and technological terms. They will have to be explained in linguistic terms, in what language makes available to the stakeholders in the artifacts in question. This calls for a dialogic, not a monologic theory of meaning.

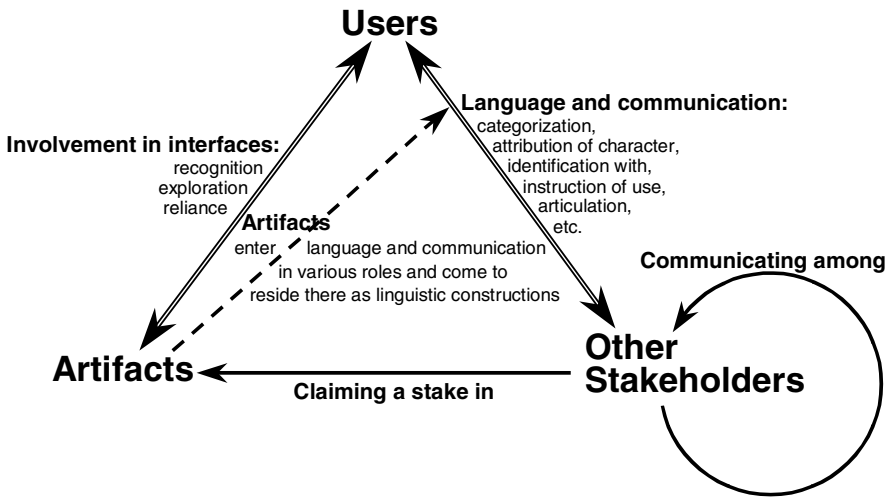


Figure 4.1 Artifacts in language and communication.

4.1 Language

Language is a complex concept. Although people are aware of using a language, they rarely know much of what is going on while speaking, listening, or reading. Writing is taught in grammar school, where students are made aware of pronunciation, spelling, grammar, composition, and rhetoric and led to think this is all there is to know of language. For the purpose at hand, this understanding is insufficient, however. To set the stage, the following distinguishes four approaches to conceptualizing language and settles on the last without much justification.

- **Language as *a system of signs and symbols***: This essentially is the Saussurean (1916), Peircean (1931), and logical positivist view of language. It considers language a *medium of representation*, takes truth as the validity criterion, and looks for references in a nonlinguistic, often physical, world.
- **Language as *a medium of individual expression***: This concept underlies much of literary theory, emphasizing authorial intentions, for example, and is fundamentally committed to individualism. It takes sincerity and consistency as a validity criterion, and looks for meanings in speakers' emotions, ideas, and cognitive structures, more generally, in the structure of the human mind.
- **Language as a *medium of interpretation***: This view looks for meanings in the possible rearticulations of forms (or texts) and relies on a community to determine the legitimacy of the interpretations offered by its members (Hirsch, 1967).
- **Language as a process of *coordinating the perceptions and actions*** of its speakers — languaging for short (Maturana, 1988): This view identifies meanings in the manner in which the members of a community live with each other: coconstructing their realities, developing an understanding of each other, and enacting their conceptions of each other and of each other's artifacts. It takes the viability of the linguistic practices within their community as criterion, i.e., whether they survive their use. Here language is dialogue, "the house of being"* (human).

A brief enumeration of several outstanding characteristics of the notion of *language as a process of coordination* will reveal its human-centeredness:

- **Language use *directs attention***, points out, warns, foregrounds, or brings forth and distinguishes. Talking of an artifact selects that artifact for attention and highlights particular dimensions and features of which there usually are many. Calling a car a big car, a designer's car, an expensive car, a lemon, an Italian car, or a sedan brings forth certain characters of a car at the expense of others. Language distinguishes between what it highlights and what it hides.

* Letter on Humanism [Heidegger (1947/1977:193)].

- **Language use *frames perception*.** Metaphors are excellent examples of the structuring of perceptions. Talking about communication in terms of the common container metaphor renders messages as having contents that authors need to enter into the message they are composing and readers need to get out of what they read. When thinking through these entailments, this might be difficult to comprehend, but the metaphor makes us think that meanings are contained in messages. By contrast, talking of communication with the vocabularies of influence, exchange, conversation, or dance has entirely different entailments (Krippendorff, 1993a). For example, in debates, the war metaphor is common. Accordingly, we attack our opponent, score a hit, defend our position, or succeed in shooting down an opponent's argument. We do not merely talk that way but also experience when we are attacked, perceive our opponents' aggressiveness, relish our victory or feel demolished when defeated — as if human communication were a war with words (Lakoff and Johnson, 1980:4). In political elections, competing candidates introduce language into the deliberations that make the other appear unelectable. BMW calling its cars an "efficient driving machine" induces a perception that is quite different from VW, which calls its car a "new beetle." People cannot help but perceive their world through the vocabularies they are using, even if only to describe it.
- **Language use *creates facts*.** Declarations, promises, requests, and acknowledgments have no truth-value. They are acts that are performed in speech (Austin, 1962; Searle, 1969) and constitute what they say (Searle, 1995). The constitution of a country is not just a piece of paper. When used, referred to, believed in, and enacted, it creates political practices, institutions, and a government, if not a country. When an insurance company assesses a car that was involved in an accident too expensive to repair, it declares it as a total loss. This declaration has very real consequences for the owner of that car. Often, declarations come in the guise of predictions that, if believed true and acted upon, create the reality they state. The economic crisis in the 1920s and the oil crisis in the 1970s are well known examples. The economic crisis was triggered by the belief that banks could not pay their customers and the oil crisis by the belief that gasoline stations would soon have no gasoline to pump. Fads and fashions are largely created by the self-fulfilling declarations of opinion leaders concerning what is "in" to wear, to use, or to say. When a sufficient number of people come to believe this is so, and act on that belief, there is a good chance that it becomes so. All facts and artifacts occur first in language, and the structure of accounts of them has much to do with what becomes real.
- **Language use *is relational***: *Everything said is said by someone and in the expectation of being understood by someone else*. Narratives are told in the expectation of being retold or at least acknowledged or responded to. They invoke second-order understanding and suggest that meaning,

whether of words or of artifacts, resides in a recursive relationship between speakers and listeners: Speakers choose their words for listeners to understand them as intended. Listeners respond to what they believe speakers had in mind. Consequently, speakers come to know what their narratives actually meant to their listeners. An example of this recursion is the attribution of interpersonal relationships, authority, for example. Authority cannot exist without some people attributing it to someone and that someone reciprocating by actually performing that role, further encouraging that attribution. Thus, language use not merely organizes people's perception of artifacts, it 'negotiates' these perceptions within human relationships, or more generally, within the very social structures in which people and artifacts constitutively participate.

- **Language use *is an embodied phenomenon***. The ability to use language entails the ability to articulate, listen, learn, and conceptualize experiences, including feelings. This ability is inseparably tied to the human body. Abstracting a system out of the process of languaging, as practiced in structural linguistics, misses what is central to understanding the role of artifacts in human communication: second-order understanding, intrinsic motivation, emotions, coordination within a community, and the enrollment of stakeholders. Meaning does not exist outside a human body.

If the fate of all artifacts is decided in language, it stands to reason that they must be designed to survive in language and human communication, in the social processes that construct, constitute, or support them, but that can also put them out of use. This elementary fact is rarely recognized among designers. A probable reason for this lack of attention is the popularity of the previously mentioned system concept of language, of semiotics, among one of its branches, which prevents seeing language as implicated in what artifacts mean in the lives of their users. Language is as complex as human nature; in fact, it is indistinguishable from human nature. The foregoing snapshots of the fourth theory of language are intended to prevent readers from falling back on the other conceptions of language. The aim here is to develop a few design-relevant language-based concepts worthy of further explorations.

4.2 Categories

In English and in most Indo-European languages, objects typically enter language in the form of nouns, which in turn invoke ideal-types of categories. From a human-centered perspective, nouns, like "chair," "automobile," "computer," "baseball," and the "Internet," are tied to conceptions of artifacts, not to the artifacts themselves. We perceive affordances, the sit-on-ability of an object, and give it a name, "chair," but abstractions like "a chair" cannot be seen; the class of all chairs, or "chairness" is not manifest in a particular artifact. The word "chair" therefore does not denote the class of real chairs, which nobody has ever seen and could not keep in mind for its

vast diversity, but it means that members of a language community expect that an artifact so named affords sitting. Naming artifacts is far from arbitrary, as some theorists of language suggest. Words draw distinctions in practices of living within a community that are appropriate in the sense of enabling its members to coordinate their perceptions and actions relative to each other and vis-à-vis what their artifacts can do for them. Here, language reveals itself not as an extrahuman system. It is what speakers do but reflects how a community uses it. Drawing interindividual meaningful distinctions and coordinating the perceptions of community members are historically rooted embodied practices. Designers may participate in such coordinations but cannot entirely control them.

As previously stated, nouns invoke the ideal types of categories, which in turn allow competent speakers of a language to distinguish among artifacts according to how typical they are to the available categories. In Section 3.3.1, typicality was defined in terms of answers to particular questions: "Which of two objects, A or B, is more typical of a particular category?" Or, "On a given scale, how typical is A to the named category?" For example, "How typical of sedans is this car?" Answers to these questions reflect one aspect of how people handle artifacts in language: "What kind of car are you driving?" "Is it a sedan or a town car?" "This looks like an unusual (meaning not typical) pickup truck." All of these assertions locate a particular car within several categories of cars or assess how close it is to the center of a particular category, with which other category it may be confused, or whether it does not belong to the category in question. When exploring the nature of a category with questions concerning the typicality of objects, one may do this in the presence of the object, with the help of photographs, or in narratives: "What kind of car breaks down as often as yours?" Without language we could hardly inquire into categories, and the insights obtainable from such experiments make sense only to those researchers who understand what people are talking about.

Categories are organized relative to each other. They do not necessarily form logical hierarchies, as is often presupposed in scientific categorizations, the elaborate systems of classification in biology, for example, part–whole hierarchies of technological systems (molecules, elementary parts, subassemblies, cars, traffic, transportation systems, countries, etc.), or generalizations that academics are good at developing. Pioneering research by Eleanor Rosch (1978, 1981) on categories has distinguished three levels of categories:

Superordinate categories
Basic categories
Subordinate categories

for example, furniture, chair, and high chair (for children), or writing utensils, pen, and ballpoint pen. Superordinate categories usually are difficult to visualize. How could one visualize the category furniture or draw a picture

of writing utensils? Basic level categories tend to have clearly visualizable ideal types. Even children can draw chairs and pens. In language, basic level categories tend have simple and short names. Adding features or dimensions to the basic level descriptions, like armchair, baby shoes, or grand piano forms subordinate categories. Common attributes of subordinate categories are kinds of users (*baby* shoes), attitudes (*sport* shirt), occasions (*evening* dress), places (*table* lamp), social classes or costs (*five-star* hotel), regions and crafts (*Shaker* furniture), sizes (*high* chair), styles (*baroque* church), technologies (*high-tech* watch, *ballpoint* pen), sources of energy (*steam* engine), and shapes (*round* table).*

Basic level categories have also been shown to be easier to recognize, requiring less time and being more reliably identifiable. Because of their visualizability, basic level categories seem less arbitrary and are more difficult to change. Superordinate categories, on the other hand, may become the target of arbitrary logical distinctions or conceptual hierarchies that reflect institutional interests. For example, a supermarket, by adding or discontinuing certain sale items, shifts the conception of what is being sold at supermarkets. Advertising has an interest in placing their products in superordinate categories favorable to sales. Even for designers it would be hard to draw superordinate categories. Corporations that offer a diversity of products and services, which customers may have difficulties conceptualizing, struggle for creating a visible identity, sometimes in the form of a logo. Superordinate categories frame basic categories. The predecessor of IBM produced typewriters. Changing its name to International Business Machines created a new superordinate category, which turned out to direct its product development.

Design cannot bypass the categorizations by the stakeholders of its artifacts, and in fact designers rarely do. Designers hear from their clients what their product is to be, the category to which it is to belong, know what stakeholders expect from a category of artifacts, and cannot help working within their linguistic understanding. The point is not to be imprisoned by categories, but to use them knowingly. Although designing something close to its ideal type may not be very interesting, it certainly aids recognition. The recognition of fire extinguishers, which need to be found under stress and without time to follow directions, is probably best served by red cylinders in a conspicuous location. Designers may have the inclination to develop an alternative form and architects much prefer hiding them in a closet reluctantly labeled "fire extinguisher." But neither may help to identify the device when needed.

When introducing a new product in an established market, whether to stay within or deviate from established categories is a strategic question. Sometimes, a new product may need to deviate from what exists when confusing it with an existing product becomes disadvantageous. Sometimes

* See Athavankar (1987). This paper also includes an excellent overview of Rosh's work from the perspective of design. A reduced but more accessible version is Athavankar (1989).

established categories are selling points, as with fake Rolex watches. Generally, new products have to be designed not to be confused with what already exists, yesterday's model, a competitor's version, or the shape of another technology. However, introducing a product that is neither this nor that may require higher advertising expenditures to inform potential buyers about what it is and what its virtues are. Whether to stay close to recognizable categories with proven qualities or pay the costs of creating a new category in the public domain needs to be examined. The SUV is an example of a highly successful crossover vehicle that combined features typical of sedans with features typical of pickup trucks. When it was introduced to the public, no such category existed. It had no name and no distinct expectations. The designation "sports utility vehicle" was too awkward for the languaging public. The vehicle filled a niche, however, now developing into a category of its own, called SUVs, a name whose etymology is lost to many. Concept cars, which are developed by car manufacturers to be shown at automobile shows, provide interesting tests for how far people are willing to stretch the meanings, dimensions, and features of established categories. The practice of designing and exhibiting concept cars provides manufacturers a test for how far they can go, heeding Raymond Loevy's MAYA principle, which suggests that designers should propose what is "Most Advanced Yet still Acceptable." It acknowledges the existence of categories and suggests that their boundaries may be movable but not without stakeholders' collaboration or advertising expenditures.

4.3 Characters

In language, nouns often occur in adjectival constructions. They permeate Indo-European languages as: <adjective–noun> forms such as "a powerful car," <noun–being–adjective>-forms such as "the car is powerful," or <noun–having–adjective–noun>-forms such as "the car has a powerful engine." The idea that objects have properties is neither natural, culture free, nor universal. They are the result of linguistic attributions. Attributions are acts performed in language, and they reflect the perceptual, emotional, or experiential coordinations (linguistic habits or conventions) in a particular community. As evident in the choice of adjectives, attributions reveal what people sense and feel, and what they believe other people sense and feel as well. Language clarifies, distinguishes, qualifies, and regulates experiences with objects. Without adjectival constructions, one would not be able to distinguish among the properties of things, the personalities of people, and what objects are said to have or not have.

Obviously, it matters a great deal whether people talk about an artifact as being beautiful or ugly, simple or complex, affordable or costly, efficient or time consuming, user-friendly or difficult to handle, safe or dangerous, advanced or old fashioned, and ecologically responsible or disastrous. Once made and reinforced by circulating in public, attributions can determine the

decisions by those who speak in their terms, are difficult to control, albeit shifting in use. Artifacts, much like politicians, are created or destroyed in language, by rumors, publicity, and mass media coverage. There are products that have become unsellable by the attribution of unfavorable characters. Advertising attempts to counter bad publicity and encourage talk about products that increase the size of their markets. Rephrasing the axiom on meaning: In the life of the stakeholders, the attribution of characters to artifacts matter more than physically measurable qualities.

How do characters differ from physical properties? Physics and ergonomics, preferring physical or physiological measures to human perceptions and talk, seek to objectify the linguistic practices of attribution by replacing them where possible with machine-centered conceptions. In that process, human senses, which ground a speaker's attributions in particular experiences, are ignored or substituted by how a mechanical device is designed to respond. Consider the experience of something being *heavy.* Heaviness makes sense only in relation to what is not heavy. The difference is experienced in lifting or carrying one thing as opposed to something else. By contrast, the weight of an object, say in kilograms or pounds, ignores references to human experiences. Similarly, the word *soft*, as distinct from *hard*, describes a quality of touch. One can operationalize "softness" by measuring the depth of the impression that an object of standard shape and weight makes on a given surface. This, however, does not explain how humans experience softness or hardness. The characters of interest to designers are rarely measurable by physical instruments. Figure 4.2 shows two characters of the Hummer, one expressing functionality, robustness, and physical power, the other sportiness, demonstrative wealth, and social power. Explaining their differences as differences in style bypasses their social meanings. Attempts to capture these differences by physical measuring devices are hopeless.

For a more common example, for a long time, Swiss watches were perceived as accurate timepieces, made by exceptionally good craftsmen. The Swatch, collectively produced in Switzerland, originally in response to severe market losses, is now said to be cool, fashionable, youthful, simple, inexpensive, and robust. Coolness has replaced the earlier character of Swiss watches, but everyone agrees that it is difficult to define. It has nothing to do with temperature, obviously. It is not manifest in particular colors, say shades of blue. The reason for escaping definition and measurement is simple. Coolness is not a physical property at all but how young people, particularly, talk about, wear, and make use of colors and shapes — and watches. Character attribution and its shifts in time are social or cultural, not physical phenomena.

Character attributions often frame measurable qualities. For example, when a product is considered *cheap* or *inexpensive*, the monetary value does not explain why people make either attribution. Cheap entails disapproval whereas inexpensive signals an opportunity — two very different perceptions. Complementary frames for objectively the same phenomena are common: The attributes *stingy* or *thrifty,* when describing one and the same

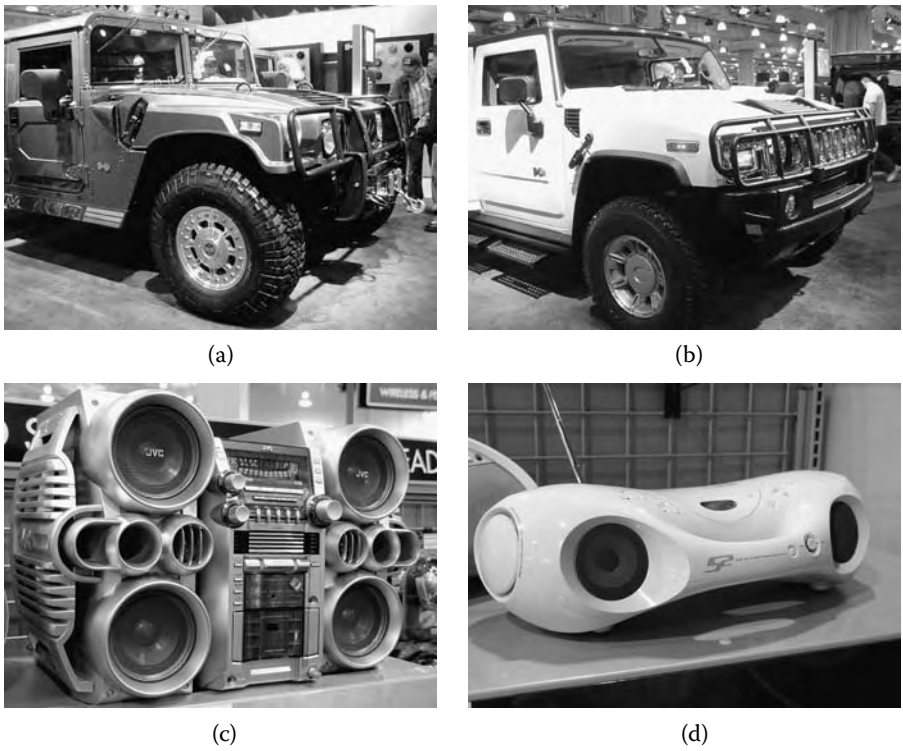


Figure 4.2 Pairs of artifacts that differ mainly in character. (a) The Hummer: military, rough, sporty; (b) suburban, wealthy, powerful; (c) a stereo system from JVC: loud, imposing, high-tech; (d) a portable stereo system from Sony: elegant, fashionable, feminine.

person's actions, express fundamentally different attitudes. Or reporting that a drug is *successful* in *90%* of all cases or that it *fails 10%* of those taking the drug profoundly affects decision making on using the drug. Research by Twersky and Kahneman et al. (1985) on economic decision-making, risk assessment, and the like, has shown that the framing of apparently objective qualities makes a great difference in how people make decisions and act. Their findings have undermined rational decision theory and earned Kahneman the Nobel Prize in economics for 2002.

By definition, the *character* of an artifact consists of all adjectival constructions that a community of stakeholders in that artifact deems suitable to that artifact. Each individual adjectival construction is called a character trait. A character trait is the relatively stable attribution of a quality by means of that particular adjectival construction. It follows that a character is composed of a set or system of character traits, which are considered appropriate and fitting for whatever reason and can be studied as such a system. For example, in different contexts, different character traits may become relevant. When an artifact is reconfigured or breaks down, character traits may be unevenly affected.

This is not to say that designers should ignore objectively measurable properties. However, they must primarily be concerned with what the stakeholders of their designs see, touch, feel, come to believe, and talk about. Since perceptions, feelings, and opinions are inaccessible to observation and measurement, designing the characters of artifacts means designing sensory experiences that elicit desirable attributes from the community of those who matter. It is the adjectival constructions that are incorporated in the reasons for acquiring, using, enjoying, customizing, or talking about particular artifacts to others. Even objective measures of product qualities, it should be pointed out, become relevant only when they enter talk about the artifacts in question.

The English language provides literally thousands of adjective pairs to describe character traits. Five kinds are distinguished below with a few examples offered for each. The last four are human-centered:

- Adjectives that are seemingly objective and potentially measurable in quantitative terms:

  Fast — Slow
  Large — Small
  Fragile — Robust
  Noisy — Quiet
  Bright — Dark

- Adjectives of evaluation, especially aesthetic:

  Beautiful — Ugly
  Balanced — Imbalanced
  Harmonious — Discordant
  Elegant — Graceless
  Ingenious — Trivial

- Adjectives of social values or positions

  High class — Low class
  Expensive — Cheap
  In fashion — Out of fashion
  Outstanding — Common
  Universal — Individual

- Adjectives of causes for particular emotions

  Exciting — Boring
  Appealing — Revolting
  Inspiring — Uninspiring
  Frustrating — Satisfying
  Disturbing — Encouraging

- Adjectives of interface qualities

  Reliable — Unreliable
  Clear — Confusing
  Difficult to use — Easy to use
  Dangerous — Safe
  Efficient — Inefficient

Some character traits on the preceding list may be recognized as aesthetic. As a theory, aesthetics seeks to generalize about the sense of beauty. The epistemological mistake of aesthetic theory is to (a) focus on a limited class of attributes, historically decontextualized elite terms, (b) search for universal propositions, ignoring the culture rootedness of aesthetic judgments and shifts in time, and (c) not recognizing that sense cannot be separated from someone capable of sensing and of using a vocabulary to describe that sense as aesthetic. Aesthetic theory is predicated on the belief that sense is caused by phenomena outside the observer and independent of the use of language and thus goes about theorizing aesthetic objects rather than the cultural attribution of qualities called aesthetic.

The more recent method of *kansei* engineering (Nagamachi, 1995) has similar epistemological flaws. *Kansei* engineering surfaced in Japan. It is ostensively concerned with translating feelings (*kansei* in Japanese) into consumer products. However, as it is impossible to observe feelings in people, much less in products, *kansei* engineering cannot avoid relying fully on the common use of what are called *kansei* words, which claim to report emotional responses (See Section 7.4.1). Despite its claims to be concerned with feelings, kansei engineering is indebted to the use of language.

The notion of character is entirely general, however. Four methods to access the characters of artifacts can be distinguished:

- **Semantic differential scales:** The first and most well-known method to conceptualize characters derives from Osgood et al.'s (1957) semantic differential.* It is a device that relies on a typically large sample of informants, subjects, or stakeholders to individually rate objects or concepts in terms of given bipolar adjective scales. These scales constitute the dimensions of a semantic space in which each object or concept becomes a point. Osgood (1974) was interested in cross-cultural generalizations, and he, along with numerous colleagues, applied hundreds of bipolar adjective scales to all kinds of things, tangible objects, people, and ideas. They found that three factors underlie most of the adjectival constructions that people use to describe their world in what he called affective meanings:

Evaluative:	Good	+3	+2	+1	0	-1	-2	-3	Bad
Potency:	Strong	+3	+2	+1	0	-1	-2	-3	Weak
Activity:	Active	+3	+2	+1	0	-1	-2	-3	Passive

As the name implies, the semantic differential enables researchers to obtain semantic differences between the characters of objects. This device

* Its title, *The Measurement of Meaning*, can now be regarded as overly ambitious. There are very many aspects of meaning that attributions cannot address and that geometrical spaces with fixed dimensionality cannot represent — see the many ways language is involved in human interaction with artifacts and how the meaning of artifacts directs the life cycle and ecological interactions with other artifacts.

enables designers to not only locate the characters of available artifacts in a semantic space but also to specify the desirable characters that the new design is to have and to ascertain which traits would need to be altered to achieve that design goal. Figure 4.3 shows a semantic space that consists of Osgood's three basic dimensions only. In this space, three artifacts are represented as points. A and B differ in only one dimension, the evaluative dimension, B being worse than A. B and C differ in all three dimensions. If many subjects' perceptions are recorded in such a space, artifacts are likely to appear as clusters of points. Those with a specific character occupy a smaller space than those whose character is ambiguous (for an example, see Section 7.4.1 and Figure 7.5).

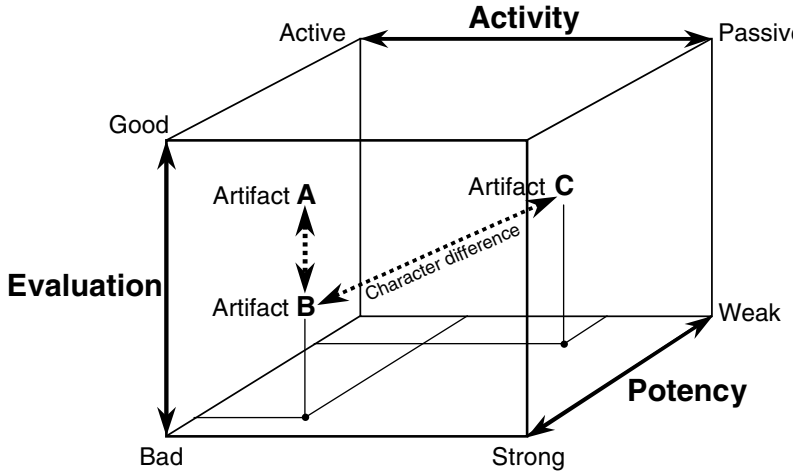


Figure 4.3 Characters of three artifacts in three traits.

Designers are rarely concerned with semantic generalizations such as Osgood's. They typically are faced with rather specific character traits that are found wanting and in need of improvement. They may be asked to revamp the character of a product to attract a new community of users or create a character that is sufficiently different from a competing product. For voters, for example, the relevant character traits of politicians may be popularity, experience, trustworthiness, being part of one's constituency, leadership abilities, and electability. For a fine restaurant, it may be the location, ambiance, clientele, menu, services, and cost. Car buyers may look for appearance, driving comfort, space, fuel efficiency, repair records, who else drives this car, and price. And consumers of soap bars may be concerned with their handiness, freshness of smell, clean appearance, and softness, to which marketers will add character traits they believe will sell. In fact, any adjective pair that is considered relevant will do.

Multiple polar adjective scales create multidimensional spaces that are difficult to conceptualize (see Figure 7.4). The statistical method of multidimensional scaling (MDS) provides ways of visualizing complex distributions in fewer dimensions.

- **Free associations:** A second method is to interview users or potential stakeholders to elicit adjectives that could be used to characterize an artifact or the practices it supports and to analyze their responses statistically. Focus groups — individuals brought together for a guided discussion of products or proposals to elicit adjectives deemed appropriate and desirable or not — are commonly used for this purpose. The data obtained by this method are especially useful for anticipating responses to issues or designs not yet on the market, using prototypes or models, for example. In this context, focus groups are preferable to individual interviews because they mimic the kind of conversations that stakeholders and users might have among themselves when confronted with a new design.
- **Content analysis:** The third method is content analysis (Krippendorff, 2004a), a research technique that is applicable to large bodies of textual matter and can focus on the kind of adjectives that surround a chosen concept. Corporations apply it to newsprint, scientific articles, Web sites, and interview data — ideally anything electronically analyzable — to monitor changes in their public image, the characters of their own products, that of competing products, and to develop suitable advertising strategies. The method is unobtrusive. It analyses what is published and read by target populations for reasons that have nothing to do with why the analysis is undertaken. By correlating the adjectives used with product successes and failures, designers can gain information about which character traits to seek out or avoid in their designs.
- **Comparisons that elicit character traits:** The fourth method can be handled more or less rigorously. It starts with a collection of existing products — prototypes of a particular category — for example, wristwatches, eyeglasses, wine bottle labels, or photographs of larger artifacts such as automobiles, buildings, or interior spaces. In a cross-cultural study, Miller and Kälväinen (2001) asked 30 subjects from United Kingdom and Finland to each sort 36 photographs of living room chairs into piles of semantically similar chairs and then describe each pile. Figure 4.4 depicts the result of an MDS analysis of the United Kingdom data, revealing the design attributes comfortable–uncomfortable and antique–modern. The phrases used to describe these dimensions are listed in this figure as well.

  In a less rigorous use of this method, more like an exercise, subjects are asked, individually or in focus groups, to arrange a set of products so that similar ones are closer together and dissimilar ones are further apart. Then they are instructed to explain these distances to the researcher (and by implication to themselves) in terms of how and by which attributes, they differ. The subjects' descriptions of these differences in

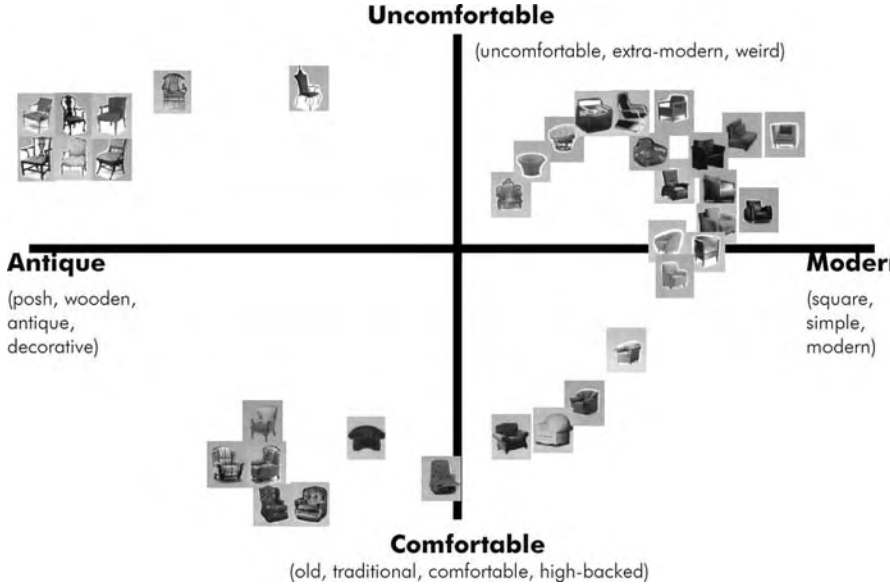


Figure 4.4 Distribution of living room chairs in a character space.

effect verbalize the dimensions of the character space for these products. Subjects are encouraged to rearrange the products and rearticulate their differences until they are satisfied. The exercise ends with the verbal articulation of the products' characters, and the visual differences among the products give an idea of how much the characters differ in the space that the subjects are satisfied with. If prototypes of new designs are included in this procedure, this gives designers a sense of where they would be located relative to what already exists or what is to be avoided. This method is limited to relatively few dimensions and is not easily quantifiable across individuals. But it nevertheless can elicit characters of which designers were unaware.

4.4 Identities

Propositions are statements, often in sentence form, that can be declared, believed, doubted, or denied. Propositions that link users to particular artifacts, categorize people according to the artifacts they surround themselves with, define who they are through what they are wearing, consuming, using, presenting, or opposing; these propositions render artifacts as means for social identification and differentiation. With such propositions, the technology of artifacts recedes behind their social meaning, behind the role they play in identifying who one is. Virtually all artifacts, besides categorizing and attributing characters to them, also come to be related to their users, owners, observers, and commentators. They support various identities. Ever

since Thorstein Veblen (1931) introduced the notion of conspicuous consumption, it is recognized that people's decisions on material artifacts are far from rational and utilitarian but better explained by efforts to preserve or strive for desirable social relationships with others within a community. This involves the ownership, use, and consumption of artifacts. Erwin Goffman's (1959) analysis of *The Presentation of Self in Everyday Life* has supported this understanding. Consequently, designers cannot limit themselves to designing artifacts that are individually usable, but must also address how their use relates to stakeholders' identities. This calls on designers to be familiar with the propositions about the identities that circulate within the target communities.

What is an identity? *Webster's Dictionary of the English Language* stresses the sameness that constitutes someone's or something's being, the essential or generic features that distinguish one person or thing from what it is not. Accordingly, an identity is relatively enduring in the face of changing circumstances. People either identify themselves or are identified by others, and in case of the latter, they may either accept or reject how they are being labeled. Once adopted, people attempt to protect their identity against reinterpretations by others by preserving its defining features. Take away a soldier's uniform, and she is someone else. Outside the hospital or a doctor's office, a physician has difficulty remaining one. In prison, people lose what they were outside of it. Someone who does not know what constitutes who he is is declared insane. There are different kinds of identity.

- *Individual identities* emerge from a person conceiving of his or herself as a self-contained individual, as a free agent, as taking responsibilities for what the person does, has, uses, or can show as accomplishments. This is manifest in propositions of who "I am," what "I like," "why I act the way I do," including what "I would never do." Individual identity is also invested in artifacts that the individual is not willing to give up for something else or does not like to see others have as well.
- *Institutional identities* derive from serving an institution in a certain capacity, occupying an office, holding a position, and exercising the rights and privileges that come with an occupation. In assuming institutional identities, a person's individual identity is replaced by the role played within that institution. This is signified by the power to command the use of specialized artifacts, wearing uniforms, occupying a particular office, and employing specific tools to accomplish institutional objectives. Institutional identities are manifest in speaking not as individuals but from an institutional position, in terms of the role the individual is assigned to play. For example, "In my capacity as…," "as a …, I own, use, have, or decide … ."
- *Group identities* derive from belonging to a group or culture, for example, having a gender, profession, ethnicity, or nationality. In assuming a group identity, a person's individuality is subsumed under what a group of individuals share or what its members are presumed to have in common.

Group identities are manifest by assertions of belonging to a group: "we like … ," "we do … ," "we have … ," and "we don't … ."

- *Brands and corporate identities* are not of individuals but of goods and services (artifacts) that are consistently provided by organizations, particular producers, or corporations. Brands and corporate identities reside in the loyalties that buyers, consumers, but also employees show for what these producers or corporations do. Brands and corporate identities are manifest when a person does not merely drive a car but "a Mercedes," or prefers "Godiva" chocolate to "Hershey" chocolate.

One important feature of all identities is their enactment in public. Identities are the subject of conversations, often negotiated or settled in the presence of others, and the markers of identity must be visible to many. Artifacts that remain hidden from view cannot serve in the formation of identities.

Another essential feature of identities is that they not merely mark individuals, institutions, groups, corporations, or brands, but they crucially distinguish among them. Individuals, for example, surround themselves with all kinds of artifacts but can establish their individual identity only with artifacts or arrangements of artifacts that are somewhat unique to that individual. Things widely accessible cannot or do not easily serve individual identification. Scarce objects, expensive goods, artifacts whose use requires unusual skills, or those that can be transmitted only by history or inheritance have a greater chance to distinguish among identities than those commonly available. One of many failed promises of the industrial era was to give everyone equal access to mass produced goods and eradicate feudal distinctions. It succeeded to destroy older social distinctions but enabled different kinds. In postindustrial society, individualism is dominant, with everyone trying to be recognizably unique. Groups continue to mark their identities, and corporations struggle to be different, outstanding. Many design problems amount to introducing features that distinguish the products of one producer from those of their competitors or must at least acknowledge the group identities of users and stakeholders as distinct from one another.

Identities do not simply exist; they are enacted and in need of protection. Much of the evidence about identities that designers need to consider stems from observing what people and institutions do to protect themselves from erosions of their identities. Consider a national flag. Owners show "their" flag to express support for their state. Burning one's own national flag in public is an identity-threatening act. A flag is not just a piece of cloth with physical properties, being flammable, for example, but it is part of a state's identity. The outrage of citizens upon seeing their own flag not merely burned but "desecrated" and the legal actions against the perpetrator they accept reveals what the flag actually means to them. Or consider a very different example, the Harley-Davidson motorcycle. By comparison with other motorcycles, this is an old and inefficient design. Members of many

motorcycle clubs in the United States ride it as the visible marker of their club identity, together with riding in groups, sharing a culture, and using a vocabulary in which fuel efficiency and ecology is far less important than the continuous demonstration of being a member of the club. Engineers and designers have developed better motorcycles since, but it rarely enters the motorcycle clubs in the United States. Where identities matter, conformity follows, and this does apply not only to riding a Harley-Davidson motorcycle, but also to the use of corporate graphics, signature products, brands, uniforms, and standards. Maintaining identities is equivalent to discouraging deviations or crossing the boundaries that identities define.

An older word for identity is style. Talking of style rather than identity focuses attention to certain formal properties of artifacts, and assigns style to periods, countries, or individuals, often from outside the sociolinguistic processes that support it as such. Abstracting formal properties out of the dynamics of ongoing identification strips artifacts of their social meanings. While style too makes sense only relative to what or who does not have it, it is a concept that bypasses issues of meaning.

It follows from the foregoing that identities provide linguistic or conceptual means for the ongoing processes of social integration and differentiation. If the objects that serve one individual's identity are used by others whom he or she does not like to be associated with, that individual may have to look for other means of identification. There are many stories of awkward situations in which a woman finds herself at a party wearing the same dress as the hostess. If one corporation discovers that another one produces a product or provides a service that is part of that corporation's identity, it may go to court and sue the competitor for industrial espionage, patent law violations, copyright infringement, or other categories of unfair practices, all of which have to do with identity. Usually, it is less costly to move on and adjust one's corporate image or invent even better technology, and this is what drives much innovation. Where there are no laws against efforts of one group to adopt the artifacts that serve another's identity, searching for new artifacts to identify with is the rule. Fashion, for example, is a system that is built on people's continuous need to reidentify themselves with new and as yet unmarked artifacts. It changes precisely when leading groups see the need to distinguish themselves from other groups that try to catch up with them. The continuous entering of a new generation of consumers into the market and its need to define itself feeds that process as well. Some products are more easily appropriated than others, and designers have to be aware that their designs enter the continuous struggle over reidentification. They can propose artifacts that fuel this process or others that resist it. Designers themselves are not immune to this sociolinguistic dynamic. They not only surround themselves with artifacts that identify them as members of their profession, they also use their rhetorical skills to impress their clients with their ability to foresee where identification is heading.

Human-centered design has to be responsive to stakeholders' multiple identities, and by introducing new artifacts into the processes of social identification and differentiation, designers cannot avoid interfering with the shifting identities they observe.

4.5 *Verbal metaphors*

Verbal metaphors are a significant part of language and fundamentally structure human perception. It is suggested, therefore, that visual metaphors (described in Section 3.3.2) may not be entirely independent of how language is used. Many visual phenomena make sense only in conjunction with listening to talk about them, certainly developmentally, but also later in life. Verbal metaphors also have a longer history of being analyzed. In the literature, verbal metaphors are simply metaphors, and this shorter reference is adopted here for convenience. For human-centered designers, metaphors are important because they can render artifacts understandable, particularly novel or complex artifacts, and bring new perspectives to perceiving them. Inappropriate use of metaphors can also obscure or hide otherwise important features. Metaphors are manifest not in noun phrases or propositions of identity but in figures of speech, tropes that invoke whole vocabularies. Designers often rely on metaphors when developing their ideas, which often occur in the realm of newness, but they should be cognizant of the metaphors that the stakeholders in their designs use to see, decide, and act.

In the literature, one can find several more or less useful definitions of metaphor and different attitudes about their use. Aristotle, who described this trope as speaking of one thing in terms of another, warned against its use because of its apparent ambiguity, recommending the use of definitions instead. This preference, shared with the discourse of contemporary positivist science, reveals a notion of language, described (in Section 4.1) as a systems view, focusing on accurate representations of objective reality as the primary purpose of language. From this abstract or objectivist ideal, it seems indeed wrong to describe one thing in terms of another. Why not deal with the object of concern directly? Ambiguity has also been attributed to analogies, which have the structure of A is to B like C is to D. A frequently cited definition of metaphor characterizes it as an "incomplete analogy," as mere poetic embellishments of prose that can only distract from the desired accuracy of propositions. This view is widespread in studies of literature and supported by theories of science writing, which discourages the use of metaphors as well. These notions of metaphor are woefully inadequate because they ignore what they do to human perception in preference of supposedly objective accounts of reality. Lakoff and Johnson (1980:19) suggest that "no metaphor can ever be comprehended or even adequately represented independently of its experiential basis." A human-centered definition of metaphor must therefore include its conceptual, perceptual and, ultimately, its behavioral ground.

Instead of providing a definition, a theory of metaphor is proposed to explain metaphors as a process that involves five steps. A simple example from everyday life will be used to illustrate these steps: the existence (or lack) of chemistry between two people (Krippendorff, 1993a,b). Chemistry is a construction in the domain of science writing about the dynamics of molecular compositions. It does not address human communication. Hence, for chemists or readers who conceive language to be a representational system, the statement "there was chemistry between us" is literally nonsense. Physical, chemical, and biological considerations may well enter human relationships, of course; they are not as determinate, however, as is molecular bonding. But in everyday life, assertions of lacking chemistry say a great deal about how their users conceive of each other and what goes on between them. A theory of metaphor addresses the question of what happens when such literally meaningless statements are encountered. Here are the five stages of the theory:

- Metaphors operate across two logically independent domains, an absent but familiar domain of experiences, *the source domain*, and a present domain in need of understanding or restructuring, *the target domain*. The metaphor of chemistry between people brings common knowledge of chemistry into the domain of human relationships.
- The effective use of metaphors presupposes the two domains to have some *structural resemblances*. Such a presupposition is often farfetched and unexamined or unconsciously assumed. Chemistry is concerned with how molecules interact and form compositions. Humans living together amounts to collaborating, communicating, coordinating individual understanding, and acting jointly. The two domains share the existence of several entities, at least two, and interactions or communication between them.
- Metaphors carry *entailments* (Lakoff, 1987:168), which are patterns of understanding embedded in the vocabulary of the source, from the source domain along the structural resemblances into the target domain, which now becomes *reorganized* regardless of what it was previously. Talk of the chemistry among people renders the domain of humans living together as the product of instant and involuntary reactions that are determined by the makeup or nature of the individual participants. "Good chemistry" attracts, creates synergy, bonds; "bad chemistry" repulses, produces difficulties, and separates the entities involved. A metaphor effectively erases alternative ways of understanding the target domain and puts its own entailments in their place, now dominating perception. Explaining human interactions in terms of chemistry provides no place for choices and absolves individuals of any responsibility for the bonding or lack thereof, which is seen as taking place on contact, and outside of individual control. "Chemistry decides for you."

- Metaphors *organize their users' perceptions* without awareness of how and in which way. When acted upon, metaphors *create the very reality* they inform. "There was chemistry between us" is a metaphorical statement, but it is not just words. Those who use it feel that way: unfortunate to have been thrown into an unworkable situation, confronted with properties that are incompatible with one's own and outside of one's control. Acting on this assessment creates or *constitutes the fact it states,* here a social fact that involves the relationship between two or more people. In the source domain, chemistry, there is no space for learning, adaptation, negotiation, or compromise. There is no concept of languaging, much less of metaphor. Accordingly, people's natures either go with each other or they do not. The whole focus of a metaphor is on the target domain. The source domain, chemistry, remains unconscious background. Users of this metaphor might not agree that human relations (interaction, cooperation, and communication) are similar to what chemistry theorizes (which would be making an analogy); but adopting this metaphor makes them see their relationships through the vocabulary of chemistry.
- *Metaphors die in repeated use* but leave behind the reality they had languaged into being. When metaphors die, their entailments become naturalized or literalized explanations. In the case of the chemistry metaphor, the idea that people have no control over their relationships has certainly entered everyday discourse. One speaks of matches and mismatches in human relationships as if this were natural. Business management then seeks to assemble people with viable combinations of personal qualities. The idea that people have invariant natures conveniently fuels research in social psychology, inquiries into leadership qualities, social and antisocial personalities, and individual behavior traits that make teams work or marriages break down.

Donald Schön (1979) gives a good example of what a metaphor does for designers. He observed a group of design researchers trying to improve the performance of a paintbrush made of synthetic bristles. They knew that the paint had to attach itself to the bristles and then be spread onto a surface. Compared with traditional natural-bristle brushes, the new one delivered paint rather unevenly, in a "gloppy" way. The designers tried out numerous ideas to correct this problem. They experimented with different synthetic materials and diameters for the bristles, and, because natural bristles had split ends, they split the ends of the synthetic bristles, without significant improvements. In effect, designers had conceived their target domain as a problem of making the bristles better and experimented with different kinds.

After many fruitless efforts, someone suggested, *"You know, a paintbrush is a kind of pump!"* He pointed out that when a paintbrush is pressed against a surface, paint is forced through the *spaces between the bristles* onto the surface. The paint *flows* through *channels* whose size is controlled by the painter's bending of the brush. Painters even *vibrate* a brush to facilitate the *flow* of paint. Talking of a brush in terms of a pump radically altered the

researchers' conception of their task. Instead of attending to its bristles, they came to observe what happens in the capillary spaces between them. They soon found out that natural bristles bend differently than artificial ones. The *tube*like gaps between the synthetic bristles turned out to be difficult to control.

Clearly, brushes are "brushes," and pumps are "pumps." People do not confuse the two domains of experiences. Painters may think of brushes as a bundle of bristles that enables them to transfer paint from a container to the surface on which it is to be spread. For painters, there is nothing wrong with this conception of what they do. But this common conception failed to lead the designers to an adequate solution to their problem of developing a new paintbrush. When the "paintbrush is a pump" metaphor entered their deliberations, the designers experienced a gestalt switch from seeing a brush as a bundle of bristles to seeing it as a system of *capillaries* that *soak up* the paint and enable the painter to squeeze it onto the surface by controlling the bending of the bristles. After reading this story, we cannot help but see painting rather differently. This does not mean that we become better painters, or that there is a correct perception and an incorrect conception. It depends on what we want to do. The metaphor changed the designers' perceptions and led to better paintbrushes and several patents.

The point of the example is that the metaphor brought a new vocabulary from a source domain, the domain of pumps (capillary actions, channel control, and hydraulics) into the conversation that radically changed the perception of the target domain. Once the researchers began to use this new vocabulary, experiences with pumps and fluid dynamics became readily available and radically altered the course of the development. The metaphor not only changed individual perceptions, occurring in language, it also coordinated the actions of the development team members, which is a social process.

Most technical innovations, it should be acknowledged, result from using metaphors appropriately. Not all metaphors work, of course, but those that do can change the world. Since metaphors cross apparently distinct experiential domains, people who can move with ease through many domains and feel free to use metaphors in conversations with others have a better chance to innovate than those who hold on to the literal language of experts in any one domain.

4.6 Narratives

It is easy to see how categories, characters, and identities appear in language as nouns, adjectival constructions, and propositions, and how perceptions and actions are altered by the use of metaphors. This section considers a larger unit of language in speech and writing: narratives or stories. Stories are outside the traditional linguistic concerns for phonology, vocabulary, and grammar. Literary scholars have studied narratives but largely as literature, not as embodied in practices of living. There has been a resurrection of the

ancient interest in narrative theory* that does address larger issues, but not specifically about artifacts or design.

So, what do narratives or stories have to do with design?

- **Narratives are human creations and so is design**. Scientific discourse is creative as well, but the theories it creates account for what objects are destined or designed to do, not how and why they come to be.
- **Narratives essentially are cooperative constructions**. They require a storyteller and listeners who may become storytellers as well. Similarly, designs may be conceptualized by someone, but produced, supported, or opposed, and used by others. Designers and their stakeholders require each other.
- **Narratives are told in the expectation of being understood**. Without someone's understanding, narratives do not exist. Storytelling implicates second-order understanding (of the listener's understanding), which is fundamental to human-centered design as well. One cannot conceive, produce, use, or develop artifacts that are meaningless or useless for lack of understanding.
- **Narratives enable their narrators and listeners to make sense of their worlds** and this includes making sense of the objects that occur in them. While human-centered design is dedicated to making sense of things as well, the objects of design are not as easily reproducible as are stories. Hence, the relation between narrative and design is somewhat asymmetrical. However, as elaborated above, technologies can also serve as source domains of metaphors that make sense of rather different artifacts — as pumps do for the design of a paintbrush, or computers do in the cognitive science's conception of human intelligence (the brain). The relationship between narrative and design is not entirely one-directional.

The central role of narrative in being human might be recognized in the following propositions, largely drawn from Larry R. Cochran (1990).

Humans live in story. Individuals are born into the stories of their family and community, and continue articulating these stories until they die. Cochran suggests that stories are metaphors of life: They have a beginning, middle, and end. The middle is the point of living. The same is true of artifacts: They are produced, are available for use, and disappear to be recycled and give rise to knowledge about their life cycle. What one knows of an artifact is the stories one has heard and can tell of it. They may be personal, as of the souvenir brought home from a trip, or the furniture inherited from a grandparent; public, as of the evolution of computation, the attribution of a chair to a famous designer, or the authorized text on a particular technology; or hidden from view, such as the lives of the stakeholders who collaborated in realizing a design. Without a story of who made one's chair or how it came into one's possession, that chair may afford sitting,

* The following digests insights from Bruner (1986), Danto (1985), MacIntyre (1984), Polkinghorne (1988), Sarbin (1986), Spence (1982), White (1981), and Mitchell (1981), among others.

but has no other significance. The artifacts that people surround themselves with mean the very roles they play in the stories in which they occur.

Humans explain through story. Individuals can observe only in the present. While a past may be remembered, what becomes public knowledge, the past and the future, is largely a linguistic construction. When an artifact breaks down, people look for a story to explain what happened and why. That story may invoke wear and tear, material defects, user negligence, supervisory ignorance, manufacturing errors, design flaws, devious intent, God's intervention, and more. Without such explanations, equipment failures are mere coincidences. The content of such explanations is secondary to their consistency with the kind of stories people already know or consider plausible. Protocol analysis, the analysis of what users say they think and do while interacting with an artifact (Ericsson and Simon, 1993), is the analysis of users' explanations. It demonstrates the close relationship between narrative structure and the makeup of interfaces with artifacts.

Humans explore their worlds through story. Requesting technical support by telephone from the manufacturer of a computer is predicated on the belief that the user could be talked through a partially incomprehensible system. A typical consultation starts with an explanation of the user's frustration. The ensuing conversation entails asking questions and eliciting answers, giving instructions and reporting on what happened when the instructions were followed. In this interactive process, a story of the problem emerges; the computer is reset according to that story, and if everything goes well, users understand a story that enables them to proceed on their own. In fact, artifacts of the complexity of a computer are understandable largely in terms of stories heard, lived through, and elaborated on independently. There never is a correct story, only stories that do or do not work.

To these propositions we need to add: *Humans design their worlds as stories*, stories they can tell others, reproduce elsewhere, combine into larger narratives, and test for their usability in various contexts (see Section 7.4.3).

We can analyze any story into smaller components, just as is possible with technological artifacts. Labov and Fanshel (1977) distinguish five components of narrative:

- **Abstract:** a synopsis of what the narrative is about
- **Orientation:** where it takes place, the principal players involved, and their circumstances
- **Narrative sequencing:** a point by point account of what happened
- **Evaluation:** the lesson learned, why the story is important, and its consequences
- **Coda:** the invitation for the listener to leave the imaginary world of the story for the present.

Most narratives concerning artifacts are describable in approximately these terms. User instructions typically start with an abstract in the form of a headline, saying when it applies, for example, "how to change the battery"

or "when to call for service." They may then orient users to the tools, resources, and competencies that could play a role in what is to follow. Orientations are typically followed by step-by-step instructions of what is to be done and what one will have to see if the steps are followed correctly. Step-by-step instructions correspond to narrative sequencing, which is the essential component of narrative and of instructions as well. Whereas in narrative, sequencing pertains to understanding by listening, step-by-step instructions usually require interactions with an artifact. In such explorations, evaluation appears in the results, in tests on whether one can adequately handle the artifact in question. Coda is manifest, for example, when the technical consultant inquires whether the customer was satisfied. It brings the user back into the present conversation. Not all elements of narrative need to be present in design efforts, but the close resemblance of narratives and design should be apparent and language is very much in the driver's seat of that process.

The interfaces with artifacts of some complexity should correlate, as far as possible, with the narrative structures of how one can explain what these artifacts can do and how users can interface with them. Users need to be able to "read" artifacts naturally and talk themselves out of trouble. To implement such a recommendation, designers might want to consider four concepts.

- **Narrative smoothing**. Interfaces are always far more detailed than their users care to recognize and any narrative could describe. For example, bicycling. One can hardly learn to ride a bicycle from a book or describe exactly and in all details what bicycling involves. The reason is that it relies heavily on tacit knowledge and unconscious routines, including the ability to balance one's body, which a bicyclist needs not to conceptualize, can take for granted, and has no reason to verbalize. Talking of bicycling merely punctuates what is involved. Similarly, comic strips consist of sequentially arranged frames that leave what happens between these frames to the readers' imagination. Creating a coherent story by filling the missing gaps is called narrative smoothing. Authors seek a balance between providing details that may be clear but tedious to read, and taking leaps that are too big for readers to follow. Interface designers need to know how much information they need to present and how much they can hide and leave to the users' imagination. Representational truth is clearly not the issue. Understanding what computers actually do is beyond most people's capability, but enabling users to create a coherent understanding of what is not seen is all that matters.
- **Narrative sequencing**. Labov suggested the rule that the order of narrating a series of events implies succession, causality, intentionality, or logical entailment. Narrative smoothing is directed by narrative sequencing. For example, the most obvious reading of the minimal succession of two propositions, "He dialed a telephone number. The train blew up," is that the dialing of the telephone number caused the explosion. By

contrast, a reversal of this order, "The train blew up. He dialed a telephone number," leads to the reading that he called to inform someone of the explosion. In the former sequence we envision a terrorist, in the latter sequence a helpful citizen. Neither is deducible by logic. Comic strips are read that way. Scenarios to conceptualize interfaces, as in Figure 3.16, are constructed according to the rule of narrative sequencing. Where a clear sequence of steps is to be taken, it is advantageous to organize an interface that way. Figure 4.5 shows one screen of an interface. Its five overlapping leaves sequences in stages the analytical work that the user is expected to follow. Each is preparatory to the next. When the upper-right corner folds over, the essential work on one leaf is done, and the user is invited to continue. Explorations of visual material do not need to follow reading conventions. Research on eye pupil movements has shown that viewers explore visual phenomena by jumping from one visually outstanding and information-rich detail to another. Eyes sequentialize their visual field, creating successions out of proximities, but are guided in this effort by a viewer's conceptualizations, which include known narratives of that visual field.

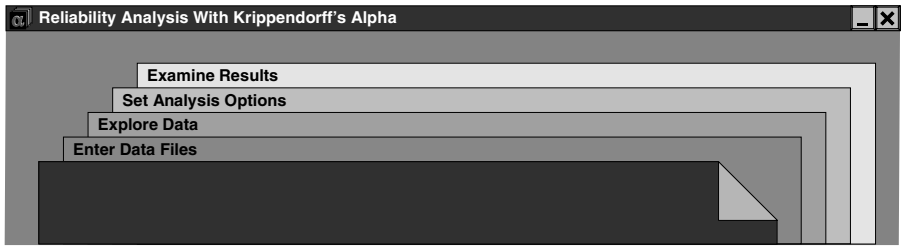


Figure 4.5 Order of tasks in narrative sequence.

- **Narrative embedding**. Although sequencing is primary, narratives also provide for embedding stories in other stories, much like embedded clauses in sentences. Storytellers and listeners have no problem keeping track of a primary narrative, while temporarily delving into an embedded one. In design, a good example of embedding is the use of semantic layers (see Section 3.4.6 but discussed there without reference to their narrative roots). People experience limits on the number of embeddings they can simultaneously track. This is why semantic layers are typically limited to a few levels of embeddings. Just as any narrative, a semantic layer needs to be understandable and coherent. As in any embedded narrative, a semantic layer needs to provide a way back to where it was opened and allow work on the superordinate layer. Clearing a copier of paper jams is embedded in the process of copying and done in the expectation of going back to the copying.
- **Narrative schemas** or semantic functions. Narratives typically rely on many simpler explanatory structures into which characters, actions, and things are entered, and comply with their logic — much like entering



(a) (b)

Figure 4.6 A billboard violating the rule of narrative sequencing. (a) Whole; (b) relevant section.

numbers into a mathematical formula. Metaphors provide such explanatory structures by importing them from an unlike source domain. In narratives, there are complementary roles: heroes and villains, buyers and sellers, or leaders and followers; casting people in these terms leads one to expect what they do. For example, the common actor–action–target schema serves to explain what people do and why, and implies intentions, attitudes, and relationships. In the design of artifacts, there are other complements: doors that are open or closed, switches that are on or off, and elevators that go up or down. Their logic is simple. Mentioning one implies the availability of the other. Where such schemas are commonly available, designers cannot avoid them and may not want to violate the logic they imply. For example, Amtrak, the U.S.-owned railway service, started to use billboards, as in international airports, to announce train arrivals and departures. In Philadelphia* and New York, they list trains as shown in Figure 4.6. This listing unwittingly violates the rule of narrative sequencing and a semantic scheme. People, objects, even rivers, and certainly trains always come from somewhere and go somewhere else. People speak of origins and destinations, in this order, and conceptualize a train as going "from A to B," not "to B from A." Although grammar affords the latter construction, it is uncommon. People also read from left to right, which leaves them to expect reading about the origin before a destination. The designers of this billboard reversed this narrative order, listing the destination of the train before its origin. As a result, passengers boarded trains that went in the direction opposite to where they wanted to go. Instead of reversing the order, listing a train's origin before its destinations, the Amtrak designers decided that the word "**To**" labeling the destination column, was too small to read and replaced it by an oversized "**TO**" and in a yellow color that signaled importance. Obviously, violating common narrative schemas can cause serious user errors. The logical solution probably does little to prevent the usual reading.

Artifacts should be designed so that their interfaces are narratable.

* As of the date of this writing, May 2005.

4.7 Culture

Culture is known by many definitions. Archeologists and material anthropologists who have not much else to go by than the physical evidence a culture leaves behind tend to equate culture with the multitude of artifacts in use at a time. For traditional product designers, this definition might be appealing, but human-centered designers would miss the users of these artifacts, the stakeholders who could further a design, and the institutions within which they work. Cultural anthropologists recognize language as the primary organizer of a culture's conceptualizations, and custom as what the members of a culture enact so routinely that they barely know there is one. Here, language is not merely narrative — making sense of one's past and future, rendering individual lives meaningful, and giving artifacts their purposes — it is a joint accomplishment by many people, and both constitutive of and constituted in culture. As such, language sustains culture and is its artifact. For human-centered designers, artifacts are not merely material products, they have social lives, as do families, social organizations, governments, and last but not least, language.

Speaking of culture acknowledges the significance of a way of living together and in that process creating artifacts that sustain its very practices of living. Being in a culture is taking part in a circular process of making sense and being made sense of by a population of people that is too vast to be comprehended. The medium in which culture is in a continuous process of being negotiated is language, conversation, and discourse. Designers may not be able to alter culture directly. This social construction is just too big. But they cannot prevent contributing to changes in culture because their proposals create artifacts that afford certain practices of living and discourage others, and enter the conversations among many. Take so-called "conversation pieces" as a start. Superficially, they have no other use than stimulating conversations. But this somewhat derogatory concept hides the fact that it is in language that artifacts come to live or disappear from view. The concept cars that automobile manufacturers display at auto shows, fashion, art forms, public buildings, and new technologies must survive in conversations to be sustained. They fascinate people at the expense of addressing other things, and they may become constitutive of social institutions. One can see this in the artifacts that constitute a department store, a church, or a private home, all of which encourage some activities and discourage others. Design can react to different cultural conditions, designing artifacts that express prevailing conditions, but it can also weaken these conditions and offer new cultural practices. This awareness is reflected in the design of artifacts in view of the conversations they encourage, in view of the consequences they might bring about, in view of a more desirable culture. The development of the personal computer in the United States was fueled by the idea, considered quite naive at that time, of making computational intelligence available to everyone and of democratizing administration. Indeed, it is not far fetched to say that the social implications of this technology

brought down planned economies, such as that of the Soviet Union, bureaucratic hierarchies weakened, and ushered in a communication revolution that is in the process of changing Western culture in fundamental ways. Not all technologies encourage cultural changes of this magnitude, but design has the potential of altering the conversations in which a culture resides, and is transmitted, maintained, and challenged.

Helene Karmasin (1998) develops some of these ideas for design. Borrowing Mary Douglas's (1996) classifications of culture, hierarchical, individualistic, egalitarian, and fatalistic, Karmasin explored the design concepts appropriate to each. For example, "(d)esign principles in (a hierarchical) culture are based on two considerations: on the one hand, design must … make visible the diverse stratifications based on opportunities and rank, indicating the value of the collective. On the other hand, this culture's design principles…make clear the reasonable, matter-of-fact and functionally-oriented use of products and to provide general rules as to the nature of good design" (Karmasin, 1998:18–19). This normative component underlies the "good form" movement in the 1950s, which ceased with the closure of the Ulm School of Design.* By contrast, design principles in an individualistic culture are informed by competition in the market, mass media consumption, the ideal product being one that is exclusively "designed for me," one that "nobody else owns." Such products "lean towards the playful, and appeal to entertainment, sensuality, experience surprise" (Karmasin, 1998:22–23) — Memphis design, for example.

One may not find any culture that is so consistent as to fall into a single category. One may also question the validity of such categorizations. For example, Douglas' categories do not cover, and Karmasin's discussion does not address, the kind of culture created by industrial mass production with its complete disrespect for indigenous cultural forms, considering the latter primitive or underdeveloped, and encouraging conversations that celebrate technological rationality. But it can hardly be denied that there are conversational patterns in which certain kinds of artifacts are more likely to emerge than other kinds, that the artifacts that do emerge, are put to use and in place are consequent to how people speak of them, and that their very use can alter the structures of society in return. The meanings that artifacts thereby acquire are not just meanings of use. They come to mean the roles they acquire in the larger conversations that drive the evolution of cultures. Nobody can control cultural processes. But designers can and should become cognizant of the conversations that enable them to create their designs, continue well after they have done their part, and ultimately impact a culture. Designers can unwittingly go with the flow. They can present themselves as prophets of cultural trends. But they can also participate in the conversations that improve that flow for their participants.

* Hochschule für Gestaltung (HfG) Ulm, Germany, 1953–1968.

chapter five

Meaning in the lives of artifacts

From a larger perspective, artifacts do not have a clear beginning and end. They have precursors and consequences, often recognizable only in retrospect. For example, jet airplanes could not have been developed before propeller airplanes worked well; and electronic computers followed centuries of work in mathematics, mechanical computing devices, and the more recent use of vacuum tubes. But neither of these developments could have been predicted from their precursors. In fact, technical rationality is somewhat blind to its consequences. For example, the failure to recognize the environmental impact of industrialization was due largely to a conception of humans as masters over nature, which could be exploited rather than cherished.

There is also no beginning and no end to design. Some equate design with problem solving, not realizing that most societal problems grow on past technological solutions, the previously mentioned blindness to ecology, for example, or the inequities caused by the costs of accessing desirable technologies. Technology-centered design especially finds itself in an endless cycle of solving problems that cause new problems to be solved. Others see design as translating ideas to change something for the better into sketches, drawings, and presentations, and as a commitment to the community of design professionals to collectively learn from what happens with these drawings for the next round of designs. Again, design is embedded in a circular process that continually generates artifacts it can subsequently build upon.

Four barriers seem to make it difficult for this larger view to take hold of design thinking.

- *Mistaking closed semantic truisms for facts*. Consider the proposition that "producers produce products." It gives the impression and only that of describing a fact, but it is nothing but a semantic tautology. Products are defined in terms of what producers do and vice versa. Definitional tautologies, while referentially closed, tend to support correlated conceptions.

For example, the producer-product truism limits the responsibilities of industry to its final products, which it can control, and absolves it of responsibilities for the cultural consequences of how people use them. It also traps product designers into the manufacturers' logic and, by doing just what they are hired to do, they take no responsibility for their designs. Despite such expectations, designers, to their credit, usually are concerned with larger contexts, with the meaning of making things happen, which is explored in this chapter.

- *Institutionalized conceptions of the players involved*. The conception of THE user as a modal target of industrial products was questioned in Section 2.4 for ignoring the great diversity of people that are instrumental to or affected by a design. To be fair, conceptions of a typical user may have a place in situations in which use is standardized, conformity can be enforced, and users have their reasons not to assume their individuality. Soldiers, for example, can be trained to use their equipment exactly as intended. Similarly, employees in manufacturing plants typically are hired to perform a particular function, at an assembly line, for instance. Institutions, by their very nature, treat people as interchangeable, as office holders in terms of which they are expected to speak and act. Research in ergonomics is mostly undertaken in military and industrial settings where performance criteria are imposed from above and complied with by employees. The theories developed for such situations are biased toward administered uses and largely inapplicable where users acquire and interface with artifacts for their own reasons and enjoy their involvement on their own terms. Institutionalizing the user of technology, including legislating the meanings of artifacts, severely limits conceptions of the larger picture. Most knowledge generated under institutionally controlled conditions is meaningless in the emerging information society.
- *Focusing on ontology instead of ontogenesis.* Science is preoccupied with constructions of ontologies — with coherent theories of what objectively exists (independent of an observer), especially natural phenomena. Indeed, not just science but much of Western culture is preoccupied with material things. But most material entities are transitory. The idea in a designer's head is materially manifest in the firing of neurons. It is an artifact that one cannot observe unless it is articulated into a defendable plan of action. The reality of such a plan can be checked out only when realized in practice. Even the proverbial products that leave their factory may first appear as goods to be distributed, then as the contents of advertisements, followed by being merchandise, acquisitions (possessions), affordances for interfaces, something in need of repair, an environmental hazard, a lesson to be learned, etc. New artifacts are always on their way to retirement. The real becomes manifest in trans-formations, in transitions, translations, rearticulations, and deconstructions.

  Each of these transformations transfers something from one manifestation to its successor. Someone capable of reading production drawings

can recognize the product it specifies, and a broken device retains many features of its working version. One can describe the genesis of technological artifacts as a network of transformations. Merely attending to material objects, to their momentarily frozen forms, severely limits understanding the meaning of artifacts in the process of their genesis.

- *Linear causal thinking*. Explaining an event by identifying its cause is favored in science and engineering. But explaining the causes, the causes of causes, etc., brings one either to an infinitely distant and meaningless past or to an ultimate mover, a god, and leaves those providing these explanations out of the picture. Linear causal explanations of the origins of ideas, problems, and design briefs, on account of which designers are believed to act, fail to recognize the social system in which these explanations are preferred, which is the system by which designers are expected to conceptualize their world so as to change it for the better. By contrast, recognizing human involvement in these processes, all explanations eventually come back to those who attempt to explain the world their way. These explanations are ultimately circular, as Gregory Bateson (1972) recognized, not linear.

5.1 Life cycles

Figure 5.1 shows artifacts in transition from one manifestation to another. The figure also suggests that such networks of transitions essentially contain circles whose beginnings and ends are analytically arbitrary. We could start tracing a circle from where an idea gets hold of a design team, leading its members to negotiate the direction of a development, working out the details to be presented to clients. This circle is completed when designers have learned from how the artifact they had envisioned passed through its life cycle, improving the designers' ability to handle their ideas next time around. We could also start this examination with a product that left an assembly line, changing its role to merchandise sold for profit, to a tool used for fun, to something to be repaired by an expert, to something being disassembled for recycling valuable parts, to being a stepping-stone to a next model. The traditional emphasis on production and consumption can be seen to be a small section of that circle, which when cut into small pieces ignores the essential circularity of an artifact's life.

The life of artifacts can become quite complex but is not entirely intractable. It includes engineers taking the designers' proposals, reworking them into production drawings that program the manufacturing machinery. It includes manufacturers who produce the parts that feed the assembly lines. Even before production begins, market researchers get involved, searching for profitable markets and advising designers, advertisers, and sales personnel on potential buyers. In sales offices, artifacts become display items, packaged to highlight their virtues and to be carried away, if needed in parts, IKEA style.

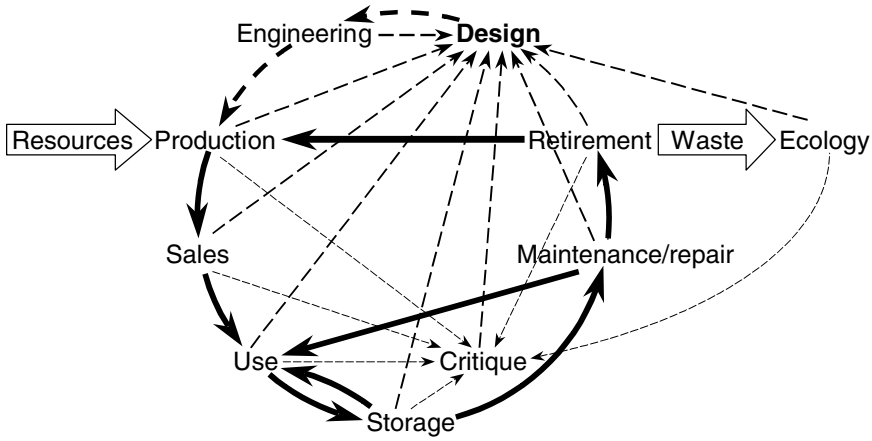


Figure 5.1 Life cycle of technological artifacts — designers' view.

Small consumable goods may be bought, used, or consumed by the same person, but larger artifacts rarely ever are. To maintain a fleet of trucks, for example, purchase agents buy the trucks for a fleet — the truck drivers drive them. Purchasers may never have sat in a truck. The criteria they apply may well consider truck drivers' opinions, but in the end, the criteria that decide are the purchasers' own. To get to the point of being driven, a truck has to survive much paperwork and pass the critical appraisals of corporate decision makers. Some artifacts are continuously in place, like alarm systems, telephone networks, or roads. Others are kept in abeyance, shelved, ready for use when needed. The interfaces with artifacts in use are very different from those kept in storage (see Section 3.3). Maintenance and repair is yet another phase in the life of most technical artifacts preceding their retirement, at which point they are taken apart and sorted into recyclable components and trash. Figure 5.1 is simplified, of course, omitting many participants: scientific consultants, design researchers, product managers, and corporate board members who control the financial resources needed to realize a design. Not all artifacts follow the same path. Consumable items are used up quickly. There are systems that support artifacts produced by others: roads for cars and software for computers. Designers need to chart the network of transitions within which their designs are expected to stay alive.

In Figure 5.1 matter-dominated transitions from one manifestation of an artifact to another appear as solid lines and information-dominated transitions as broken ones. One could say that artifacts never actually cease to exist as long as there are enough people that keep them in one material form or another. When one of its materializations fails, it continues by another name.

5.2 Stakeholder networks

The stakeholder concept (introduced in Section 2.4) expanded the stereotypical user that industrial designers thought they had to satisfy to the great

diversity of people that claim their stake in a design. This was not merely to highlight the empirical fact that people differ in individual capabilities, resources, and intents, but to acknowledge vastly different interests in a technological development, in a design, and in getting an artifact through its life circle or preventing it from being realized. Each makes a living from transforming one manifestation of artifacts into another. Designers need to understand the interdependencies of these stakeholders' worlds.

Designers should not forget that they are stakeholders in their own designs as well, and so are the representatives of the designers' clients who designers rarely underestimate regarding their ability to support a design or prevent it from going forward. Yet, the representatives of designers' clients rarely speak entirely for themselves. They may have other stakeholders to satisfy — bosses, corporate interests, and the public — and act according to what is at stake for them. In fact, the people to whom designers present their proposals are often mere points of contact within a vast network of stakeholders who need to cooperate to bring a design to life. Users are parts of this network but — save for participatory design and testing — they may occur only in the conversations between designers and their clients. Given this complexity, what are the characteristics of stakeholders that designers need to consider?

- **Stakeholders are political actors in pursuit of their own agendas**. They have interests, know what is at stake for them, can voice their support or opposition to what is being proposed, and act according to what the prospect of realizing or using an artifact means to them.
- **Stakeholders attempt to alter the manifestations they have access to**. Designers work to convert problems or ideas into compelling proposals. Their clients take the designs they like and distribute them with their stamp of approval to those who matter. Engineers convert agreed upon functions into the production drawings of working mechanisms. Manufacturers use these drawings to convert available materials and parts into tangible products. Salespeople treat these products as merchandise and convert them into deals. Consumers utilize goods, much as users transform new artifacts into used ones. Recyclers take retired artifacts apart and profit from knowing what can be recycled. Advocacy groups publicize, criticize, or judge artifacts and their uses with the aim of changing public perceptions of them and the processes of their genesis. In effect, all stakeholders respond to manifestations of artifacts and transform them into other manifestations, whether they merely approve or disapprove, or make something physically different out of them.
- **Stakeholders act in their own world, but in the awareness of different ones**. This is to say that stakeholders pursue their own criteria for participating in the process but can hardly prevent affecting each other, even without intent, by the very fact of enacting their expertise and involvement. As a rule, stakeholders are aware of each other's stakes. When stakeholders act in the awareness of each other's interests, resources, and

public standing, the genesis of technological artifacts enters the often-dreaded domain of politics. In fact, the network of stakeholders that designers have to deal with is a network of political expectations. Sometimes it pushes a design through its life cycle; sometimes it makes realizations extraordinarily difficult.

- **Stakeholders emerge whenever desirable possibilities or undesirable prospects become apparent**. For example, the Coca Cola Company once announced that it would discontinue production of its traditional formula. Stakeholders popped up everywhere and forced the company to keep the beverage on the market, now called Coca Cola Classic. Having accomplished their mission, the stakeholders disappeared. As noted by Rittel and Webber (1984), stakeholders rarely are rational problem solvers. They tend to be aware of the political nature of the networks in which they play their part and as such are concerned more with the outcome of a design, what it means to them, than with the problems that design is to solve. Stakeholders create conflicts in view of desirable resolutions, and in that process preserve or enhance the possibilities they value. This motivation applies to the spontaneous emergence of stakeholders but also to the employees in a company who are more likely to facilitate a design if it creates possibilities of advancing within that company and resist it when their position is threatened. Designs that call for mutually beneficial partnerships, cooperation, or larger visions are more likely to succeed than those that need to be pushed through by someone in charge.
- **Stakeholders, by communicating their actions, create other stakeholders**, often unwittingly. Within traditional corporations, the network of stakeholders is usually well defined. Employees know each other's stakes. But in public life, stakeholder networks can grow and shift in unpredictable ways. For example, if a product comes to be known as unsafe or a health hazard, true or not, people who are concerned about these issues may come forward, assert their stake in that matter, enrolling still other stakeholders into that network, and ending up making that product difficult to sell without modifications. This is the way that consumer-advocacy groups become effective voices for change. It is also the mechanism through which companies can grow loyal user groups.
- **Stakeholders organize themselves into interest groups with compatible political agendas**. Examples are the ecology movement, which consists of numerous groups variously pushing for the use of biodegradable materials, wind energy, pollution restrictions, and more. A different example is the National Rifle Association in the United States. It serves a diversity of interests, from gun collectors, hunters, rightist militias, to federalists, principally opposed to government regulations, and is focused on preventing legislation that limits personal possession and use of firearms. In interest groups, stakeholders become more effective not only by being able to pool their resources, but also because groups are more difficult to address and hold responsible for their actions than individuals.

- **Designers are but stakeholders as well**. Designers cannot mastermind the realization of artifacts. A design can succeed only when it provides possibilities for others to participate in realizing its life cycle. When creating possibilities for some stakeholders but not for others, design cannot avoid its politics. In a postindustrial society, more than before, designers can no longer hide behind powerful manufacturers; they have to become as knowledgeable about processes of genesis as the stakeholders that their design brings forth. Good designs embrace the stakes of its diverse stakeholders, not just the designers'.

In other words, designers need to propose projects that attract others to participate — but what are projects of which Section 1.2.5 spoke?

5.3 Projects

Not every design is so new as to require a realignment of its stakeholders. The redesign of familiar artifacts — cars, appliances, or airplane cockpits — not only draws on a respectable history of such artifacts, but also on stakeholder networks that have been institutionalized into structures of employment within which designs are implemented along well-established paths with only minor modifications. Under these conditions it may indeed be understandable that manufacturers encourage designers to focus on the users of their products who are the relatively uncertain parts of a stakeholder network. Under these conditions, the motivation to promote a design is often regulated by professional obligations and financial rewards. However, these also are the circumstances in which design is relatively insignificant, offering only small variations of the already known.

When technologies are developing or are acquiring new uses, the role of stakeholders other than users becomes important. People do not just do their job, they have opinions about what they do, claim their stakes, and want to be heard. Under these conditions, designers cannot limit themselves to making drawings or models, writing perfect specifications like engineers do, or claiming knowledge of the stereotypical user. Instead, designers must provide spaces for other stakeholders' participation. This means trading the specificity of a proposal for the ability of other stakeholders to have their say as well and participate in bringing a design to fruition. The real challenge for postindustrial designers is to enroll stakeholders in their project, to convert opponents into supporters, and to create the resources needed to realize what they like to see. Projects inherently inspire cooperation among stakeholders. Beyond Section 1.2.5, one can say:

- **Projects energize the network of stakeholders** in a technology or artifact.
- **Projects keep spaces open for competent stakeholders to feel invited**, appreciated for what they could contribute, and respected for whatever brings them into the project.

- **Projects have a "point,"** a narrative that participants can affirm to each other, offering a sense of purpose or vision. **It directs the attention of participants**, coordinates their actions, channels their resources, reduces dissention and conflicts, and enables collaborators to know when their project is concluded.
- **Projects cannot be controlled by any one member**, much as in everyday conversations. What is on the table, a design, maybe perceived variously by the stakeholders involved and what results from this cooperation may not be what either participant had envisioned. To preserve the cooperative nature of a project, nobody can dominate it, not even its designer.
- **Projects may transcend the tenure of its members**. Stakeholders may move in and out of a project. They make their contributions, different ones at different times, on which subsequent participants can build. The viability of a project does not depend on particular individuals but on the rearticulation of its essential narrative by its current participants — despite changing membership, unanticipated circumstances, and progressing states of development.

Thus, the initiators or "designers" of a project will have to be concerned with:

- *Its social viability:* its ability to continuously attract those capable of contributing to it and inspire cooperation among them.
- *Its directionality:* its ability to focus participants' attention on the point of the project.
- *The commitments* of those involved to the project: their willingness to provide energy beyond the tangible compensation for their efforts.

5.4 Genetic meanings

In the context of use, meanings guide a user's interface with an artifact, ideally getting that user from recognition to exploration and to reliance (Figure 3.3). In the context of language, meanings coordinate the perceptions and actions of a community of speakers regarding particular artifacts. In the life cycles of artifacts, by contrast, meanings direct the genesis of artifacts, what their present manifestations can give rise to or will become. Hence, what is important here is not the form of an artifact but the sequence of its trans*form*ations from one form into another; not its materiality but its transition from one materiality to another; and not its use but its translation from one use in one context to another use in another context. Within life cycles, the genetic meanings of an artifact guide its passage through the very network of stakeholders it brings about. At any one moment in such networks, what can be seen, what can be photographed, is always only the snapshot of an artifact in transition, not the artifact's life.

Thus, genetic meanings provide the directionality for an artifact to mature and move on. In the context of their genesis, a successful product is

one that moves easily through its market, to users and beyond, and is succeeded by a newer model, a more advanced design. A successful presentation of a design cannot merely present the model of an artifact, it must communicate to clients the possibilities that a design opens up for them, possibilities that may not have been anticipated and are now found inspiring. Semantically, "success" entails a "successor." Hence, genetic meanings link artifacts at one point in time, and perhaps to a particular stakeholder, to their successors at a later point in time. The genetic meanings of artifacts inform stakeholders what can become of the artifacts. A genetically meaningless artifact has no discernible future other than that predicted by physical laws; for example, the second law of thermodynamics suggests that everything is subject to entropic decay. Genetic meanings enable stakeholders to do something other than what natural laws could explain.

In the life cycle of artifacts, genetic meanings pertain not only to designers' visions. Designers may see possible futures that others can't, and the designer's problem is to pass these possibilities on to others who could realize them. Often these are clients. However, clients rarely speak for themselves. Typically, they become convinced of the value of a design only if they can see a way to pass its possibilities on to still others who matter, but for their own reasons. A successful design is one that is passed on, opening exciting possibilities at each point. Engineers may consider a design successful if it entails technological challenges they can handle. Advertisers may consider a design successful if it has features that can be used for promotion and allow it to be moved through a market with little effort. To environmental groups, a successful design is one that sustains ecological diversity. All of these meanings enable stakeholders to proceed, to recognize their own contributions, and to claim success for their mission.

The genetic meanings that designers need to consider could be conceived of as an onion with multilayered meanings that can be peeled off one by one as it passes through a stakeholder network, or as a story that, while addressed to one, embeds stories addressed to successive others. Each version of a designers' proposal ought to address the hopes and aspirations of its listeners. Each version needs to speak of the possibilities that participation in the project opens up to them, and each should encourage others to become enrolled into the project. The design narratives that are to be passed on with each *trans-form*, need to accomplish four things.

- *Establish the source's credibility.* All stakeholders need to demonstrate their competence in participating in the genetic process. Without credibility few will listen, the genetic meanings being passed on may lose their appeal, and a design may stop progressing through its life cycle. Human-centered designers need to substantiate their semantic claims, what their proposal can and will do, or cede to those who will test them on their own, including proving them wrong.
- *Address sequentially relevant stakeholders* in a language familiar to them. Like a chain letter, the genetic meanings of an artifact must direct it

through a network of stakeholders and in the order in which their contributions are relevant. Such sequential addresses may use the stakeholders of similar artifacts, ideally, but when such networks do not exist, genetic meanings need to create them.

- *Motivate individual stakeholders* by making the possibilities that an artifact offers to them recognizable and attractive while *energizing the stakeholder network as a whole.* The life cycle of artifacts does not only involve individual stakeholders but also those who represent organizations, businesses, or have the well-being of particular institutions in mind.
- *Inform what could be done*, where to find the resources needed to proceed, and how to realize the artifact's manifestations in the context of its life cycle. This includes communicating whatever it takes to communicate with subsequent stakeholders (establish their own credibility, address relevant other stakeholders, motivate others to contribute by informing them of what they can do), ideally preserving the point of the project.

To create design proposals that provide the genetic meanings for artifacts to succeed through its life cycle is not an easy matter. But design is never easy. Designers have written manifestos and position papers, and created compelling visualizations of the kind of futures that their designs would encourage. Sketches of products alone rarely communicate possibilities.

One may be reminded of the etymology of "success." It derives from "succeeding," from the ability of something to survive successive transformations. *Prospectively*, a successful design affords the meanings of all those stakeholders who can move an artifact through its life cycle, making it part of a social process. *Retrospectively*, a successful design is one that came to fruition with the help of many and gave rise to further developments.

In a postindustrial society, it is nearly impossible to devise a plan that would address all possible contingencies that an artifact may come to face. Unlike in highly structured situations, during the industrial era, for example, it is now less likely that projects succeed as intended. The need to enroll stakeholders into a project almost always amounts to delegating parts of a design to be filled in by its participants. In the end, what an artifact becomes is what its network of stakeholders makes it to be. This fact is reminiscent of D'Arcy Thompson's (1952) Law that states: *"Everything is what it is because it got that way."* In analyzing why things "got that way" proactively, one is led to yet another theorem of the axiom of meaning:

No artifact can be realized within a culture without being meaningful to those who can move it through its various definitions.

Clearly, a design that does not compel a client to consider it will die at presentation. A design that does not allow engineers to see ways of solving the technical problems it entails will not reach production. A design that salespeople cannot praise is not likely to be sold. A design that is not

understandable for what it is or can do is not likely to be used. Meaning drives every phase.

According to Ken Friedman,* "One study suggested that of new product ideas that move beyond the proposal stage, 57% achieve technical objectives, 31% enter full-scale marketing, and only 12% earn a profit (Mansfield et al., 1971:57). According to some experts, over 80% of all new products fail when they are launched, and another 10% fail within 5 years" (Edwards, 1999; Lukas, 1998; McMath, 1998). It is not too clear how current these figures are. They suggest, however, that many artifacts do not have the meanings to enroll the stakeholders needed for artifacts to complete their life cycle.

Theodor Ellinger (1966) was an early proponent of the idea that meanings drive a product to its realization. He discussed what a product must convey within a market to reach its intended consumers. He suggested also that it tell about its intended use (see Chapter 3), which is implied in the last of the four components of a design narrative. For the semantic turn, market considerations are important in the life of many artifacts, but they are surely not the only concerns. Ellinger's conception of meaning is limiting in this regard. Advertising expenditures suggest that products alone may not provide the needed information on their own, but the narratives surrounding them, whatever channels carry them, may support this effort.

5.5 Critical sizes of supportive communities

Industrial designers had been trained to see only the users of their designs. Indeed, during the industrial era, scarcity of resources, and the institutionalization of production and consumption took care of most uncertainties, leaving the user as a variable. The idea that artifacts should be designed to be good for everyone did not only respond to industry's need to expand its markets with mass produced goods, it also fueled socialist ideals, which grew out of industrial-era thinking. Recall the dress code in Maoist China, which unified the country by suppressing cultural differences, leaving little to self-expression. Strangely, the idea of universally useful artifacts is also in tune with Western notions of democracy of everyone's vote carrying the same weight, compulsory education and standardized tests, the mass media, universal telephone service, universal health coverage, but also the conception of "gaps" in usage that need to be closed where universality is not achieved. Numerous studies have inquired into the "information gap" between information-rich and information-poor populations, and the "digital divide" between those who use information technology and those who do not. As attempts to overcome such apparent gaps are made, voices lamenting the loss of culture, traditions, and ethnicity are heard. Universality is a value that is reminiscent of the industrial era but no longer meaningful in a postindustrial world.

* Posting to Phd-Design@jiscmail.ac.uk on September 22, 2004.

As far as the use of technology is concerned, sameness and equality is not only unachievable but also socially stifling. Fashion, for example, necessitates differences between groups that lead and groups that follow. Professional expertise derives from taking unequal educational paths. Training in the art and science of medicine differs from that for engineers, naturally. In establishing group identities, differences between "us" and "them" are often manifest in the selective ownership and use of artifacts, from having pierced ears to owning a Porsche. Maintaining a group's identity means preserving the gap between insiders and outsiders, even if it requires moving to new means of identification when the difference is seen as eroding. Gaping differences in the use of technology are not only inevitable, they reveal themselves as the driving forces of technological developments. Here are three propositions for contemporary designers to consider:

- *All technologies reside in particular communities* that are capable of moving them through their life cycle. There are bicyclists, but not everyone needs to ride a bike. There are avid readers of books, but not everyone needs to do the same. There are people who drive cars and others who do not. There is little, if anything, that has the same meaning to everyone.
- *To be viable, every technology needs to exceed a critical mass in the minimum number and kinds of stakeholders* enrolled in the project of that technology. For producing simple items, like bread, that critical mass is low: a baker, a number of customers large enough for the baker to make a living and pay suppliers of fuel and flour. For complex technologies that critical mass tends to be far higher. A project for a multiuser system may require high startup investments, expensive publicity, and long-term commitments on the part of its early adopters. When changing automobile production to a new model, designers can fall back on familiar networks of experts for whom change within certain parameters is routine. In developing a new program of graduate education, overcoming the critical mass is not easy. Artifacts with larger critical thresholds may rely on incubators, established institutions that can afford to nourish a development until it can stand on its own. For example, to start a graduate program in design, one may initiate it in a different department, offer some courses, and borrow professors from related disciplines until the program has enough students, qualified faculty, and staff to function on its own. The most common way to introduce a new technology is to piggyback it on to one that already has enough stakeholders. The availability of a community is always critical.
- Some technologies are *confined to a designated community*, medical equipment, for example; others *become more useful as more people participate*: the telephone, the mass media, and the Internet, for instance. Still others *thrive in being passed from one community to another*; fashionable products are one example. They may start with a few technology buffs, enter the class of people whose identity depends on being early adopters, and end being used widely and no longer perceived as outstanding. Here, the

rule of the critical mass applies as well but changes over time. Universality is not an issue. But when stakeholders no longer transform an artifact within a life cycle, it surely disappears.

- Technologies may well be designed to exclude people for legitimate reasons, such as preventing children from opening medicine bottles or unlicensed users of handguns from firing them (see Section 3.4.2). It is an ethical proposition, however, not to deliberately exclude populations of stakeholders that are knowledgeable, capable, and willing to use a technology without harming anyone. It entails making special efforts to include people that are socially disadvantaged or have various physical disabilities.
- The final proposition on the critical sizes of communities offered here is ethical as well. It concerns stakeholders who oppose a project for the technology it aims to materialize and put to use. For entirely practical reasons, projects always have to be defined to convince enough stakeholders to come on board, but they should not be defined to disable legitimate opponents of a project and the technology it is to bring it about — unless the lives of people are affected. Drilling oil in Alaskan wildlife reservations is a matter of public debate. Designers' ingenuity ought to be able to circumvent most opposition. But the design of spyware that enters someone's PC without knowledge, permission, and defense against transmitting personal data to an unidentifiable party exemplifies an albeit minimal case of what human-centered designers might not want to pursue on principle.

5.6 Whole life cycle accounting

Engineers tend to be committed to achieving measurable performance criteria for their designs: Bridges have to be designed to withstand winds of a specified speed and hold up to use by the heaviest vehicles on the road, tanks for example. Transformers have to perform under conditions of a specified maximum load. Computers are expected to work at a specified speed. Such technical specifications, often considered bottom line requirements, nevertheless focus the concerns of engineers to what they can test and control, usually by mechanical or objective measurements. As such, they conceptually exclude concerns of the life cycles of technologies and are the cause of many environmental problems.

One barrier to considering the larger perspective developed in this chapter can be found in adopting efficiency criteria for designs. Much like efficacy and utility, efficiency is the proportion of desirable effects to efforts expended to obtain these effects or desirable output to necessary input. It comes in various forms but is most typically measured in terms of energy, time, and money. For example, for cars, energy efficiency is expressed in miles driven per gallon of gasoline (or liter of gasoline per 100 km) consumed. Ergonomics has embraced efficiency criteria by measuring human performance, for

example, by the number of units produced in time. Taylorism is an effort to organize the workplace for the sake of efficiency. Cost–benefit analysis feeds into efficiency concerns by using money as the unit of measurement. In gauging usability, error rates, the number of steps needed to reach an objective, the time taken to learn an interface, all measure the lack of efficiency. The whole idea of optimizing, maximizing, or satisficing a scarce resource has a long tradition in northern European culture, in which it is equated with (technological) rationality, with uniform standards applicable everywhere and to everybody. The industrial era was dominated by criteria that served the industry more so than users.

Norman's (2004) concept of user-centeredness expands the usability concept to include likes and dislikes, emotions as he calls them. He recognizes that attractive artifacts are more likely learned and work better than unpleasant ones and then proceeds to explain what makes things more pleasurable. While correctly distinguishing between technological rationality and psychology, Norman's conception of meaning invites three kinds of criticism. The first is his focus on psychology. True enough, aesthetics, fun, and enjoyment matter, but these are largely socially defined and linguistically established within different communities. What is pleasant in one community may be rebuked as self-indulgent in another. Evidently, language matters, but the language of psychology does not reflect on its own individualist biases. Second, by framing these considerations in seemingly objective terms — more pleasurable things are better to sell, more likely used, and cause fewer distractions — Norman wants to offer designers efficiency measures of a psychological (emotional) kind. That many Americans are willing to drive gasoline guzzlers when they like a car certainly supports his contention that technological efficiency measures are less meaningful* but not that other generalizations are better. Some people disobey generalizations about their use of artifacts, whether these generalizations are proposed by scientists, legislated by authorities, or encouraged by majorities. This is why fashion develops in defiance of easy generalizations. Third, and most importantly here, Norman's conception of meaning does not extend beyond use. Efficiency measures are not generalizable to the life cycles of artifacts, which depend on a network of stakeholders, each applying their own "rationalities" to their areas of interest. Efficiency in sales may have no impact on the efficiency of use. Efficiency of production may not sustain ecological balances. Money is always shortsighted. Long-term costs are especially difficult to assess, for example, the cost of using up nonrenewable oil reserves, or cutting down the Brazilian rainforest. Stakeholders resist being forced to adopt the efficiency criteria imposed on them by others.

It follows that in complex stakeholder networks, accounts of what a technology is to do cannot be limited to any one participant, one unit of measurement, or one perspective. For example, solar energy is daily available,

* A "pleasure-to-cost ratio" has been suggested as a psychological efficiency criterion. It attempts to preserve the efficiency concept without a clue to how pleasure could be generalized.

considered totally clean, and the right thing to use. However, by limiting their concerns to their immediate worlds, environmental-friendly citizens may not realize that what they pay for when acquiring the collectors and batteries pays for the mining of the needed chemicals and the use of fossil fuel for their manufacture, which in turn adds the very pollutants that these well-intended citizens sought to keep away. Or, making government resources available for the development of hydrogen-powered cars, promising cleaner air, turns out to be mere window dressing as long as the needed hydrogen is generated by using fossil fuel. Here, the oil industry benefits from the limited scope within which hydrogen-powered cars make sense.

The same can be said when imposing one stakeholder's cost–benefit analysis onto others who may consider different units of measurement relevant. For example, the global benefits of emission restrictions for individual cars in one country maybe be wiped out by the unrestricted number and size of the cars used in that country. This is a source of disagreements between European and North American environmental policy makers. Western Europeans live with less stringent emission standards but drive smaller and fewer cars than the Americans do, so the United States contributes most of the world's air pollutants. Similarly, the efficiency of atomic energy is much debated with proponents measuring efficiency at the power plant and opponents coming to vastly different conclusions by including the environmental impact of mining and refining uranium, and recycling spent rods, not to forget the unimaginable consequences of storing atomic waste in the vicinity of which humans cannot live for generations to come.

In sum, while the life cycles of artifacts are designers' foremost concerns, designs come to fruition only when they can enroll diverse stakeholders into networks of cooperation and enable them to play constructive roles in their worlds. Life cycle considerations must not be dominated by any one group of stakeholders, including designers. Uniform criteria almost always create technological monstrosities that can haunt generations of people. Unfortunately, the environment of humans has no voice of its own. By relying on stakeholder networks within which competing conceptions, including of the environment, are negotiated and resources are traded for the commitment of participation, there is a chance for a fairer treatment of what is otherwise difficult to understand.

chapter six

Meaning in an ecology of artifacts

The three preceding chapters developed notions of meaning for artifacts by putting them in different contexts: in the context of individual human use; in the context of language, human communication, or social use; and in the context of their life cycle, their genesis, or the process of their becoming. These contexts are quite different from one another. The thread that ties them together is the concept of meaning, human involvement. This leads to the current chapter, which addresses what is still missing, how artifacts relate to each other meaningfully, and what they do to each other consequent to how humans conceive of them. These are the meanings of artifacts in an ecology of other artifacts, in the dynamics of technology in use, in designers' minds.

6.1 Ecology

Generally, ecology is a theory of how large numbers of species of organisms interact with one another and in that process feed on each other, reproduce, proliferate, find their niches, or die. Traditionally, ecology concerned only species of animals and plants living in environments of nutrients and hazards. Because it is not easy to keep track of very many species, early ecologists limited their theories to small ecosystems, such as particular lakes, forests, or deserts.

Although rarely acknowledged by ecologists, humans necessarily are part of any ecosystem for two reasons. First, even small ecologies usually are constructed with the intent to intervene. By intervening, humans alter the very ecosystem they try to understand, thus entering that ecosystem as participants. Second, large ecosystems and especially global ones, to be fair, can hardly exclude humans as a species that lives with other species, feeds on many, encourages some, or seeks to extinguish those considered harmful to them, deadly bacteria, for instance.

Environmental action groups have identified many global ecological issues, most of which depict technology as disturbing ecological balances. Agriculture, for example, is seen as reducing the diversity of plants; road systems restricting the movement of animals; industrial waste killing marine life in rivers; hunting rifles reducing the population of birds, rabbits, and deer; and clear cutting of trees threatening the habitats of species living in forests. Such characterizations, true as they are, tend to ignore the effects of acting on such ecological issues. For example, the idea of preserving endangered species, of keeping an ecosystem in balance, while well intended, preserves not nature but human conceptions of nature. In an ecosystem, the nonhuman species do not have a concept of ecology and do what they do as species, each for itself, preying on some, cooperating with others, expanding or declining in numbers, and evolving in interspecies interactions — just as ecology theorizes.

However, as a human artifact, ecology participates in the ecosystem it claims to describe in most unecological ways: supporting conservative and romantic ideals of nature, which is not to be criticized but to be acknowledged as such. In global ecologies, it is the use of technology that has made the human species the dominant one. Yet, there is little published on how species of artifacts interact, what they do to each other, and what guides these interactions. The following explores the parallels between how biological species interact and how species of artifacts interact with the aim of presenting an ecological conception of meaning that designers are advised to consider.

6.2 Ecology of artifacts

As Kenneth Boulding (1978) has pointed out, humans know far more species of artifacts than species of living organisms: shoes, bottles, breads, books, automobiles, tools, fasteners, furniture, paintings, airplanes, buildings, coal mines, streets, and communication systems. Department stores, mail order catalogs, and the Internet display only a small part of this diversity. A dictionary lists the names of many more.

Species of artifacts also cover far greater ranges than biological species do. Skyscrapers are larger than whales. Artificial molecules are smaller than bacteria. Modern computer networks have more memory than any single living organism, although of a very different kind. Within the few years of its existence, the Internet is making far more images available than any one human being could read in a lifetime. People use the metaphor of a spider web to conceptualize the World Wide Web. A spider web is several times larger that the spider that builds it, but the World Wide Web spreads throughout the world. Its size dwarfs its builders and users. Space ships can move further and faster than birds or insects. Cities outlive their inhabitants, and archives and museums can keep artifacts alive far beyond their usefulness to users. The role of artifacts in museums is different from the role they play in everyday life. In fact, the same artifact may pass through different uses

with different generations of users (Krippendorff, 2005 in press). Biological species, by contrast, change their behavior only slowly if ever, at least during human times.

Most species of artifacts are mass-produced, although it may not be clear whether the series of industrial products exceed the mass production of plants, bees, ants, or bacteria. When artifacts are assembled into transportation systems, corporations, or networks, they create systems of artifacts whose complexities far exceed the complexities of forests, ant colonies, and beehives. Ant colonies and beehives hardly cooperate and perhaps don't even know of each other. Technological artifacts do not know of each other either but interact with each other on account of designers' specifications and/or users' desires to connect them. Unlike biological species, which need a minimum number to be viable, complex artifacts may well survive as unique artifacts.

This is the crucial difference between ecologies of biological species and of artifacts: *Biological species interact on their own terms; artifacts interact on human terms*. Whenever people arrange their furniture at home, wire pieces of computer hardware together, or install something for the benefit of their community, they display some ecological understanding of how artifacts can work together. Designers demonstrate this very understanding when claiming that their design can play particular roles relative to other artifacts. But the "human terms" that people bring to how they connect artifacts pertains to a highly localized understanding. This understanding does not embrace the whole ecology, and it is not generalizable to it. It is distributed, not shared. Whereas ecologies of biological species result from the multiplicity of local interactions between the organisms of different species, ecologies of artifacts result from enacting the multiplicity of local ecological understandings.

Thus, ecologies of artifacts, even of only moderate complexity, escape any one individual's understanding. To cope with this complexity, we may have to be satisfied with partial theories of how artifacts interact. The two established ways of carving a manageable section out of an ecology of artifacts are:

- **Diachronic accounts** of one or few artifacts, tracing their evolution, much like the lines of biological speciation, through time with attention paid to their changing roles and frequencies in the interaction with other artifacts.
- **Synchronic accounts** of a particular subsystem in an ecology of artifacts, for example, of how communication technologies hang together, and tracing their interdependencies just as one would study food chains in an ecology of biological species.

For a *diachronic account*, consider the evolution of the telephone. Shortly after inventing the telephone in 1878, Alexander Graham Bell introduced it as a wooden box, which hid its (rather simple) wiring but displayed a surface

mounted bell (the ringer), a fixed mouthpiece (microphone) to speak into, a crank to generate a signal, and a detachable earpiece to hear the faint voice of the person on the other end of the line. During that time other electric inventions flourished, and new kinds of devices entered nearly all spheres of daily life. This flooded the ecology with numerous appliances and instruments in increasingly distinct forms. Keeping up with these evolving species, the ringer of the early telephone disappeared from view. The mouth- and earpieces merged into a single handheld receiver. A dial appeared that allowed users to enter numbers instead of asking telephone operators to arrange for connections. Several distinct families of telephones emerged. However, in the 1930s, the telephone had become comparably stable in appearance. Its form became almost universally recognizable and had few competitors. Designers introduced only miniscule variations. When buttons replaced the dial, the phone did not change, at first. But then the Trimline phone began to house the dial (or push buttons) in its hand piece, shrinking the body to a vessel for that hand piece. Yet, the telephone of the 1940s continued to be reproduced. This was because AT&T in the United States was a monopoly, and the postal and telegraph services in other countries were mostly state owned and inherently conservative. National monopolies dominated telephone development until 1996 when the U.S. Supreme Court rendered its decision to break up the AT&T Bell System. This opened the market to competing manufacturers who now had to distinguish their products from their competitors. There were plenty of ideas available to draw from: military technology (field phones), science fiction movies (portable communicators as used in the television series, *Star Trek*, for example), and headsets and microphones (from telephone operators). Without significant changes in telephone use, competition produced a host of different shapes, displacing the telephone icon of the 1940s, which incidentally has survived in telephone signage. The telephone companies that emerged after the breakup of telephone monopolies have given up the hope for a new telephone archetype. Numerous technologies fused, combining image and sound. The early videophones failed because people did not like to be seen by those they wanted to talk with. The merger of telephone and facsimile transmission was successful but produced no new compelling form. The answering machine cooperated with the telephone for a while but then evolved from a separate technology into a service provided by telephone companies. A technological fusion of the telephone with computers, writing, conferencing, and television, e-mail has produced a myriad of new gadgets. The dominant form now has become the wireless "cell phone," its merger with digital cameras, but most importantly with handheld personal organizers, palm-sized computers that provide all kinds of functions of which voice and e-mail contact with other people is only a part. The telephone as a separate interface is dying. Its identity is fading into the shapes of pocket-sized or wristwatch-like communication devices. The very word "telephone" may be replaced, retired, or restricted to a historical artifact.

This evolution made it difficult for telephone companies to identify themselves with a clear product.

In short, diachronic ecological accounts trace the evolution of one or a few artifacts through time and in the context of other technologies, institutions, and social problems. In biological evolution, there is mainly speciation, the splitting of one species into two separate ones, often depicted in tree diagrams. In the evolution of technological artifacts, many more processes are evident. There is speciation but also the merging of several artifacts into one. There is the migration of features from one species of artifacts to another. Keeping a species of artifacts alive means reproducing its kind in large numbers. However, it is "imperfect reproduction" that increases the chance that a species will survive the interactions with other simultaneously evolving artifacts. Species of artifacts succeed by being improved, even if it means being absorbed in another technology or becoming invisible.

Synchronic accounts, by contrast, describe the network of *concurrent connections* between artifacts that codetermine their use. Continuing with the telephone example, albeit superficially, these connections can be described as being of four kinds.

- **Causal connections** link artifacts with each other physically. Without a network of wires, radio transmitters, or satellites, a telephone would be useless. Without a modem, a computer could not connect to other computers via the Internet. All causal (non-human) interfaces are based on mechanical connections or various ways to transmit signals through wires or wireless connections. The design problem typically turns around interface standards, which can emerge in the history of use or result from negotiations among manufacturers. Once in place, causality runs its cause without much human intervention.
- **Family resemblances** are established by conceiving of different artifacts as belonging together, as cooperatives or families. Telephones, telephone books, answering machines, and telephone booths for privacy do not cause each other but users conceive of them as conceptually connected. Tableware, flatware, and dining room tables constitute a family in the sense that they are all used together, purchased in view of going together aesthetically, and supporting each other in the practice of dining. Family resemblance can also derive from part–whole relationships, the concept of a system with parts. Systems, family resemblances, belongingness, it should be emphasized, are purely conceptual, often normative, and thus distinct from physical connections. When parts are expected to play particular roles within a larger system, it is always someone whose conception connects them, someone who sees family resemblances, including parts as serving functions for a larger whole. Artifacts do not know how they are conceptualized and what is expected of them. There is nothing physical here.
- **Metaphorical connections** are conceptual as well. Metaphors carry meanings from one species of artifacts, usually from a more familiar one,

to another species (discussed in Sections 3.3.2 and 4.5) and thus create relations between them. Describing a new device as an electronic book builds a bridge between two species of artifacts. Thus metaphors participate in organizing an ecology of artifacts by transferring knowledge from one species of artifacts to another.

- **Institutional liaisons** are formed when different institutions find themselves dependent on the same species of artifacts or practices. So, television stations, equipment manufacturers, studios, producers of news and entertainment, advertisers, cable or satellite service providers, all thrive on people watching television. Television would become useless if the institutions involved no longer see their interests as joint and fail to cooperate for a sufficiently long period, just as this institutional liaison would collapse if people stopped watching television altogether. The liaison is implicit in the joint commitment to keep a species of artifacts alive or to developing them their way. Another example is the automobile, which is supported by a complex system of institutional interests that are geared to keep cars usable and on the road.

Designers can hardly ignore any of the ecological interactions mentioned here. Their designs will enter into relations with other artifacts and must be designed in view of their ability to survive such ecological interactions. Diachronic accounts of an ecology of artifacts provide an understanding of the interactions that shaped the evolution of a species of artifacts over time, which a particular design may be designed to continue into its future. Synchronic accounts describe the network of artifacts that a design will have to face at any one moment in time. The stability of an ecology depends on the strengths of such connections. However, by going through its life cycle, any innovative design is likely to disturb existing stabilities, creating or closing options for further design, hence the need for understanding the ecology into which it enters.

6.3 *Ecological meanings*

In an ecology of artifacts, *the meanings of an artifact consist of its possible interactions with other artifacts*, both, with its own kind, but more importantly, with artifacts of other species. Ecological meanings, where recognized by individual stakeholders, guide artifacts to their energy sources, set them in motion, bring them into interaction with various other artifacts, and have global ecological consequences. An ecology of artifacts is the net effect of what very many stakeholders do locally.

Any ecology recognizes three kinds of interactions: cooperative, competitive, and independent. Since ecologies are concerned with interactions among species, with populations not with individual members, the numerical size of species are a critical variable in describing how species of artifacts affect each other. For any two species A and B, we distinguish:

- *Cooperative* interactions, when an increase in the numbers of A increases the numbers of B.
- *Competitive* interactions, when an increase in the numbers of A decreases the numbers of B.
- *Independence*, when the change in numbers of A are not correlated with the change in numbers of B.
- The size of any one species may grow or decline in terms of these interactions but also on account of the proportion of its rate of *reproduction* and its rate of *retirement* (death).

In ecologies of biological species, every species is essentially on its own, reproducing its kind, unaware of the global consequences of the actions by its individual members. If the concept of ecological meanings would have to be applied there, it would have to reside within each species. Ecologies of artifacts, by contrast, are put together by the human species, and ecological meanings derive from the composite effects of numerous individual decisions concerning the deployment of technology. Ecological meanings have numerous analogies in phenomena we know from the use of language, which points again to the importance of linguistic models of understanding.

Two artifacts can be said to be *synonymous* in meaning if one can take the place of and assume the role of the other. Synonymy of ecological meanings is taken for granted for members of the *same* species of artifacts, and it is also the ecological reason for their mass production. Mass production, whether of ants or of automobiles, assures the continuation of a species in view of certain deaths or losses. Given the rate of an artifact's failure, consumption, or retirement, with all other conditions assumed unchanged, ecological stability is sustained when species are reproduced at the same rate as they are removed from the ecology. This is the most general ecological law. But stability in an ecology of artifacts is a rare and probably the least interesting case.

Near synonymy of artifacts from *different* species creates competition. Consider horses and tractors for example; both can pull plows and carts, and are capable of transporting riders. The fact that tractors started to replace horses on farms beginning in the 1930s is due largely to the advantage of tractors, which had to be fed only when in use, were more powerful than horses, and easier to control. This did not extinguish horses, however. Horses found another niche in sports, where they acquired meanings that were not prominent when they worked the fields. A similar competition existed between typewriters and computers. Typewriters did not fare so well, however. After giving computers their keyboards, they currently survive only in technical museums, having become ecologically almost meaningless wherever computers took over their job of putting type on paper. Characterizing a computational device as an "electronic book," which is a metaphor, brings it in direct competition with the traditional paper book.

As already argued, artifacts do not interact like biological species do. Rarely does one artifact gobble up another, which gives rise to food chains in biology. Artifacts cooperate or compete based on what they mean, and at the point where they are compared with one another, for example, when users decide to buy, install, or use one artifact at the expense of an ecologically similar other artifact. Competition between phones and instant messaging devices happens at the point of purchase. After one is adopted, they need to cooperate with other artifacts. Such decisions are influenced by perceived differences in how they function technically, how they relate to other technologies already in place, and the opportunities they offer for the decision maker. Designers of competitively successful artifacts must provide the ecological meanings that swing decision makers to their side. Ecological meanings may suggest causal links, family resemblances, metaphorical connections, or be based on institutional liaisons — as discussed previously. However, their ecological net effect is always either cooperation, competition, or independence.

The sociology of technology has contributed several diachronic studies of particular artifacts, for example, of the bicycle (Pinch and Bijker, 1987). Generalizing, Pinch and Bijker observed that technological developments proceed in phases from an invention that affords elaborations in design, has different meanings for different interest groups, through struggles among stakeholders as to the problems the artifact is to solve and how it is to be interpreted, to a point at which the artifact exhibits stability in design (structural stability) and meaning (interpretive closure). Then, technological changes consist of only minor variations of the artifact's ideal type, and diverse stakeholder groups have come to some kind of consensus as to what it means, how it is to be used, when, and by whom — diminishing what the authors call its interpretive flexibility — and the problems that gave rise to the artifact are seen as solved. At that point, the artifact has also replaced all the artifacts it could, its ecological meanings are institutionalized, and its interactions with other species of artifacts have become relatively settled. Ecological stability is probably rare, temporary at best. But the social nature of ecologies of artifacts, the role of artifacts' meanings for different stakeholders, is well demonstrated.

By cross-tabulating cooperation, competition, and independence, Boulding (1978:77–88) found seven types of interactions between any two species of artifacts: three symmetrical interactions: mutual cooperation, mutual competition, and mutual independence; and four asymmetrical interactions: parasitism, predation, dominant-cooperation, and dominant-competition. These provide a vocabulary for aggregating the effects of meanings on ecological interactions, here between two species.

- **Mutual cooperation**, the ability of a design to support another species of artifacts and in turn be supported by that species. Cars and roads mutually cooperate. More cars demand more roads, and more and better roads encourage more and better cars. In a similar way, the increasing

complexity of computer software drives the memory capacity of personal computers, which in turn enables more complex software to be developed and installed. Apple's iPod mediates a mutual cooperation between several new technologies pursued by the music industry, artists, and listeners with highly individual tastes for music, allowing them to download the music they enjoy. This cooperation grew on top of the record, tape, and CD industry, and now provides a technology that is even easier to distribute and use. Designs that encourage mutual cooperation among different technologies increase the others' chances to survive and often end up in symbiotic relationships.

- **Mutual competition**, the ability of a design to compete with an equally competitive alternative. Today, individual cars and public transportation systems are so related. Designs for public transportation — rail, city buses, and trolleys — will have to consider what would make public transportation more attractive than driving individual cars. The interaction between public and individual means of transportation may not end in the victory of one over another, but could drive one of the two into an insignificant niche.
- **Dominant-cooperative**, the ability of a design to take advantage of another species, depending on it as a leader or host, while increasing the numbers of both. A popular new technology tends to encourage numerous so-called secondary gadgets. The advent of digital photography was followed by software for manipulating photographic images, photo quality color printers, and Internet-based services that store one's personal photographs and reproduce them for a price. In fact, it was these gadgets that made digital cameras popular in turn. They would disappear when the dominant species, on which they depend, digital cameras, became obsolete. Shopping malls, mail order catalogs, and advertisements in popular magazines are full of secondary gadgets: to make air travel more convenient (suitcases that fit under the seat of an airplane, neck support for sleeping on uncomfortable seats), to make food preparation at home easier (using electric can openers rather than mechanical ones), to reduce mothers' baby chores (audio-visual crib monitors), to organize tools more efficiently, decorate one's home, make one's car appear fancier, etc. Designers of secondary gadgets tend to capitalize on minor shortcomings of the dominant technology.
- **Parasitism,** the ability of a design to thrive on another species of artifacts without increasing the latter's numbers. Once telephone lines are in place, corporations may rent them at discount rates from their owners and sell their services to individual customers, shifting controls from owners to service providers, without necessarily increasing the number of telephones or calls. Spam (unsolicited bulk e-mail) on the Internet is a parasite that is difficult to eradicate, given the open nature of the Internet. Its increase does reduce the benefit of e-mailing on the Internet. Parasites that kill themselves with the host technology are called *predators*.

In an ecology of artifacts, parasites and predators are not necessarily bad or unethical.

- **Dominant-competitive,** the ability of a design to piggyback on a dominant species in order to compete with another species. This is what the Microsoft Corporation was accused of doing by building its own Internet browser into its widely used Windows operating system, making it difficult for other browsers to be installed. It is unclear whether personal computers, the dominant species, are numerically affected by this measure. But there was sufficiently widespread suspicion that this competitive relationship constrains Internet use.

It should be reiterated that ecological meanings, while guiding particular users' choices, fundamentally concern the net effect of many users' actions on how species of artifacts interact, grow, or decline in numbers, and successively evolve. Ecological systems are complex, largely exceeding an individual participant's understanding. The ecological meanings that the stakeholders in technologies recognize are certainly predictive of how they act, but their actions often have entirely unanticipated consequences. Designers who can handle the ecological meanings of their proposals have a better chance of keeping their designs alive.

6.4 *Technological cooperatives*

Artifacts compete wherever alternatives are available. But most species of artifacts end up engaging in mutually cooperative relationships. This may have something to do with designers' preference for *coherence* and users' inclination to rely on *common* sense — the presumption of a shared rationality or techno-logic — as opposed to facing the uncertain perturbations of radically new alternatives. The result of acting on such preferences is the growth and increasing refinement of complex networks of artifacts: technological cooperatives.

Some artifacts grow to dominate technological developments; others become marginalized in that process. For example, a hundred years ago, the automobile set an ecological dynamic in motion that ended up transforming Western culture. Cars necessitated the creation of a network of roads, an industry that has become a major economic engine and numerous subsidiaries; the oil industry; gasoline distributors; repair shops; and roadside fast-food chains. The automobile caused the institution of traffic laws and agencies to enforce them, which are manifest in systems to license drivers, inspect vehicles, and tax citizens for the use of roads. It produced suburbs, transformed the architecture of cities, made oil an increasingly scarce commodity, led to wars, and transformed everyday life. Species are dominant in their ability to knit together technological cooperatives that nobody could have designed as a whole but in which everyone participates by enacting the ecological meanings of its individual pieces. Dominant species are likely

to be relatively stable in design and meaning. The automobile still has four or more wheels, an engine, one driver, and passengers, and serves as a means of transportation. It sidelined other means of transportation, horses, dog-drawn sleds, and bull carts. In the United States, even trains are badly affected.

A more recent technological cooperative is emerging around the electronic computer. The convenience of typing, sending messages, and accessing information more easily is of local importance. But this convenience drives the networking of computers into larger cooperatives, the Internet being a space that has attracted numerous species of artifacts and institutions that cooperate with and further develop this technology while competing for shares. Computers have entered virtually every home, business, industry, social organization, and governmental agency. They have aided the trading of stocks and bonds, selecting trial juries, scheduling air traffic, and creating entertainment, even regulating fuel injectors in automobile engines. Computers have changed the way people relate to each other as members of communities and the way government is performed. Some political practices have become less likely, dictatorships, for example, or central planning. Others need to be watched, such as the hidden monitoring of consumer or political behavior and the decline in the validity of available information on the Internet, for example.

Technological cooperatives are held together by collective human actions, often coordinated by social institutions. They have histories and develop over generations in time. They have no parallel in ecologies of biological species. They do not merely aid social life; they can also shift what it means to be human. Thus, the interaction patterns that sustain or drive technological cooperatives easily escape traditional design considerations. The ecological meanings of artifacts that designers have the option to encourage or omit can make a difference in how large technological cooperatives develop. Supporting, undermining, or gradually redirecting them by design is a possibility that the semantic turn offers.

6.5 Mythology

Notwithstanding common beliefs in technological determinism, which attributes autonomy to technological cooperatives and depicts humans as having to adapt to them, we need to be reminded that ecologies of artifacts result from human participation in those ecologies. The second law of thermodynamics, according to which all closed systems run eventually down into chaos, is not (yet) noticeable here because an ecology of artifacts is not closed. It is constantly attended to by design. Witness the exponential growth in the diversity of artifacts and the emergence of huge technological cooperatives discussed previously. What holds an ecology of artifacts together, what makes it grow beyond the ecological meanings of participating artifacts and their stakeholders is mythology.

Granted, all people have their own rationales for handling artifacts — replacing furniture at home, buying new cars, installing more powerful software on their computer, participating in Internet communities, and improving their practices of living. However, such local world constructions, their mediation by communication, are never sufficient to understand an ecology as a whole. It is not surprising, therefore, that the ecological arguments of successful designers, who are known for their keen sense of what fascinates others and where a culture is going, tend to be vague but celebratory of newness and connectedness in ways difficult to explain. Prophets of the emerging information society, for example, William Gibson (1984/1995) with his *Neuromancer,* fascinate by their poetry. His novel gave cyberspace its name if not a direction. Isaac Asimov's (1974) science fiction novels are easy to imagine, difficult to translate into working artifacts, but they touch something that readers would not mind experiencing. Reading the not particularly poetic Bill Gates (1996) on the future of technology, one wonders what he really says; yet his corporate decisions demonstrate that he does have his hands on the pulse of our time, much of the time.

Except for the focusing of a particular species of artifacts and describing it diachronically or synchronically, languaging on the ecological level taps into the very mythologies that enliven a culture and fuel its technology. A mythology is the largely unconscious narrative that underlies human thought, stories, and collaborative practices. A culture can hardly exist without its myths, without its metaphysics, and without its visions, even if only poorly articulated. In some cultures, mythologies are encoded in ritual practices. In others, they surface in the telling of stories of immense and untapped powers, or of supernatural beings who need to be pleased in order to set things right. During the West's industrial era, it was the conviction that technological progress would improve life for everyone, the Bauhaus' vision of marrying art with socialist ideas of mass production. In the postindustrial West, mythologies are hidden in narratives of personal successes, of the opportunities available to go where one wants to be, but also of the frustration of those deprived, their noble effort to set things right, and of the dangers of big institutions going too far. A contemporary myth is that of choice, the more the better, more programs for running a dishwasher, more TV channels to select from, and a bigger library of books at home. People pay for having these options without actually exhausting them. Fictions of future technologies — from George Orwell's *1984* and the television series *Star Trek*, to films like *Star Wars*, *The Terminator*, and *The Matrix* — are important media for articulating the myths that render human participation in the ecology of artifacts meaningful and exciting. They also fuel the imagination of designers. Mythologies give coherence to a culture, legitimize design efforts, assign ecological meanings to artifacts, and direct human individual and community involvement.

Designers always straddle between what can be said and what can only be felt or intuited yet needs to be attended to. By acknowledging ecological meanings beyond rational reach, the semantic turn recognizes mythology as

the ultimate driver of an ecology of artifacts. It is located in or should occupy designers' minds (Krippendorff, 1990:a21)* because:

Design strategies that go against the ecological wisdom of a culture are likely to fail.

* The concept of mind as a recursive process in which the human brain merely participates is due to Bateson (1972).

chapter seven

Design methods, research, and a science for design

The preceding chapters developed concepts that expand the design discourse of Section 1.4 along numerous dimensions. They enable designers to reestablish themselves as experts in the meanings of artifacts, which is an exciting empirical domain of considerable economic and cultural significance. This domain is one that no other profession has explicitly claimed for itself. In contemporary technology, the semantic turn has become so important that if designers do not stake out this territory for themselves, other professions will. Human-centeredness, its focus on the understanding that guides human interfaces with all kinds of technologies, including with other human beings, offers designers a way to reinvent themselves.

Although the semantic turn is new, similar concerns have always been part of design. For example, architecture has a long history of interest in appearances, monumentality, celebrations of rulers' might, and symbolic expressions of functions — churches, bank buildings, schools, military structures — signaling their expected use. But it dealt with these issues largely in terms of representations (e.g., Eco, 1980: Broadbent et al., 1980), somehow bypassing the humans involved. Consider that most books on architecture show buildings without people. Chapter 4 reframed what architects discuss as style, symbolic, or aesthetic in terms of the characters of artifacts, locating these phenomena not in the form of buildings but in how their stakeholders speak of them. From this new perspective, public spaces that afford multiple user perceptions — finding one's way in cities and on highways, identifying the purpose of buildings, memorializing victories or tragedies in the form of public monuments — are seen less as representative of reality than as providing a means of orienting users' attention and actions. Architectural knowledge about these issues was largely restricted to critics and historians. The semantic turn renders these phenomena available for systematic consideration and opens the way to a science for design.

A second example shows the semantic turn as redrawing boundaries among designers. Graphic design, for example, has always been concerned

with creating two-dimensional messages that inform, compel, or fascinate. It balanced its concerns between aesthetics and content — aesthetics as a generalization from what large numbers of viewers or readers consider pleasing (a character trait as described in Section 4.3) and content as a misleading metaphor — as if graphic material were a container of information. The new media make the work of graphic artists increasingly indistinguishable from that of product designers. Industrial designers in particular have made inroads into graphics by adding the interactivity and informative richness of interfaces to what graphic designers did before. The boundaries between these two design professions are dissolving, not by intention but by the recognition of common design concerns and knowledge of what people do with the artifacts they design. Two-dimensional artifacts must be read just as three-dimensional ones are. Theories of meanings, being concerned with people (see Chapters 3 to 6), do not distinguish artifacts by their dimensionality but by what people see they can do with them. The semantic turn implicitly dissolves these and other boundaries.

A third example suggests that the semantic turn expands traditional boundaries of design. Consider the design of museums, exhibitions, and stages, all of which are recognized design specialties. Often, such public situations are treated as communication problems and solved by graphic displays. They are not merely communications to the public; however, they are institutional events, multiuser systems that would not come to be without collaboration among stakeholders. In designing such situations, stakeholders cannot be considered individuals, much less typical users or consumers, but must be seen as active and intelligent members of organizations, speaking and acting on behalf of absent others, in the name of institutions or missions. Designing for public situations is not essentially different from designing an election campaign, launching a new line of fashions, developing a corporate strategy, or "reinventing government." The knowledge needed is essentially social, political, organizational, and economic, and the design may not promote a particular technology but create the conditions for certain technologies to arise. Projects (Sections 1.2 and 5.2) may be less controllable by any one designer, but can lead to far more consequential changes in the world. Thus, by focusing on the kind of knowledge that designers need to bring to a situation, rather than the traditional specializations, design can expand into new areas that badly need responsible attention.

The semantic turn that foregoing chapters have presented is just a sketch. It amounts to redrawing the boundaries of design, radically embracing human-centeredness, and fueling inquiries into various meanings, design methods, and a rhetoric that enables designers to collaborate with other professions on a more even footing. Much needs to be done. It is part of the discursive nature of design that the project of design can never be complete because it is embedded in culture's shifting conversations and technological developments.

7.1 A new science for design

Summarizing the previous section, we could also say that the semantic turn creates a space for explorations heretofore ignored or relatively unattended. The entry to this space is a new vocabulary that speaks and writes to the meanings of artifacts, interfaces, users' conceptual models, stakeholders, affordances, languaging, and the like; in other words, it provides a discourse for design. It allows for asking new questions of interest not merely to inquiring scientists but foremost to practicing designers. Inasmuch as these questions are not dealt with in other disciplines, of which more is said in Chapter 8, the semantic turn in design calls for a new science *for* design.

Recall Nigel Cross' distinction elaborated in Section 1.4.3 between:

- "*Science of design*, … that body of work which attempts to improve our understanding of design through 'scientific' (i.e., systematic and reliable) methods of investigation" (Cross, 2000:96).

To sharpen this definition, we need to add that these investigations are undertaken in terms of and from a position outside the design discourse. From this perspective, design is the object of research, undertaken by various academic disciplines, yielding knowledge *about* design.

- "*Design science*, … an explicitly organized, rational and wholly systematic approach to design; not just the utilization of scientific knowledge of artifacts, but design in some sense a scientific activity itself" (Cross, 2000:96).

The semantic turn, by contrast, leads to a

- **Science for design**, a systematic collection of accounts of successful design practices, design methods, and their lessons, however abstract, codified, or theorized, whose continuous rearticulation and evaluation within the design community amounts to a self-reflective reproduction of the design profession. It also includes ways of drawing on related knowledge bases in support of particular design decisions, collaborating with diverse stakeholders, doing project-specific research and, most important, providing methods for validating designs. Its aim is to keep design discourse viable and productive.

Unlike a science *of* design, which aims at accurate representations of its object — design practices, designers, institutions of design, aesthetic conventions, or history of particular designs — applying various scholarly perspectives, and using the natural sciences as a paradigm for theory construction, a science *for* design is proactive. It cannot be limited to theories of what exists, to patterns that were observed in the past, and to generalizations of the limits on what can be done. It must provide the intellectual tools needed to realize what did not exist before, to introduce desirable changes in the world, to project the technological, social, and cultural consequences of a design into the future, and, above all, to provide compelling justifications of

designs to those affected by and needed to bring about these futures, their stakeholders. Therefore, a science for design simply cannot follow in the footsteps of the natural sciences as so many new scientific disciplines try to do. It must pursue its own paradigm of inquiry and ways to generate practical knowledge. If designers do not construct a viable science for design, nobody else is likely to do it for them. Here are five outstanding features that a science for design needs to acknowledge and address.

First and foremost, *designers essentially are concerned with artifacts*, products, and practices that do not yet exist and could not come about naturally (Section 1.4.1). A science for design, therefore, cannot be built on or be limited to propositional knowledge, statements of observer-independent facts, or generalizations from previous observations, for example, the belief in the continuity of technological progress. Phenomena that are caused by nature cannot guide designers' attention. Whereas scientific theories are abstract, ideally mathematical and descriptive of repeatedly occurring phenomena, designers are concerned with variation, with what does not happen often and does not conform to theory. In fact, design is predicated to disrupt present stabilities, to challenge common assumptions, and to overcome seemingly insurmountable barriers. Design is inherently revolutionary or at least evolutionary and in any case unpredictable by natural laws. Design is a discipline that cannot be disciplined by what worked previously, by conventions or rules established by an authority. Design is fundamentally concerned with innovation, with making changes happen, and designers are especially challenged by common beliefs in what cannot be done.

An important occupation of a science for design, therefore, is to search for variability. Literally, variability means "able to vary," creating spaces of possibilities in which people can move, act, and live with technology and with each other, including languaging their worlds. Because a science for design searches for variability, for spaces in which actions affect futures, it also needs to inquire into the sources of resistance to change and create ways to remove or circumvent them if they prove to be barriers to desirable futures. A science for design certainly respects the re-search traditions of the natural sciences — repeatedly searching for generalizable patterns that existed in the past, theories based on experiments conducted yesterday — but it cannot treat these findings as anything other than providing the backdrop against which variability needs to be recognized.

Second, while variability is the condition for design to take place, *designers also need to have a sense of which futures constitute improvements* and which do not — for whom, within which time frame, and by which efforts. Inasmuch as designers rely on stakeholders to realize their proposals, support communities of users with their designs, and live with these others in worlds they help to bring about, their own visions of desirable futures must acknowledge the visions of others, those who come forward as stakeholders of designers' proposals, and those who are inadvertently affected by the consequences of their designs. Whether visions originate in someone's head or emerge in conversations, they certainly surface in language, not in observations. Hence,

a science for design must speak a language of possibilities — not just of the designers' — but foremost of those for whom a design is intended and those whose life is affected by it. A science for design, therefore, has to be a second-order science, one that enables designers to listen for the futures that others consider desirable, even when designers consider their futures limited or disagreeable. As such, a science for design has to be a social science, one that is open to technological considerations, to the multiplicity of understandings by users, actors, or commentators on technologies, and can cope with change. A science for design must continually prove itself in communication with those it claims to serve.

Third, given a more or less detailed understanding of stakeholders' understandings, there arises the issue of *how these understandings of understandings can inform design decisions*. Engineering has it easy. It is concerned with the functional aspect of technology, with mechanisms that do not understand the function they are intended to perform. Their criteria — functionality, economy, efficiency, and durability, to name just a few — are all defined without considerations of meaning. Methods in engineering therefore involve first-order understanding. Methods in design cannot be so limited. They have to be sensitive to how users understand and learn to handle a design, how the interfaces with them evolve, what stakeholders will say about it, which communities are sure to embrace it and which might not, how it affects the future of the industry, and how it alters the ecology of artifacts. This may appear to be a tall order for a science for design. Six proven human-centered design methods that respond to diverse stakeholder conceptions are presented further on. The first two of these were developed under the aegis of product semantics but turn out to have wider applications and are known by different names as well.

Fourth, the reputation of any discourse stands or falls on the acceptability of its claims. The natural sciences rely on observational evidence gathered in support of propositions, theories, and models of reality. Their primary criterion of validity is the correspondence between statements and observations: truth; including the correspondence between generalizations and the observations they entail: prediction. In engineering, predictions and constructions are coupled and the validity of its claims pertains to how proposed machinery is expected to function in a specified context. Ergonomics parallels this functionalist ideal by qualifying these either/or propositions with measures of the efficiency of human performance. Marketing supports its generalizations by using inferential statistics, the theory of large numbers, in particular.

Historically, presemantic industrial designers stood comparatively empty handed. Famous designers could rely on their reputation and on the eloquent use of a vocabulary that impressed with designers' aesthetic sensitivities, claims to know what THE user "really" wanted, and predictions of cultural trends. It also relied on the functionalism from engineering. As suggested earlier, this vocabulary has lost much of its rhetorical appeal, especially when facing other disciplines that have developed a more

compelling rhetoric and can provide empirical evidence for their assertions, for example, in quantitative terms. Today, substantiating designers' claims is all the more important because the risk of failures has considerably increased since industrial era manufacturing. Corporate decision makers are compelled by the most convincing arguments for a design.

Regrettably, to gain credibility in struggles over acceptance of their proposals, designers have been borrowing validity criteria and associated concepts from disciplines that have been more successful in arguing their cases: user expectations from market research (as if all stakeholders could be reduced to buyers), physiological efficiency measurements from ergonomics (as if the users' own criteria of usability would not matter), costs–benefit analyses from economics (as if there were a universal calculus for benefits), mechanical functionality from engineering (as if meanings would not matter), and so forth (see Chapter 8 for details). However, these widely popular criteria and concepts barely go to the heart of designers' work. Blindly adopting them in design discourse effectively surrenders the uniqueness of design to the aims of the dominant disciplines, forcing designers to play subordinate roles in joint technological developments.

Realizing that there are many paths to validity, an important component of a science for design, therefore, is *to provide ways for designers to substantiate the claims made for their designs*. Since human-centeredness moves second-order understanding, meanings, affordances, and interfaces with technology into the center of design concerns, designers must substantiate what are essentially semantic claims. Unlike in science and engineering, substantiation of semantic claims cannot bypass the stakeholders involved in a design.

Fifth and finally, to remain viable in the face of inevitable challenges, *a healthy design discourse must examine itself, correct its misconceptions, amplify design successes, and continuously expand its vocabulary.* Individuals, who speak as professional designers and thereby take advantage of their membership in a design discourse community, ultimately are responsible for maintaining the respect this discourse enjoys in public. It can be ruined by manifest incompetence and unethical behaviors, undermined by carelessly adopting the language of other disciplines, but it can also be enhanced by outstanding accomplishments.

In other words, a science for design, while investigating design from inside its boundary and supplying the design community with reliable concepts, methods, and knowledge, has to sustain the viability of its own discourse.

In academia, a distinction is drawn between a science and the philosophy of science. These are different areas that are attended to by different scholars. The latter investigate how scientists define their objects, reveal the epistemological assumptions underlying scientific work, and attempt to generalize a presumed common logic of inquiry (e.g., Popper, 1959). Practicing scientists need not be familiar with the writings by philosophers of science, and the latter's influence on ongoing research is small. A science for design, having to be supportive of the work that designers do yet critically investigating it,

cannot afford to make this distinction. It is both, *a science of making and a philosophy of realizing artifacts with and for others*.

Recall Simon's (1969/2001) proposal for a science of the artificial. A science for design embraces his pioneering proposal with few but crucial exceptions. As a significant early contributor to computer science, he conceptualized the interface in engineering terms, not anticipating how contemporary human interfaces with computers have redirected, almost forced, the attention of design toward human-centeredness. His science of the artificial put problem solving into the center of his project. He conceptualized problems and their solutions from the perspective of computability, even in such domains as management and organizational design, using technical–rational criteria that can now be seen as void of second-order understanding. As Rittel and Webber (1984) showed, when multiple stakeholders are involved, defining a problem is part of its solution, which challenges the vocabulary of problem solving where multiple players are involved. The following proposal for a science of design starts differently, with an exploration of what can be done and draws on stakeholders to answer the question of which changes are desirable. A "second-order science of the artificial" was endorsed at the earlier-mentioned NSF workshop on the future of design (Krippendorff, 1997). The semantic turn has opened exciting opportunities for exploration. The following sections outline three key methods of the science for design, raises the issue of validation of semantic claims, and concludes with a meta-perspective on this science.

7.2 Methods for creating spaces of possible futures

Psychological theories of creativity celebrate imagination as its source. This may well be true, but it has been also been shown that imagination is guided by categories in language and the absence of various inhibitions: the fear of failure, of being ridiculed in public, of losing respect within one's community, and of retribution for offending an authority, client, or friend. These fears are social in nature. Incompetence, mockery, loss of face, and demise of cherished relationships, all have to do with people who matter. Inhibitions are also learned, often in early childhood, for example, in response to being punished for trying something unexpected and thereby not conforming to what everyone seems to agree everyone should be like. Children are not born with such inhibitions, and adults may have difficulties realizing them after they became omnipresent in their lives. It then becomes difficult to be receptive to new ideas, to reconceptualize the known world, and to see the very possibilities on which design depends. There are several ways to encourage the creation of spaces for design.

7.2.1 Brainstorming

Brainstorming and numerous versions of this method aim to provide a social environment that is conducive to generating radical ideas. They capitalize

on the ideal of conversation in which participants respect each other for what they say, topics are allowed to emerge unpredictably and seriously considered within the conversation. People may be inhibited, but their capability of engaging in an open conversation is nearly universal. A project for developing teaching tools (McDowel, 1999) recommends the following rules for conducting group brainstorming sessions:

1. In a small or large group select a leader and a recorder (they may be the same person).
2. Define the problem or idea to be brainstormed. Make sure everyone is clear on the topic being explored.
3. Set up the rules for the session. They should include
   - Letting the leader have control.
   - Allowing everyone to contribute.
   - Ensuring that no one will insult, demean, or evaluate another participant or his/her response.
   - Stating that no answer is wrong.
   - Recording each answer unless it is a repeat.
   - Setting a time limit and stopping when that time is up.
4. Start the brainstorming. Have the leader select members of the group to share their answers. The recorder should write down all responses, if possible, so everyone can see them. Make sure not to evaluate or criticize any answers until done brainstorming.
5. Once you have finished brainstorming, go through the results and begin evaluating the responses. Some initial qualities to look for when examining the responses include
   - Looking for any answers that are repeated or similar.
   - Grouping like concepts together.
   - Eliminating responses that definitely do not fit.
   - Now that you have narrowed your list down some, discuss the remaining responses as a group.

What these rules do not spell out is the desirable composition of the brainstorming group. Supervisory relationships within the group often restrict the open expression of ideas. Sexism, from posturing to denigration of the other gender, tends to get in the way of successful brainstorming sessions. The perception that individual members need to score points or deserve credit for their ideas is a barrier as well. The best outcome of a brainstorming session is a number of previously inconceivable ideas or approaches for which the stimulating process deserves credit, not particular participants.

7.2.2 Reframing

Reframing is a method similar to brainstorming, but it focuses on multiple paths to understanding an often-intractable situation. Reframing asks

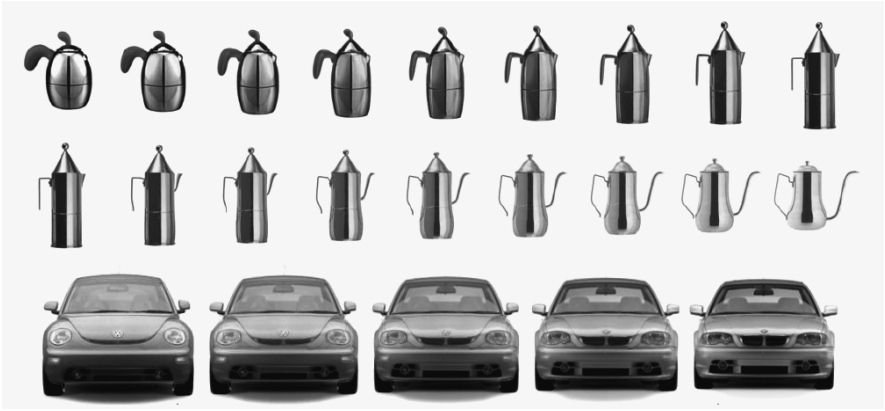


Figure 7.1 Morphing: generating the forms between known characters.

designers to try out one or more of the following cognitive devices until something is found that is worth pursuing further.

- **Transforming the known**:
  - Modifying its dimensions to their limits (like shrinking an artifact to its interface, see the data glove in Figure 1.3, or systematically distorting it up to where it can no longer be recognized as such and evaluating what lies within these limits).
  - Morphing, finding all possible gradually intermediate forms between two or more known ones. Figure 7.1 shows the results of using computer software (Chen and Liang, 2001). Morphing does not need to stay within the same category as in this image, however. Think of all possible means of personal transportation between a unicycle and a multiwheel vehicle. It reveals different categories of vehicles, their recognizable boundaries, a rickshaw, for example, but also many not yet realized artifacts.
  - Turning it inside out (like at the Pompidou Center in Paris).
  - Unifying (as in organizing disparate elements under a common perspective or control) or decentralizing its organization (as in moving from centrally controlled to parallel computing).
  - Removing as many of its features as possible (going to its essence) or adding as many features as can be thought of (like a car with many wheels).
  - Making visible what it covered (like the Accutron watch) or hiding what it showed.
  - Putting it into the background when it is prominent or into the foreground when it is hidden.

  - Modularizing it into freely combinable components or fusing multiple technologies into one whole multipurpose unity.
  - Turning a continuous flow into manageable steps (as in digital photography) or vice versa.
  - Realizing what is hardware in the form of software or vice versa.
  - Reconceptualizing the direction of causality of a phenomenon.
  - Replacing its materials by yet untried ones.
  - Placing it in new contexts to find new uses, etc.
- **Using alternative metaphors**: Metaphors change perceptions; talking differently about a problem or possibility, and collectively elaborating the metaphorical entailments can offer new insights. In physics, light can be conceived as a wave and a particle; both are metaphors proper. Most phenomena are describable in terms of several metaphors. Each has different entailments, some of which may be productive.
- **Finding analogies of a given situation**: Objects are similar when they have something in common. They are analogous when they agree in certain relations among their components. The minimum analogy is described by "A relates to B as C relates to D." The power of analogy is that it makes knowledge of the relationship, say, between A and B, available for conceptualizing the relationship between C and D. It invites replacements of one technology by another, and directs technology transfers from known to new artifacts
- **Employing various theoretical perspectives**, especially taken from different disciplines: For example, an interface may be seen from the perspective of psychology, cognitive science, mathematics, ergonomics, information processing, communication theory, or graphic arts; each offers different vocabularies, concepts, and explanations, jointly offering new possibilities and revealing the constraints of each comparatively.
- **Eliciting various stakeholders' conceptual frameworks**: Besides users, there are financiers, engineers, marketing experts, distributors, salespersons, cultural critics, repair shop owners, environmental action groups that distinguish themselves by having different stakes in a development and bringing different conceptual frameworks into play. There are also visionaries, political strategists, artists with open minds, and children with refreshing naiveté. Attentively listening to different voices without dismissing differences out of hand is a great source of variety.
- **Translating a problem or design into different media for conceptualization**: Traditional designers tend to express themselves by using sketches, three-dimensional models, and photographs, which are biased in favor of static appearances. Narratives, diagrams, scenarios, and computer simulations can emphasize processes but may be short on specificity. Mathematical equations impose a rigor that may not be warranted but have the advantage of the absence of physicality. Computer aided design (CAD) informs visually but at the expense of all other senses, touch and

> kinesthetic senses being the most important omissions. Creative mathematicians are often capable of translating one formal system into another, going from set theoretical propositions to those in matrix algebra, to those in probability theory, to those pertaining to color maps. All media of representation enable recognizing some underlying structures while rendering others intractable (Simon, 1969/2001:85–110). By translating a problem, a possibility, or a design from one medium to another, the designer is usually led to new insights, a richer understanding.

Reframing makes visible, solvable, or understandable what any one medium of representation hides. All devices suggested above amount to journeys that enlarge the conceptual spaces of designers. Jointly, they provide an understanding richer than either method alone. Their skillful use forces us to not get stuck or prematurely settle on one idea at the expense of better ones — a cognitive trap in which students often find themselves.

7.2.3 Combinatorics

Combinatorics, by dictionary definition, is a branch of mathematics that studies finite collections of compositions, of objects that satisfy certain criteria. One of its branches is concerned with the number of possible compositions in those collections — the creation of a space of possibilities. Figure 7.2 lists all possible binary decision trees among seven alternatives.* Another is concerned with deciding whether certain "optimal" compositions exist. A third evaluates which of the possible compositions — combinations of objects, parts, features, or functions — have certain properties, for example, of being combinable or able to work together, and in the case of artifacts, of forming viable systems. The approach presented here stems largely from engineering, but it is of interest to designers' investigations of suitable combinations of technologies.

The design of a new car may serve as an example of how combinatorics may be used. A car can be regarded as a part of the transportation system and in turn as composed of subsystems like engine, wheels, brakes, transmission, steering wheel, electric system, and passenger cabin. The larger system of which the car is a part is taken for granted. Its conceptual decomposition reflects what a designer considers important. Engineers who use this method might start by listing known kinds of engines, wheel systems, and brakes, perhaps in the form of a cross-tabulation, which would then enable them to explore systematically all possible combinations in search of viable compositions. The list of alternative subsystems does not need to be constrained by what exists or has been tried. For example, a car need not have two pairs of wheels. It could have any number above three. And steering does not need to be limited to two front wheels; it may be distributed over all wheels, allowing the car to turn around in one place and move sideways, for example, into parking spaces.

* Enumerated by Robert M. Dickau of the Math Forum @ Drexel University, Philadelphia. http://mathforum.org/advanced/robertd/catalan.html and http://www-history.mcs.st-and.ac.uk/~history/Miscellaneous/CatalanNumbers/cattree7.gif as of May 13, 2005.

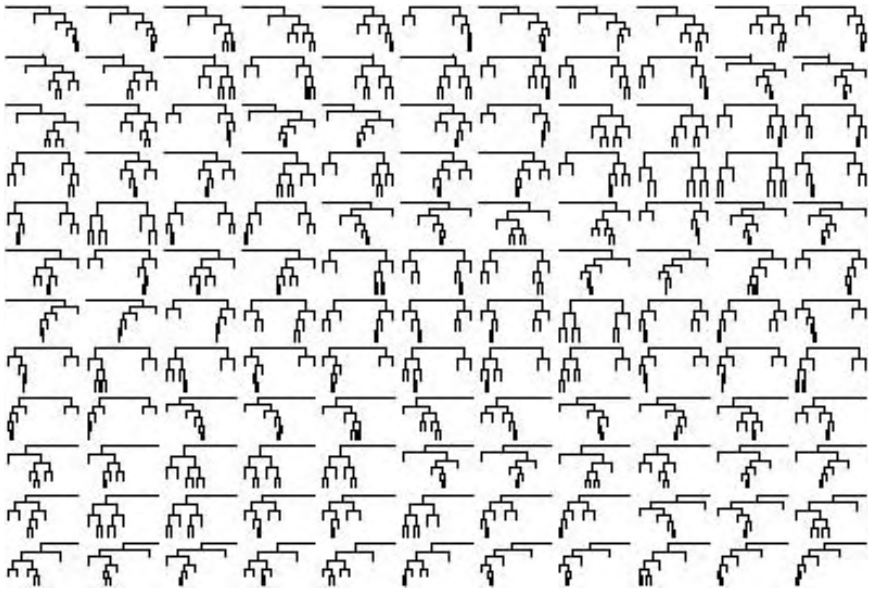


Figure 7.2 Combinatorics: possible ways of deciding among seven alternatives.

Combinatorics encourages designers to generate combinatorial possibilities that may not have been envisioned without that method and evaluate them one by one. Sometimes the space is too big to be fully explored, in which case designers may have to be content with alternatives that, following Simon (1969/2001:25–49), are satisfactory, not optimal. The key to combinatorial innovations is the resolution of possible incompatibilities between the subsystems to be combined into a coherent "collaboration."

Genrich Altshuller (Altshuller et al., 1997), the founder of TRIZ, a Russian acronym for Theory of Inventive Problem Solving, came to his theory after analyzing thousands of patents worldwide. He distinguished five levels of innovation:

Level 1: Simple improvements of artifacts already known by the engineer. They amount to altering some dimensions of what is known and works today.

Level 2: Innovations that combine technical knowledge of different aspects of an artifact, and most importantly, resolve a technical contradiction. A technical contradiction exists when the effort to improve one characteristic or parameter causes the deterioration of one or more others. For example, increasing the power of a gasoline engine makes the engine heavier and causes gasoline consumption to rise. Technical conflicts are resolved by compromise.

Level 3: Inventions that combine knowledges from different areas in which an artifact is to operate, resolving so-called physical contradictions. The latter exist when a part or function must be both present and absent. Physical contradictions cannot be resolved by compromise. For example, the landing gear of an airplane is both needed and not needed. One resolution is to separate the two conflicting states: extending the gear during takeoff and landing and retracting it during flight. A pocketknife resolves the requirement of having the blade available when needed for cutting and protecting the user from it when it's not in use by making the blade fold into the body of the knife, physically separating the two states.

Level 1, according to Altshuller, does not amount to a technical innovation although it is a common aim of design. Levels 2 and 3 present resolutions of contradictions that did not exist before, are innovations by definition, and add to the space of possibilities that a designer many wish to explore further.

Level 4: Innovations that implement a breakthrough technology require knowledges from different fields of science. This level also includes mere improvements but does so by establishing analogies between an existing technology and a new one. For example, switching from using vacuum tubes in radio receivers to transistors did not change the structure of the receiver but allowed it to become smaller and more reliable. Or switching from typewriters to computer word processing did not eliminate the use of written messages, but eased editing and linking with networks of communication and printers.

Level 5: Invention of a structurally new phenomenon that pushes an existing technology to a higher level. For example, following ENIAC, the first working computer at the University of Pennsylvania, which had to be programmed by hand, von Neumann's proposal to store computer programs as data in the very computer memory to which they were to apply was the key to the distinction between software, hardware, and data and amounted to a revolution of computer technology.

Altshuller and collaborators developed 40 principles for technological development that designers may wish to consult in order to solve technical problems and expand the space of possible artifacts. Combinatorics underlies most of them. Here are the most relevant ones, although not stated in TRIZ' terms:

- Try making existing artifacts lighter, smaller, faster, more efficient, cheaper, or more usable.
- Expand the functionality of existing artifacts by letting them embrace correlated functions. Affixing an eraser to the end of a pencil is a simple example. Adding redialing, conference calling, recording, and forwarding options to the telephone is another, all of which tend to be used concurrent with, in preparation for, or subsequent to telephoning.
- Fuse different technologies with sharable features into new artifacts. A handheld device that combines a telephone, notepad, address book, voice recorder, e-mail, personal organizer, and digital camera into one artifact demonstrates this principle at work. These fused technologies share the need for electronic memories, small amounts of computation, and similar user-friendly interfaces.
- Allow the structure of an artifact to adapt to a diversity of situations rather than making it relevant to only one. Software that recognizes the recurring practices of computer users and makes interfaces easier as time progresses (e.g., by learning to recognize users' voices, minimizing searching, clicking, and showing rarely used options) demonstrates adaptability.
- Convert a repetitive activity into a continuous process. Instead of cycling through the four stages (intake, compression, combustion, and exhaust in two revolutions) of a piston engine for taking in cool and emitting hot gases in rapid alternation, reconceptualizing these steps as a continuous process yields a turbine.
- Automate what users do not like to do, cannot do reliably or efficiently, and hide irrelevant features behind relevant ones. Interfaces with personal computers exemplify the use of this principle.

TRIZ's "law of ideality" states that all technical artifacts, in the course of their development, tend to become more reliable, simple, effective, inexpensive, smaller, in other words, more ideal. This law may well be recognized in the trajectory of artificiality (Section 1.2). TRIZ researcher Lev Shulyak (undated) asks: "What happens when a system reaches ideality?" and answers "The mechanism disappears, while the function is performed." … "The art of inventing is the ability to remove barriers to ideality in order to qualitatively improve technical systems." In semantic terms, the evolution of artifacts toward ideality is a move toward systems that are increasingly languagelike, become so natural that they can be taken for granted, are used by their many users, and in each others' view. They become closer to their ideal-type. It should be recognized, however, that ideality is not a property of technology but of the relationship with the community of its users.

Altshuller's examination of patents yielded another interesting statistic. The vast majority of inventions, he observed, consisted of solutions to problems solved elsewhere, in an empirical domain outside that of the inventor. These so-called inventions merely solved the problem posed by inaccessible

information. Project research, designers' search for ideas in design magazines, typically imports spaces of alternatives into a design process from elsewhere. Inventions on Levels 3 and 4 overcome combinatorial incompatibilities. They do not emerge from systematic research.

To sum up, combinatorics is a method for exploring compositions, combinations of features, functions, and systems, which could result in new artifacts. In the hands of engineers, it concerns technological developments, not human–machine interfaces. In fact, it has made no room for the voices of individual users or stakeholder communities to be heard and enter the process. TRIZ's law of ideality is telling. It assumes universal consensus on what is good and desirable, which the premise of the semantic turn cannot support entirely. Moreover, TRIZ's hierarchical concept of technology still preserves a conservatism that designers need to overcome. In TRIZ's terms, "problems" always arise in the larger system and are solved by addressing the components of the problematic parts. Problem solving grows out of the kind of rationality described in Section 1.2 as technical — when one starts with the concept of car and varies its known components, one always ends up with car, albeit a new one, but a car nevertheless. Yet, many design issues start with seeing possibilities, inventing a desirable future, or reframing what everyone else considers a problem into an attractive project, not by solving a problem as stated.

Combinatorics is a powerful tool for generating possibilities. Its limitation to first-order understanding, to solving technological problems of artifacts, is probably not so much a structural fault of the method but related to its mathematical origin, which is attractive to engineers. Combinatorics, reframing, and brainstorming may all be used to create spaces in which stakeholders in a technology can be heard. How to listen to the voices of interested others is addressed in the next section.

7.3 Methods for inquiring into stakeholders' concepts and motivations

As stated above, designers can succeed only when their proposals are understandable to those who matter, when the stakeholders in their design are inspired by the futures it promises, when the design attracts the requisite resources, and garners the social support needed to succeed in the life cycle of artifacts. Designers can benefit from several levels of this second-order understanding.

- Understanding stakeholders' desirable futures that they believe technology could bring forth and those they fear or would rather avoid.
- Understanding the possible concepts and meanings with which users approach artifacts of the kind to be designed, what they are accustomed to doing and confident in being able to handle.

- Understanding how interfaces with the proposed artifacts might evolve, what users are eager or resistant to learn, to create, and to embed into the practices of their community.

7.3.1 Narratives of ideal futures

Humans are very much at home in the language they are accustomed to speaking and consequently use the concepts that they can talk of easily. Newness is mostly born in language, in new ways of narrating the world, particularly of future worlds — not just among designers, but equally among those on whom the realization and use of a design ultimately rests. Probably the most important source of alternative futures is myth and fiction, which are limited only by human imaginability. The dream of humans being able to fly comes from Greek mythology. The word "robot" comes from a play, first performed in Prague, in 1920 but based on the sixteenth century myth of the humanoid golem. The humanoid golem was feared, but intelligent machines were faked for hundreds of years before Isaac Asimov developed his idea of robots. It has influenced artificial intelligence research and development and, above all, it made this academic enterprise acceptable. Voice recognition and communication by computers is just part of it. Although modern robots in industry, in airplanes, and in the stock market do not look like those described in Asimov's novels, they increasingly do things that humans do not like to do. George Orwell, in his science fiction novel *1984*, a year long since past, described a possible future that has scared some and taught others. Now, "Newspeak" is practiced in government and politics. It also made a difference in social legislation. Entertaining movies feature cars that can fly, people that can be beamed from one place to another, laser guns and light swords have given rise to the Star Wars defense system in the United States. In his futuristic novel *Neuromancer*, Gibson (1995) coined the word cyberspace and gave it meanings that are fueling the development of global information infrastructures. The "information superhighway" has become a political issue.

All of these ideas were far ahead of their times, occurred in the words of novelists and poets, and entered the imagination of their readers, including that of futurists, engineers, and politicians. It created futures that were at least imaginable, more or less realistic or possible, but it also enabled people to ponder their desirability. People who can see themselves living in these futures can generate the enthusiasm that fuels the work of many and make these futures real — starting with using artifacts that lead in this direction.

Analyzing science fiction and prophetic descriptions of technological developments can be an important source for understanding possible futures — not because poets imagine them but because their works are known to introduce vocabularies that inspire many more people that read these poets, eventually reaching those willing to enact their visions and to change the world for the better.

7.3.2 Surveys and structured interviews

Surveys and structured interviews probably are the least informative methods for gaining insights into stakeholder concepts. Because surveys tend to be applied to large numbers of interviewees — potential buyers, users, or voters — who are drawn from even larger populations, for such survey data to be analyzable, questions and their possible answers must be standardized, i.e., same for everyone. Under these conditions, the questions that the interviewees are answering always are the questions that the questionnaire designer — the market researcher, pollster, or their clients — wants to see answered, not what the interviewees have in mind and want to talk about. Surveys always reify the researchers' concepts. All that the interviewees' are granted to influence are the frequencies that fit in the researcher's categories. Save for these frequencies, surveys cannot provide anything new or unexpected. Nevertheless, when the frequency distributions in the researchers' terms are a design issue, surveys may help. Surely, if only a few find a particular product attractive, designers may need to worry, but such a finding may not answer the designers' question concerning where they need to carry a design.

Besides severely restricting the voices of interviewees, surveys have other biases. Interviewees tend to answer questions not according to what they know or believe but in terms of what they expect the interviewer wants to hear. Interviewees generally avoid the embarrassment of admitting not understanding a question or not knowing the answer but give one anyhow. When interviewees do indeed give a well-reflected answer to a question that does not fall into the predefined categories, their response is recorded as "NA," "not applicable" or "not answered." Designers who are confronted with survey results need to know how to interpret them or radically redesign them to generate the needed diversity.

7.3.3 Unstructured interviews

Unstructured interviews seek to avoid the conceptual imposition by standardized interviewers, which is typical of survey research. Here, interviewers engage interviewees in a focused but otherwise natural conversation. Interviewers may follow a rough outline of the kind of questions they like to see answered and try to steer the conversation in their direction, but remain open to unexpected answers. Such interviews take more time, prevent interviewing very many people, and the unstructured richness of the process creates data that are not easily comparable across interviewees, and not easily summarized either. Taped conversations and interviewers' notes taken during a session have been subjected to content analysis (Krippendorff, 2004a).

But what are the questions that can lead to a space of the available conceptual possibilities? If a designer seeks guidance for developing a new product, not surprisingly, the least informative questions concern user

preferences, how that product should look, what features it should have, and why people would use it. Answers to such questions usually reproduce what people know already exists, what they have seen advertised or used by others. The frequencies of such answers may be of interest to market researchers but do not reveal anything new. Creative answers are rare. Questions concerning existing practices are stymied by the common phenomenon that what makes people rely on an artifact, what works well, tends to be taken for granted, not attended to, and rarely verbalized. Think of bicycling, which would be difficult to describe, or character recognition, which is done routinely, reliably, but without knowing how it works. Surgeons have no difficulty speaking about what their instruments are for but may not be able to say how they handle them during an operation, which is precisely what a designer may want to know.

The questions that turn out to be most informative are questions concerning the problems that people have with particular artifacts. Disruptions from reliable use is what people become aware of, remember, and talk easily about with others, whether complaining, requesting advice, or giving reasons for abandoning a practice. For designers, stories about what got people into trouble are stories about how users' conceptual models do not fit the technological artifacts they are applied to; they are stories of the discrepancy of users' meanings and designers' meanings.

It should also be noted that problems and solutions are conceptual pairs. When people speak of problems, they already anticipate a direction in which it might be solved. Talking about problems therefore signals users' willingness to change, to learn, or to let go. Thus, stories of problems with technology indicate to designers the range of artifacts that users would readily accept. A prerequisite for eliciting such stories is to have the interviewee's trust in not being blamed for revealing their failures. Assuring interviewees that they are being listened to, not judged, is consistent with the premise of human-centered design to materially support users' concepts, not force people to accommodate artifacts that are not designed with human interfaces in mind.

7.3.4 Focus groups

Focus groups are favorites in market research and when product developers are at a loss as to how their own and their competitors' products are perceived. Typically, a focus group consists of 8 to 10 stakeholders who are invited (and paid) to talk about a product, advertisement, practice, or problem. The discussion is unstructured — for reasons similar to those in favor of unstructured interviews — except for a moderator who leads the discussion and pursues an agenda. Focus group sessions typically take a few hours and are taped for subsequent content analysis (Krippendorff, 2004a) or carefully observed through a one-way mirror.

The advantage of focus groups over individual interviews lies in the social stimulation they supply quite naturally. In unstructured interviews,

all relevant ideas, observations, or complaints are registered and taken as data. But in focus groups, anything said can stimulate discussion that can move in unanticipated directions, invites elaborations, dismissals, and retractions, and thus generates data that have passed a social reality test. Again, stories of difficulties are often more instructive than demonstrations of knowledge. For example, in a focus group on fashion, members could relate many embarrassing experiences and, unexpectedly reveal to the researcher the existence of loosely organized advice networks, especially among women on what to wear and when.

Focus groups entail the danger of eliciting opinions that may not matter when actions are the issue, for instance, buying, using, or supporting something. Opinions are easy to assert but may not indicate what people would do, especially when actions require efforts. For these reasons, the more successful focus groups are given something to see, touch, or use: worked out scenarios, video presentations, competing products, or prototypes.

Since focus groups have to be small, a persistent problem is to assure that they fairly represent the community of interest. Focus group researchers prefer to invite so-called lead users, those who would adopt a technology before others do. But they are difficult to identify. Another problem is that the data that focus groups generate are largely unique to the composition of the group, easily influenced by the moderator, and therefore difficult to generalize to a community. The consistently best use of focus groups is to check for designers' misconceptions about users and unintended uses by different stakeholders.

7.3.5 Observational methods

While surveys, interviews, and focus groups operate largely in language, there is much virtue in going to the other extreme and observing what people routinely do before or after a new artifact enters their lives. Focusing a video camera on a workplace for which computer support was to be designed provided unexpected insights about how workers managed their work, the sequence of steps they followed, the scenarios of their work, and what happened when work flow was interrupted by telephone calls, missteps, tracking their own work, reporting to supervisors, maintaining good relations with colleagues, and taking time out for various personal matters. Designers unfamiliar with the environment that their artifacts are to enter are often surprised by how people juggle their artifacts, their obligations, and their coworkers.

To better analyze what is happening, observers may record events by a variety of methods: paper and pencil records, audio and video tapes, or collecting the by-products of the interactions of interest: URLs visited, hypertext links used, e-mail messages sent and received, operative notes generated during surgery, or insurance claim forms filed and processed, all of which are observational records of sorts.

Observational records, more or less systematically gathered, can be examined in considerable detail. By analyzing such equally unstructured data, which are typically limited to a small number of cases, designers may get considerable insight into unconscious habits, nonverbalizable routines, and conceptual models in action, particularly when disruptions are observed. They may give designers suggestions for needed improvements but can reveal little about peoples' conceptions and the possibilities they would accept.

7.3.6 Protocol analysis

In pursuit of a concept of human cognition called information processing, Newell and Simon (1972) pioneered and championed the use of verbal protocols for understanding human problem solving (further developed by Ericsson and Simon, 1984). Their intellectual efforts were paradigmatic for subsequent artificial intelligence work, especially in the design of expert systems, and left behind a method that has since been used to understand how users interface with their artifacts. Users are asked to "think aloud" while performing a task. The data thus obtained provide the analyst with records of the correlations between verbal accounts of what users see and think and analyst observed interactions, unmistakably demonstrating the users' operational meanings of what they do. By not merely observing the dynamics of human interfaces with given artifacts, but also recording how users say they think about their interactions, valuable specificity is added to the otherwise "meaning-dry" observational records on the one hand, and abstract verbal accounts obtained during interviews on the other.

One recognized limitation of asking people to talk about what they do while doing it is that method marries two usually separate modes of interaction. We often talk while doing something, for example, while driving a car or while jogging with a partner, but rarely about what we are doing at that moment. Expert performers, whether they are piano players, sailors, jugglers, or experienced computer users, are presumed to enact conceptual models of what they do, but it is well known that verbalizing these actions with the desirable level of detail may not always be easy. When such experts are asked to explain what they do while doing it, they tend to slow down what they do or can't even do it anymore. In pursuit of a logic of human problem solving or information processing, Simon and Newell had no reason to worry about this problem. Designers, however, may be interested in conceptual phenomena that are less governed by computational logic.

7.3.7 Ethnography

Ethnography is an eclectic observational technique that was developed by anthropologists. Literally, the systematic description of culture, ethnography employs a naturalistic perspective by relying on first-hand experiences of fieldworkers in naturally occurring settings, rather than in artificial or

experimental situations. The most important commitment of ethnographers is to account for people, their artifacts, their cultural practices, and their environments from the perspective of the people who inhabit these environments — providing second-order understanding in effect. Moreover, by including themselves in the ethnographic accounts they provide, the method aspires to be self-reflexive. Ethnography is holistic in analyzing situations with a range of embedded units of analysis, from micro actions in day-to-day human interactions with artifacts, to how values and worldviews are negotiated and enacted. Each unit is considered a constituent of a larger unit and provides the context for its constituent parts. Ethnographic fieldworkers typically spend quite lengthy periods at a study site. They may use any means of recording their observations and interactions with informants, including basic quantifications where they matter. Their relationship with the people they study is collaborative and relies heavily on the rapport that develops between them.

The principal virtue of ethnography is its ability to make visible the "social world" aspects of physical settings. It enables inquiries into many contexts and technologies; can disclose the scripts, plans, concepts, motivations, and myths that the informers bring to a situation; and offers descriptions of how a technology is integrated into everyday life and use. It has been most effective in relatively confined environments, such as public spaces, air control rooms, surgical operating rooms, ceremonies, and therapeutic interactions. In such settings, participants, typically few in number, have a clear focus of attention and cooperate in practices that complement each other; artifacts have meanings that are related to these practices, and what takes place is understood, not necessarily shared, as part of the culture that provides the context of what takes place.

Ethnographic inquiries focus either on current practices, which designers intend to change, or on practices that emerge after a design has been introduced. Although inquiries into aspirations and desirable futures are not excluded, predicting the use of new technologies within established cultural settings is difficult. The most important contribution that ethnography can make to design is to go beyond accounts of the existing cultural nexus and inquire into what people are willing to abandon, where openings for newness exist, and which resistances need to be overcome and how. Culture is a process of continuous learning, of inventing new artifacts that in turn change the conceptions of the world of which they have become a part.

*7.3.8 Triangulation of methods**

No single method can provide everything that designers may want to know about their stakeholders. Protocol analysis is a powerful method for accessing how user concepts unfold, step-by-step, into interfaces with artifacts, but interfaces usually evolve faster than people can explain themselves.

* See Krippendorff (1990).

Videotapes can visualize how technologies are integrated in a work environment, but are silent about the concepts that workers bring to bear on that integration. They often dupe their viewers into believing that they could see what those on the tape would see as well. Ethnography's second-order understanding is favored by many but requires considerable effort and presupposes academic training. The real challenge lies in letting different methods collaborate to answer the crucial design question: Where would people be willing to go if artifacts could be made available that could change their future?

For example, a study of midwifery practices would have benefited from a protocol analysis of the interactions between midwives and their clients. But "thinking aloud" would surely have compromised the very communication with clients on which much of the work of a midwife rests. Acknowledging this limitation, interactions were taped, and instead of using a researcher-driven content analysis of these tapes, the midwives were asked to narrate their tape, explaining to the analysts how their concerns shifted from moment to moment, what they noticed, why they said what they said, and what they hoped to accomplish. The tapes could be slowed down, sometimes surprising the midwives with what they had not noticed while absorbed by another focus. The data thus obtained could be analyzed as a protocol, as an unstructured interview, or by means of content analysis — in any case, making good use of different methods that overcame each other's weaknesses.

Another example is to turn the artificiality of creating verbal or behavioral protocols into a natural situation, for example, into a training session. The questions that real trainees are asking and the explanations that experts are giving until trainees have mastered a procedure provide an important window into users' conception. Again the uncertainties caused by disruptions, while to be avoided, prove to be the most valuable source of information of what designers needed to know about stakeholder conceptions. The analysis of such data combines protocol analysis with ethnography, and incorporates observational methods.

7.3.9 Stakeholder participation in the design process

One of the most striking limitations of the methods reviewed in the previous sections is their preoccupation with what is known about available interface practices, whether the researcher focuses on disruptions or problems, or elicits comments about competing products, models, or prototypes of new artifacts. Inquiries of this kind can point to problems and confirm or disconfirm expectations. However, they do not offer much guidance as to how to go from one practice to another, from handling an existing technology to handling a more advanced one, or from accustomed interfaces to new kinds. Yet, what would clearly be informative for designers would be to learn how users handle newness and how much energy they are willing to invest in learning something new. Learning to interface with a new technology means

stepping out of the confinement of existing conceptions. Merely talking about desirable futures is different from participating in bringing it about. While ethnographers pride themselves on being able to describe how a technology is acquired, they can do so only if this technology is available, at least to a few informers. This is a fundamental limitation of all naturalist inquiries that a science for design needs to acknowledge and overcome.

An attractive proposal is to involve the stakeholders in a technology in the process of its development, learning from their learning processes as the designers' work unfolds. Frequently asserted as highly desirable, stakeholder involvement is rarely practiced, however. There are several models that are billed as stakeholder involvement. Most fail in crucial respects.

- **Concurrent marketing research:** This technique can give designers feedback but only in the terms chosen by market researchers. It literally uses its interviewees without offering them a voice of their own (Section 7.3.2), and because it is the market researcher who defines the terms of interest, enthusiastic and knowledgeable stakeholders are reduced to mere respondents. Marketing research represents the wants of its clients, which rarely overlap with the desires of existing stakeholders.
- **In-house usability laboratories:** This method is an organizationally convenient way to address usability issues. Such laboratories follow the tradition of ergonomics and bring so-called usability experts into one department, which typically evaluates all of a company's products that have user implications. Such laboratories legitimize experts on users that they know mainly through generalizations from test results. The narrow focus on users tends to exclude other stakeholders outside a company's market interests and amounts to the development of blinders against recognizing deviant meanings.
- **Outside consultants:** Consultants have been hired to explore what stakeholders have to say about critical design decisions. The use of focus groups, observational techniques, or ethnographic field methods is one way of preventing the above-mentioned blinders to determine corporate policies. It has been noted that consultants, not being part of a design team, may well bring new perspectives to bear on a problem but they have only a partial understanding of the products in development. Also, being located elsewhere and called upon when needed, consultants are not as easily accessible as in-house researchers are (Buttler and Ehrlich, 1994).
- **User-aided design group**. In contrast to these three traditional approaches, Apple tried another approach when they developed their notebook computer: a user-aided design group.

  Apple's idea was to bring on board a small group of consultants who had real life experiences with the kinds of products that Apple was developing. They could work together for longer periods and learn to use the new technology as it emerged. The group was composed of

individuals with multidisciplinary skills, with sufficient levels of education, experience in testing, ability to observe and to participate in studies of interfaces and any materials, such as instructional resources, that link Apple to its stakeholders (Gomoll and Wong, 1994).

In a development project for Philips, Eindhoven, the Exploratory Design Laboratory (EDL) at then RichardsonSmith expanded stakeholder involvement in the design process by including software engineers, marketing researchers, CEOs, and subject matter experts, who came closest to users. EDL prepared questions and organized workshops during which design decisions were made jointly. The most rewarding response to this cooperation came from a participant who, after the project was almost completed, wondered why EDL was hired when those involved in the workshops were in fact making all the decisions (Section 7.4.5). Stakeholder involvement is intended to weaken the distinction between single-minded designers or cohesive design teams, and those affected by a new design, to shape technology according to what it means to the communities affected by it, and to assure moving it through its life cycle.

7.4 Human-centered design methods

The methods sketched below share three features.

- **They are design methods**, which entails that they systematically expand spaces of possibilities and then contract them to arguable proposals for artifacts that promise to bring forth desirable futures or prevent undesirable ones from occurring.
- **They are concerned with how stakeholders attribute *meanings.*** Because meanings are acquired in use, not designed, *decisions on meanings ultimately cannot be taken away from those who are affected by a design*, its stakeholders. Human-centered design methods, therefore, weave available knowledge of how meanings arise within relevant stakeholder communities into the design process in order to assure that a design *encourages the meanings that lead to reliable interfaces* while discouraging those that cause disruptions, disappointments, breakdowns, and harm.
- **They render design proposals *empirically testable,*** at least in principle. Because a projected future cannot yet be observed, they *provide arguments, demonstrations, if not tests for the projected reality of a design*. The succession of generating ideas and then evaluating them is characteristic of design generally and build into the steps taken by any particular design method reviewed below.

Regarding meanings, designers need to realize that they cannot possibly "build" meanings into artifacts or "force" others to see the meanings their way. Meaning is not a matter of "adding content" to an artifact so that the users can understand what was intended by the designer. By the same token, meanings cannot be "conveyed" either. Similarly, it would also be a mistake

to consider meanings as subjective and the physicality of artifacts as objective. This attitude privileges one's own reality and dismisses that of others — as if designers could have access to what is real, whereas their stakeholders had only inferior perceptions (see Section 2.5). As discussed in Section 2.3, meanings arise in use and direct the unfolding of one's interfaces with an artifact. What designers can do is to limit the usability of an artifact (Section 3.4.2) and prevent certain meanings from being created and enacted. Usability may be restricted to protect users against locking themselves out of their cars, prevent children from opening a medicine bottle, and make it impossible for electronic devices to be plugged into the wrong receptacles. Computer interfaces are full of useful restrictions and warnings. A hammer, by contrast, can be used in many ways, to build something and to harm somebody. Where the usability of an artifact cannot be constrained physically, designers will have to rely on the meanings available within a community of stakeholders. There are three general principles concerning design semantics:

First, the meanings that a design invokes within a stakeholder community must be consistent with what the designed artifact actually affords doing. A design should not semantically encourage more than it can deliver. Promising less is preferable to promising too much.

Second, just as in interpersonal communication, advertising messages, brands, public spaces, and systems of traffic signs, artifacts should be designed to not merely encourage meanings that further stakeholders' objectives (their desirable futures), but simultaneously to discourage interpretations that, when acted on, could lead stakeholders into difficulties, wrong directions, and ultimately into harm's way. In design, semantics must not only direct users' attention but also discourage paths that would get users into trouble or are not desirable.

Third, learning is being human. Indeed, meanings shift in use, continuously one may add. User concepts never are as stable as mechanisms are. It would follow that, unlike engineering, design semantics cannot limit its concern to meanings that are presently invoked and stereotypically enacted within stakeholder communities. It must leave open sufficiently large spaces for new interpretations to emerge, for novel interfaces to surface, and for different conceptual dynamics to unfold — within the limits of what can be foreseen as desirable.

The following sections sketch five proven methods among many others that still need to be formulated and tried.

- (Re)designing the characters of artifacts
- Designing artifacts that are informative (expressive) of their workings
- Designing original artifacts, guided by narratives and metaphors
- Designing design strategies
- Dialogical ways to design

7.4.1 (Re)designing the characters of artifacts

(Re)designing the characters of artifacts (Section 4.3) is a method pioneered at The Ohio State University. Its principal proponent is Reinhart Butter (1989, 1990). The character of an artifact consists of a set of traits, each corresponding to an adjective or adjective pair that attributes a quality to artifacts or describes a feeling invoked by them. It can but does not need to make use of Osgood et al.'s (1957) semantic differential scales representing the character traits that stakeholders recognize. The method

- Is applicable to a conceptually unproblematic kind of artifact or concept of something already known, at least by name (automobiles, truck cabin interiors, wristwatches, perfume containers, brand images, but also politicians — all forming a particular category).
- Requires an initial definition of the characters to be achieved and those to be avoided. Characters may range from aesthetic qualities and the summoning of feelings to group identities. These are given in terms of two sets of adjectives, usually supplied by a client, sometimes developed in cooperation with that client. As a rule, the desirable characters are easy to state. For example, one may want a car to look powerful, a wine glass elegant, the wearer of an evening dress outstanding, a gift personal, and a medical instrument professional. The characters to be avoided are those that users find ugly, discouraging, or repulsive, but also those of the products of a competitor against which a client's is to be distinguished, and those triggering the objections of powerful political, ethnic, or ideological groups. The motivation to redesign characters may also stem from finding that an existing product has been given attributes that are detrimental in some way, and the task then becomes one of moving the character away from it. Fashion calls for such redesigns.
- Requires the designation of the communities of users, stakeholders, or critics whose attributions matter. Characters do not exist in the abstract. They result from the language habits of particular communities of people that need to be identified to substantiate a design. A designer cannot be the only judge of a design's character traits.

The method involves five reasonably general steps*:

1. **Elaborate or expand** on each of the contrasting sets of initially given verbal attributes to include synonyms or adjectives with similar meanings, ideally representing all the verbal characterizations that stakeholders might make within the empirical domain of the artifact to be redesigned. To obtain an expanded set of attributes, designers may consult

* Butter (1989) has described the design of characters in eight steps. His are smaller than the ones described here but cover essentially the same ground, beginning with specifications of desired character traits, and ending with an assessment of the compatibility of the manifestations in a design, here step 4.

a thesaurus, use focus groups, ask linguists or experts on the subject, interview competent practitioners, or content analyze relevant literature, product descriptions, reviews, consumer reports, or critiques.

2. **Analyze and group** the two expanded sets of attributes, ordering them by how familiar they are within the intended community of users (discarding unfamiliar or idiosyncratic ones), clustering them according to their similarities and differences (highlighting central ones and downplaying peripheral ones), and extracting underlying dimensions (using multidimensional scaling, MDS-techniques, or factor analyses [Osgood and Suci, 1969], for example). Always preserve the contrast between the attributes that are desirable by the stakeholders and intended by the designer, and those to be avoided (using a discriminant analysis, for example), to exclude nondistinct and ambiguous attributes. When in doubt, test the clarity of the distinction with competent users of the technology. The surviving attributes guide step 3 and provide criteria for step 5.
3. **Generate sensory manifestations** of the chosen character traits. Character traits are physically manifest in features of existing artifacts or objects of nature that suggest a certain attribute to be their appropriate characterization. In this step, all conceivable manifestations of each of the desirable character traits are collected. For example, if the appearance of fastness is the issue, one would search for ways that the adjective pair "fast" and "slow" could be visualized. If gender were the issue, one would sample shapes that indicate "femininity" as opposed to "masculinity," and if durability is wanted, one may collect photographic images that demonstrate "sturdiness" as contrasted with "fragility." An important source of such manifestations is stakeholders of various kinds who could be asked to associate objects, images, or examples of how the attributes in question are manifest. Competing products, factory samples, clippings from magazines, sketches, and drawings are all appropriate as manifestations and are, in fact, widely used. The manifestations of the desirable character traits have to be clear (i.e., sufficiently distinct from the manifestations of the undesirable character traits), and ambiguous ones (i.e., being located in the middle of a scale between desirable and undesirable character traits) discarded. Surrounding oneself with pertinent images of these manifestations, so-called mood boards, is common. However, the purpose of this method is not to get designers into a creative mood but to provide a large collection of manifestations that, when used appropriately, can assure that a design invokes the intended meanings and avoids undesirable ones.
4. **Reconcile incompatibilities** in these manifestations. The challenge and creative part of the method is to explore which manifestations are combinable and can inform the structure of the design. Some manifestations are independent of each other, like surface appearances (e.g., cool colors) and smells (e.g., leather seats in a car). They tend to pose no serious

design problems. Others concern arrangements (e.g., the informative differentiation of multiple controls) or styles (e.g., high tech, art nouveau, punk, Japanese), both of which require designers to make compromises in favor of the unity of the arrangements (e.g., using similar shapes, fitting push buttons into a grid). When one manifestation precludes the use of another and both represent desirable characteristics, incompatibilities appear (Section 7.2.3). Designers may then go through the list of alternative manifestations of the desirable character traits and search for compatible ones and synthesize them into a new unity that evokes the character in question. Incompatible manifestations are typical and challenge the designer's creativity.

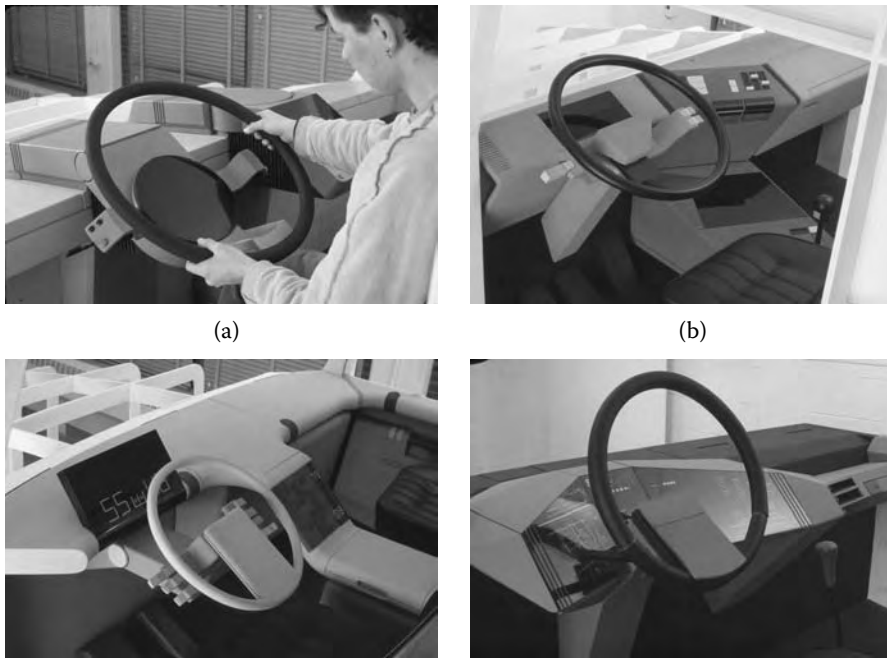
Resulting from this step is a representation, model, or prototype that is sufficiently realistic to serve in a test of the degree to which the result approximates the desirable character traits and avoids the undesirable ones for the stakeholder communities of interest.

5. **Test** the design that has emerged from step 4, one character trait at a time, against the attributes developed in step 2. The character of the proposed design should deviate as little as possible from the desired character and be as far apart as possible from the undesired character. Suitable evidence may be obtained in various ways: (1) Asking stakeholders to characterize a design in their own terms, and then content analyze their responses in terms of the attributes in step 2, or (2) perhaps less desirable but more easily analyzable, using semantic differential scales, one or few for each character trait, and have subjects rate the design in terms of its polar adjective pairs.

Step 1 clarifies the terms in which the design is to be perceived. Step 2 roots these terms in a community of stakeholders. Step 3 combinatorially expands the range of options beyond an unaided designer's imagination but is guided by how stakeholders conceptualize the character traits at issue. Step 4 involves what designers traditionally do to synthesize superficially disparate elements into a new unity. Step 5 brings the stakeholders back into the process to test the extent to which the design meets the expectations set out in step 1.

This method has been successfully employed in a variety of situations. Using it as a guide is not a straightjacket; on the contrary, it provides for a systematic expansion of designers' imagination, creativity, and sensitivity, and then reduces the space of possibilities it creates to a realizable proposal. Its human centeredness is assured by consulting stakeholders; potential users, including linguists, experts, critics; and other interested parties, finally assuring that a design's character is identifiable. It must be reiterated that the character of a design is manifest in the use of language, which reports on perceptions and feelings that a design ultimately aims to assure.

A good example is the design of the interior of a truck cabin at The Ohio State University (Butter, 1989). Figure 7.3 depicts four results. The manufacturer, Freightliner, provided the specifications of a 40-ton truck cabin. Butter's students built a full-scale model of the cabin's interior space for entering their designs. In agreement with the manufacturer, each of five teams pursued a different character: high tech, low tech, contemporary, functional, and futuristic. Steps 1 and 2 involved generating adjectives that could be considered the traits of the desired characters. For simplicity, the traits to be avoided were those taken by the other four teams, the aim being that the five characters to be achieved should be distinct, not to be confused across teams.



(a) (b)

(c) (d)

Figure 7.3 Contrasting characters of truck cabin interiors. (a) High-tech; (b) functional; (c) contemporary; (d) futuristic.

For the *high-tech* cab interior the following character traits emerged from step 2:

Electronic
Modular
Interactive
Quiet
Exposed

The manifestations found for these traits in step 3 were too numerous to be listed here but fell into these categories:

Transparent technologies
Building-block architectures
Adjustable components
Rectilinear geometries
Ergonomic color schemes

The results of integrating compatible manifestations into one testable proposal, step 4, may be seen in Figure 7.3a. This truck cabin resembles a mobile electronic office and the control center of a transportation system of which the driver is a part.

The character traits for the *functional* cabin that resulted from step 2 were:

Simple
Strong
Industrial
Hygienic
Responsive

And the manifestations of these character traits that resulted from step 3, again many in number, were summarized in these terms:

State-of-the-art technology
Hidden sophistication
Water repellent geometry
Spacious dimensions
Dirt-disguising colors

The result of step 4 can be seen in Figure 7.3b. It reveals the technical-purposeful nature of the truck and highlights simplicity and ruggedness without playing on the electronic sophistication of the high-tech design. The photograph does not easily convey the spaciousness experienced when inside the cabin. Interestingly, during step 4, an unexpected character emerged — also in the *low-tech* character, not shown here. The manifestations included many from the military: ruggedness, monotone (colors), absence of concern for comfort, and mechanical controls. To prevent this from being confused with the character functional, the designers added these undesirable adjectives to list of character traits to be avoided and ended up with a rather unintimidating, familiar, operationally simple, but also exciting design.

The character traits of the *futuristic* cab interior were analyzed to be:

Sophisticated
Precise
Organized
Alert
Serious

Generating manifestations, such as:

Space-age engineering
Cockpit layouts
Office type hardware
Crystalline shapes
Material paint scheme featuring sparse colors

Reconciling these manifestations in step 4 produced a startling sculpture with unusual qualities, clean and simple interface surfaces, completely surrounding both driver and passenger, as shown in Figure 7.3d.

Although none of these proposals was formally tested, asking individuals who were available to describe these interiors resulted in lists of attributes that could then be compared with the attributes of the intended character. The futuristic design matched the intended character most closely (Butter, 1989).

Figure 7.4 shows a multidimensional analysis of the manifestations of seven character traits, here of cars (Chen and Liang, 2001:533). The adjectives used by subjects were factor analyzed to extract the underlying dimensions of the attributions. This technique was used to obtain measures of the semantic differences between existing cars, but it could also be used in step 2 to find a simplified set of character traits, or to measure how a proposed design deviates from those on the market.

Two additional examples may demonstrate the diversity that this method encourages. Figure 3.10 realized desirable characters of elevator interiors by means of metaphors. The aim of the project was to ease the psychological problems some people experience when confined in small spaces.

Figure 7.5 demonstrates the use of graphics to alter a character. Here, Butter's students at The Ohio State University had designed water boards, 2- to 3-foot long floating devices for exercise and fun. One version was used by students of Hans-Ruedi Buob at the Schule für Gestaltung, St. Gallen, Switzerland, who explored semantic principles. The image demonstrates how the character *lightweight*, shown in Figure 7.5a in the form of a feather, became successively more abstract in Figure 7.5b and c. Figure 7.5d shows the version chosen with one of a different character. Figure 7.5e illustrates the systematic

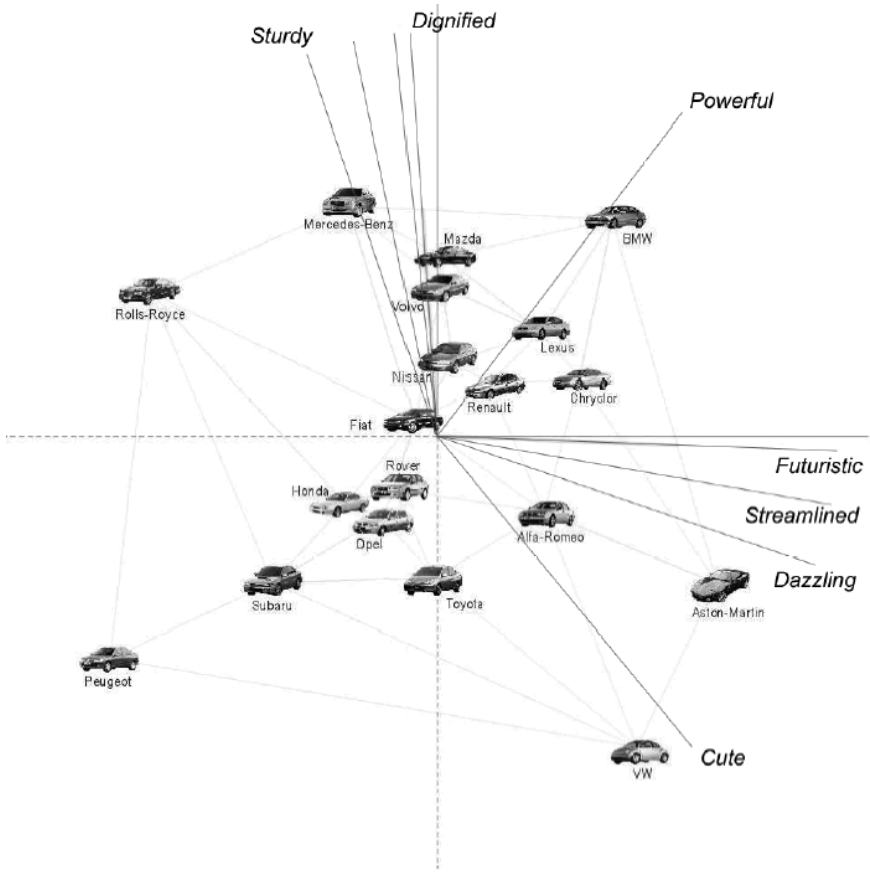


Figure 7.4 Multidimensional character space for cars.

explorations of different manifestations, and Figure 7.5f shows nine semantic differential scales, the last of which measures *light* versus *heavy*. Accordingly, lightweightedness is arguably clear. It is also correlated with *young*. The other traits are more in the center of the scales, more ambiguous, thus satisfying the assignment. Abstract representations, like the version finally selected, are not necessarily clearer than detailed ones. In the end, tests with the appropriate stakeholder communities are decisive, especially when unexpected.

The method of (re)designing characters embraces a Japanese system for evaluating designs, called "Kansei Engineering." It is concerned with the characters of artifacts as well. It uses a slightly different but partially intertranslatable terminology, makes different epistemological claims, omits a crucial step, the reconciliation step 4, but also relies on adjectives, including Osgood et al.'s (1957) semantic differential scales. The latter enables quantitative analyses, which *kansei* engineers take seriously. In Japanese, the word "*kansei*" covers several English meanings such as sensitivity, sense, sensibility, feeling, aesthetics, emotion, affection, and intuition — all of which apparently are conceived in Japanese as mental responses to external stimuli, often



Figure 7.5 Exploring alternative characters by graphical means.

summarily referred to as feelings. In Japanese, *kansei* contrasts with "*chisei*," knowledge of external facts (Lee and Stappers, 2003). Kansei researchers assume that all humans have their own *kansei* in response to external stimulus conditions.

According to Mitsuo Nagamachi (1995, 1956), the originator of Kansei Engineering, the method involves four steps. It starts with collecting appropriate *kansei* words from the users of the technology in question — or a particular class of adjectives. Second, correlations between product features and *kansei* words are established — or manifestations of the attributes are found. Third, a data bank of these correlations is searched for appropriate *kansei* words, using semantic differential scales to represent them, and analyzed, typically applying factor analysis to reduce their often large numbers to a manageable set. This expertlike computer system is the heart of *kansei* engineering. Fourth, potential users judge the new design in terms of *kansei* words to establish how close it is to the ideal, which is exactly step 5 above.

Accordingly, Kansei Engineering assumes a very definite relationship between product features and feelings. If these features could be controlled, which is what designers are expected to do, and if the feelings of users could be predicted from the features they face, then feelings could be engineered to improve sales, usability, and users' satisfaction. However, *kansei* engineering cannot predict feelings. *Kansei* words cover far too many, diverse, and mostly socially constructed phenomena. By starting with collecting words and ending with rating the words that subjects consider appropriate for describing a particular design, user interfaces, and meanings, Kansei Engineering keeps both feet in language. It may predict the *use* of *kansei* words, which undoubtedly is mediated by user concepts, expectations, feelings, and, perhaps more importantly, governed by prevailing linguistic conventions. But feelings remain merely implicit in this use of language and this method of evaluation.

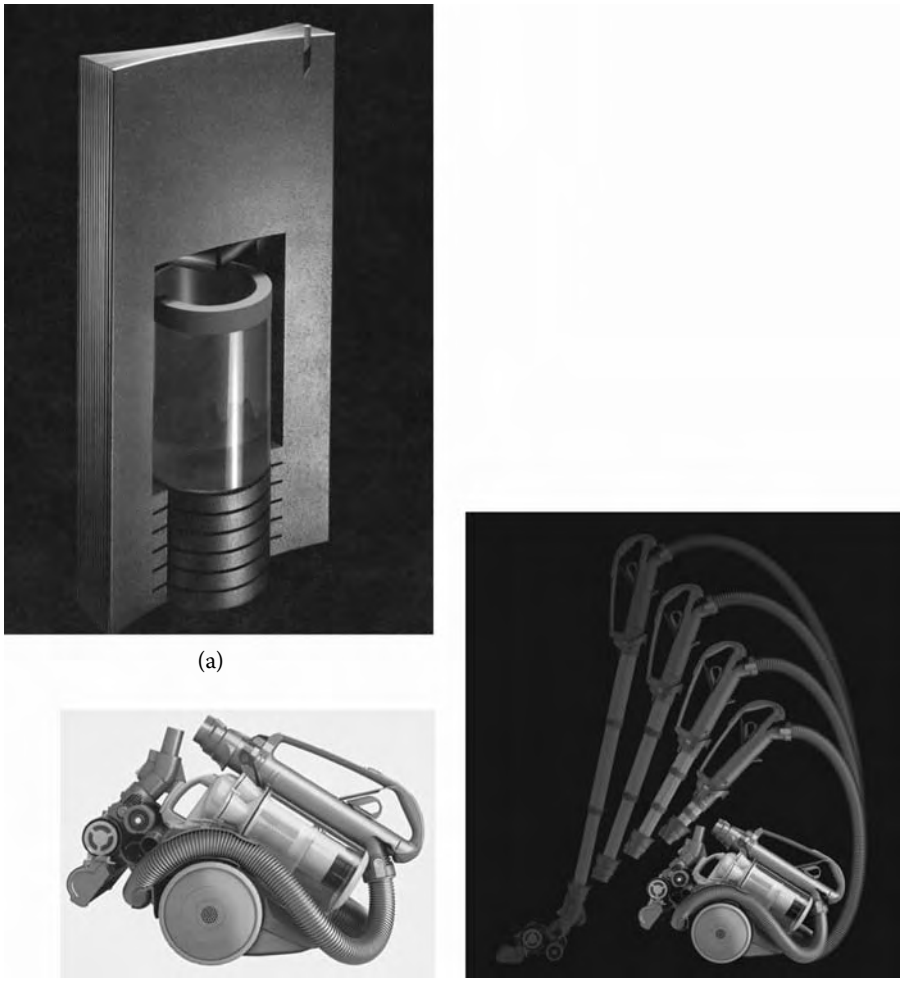
As already suggested with Figure 4.1, artifacts enter the domain of language and communication among stakeholders, just as interfaces, concepts, or feelings do. It is in language that artifacts become characterized for what stakeholders believe they are, using multiple adjectives and other linguistic constructions (Section 4.3). Adjectives attribute character traits to features of a design but only if they are perceived, experienced, felt, and understood according to a community of stakeholders. Thus, the method for (re) designing characters, including Kansei Engineering, cannot escape the cultural conventions of a language community for the appropriate use of these attributes. The qualities of feelings, therefore, are not universal, and certainly not literally caused by particular forms. Although the word "*kansei*" may be clear in Japan, it is not easily mapped into concepts common in Indo-European languages. Human-centeredness entails the respect for cultural relativity.

7.4.2 *Designing artifacts that are informative (expressive) of their workings*

This design method seeks to highlight the understanding of what an artifact does, its composition and function, ideally making its workings self-evident. The method is as follows.

- Is applicable to a category of artifacts whose technology is relatively unproblematic, essentially solved from the perspective of engineering, like air humidifiers, espresso machines, personal computers, telephones, but
  - Their technology may not be widely understood as such by their users.
  - New features have been added to a known artifact, and it is desirable to distinguish it from similar artifacts.
  - Current artifacts of this class have become so mindlessly unexpressive that a celebration of their functional composition could become a

source of visual delight. For example, current office equipment, fax machines, copiers, printers, and shredders have become so similar in shape and color that they hide their differences. A redesign may aim to distinguish among them according to what they do and how.



(a) (b) (c)

Figure 7.6 Designs that are expressive of their workings. (a) Room dehumidifier by Paul Montgomery, 1987; (b) and (c) cannister vacuum cleaner by Dyson.

- There must be some virtue in foregrounding the design — aesthetic, branding, use, user identity — as opposed to blending it into the background.
- Requires some knowledge of the communities of users, stakeholders, or critics who are expected to make sense of how the artifact works or experience pleasure in seeing its technology revealed.

Although Peter Stathis and I have formulated, taught, and applied these steps, the principal proponent of what it intends to produce is Michael McCoy, who taught at the Cranbrook Academy of Art for some time.* McCoy sought to overcome the meaninglessness of universal design, which had the effect of putting different artifacts in similarly colored boxes. He did not want to return to the functionalist celebration of technology, as in the 1960 Accutron wristwatch, which showed its novel technology through its face (and made the reading of time extraordinarily difficult), or by exposing the ordinarily hidden infrastructure of an artifact, as in the Pompidou Center in Paris. McCoy sought a middle-way between reducing an artifact to its interface and finding aesthetically pleasing forms, by demanding that products express what they do in the context of its use. One example from his Cranbrook period is Lisa Krohn's phonebook in Figure 3.8. Its pages can be turned like a book, changing the functions of the buttons on each turn, making it a dial phone, a phone book, a message recorder and replayer, a record keeper, and more. Its components reveal its usability.

The method involves five steps as well.

1. **Decompose** the proposed working of *a design into networks of conceptual components* whose relations are believed helpful in understanding the artifact as a whole. A wiring diagram is an example of how a complex function can be decomposed into a network of component functions. Even in electrical engineering, there are usually several ways of decomposing the same function into conveniently conceptualizable parts. Designers are encouraged to explore as many decompositions as practical before settling on particular components. Some networks may result in components with purely technical definitions that users may not (need to, or cannot) understand, which are to be avoided. Some may result in so many components that users could become confused by or scared of the complexity they would be facing. The point is to find networks whose components are separately understandable and allow the arrangement to tell the story of how they work together.

   We can only imagine the networks of components that the designers of the Dyson vacuum cleaner in Figure 7.6b,c may have considered. Classical vacuum cleaners connect replaceable suction attachments to hoses to containers with filters to exhaust fans. They have handles to control the unit, on-off switches, and electric cords. These components are obvious; their names appear in a dictionary, but there are different ways of combining them to form a whole. This makes for upright, canister, handheld, robotic, central, and industrial vacuum cleaners. Dyson used an industrial technology — separating dirt from the air by cyclone action rather than filter bags — which encouraged special kinds of networks. For another example, chairs have a longer history and their

* Aldersey-Williams et al. (1990) present a retrospective of work done at the Cranbrook Academy of Art. Michael McCoy's influence on the 3-D department is evident in this volume of outstanding examples.

component parts are better known than those of computers. Although the number of parts of chairs is pitiful by comparison with those of computers, there are far more decompositions of chairs than of computers — but this may all change. The point of this step is to get to as many networks of components as seem worthwhile to explore. There is a rule of thumb, not to be taken too seriously, that suggests a cognitive limitation: on any one level of decomposition, the number of component parts should not exceed seven plus or minus two (Miller, 1956).

2. **Generate** lists of *ways that each component could be realized and rendered familiar*. While decomposition is an analytical task, generating lists of familiar realizations expands available options. Exploring alternative technologies is an important source of variability. Designers may also scan the world of the users' artifacts for clues of what makes sense to them, explore the availability of user conceptual models (UCMs) for each, or search for metaphors that could shed light on what each component does. A component that is too hot to be touched may be shown with protective shields. One that contains fluid may be conceived of as a cylindrical container. The housing of a set of batteries may express the power that the battery provides. This is not to say that every part of a device needs to be highlighted. Not all components are equally important, attractive, or meaningful. A component with a technology that is difficult to understand or repulsive may be packed in one that is neither. Dyson's vacuum cleaner used an uncommon technology, a cyclone system, that the company wanted to emphasize as its distinctive feature. The point of this step is to have a large number of conceptually comprehensible components available.
3. **Explore** how *the design could fit into the users' environment of other artifacts*. This environment may contain features that support a design — a surface to stand on; other devices it needs to work with, or should not come in contact with; ways of keeping it when not in use; semantic layers; or what it is to replace, an older technology, for example. Here, one could treat the relevant features of the environment of a design as other components, similar to those considered in step 2, except that they are taken as given, not as the target of design. The importance of step 3 is to find the contexts of an artifact's use and rely on that knowledge in order to limit the number of realizations envisioned in step 2. Paul Montgomery's celebrated humidifier (Aldersey-Williams et al., 1990:94) in Figure 7.6a was meant to be freestanding, a sculpture in an elegant home. Dyson's vacuum cleaner had to be able to be rolled and stored in a small place.
4. **Reconcile the incompatibilities** arising from arranging the components — envisioned in step 2 according to the conceptual networks developed in step 1, and satisfying the criteria in step 3 — into a *coherent composition*. The first and most significant problem is to evaluate the technical realizability of the up until now merely conceptual decompositions. Fluids cannot be screwed to sounds, and sitting surfaces need something to

hold them up — notwithstanding the ideal of floating in space. To preserve the understandability of the components, one may have to create transitional shapes between them or find an overriding metaphor that suggests coherent interpretations for a majority of the components. A transitional component is exemplified by the use of bellows to connect a height-adjustable seat of a chair to its stand. One example of the use of an overriding metaphor is Krohn's phonebook in Figure 3.8a. Another is Montgomery's dehumidifier in Figure 7.6a. The parts of the latter consist of air intake, a container for collecting the extracted water, the cooling machinery, and the on-off switch. The dominant shape suggests that humid air is condensed as it enters the device from its sides and reduced to water at a point where a container resides for receiving it. Its shape metaphorically suggests what a dehumidifier does. It does not matter that dehumidification in fact works quite differently. Yet the idea of squeezing water out of the air is understandable (and would not get the user into trouble). The designer of Dyson's canister vacuum cleaner went into another direction, exposing all the parts that its users need to handle and understand, and then packing them into the smallest space possible. Here each part has to tell what it is for and find its place when not in use. Note how the flexible tube straps around the wheels, and the handle telescopically extends as seen in Figure 7.8c.

Step 4 is, of course, where creativity is central and designers can excel. The method described here merely provides a way to explore many compositions before settling on one that hopefully is most clearly expressive of the working of an artifact. Figure 7.7 depicts one of Butter's student's efforts to reconcile alternative ways to operate a drinking fountain for exclusively cold water.

5. **Test** whether and how easy (natural) it is for users (a) to distinguish the design's components, for example, by naming them or by identifying what they do or how one could handle them; (b) to figure out from the arrangement what the whole is meant to do; (c) to find places in their environment where it finds its place, where it belongs; and (d) whether the design compares favorably with existing artifacts. Such tests may rely on observations of how users actually handle the artifact without instruction, or on narratives obtained by asking users to explain what does what, when, and where. The test criterion is straightforward: the artifact should afford the actions resulting from any user's "reading," while expressing the essence of what it does.

Again, steps 2 and 3 of this method encourage the creation of a space of possibilities beyond the unaided imagination of designers, and step 4 calls for reducing this variety to something new and better. When testing prototypes or models in step 5, stakeholders are indispensable. But they can also participate in step 2, generating alternative ways of realizing understandable components, and in step 3, exploring the environments in which that design maybe put to use.

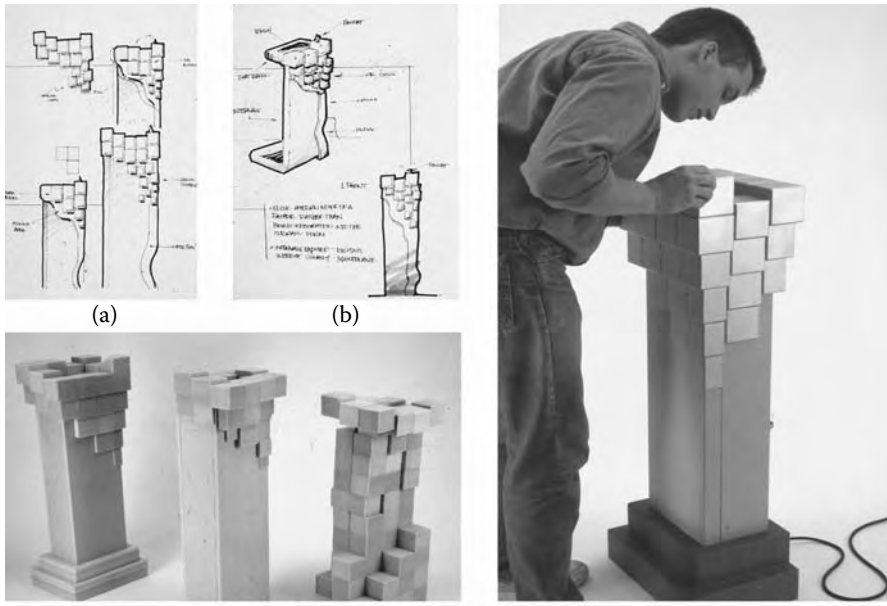


Figure 7.7 Exploring ways to synthesize manifestations of coldness for a water fountain.

Stating this method is not to claim that the previously mentioned designers followed it, or that McCoy subscribes to its formulation. It turns out to be a useful pedagogical device and a guide to creating artifacts that express something about what they do in an environment of the users' choice. It is safe to say that designers with such aims go through at least some of these steps intuitively.

7.4.3 Designing original artifacts, guided by narratives and metaphors

Some designers like to think of fresh ideas as the source of innovation and the drivers of technological development in general. This may well be an individual designer's experience, but ideas, intentions, mental imagery, and visions cannot inspire anyone else unless they are made clear, sketched, narrated, or demonstrated in various forms. Without articulating their "fresh ideas," whatever they are, designers could not possibly work in teams, consult experts, convince clients, and enroll stakeholders. Moreover, most ideas occur to people in conversation, during brainstorming sessions, for example. Avoiding mentalist constructions, Chapter 4 made clear the central role of language in making artifacts meaningful. Taking this to heart, we have to recognize that a good deal of design is accomplished by narratives, in conversations, that is in language, broadly conceived to include visual

aids, presentations, and performances. Design proceeds from elusive narratives of desired futures and metaphorical redescriptions to specific proposals for realizing artifacts in support of these futures. So conceived, design begins with one kind of narrative and ends in another, a proposal, for example, which is a narrative that is capable of realizing and accompanying a design through its life cycle with the help of others.

The method suggested here takes the role of language in design seriously. It encourages awareness of verbal metaphors (Section 4.5) and narratives (Section 4.6), draws on the enormous creativity that language has to offer, and squarely embeds design in these forms of languaging. This method is indebted to the late John Rheinfrank, who led design teams by injecting into the deliberations rather vague concepts that had the ability, however, to provoke everyone onto a different plane of languaging. He worked with this author on several projects at the already mentioned EDL, developing new interfaces for multiuser systems, formulating corporate design strategies, creating educational opportunities, and narrating systems of novel products into being. Rheinfrank was influenced by Chistopher Alexander et al.'s (1977) *Pattern Language*. It suggested a path from examples in a proto-language to buildings and towns. The steps outlined further on are not Rheinfrank's formulations, but the practices they describe clearly are. They have proven successful in a variety of situations. Three virtues may account for it.

- **Implicit understanding**. The method's human-centeredness is grounded in the near universal fact that everything said is said in the expectation of being understood. It is difficult to tell a story without anyone listening, knowing that those present attend to other things or are incapable of understanding the narrator's language, just as it is difficult to hear someone speaking and not making any sense of it. Talking about artifacts is subject to the same considerations. Artifacts that occur in stories are always narrated to be understood. Understanding becomes evident when listeners are able to rearticulate what they heard or do something with it, including answering pertinent questions about it. Thus, the very act of narrating artifacts implicates second-order understanding. Coherent talk is a social phenomenon, and narrating artifacts is what renders them *meaningful* to those involved.
- **Variability and creativity**. Rearticulations of narratives rarely produce exact copies. This is lamentable only to technological determinists who would compare the retelling of stories with what tape recorders do. In rearticulating stories, people cannot avoid adding their own perspectives and knowledge of the subject matter and weeding out elements that are meaningless to their listeners. Variability is expected, natural, and highly desirable because it makes a story more understandable, useful, and efficient. Thus rearticulation increases the narratability of a story, not necessarily its content. It follows that artifacts, when part of a story that is passed on from one person to another or regurgitated by someone,

naturally improve their narrative makeups toward greater understandability. This is the core of this narrative design method.

- **Coordination, not sharing**. Finally, implicit in the two previous points is the recognition that coherent talk is a social phenomenon, that artifacts are always culturally defined, and that design is at its best when done in communication with collaborators as well as with opponents. The method does not prescribe how artifacts are to be narrated but provides a handle for utilizing both the natural creativity that the collaborative use of language assures and the equally natural tendency of rearticulations to converge on proposals that become increasingly acceptable to those involved, sufficiently useful, and real.

Superficially, innovativeness and understandability may seem incompatible. Yet, even novel narratives are understandable as long as they rely on a sufficient number of reasonably well-known concepts. The same is true for novel artifacts. The involvement of different stakeholders in narrating a design into being adds another important feature to narrative theory: The understanding of narratives need not be shared across different stakeholder communities. The narratives of users may go only as far as they have to. The narratives of engineers too do not need to go beyond their technical expertise, and this is true for all other stakeholders. An artifact has the chance of becoming real when narratives fuel interactions within the network of stakeholders that can make it happen.

The method:

- Is applicable to the design of novel artifacts, systems, procedures, services, or strategies for which workable concepts or reliable exemplars do not yet exist.
- There are exciting but elusive narratives of desirable futures, compelling metaphors with little reality to back them up, concepts or artifacts in search of a reality, ideas considered fertile or intriguing, perhaps contained in the stories told by visionaries and dreamers.
- It requires some knowledge of the technological possibilities that could make these elusive stories more real, but more importantly, of the metaphorical capabilities and material resources available in the communities that are expected to become producers, users, stakeholders, beneficiaries, or critics of an anticipated design.

The five steps comprising the method may not necessarily happen in the order listed.

1. **Collect or generate relevant narratives and metaphors.** This step is to expand the designers' narrative horizons by getting to as many sources of conceptualizations as possible. A designer's clients are the most obvious start. They usually come to a designer with a story of what they envision. To expand on such articulations designers may want to become acquainted with the fictional literature around that vision

(Section 7.2.1) or surround themselves with creative writers. To fight terrorism, the U.S. military employed Hollywood writers to help the military invent scenarios they could not imagine on their own. In any culture, there are pockets of fertile but unrealized stories. In the West, science fiction has been the driver of new technology. But there is also literature on the latest technical innovations, prophesies of futurologists, and pronouncements by industry leaders — not to believe them, but to mine their metaphors and concepts. Talking to leaders in various technical disciplines that could frame the same phenomenon their own ways is another way to connect with alternative stories. A more conservative approach is to elicit the stories that people can tell about the opportunities and fears of the technology they already know. But, the point of this step is not to create an inventory of existing narratives concerning the technology of interest, but to find alternatives to what is known or unusual metaphors whose entailments could drive a design into uncharted directions.

2. **Analyze narratives into scenarios and the semantics of components.** One principle of understanding stories is narrative sequencing: Telling A before B implies that A happened before B did. Stories describe artifacts in an order in which they can be understood, which often coincides with the order in which they are approached and used. This principle justifies mapping stories into scenarios (Figure 3.16), which can eventually be translated into interfaces. Materially supporting narrated sequences almost guarantees understanding.

   Not a principle but typical of European languages is describing objects in terms of nouns and relations between them in terms of verb phrases. The noun structure of a narrative can give a clue to a comprehensible physical composition of the artifact in question — not how it is actually built but how it is being understood. Nouns are not simple, however. There are abstract nouns that refer to processes, such as "production" or "flow;" there are nouns for relationships, such as "interaction" or "interface;" and there are nouns for feelings, such as "exhaustion" or "excitement." What is needed here is a semantics for mapping narrative structures into understandable compositions of material artifacts. Language is full of folk schema that may be utilized. Anthropologists have studied them for years (e.g., Thompson, 1932). For example, there is the common actor–action–target–means–effect scheme for describing what people do with the help of technology, or "who, says what, to whom, with which effect" (Lasswell, 1960). There is the producer–product scheme, sometimes with and at other times without implied intention. There is the scheme according to which material things have properties (fixed attributes), dimensions (in which they exist but along which they may vary), and features (that they may

or may not possess). Relations can be conceptual as in similarities or being part of, which need to be noticed. They may be causal as when one component affects another, interactive when components affect each other, or transactive when the components are being changed in the process as well. There is no single semantics for all artifacts.

Even these not particularly poetic accounts are full of metaphors, for example, that material things possess properties — as if they could own anything, that artifacts have abilities — as if they had the choice of doing or not doing something. But the metaphors that are most instructive about how people understand technology are those that have complex conceptual entailments. For example, when attributing intentions to computers, users find it natural to become angry with them when they do not cooperate, or when saying that a software "mines" textual information, text analysts are lead to believe that the software is an actor and information is a scarce but intrinsic quality of text (see Sections 3.3.2 and 4.5).

3. **Explore available technical details.** Exploring technical details augments the scenarios and networks of components by suitably specialized narratives. What makes a mapping of narrative structures into material structures difficult, however, is another narrative principle: cognitive smoothing. A good story omits the details that listeners can be expected to supply as being obvious. Asserting that "a particular icon was clicked and a window opened" does not say anything about the underlying mechanism, but by the principle of narrative sequencing, causality is presumed. If this is part of how an artifact is narrated, the designer has the freedom to invent any mechanism provided it conforms with this narrative structure. Doing so would preserve the coherence of the narrative without having to reveal how the effect is implemented. Unlike narratives, design needs to solve the details of an artifact to the point that it works. This step needs to create a space to detail what is left unsaid, taken for granted, or is of no expressed concern but must nevertheless work for an artifact to enter its life cycle.

   Cars that also fly, time machines, devices that communicate across languages or beam people across space, and perpetual motion machines are the stuff of narratives. They reside there because narratives can easily cope with abstractions and gloss over details, and they are compelling as long as listeners do not care to get to the details. One should not therefore dismiss them as unreal. History has shown that myths can become real, like the ancient dream of humans to fly — not quite as envisioned in Greek myths, but flying nevertheless. Or the medieval dream of machines that can think and play chess — first presented as a magician's trick with a human hidden under a chess table, now a reality. Narrowing the gap between narratives and working artifacts consists of narrating mechanisms that provide the missing

details. Narratives that can do that are typically disciplined by specificity, physicality, and economy, and include practical steps that people can take to bring them about.

At this step, then, the designer can explore the many ways this gap could be bridged, searching among alternative materials, examining technologies being developed in unfamiliar empirical domains, and probing new theories. Since the designer is concerned with gaps in otherwise compelling narratives, there is less of a need to require that these details be comprehensible to all stakeholders involved. There is the possibility of relying on experts in particular gaps.

4. **Synthesize and realize.** This step explores ways of rearticulating the narratives of artifacts that the previous steps made available, putting them together into an ideally coherent working whole, including sufficient details to be realizable as far as their stakeholders are concerned. The previous step provides numerous narratives from different people, ranging from dreamers to experts in details, typically contradictory, not necessarily describing the same possibilities. Only some of these warrant attention; most others may have to be dismissed as sheer fantasies or be considered later. Dismissing narratives as unrealistic is too easy and should instead be carefully considered. Fantasies have a reason to be compelling, and it is often possible to find metaphors that make a seemingly unrealistic narrative come true. The dream of humans being able to fly was mentioned. A perpetual motion machine, which contradicts accepted laws of thermodynamics, may be physically impossible but reframing the task to design a machine that requires no costly energy may well yield a solution, relying on solar power or fluctuations in the earth's magnetic field, for example. Time travel, another physical impossibility, yet an attractive idea, may be realized not quite as in fictional narratives, but by enabling someone to travel through the recordings of her own past and relive and alter her experiences of it. Metaphors are powerful devices, used here to reframe the narratives of a vision, rearticulate a familiar but unworking story, and reconceptualize an unrealistic goal. Therapy is full of examples of patients living for years with a depressing story that could be retold or synthesized differently and become enabling. Overarching metaphors that reframe the narratives in which an artifact occurs can provide the synthesis needed here. If a grand metaphor cannot be found, the metaphorical entailments of its components should at least not contradict each other (not be the result of mixed metaphors).

   For syntheses to be realizable, artifacts must afford several simultaneously plausible narratives. If it is a material artifact, it must be possible to convincingly narrate its geometry, its physics, and its mechanical functioning. If a narrative is defective in this regard, one has to find the missing detail or abandon it. If it is to be used by someone, it must be possible to narrate how people can interface with it in ways

that are intrinsically and/or extrinsically motivating. If it is to be sold in a competitive market, it must be possible to narrate its ecological meanings and its benefits for those who decide on its fate. If it is a public monument, it must be possible to tell its story without offending critics and powerful interest groups. If it is an organizational practice, it must be possible to bring the needed participants on board, etc. Narratives that cannot convince the stakeholders of a design may need to be discarded, at least until other stakeholders, a different cultural climate, or new technology make them viable again.

5. **Test.** Test whether the narratives that relevant stakeholders offer when facing or interfacing with a model or prototype of the proposed design are indeed afforded by that design (promises what it can deliver, would not have undesirable consequences, or lead to disruptions of the interface). For such narratives, see Section 7.3.6 on protocol analysis, but extended here to other stakeholders. Tests may also include ecological considerations: what stakeholders would do with other artifacts that compete or cooperate with the design.

Two examples may illustrate this method. One is the development at EDL of a computer system to support the work of insurance companies. This assignment came in the form of a conveniently general narrative with ample room for elaboration, except for the requirement that the company's newly developed image processing technology had to be used. The design process began with collecting the narratives of those involved in insurance claim handling: stories of how case workers were expected to proceed, the inconveniences, difficulties, disruptions, and failures they experienced; interviews with supervisors; examinations of case files that contained the forms and documents that workers had to work with, including the log of actions taken and notes to follow up on these actions; videotapes of case workers in their environment doing their job; and their comments when they saw themselves on tape.

Analyzing these narratives into scenarios, the second step, was not too difficult because there were administrative rules that everyone had to follow: some legally required, others instituted by supervisors. But there were also many habitual practices that were attributed to experience and skillfulness. A distinction was made between regular procedures, which could be automated, and responses to situations that arose unexpectedly, derailing the routines for a while by requiring intelligent attention. Charting these, one could see redundancies and possible shortcuts.

The exploration of available technologies, step 3, included the technology that the client hoped to market but soon expanded to the big field of information technology. At the end of this step there was much consensus on what would be desirable, the space of possibilities, but not yet on how the system could be put together.

Step 4 was dominated by the search for a guiding metaphor that could pull together all the practices that case workers had to manage but now in

a new information environment. The interface had to be plausible to the system's users. The client was a Dutch company, and their guiding concept was a windmill that goes around and around, doing the same thing over and over again. If this were to be built into the interface, it would reduce the caseworkers to meaningless machine parts, filling out the same sequence of forms over and over again, much like in big bureaucracies. The stories of claims handling thus elicited were totally uninspiring. Among the many metaphors explored for organizing the narratives was walking through a house from room to room, each dedicated to a particular task and equipped with different tools. Another one was an obstacle course with different milestones to accomplish, and sailing an ocean with harbors to visit and do different kinds of business. The team also looked into concepts from communication theory, ancient travel maps, board games, architectural structures built on a "factual" foundation, until the metaphor of detective work emerged. This rearticulated the caseworker task most inspiringly: evaluating documents for their authenticity, scouting for hidden evidence or lies, making a case in the face of adversity, helping victims and trying to be fair, until the facts were sufficiently clear and the claim could be settled. This metaphor rendered casework stimulating and intellectually challenging. It is fun to be Sherlock Holmes when doing work. This metaphor informed the interface that EDL finally proposed. (There is more to this story in Section 7.4.5.).

A second example stems from a workshop at the Design Science Department of Musashino Art University in Tokyo, led by this author. The assignment was to design "a new kind of water faucet; one that did not resemble anything currently available yet had to be clearly understandable in the context of its use." The undergraduate class was divided into 13 groups of 6 or 7 students each. Step 1 was to find and list all conceivable ways humans can control the flow of water. In Japan, handling water has a long history and a rich vocabulary that is part of its cultural heritage, which this method intends to mine. The large number of students created an enormous collection from damming a lake, canalizing a river, stepping on and off a garden hose or kinking it, pouring tea from a kettle, taking showers, tilting a water-filled bottle or bucket, opening and corking a barrel, using barriers to redirect a flood, scooping water from an open container, drinking from a water fountain, squeezing a sponge, wringing out soaked cloth, mixing the water at the confluence of two streams, and, of course, turning a faucet, which was to be avoided. Each example brought its own vocabulary into the narratives that fueled the design process.

Step 2 was to analyze some of these examples into scenarios. For whatever reasons, students had difficulty grasping the notion of narrative and trouble analyzing the entailments of metaphors. However, everyone was familiar with comic strips, which are visual narratives and scenarios of sorts, and in these terms, students had little problem conceptualizing different means of controlling water flows in the contexts of their choice: ritual baths, showers, kitchens, public parks, and so on. A few drawings from one group, shown in Figure 7.8, give a glimpse of the range of instances they explored.

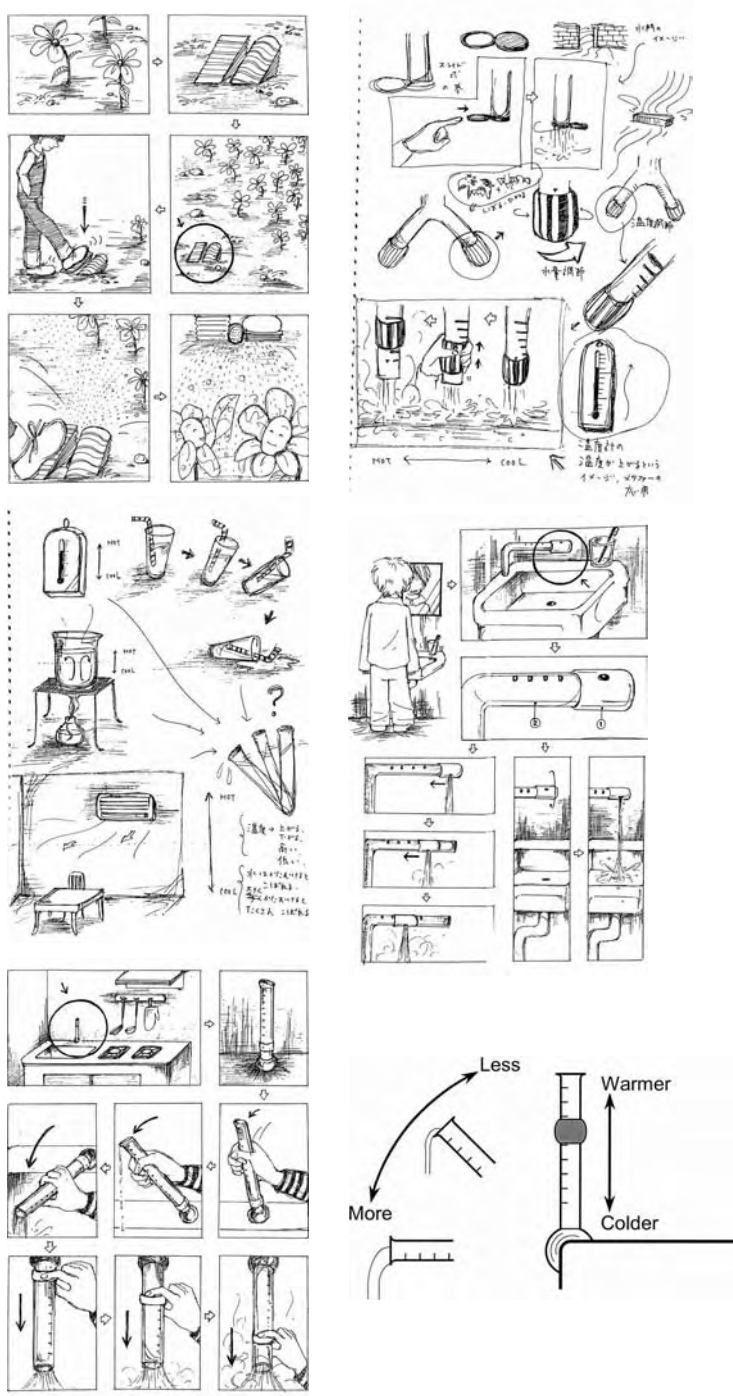


Figure 7.8 From narratives to metaphors to innovations: a faucet.

Because water faucets are essentially low tech, tubing, heat-sensing and controlling technology, and mechanical connections are relatively unproblematic, students were asked to skip step 3 and built testable models instead. Step 5, testing for understandability, involved presenting the models to nondesigners and asking each informant three questions: "What is it for?" "How would you handle it?" and "Would you prefer this to what you already know?" The answers to these questions, while neither systematic nor conclusive for an exercise like this, adjudicated the design's success.

Among the many outstanding proposals that resulted from this workshop, one group's* design, together with a few of its exploratory drawings, is shown in Figure 7.8. The first drawing analyzes one way of controlling water in the form of a sequence of cartoonlike images, a visual narrative. The second explores alternative ways to open and close a tube providing water, one for hot and one for cold, but there already is the noticeable emergence of using a thermometer as a metaphor for setting the temperature. The third looks at pouring water as a possible path to the new faucet, which eventually found its way to an ingeniously simple device in the last two drawings. They show an open cylinder on a ball that invites pivoting it in several directions. It utilizes the culturally well-established concept of pouring water — out of a glass or teakettle — the more horizontal the larger the volume. By swiveling it to the user, it metaphorically brings the water to her or him. Scalelike indentations on the cylinder encourage the perception that the ring around that cylinder is movable to different settings. The motion of that ring is linguistically and metaphorically correlated with thermometers: the *higher* the warmer, the *lower* the colder. The color of the ring was brown. Red might have brought it closer to familiar thermometers whose alcohol columns are typically red. Cursory tests proved the use of this rather unusual faucet to be obvious and self-evident. Owing to the metaphorical relationships of how people talk of pouring and temperature, there was no confusion.

Novelty and obviousness seem two opposing qualities of artifacts. If novel artifacts can draw on familiar terms and use metaphors that tap into the cultural heritage of a community of stakeholders, then it is quite possible for them to have the meanings they afford, instantaneously and reliably.

7.4.4 *Designing human-centered design strategies*

Designing design strategies involves creating the conditions for generating human-centered artifacts more effectively. This is not to be confused with meta-design, with a concern for abstractions or generalizations, rather, it is the design of the organizational, economic, or cultural contexts that nourish design culture. Some typical examples of such design efforts are:

* Group 9, whose explorations are shown here, consisted of Syugo Yamamoto, Mizuri Sekiguchi, Etsuko Hikichi, Satomi Usuki, Takeo Saito, and Akiko Hara.

- **Managing design.** Aligning the management of design departments with the strategic objectives of a corporation, already mentioned (Section 1.3.5) as a distinctly new form of design, which aims to make design activity more efficient and purposefully directed.
- **Designing intuitively obvious design conventions** for a larger class of artifacts. For example, in the 1980s, EDL developed a collection of conventions, a design language as it was called, including preferred metaphors, a system of icons, and rules for Xerox's design of various copiers. These conventions incorporated technical knowledge and cultural foresight, and left ample room for innovation. Among the many recommendations was an elaboration of the already discussed desktop metaphor for document handling, allowing users to follow the path of the paper flow and to undo paper jams, and a system of icons for directing users' attention to what could be done. These conventions saved time by not having to invent novel ways to indicate various activities and provided consistency over a range of Xerox products, which in turn increased Xerox's brand recognition. And maintaining consistency over time also eased users' ability to transfer their interface practices from one generation of copiers to the next, enhancing brand loyalty. These conventions were so successful that they have been adopted by virtually all other office equipment manufacturers; see the example of a Canon copy machine in Figure 3.15.
- **Designing rich combinatorial (interface) standards.** Highly integrated systems tend to consist of unique parts with distinctive interactions, serving specialized functions, as in a mature living organism. A significant failure in one typically fails the whole. Self-organizing systems, by contrast, have enough uncommitted variety (redundancy) to reorganize themselves in the service of preserving their identity. A key feature of such systems is the use of standards to replace components or to network many similar components into different kinds of parts. Lego, for example, is based on a simple standard for attaching building blocks to each other. Once in place, numerous combinations can be realized without being anticipated in detail. In the 1970s, automobile manufacturers began to modularize automobile design. New models could be designed from more readily combinable and separately produced components. It increased the efficiency of the design process, shortened the time to reprogram assembly lines, and allowed car manufacturers to be more responsive to market fluctuations. The Internet, essentially consisting of protocols for the transfer of electronic files, documents, and messages, demonstrates the generative capabilities of interface standards. They allow autonomous developers to cobble numerous technologies together in ways exceeding anyone's anticipation, which accounts for its enormous usefulness and growth. The grammatical rules of natural language, while not designed as such, allow for a nearly infinite number of expressions to be formed. Designing acceptable combinatorial standards allows

design to be delegated and amplified, and can provide the ground for technology to grow far beyond any single designer's imagination.

- **Redesigning inspiring organizational cultures.** Generally, creativity emerges when constraining bureaucratic structures and traditional authorities are removed in open conversation. People get inspired when they see their work as contributing to something larger than performing a function for an employer, or doing a job. A culture has much to do with how its members talk about each other, their worlds, and why they do what they do. An organizational culture renders work within an organization meaningful, and this can be more or less conducive to design, including for the well-being of the organization. In the 1960s, Braun A.G. was a pioneer in design, and designers were excited to work there. EDL was an incubator for human-centered technology, and it provided a climate of mutual respect in which creative ideas could flourish. There are no simple prescriptions for how to change an organizational culture except to start talking differently, talking about possibilities rather than obstacles, including different voices rather than dismissing them, reassuring the vision of a project, rather than performing one's function. To establish a creative climate, Xerox Corporation instituted the Palo Alto Research Center (PARC) to conduct pioneering interdisciplinary research in physical, computational, and social sciences. It became the birthplace of several human-centered technologies. The computer company Apple created a usability lab within its corporation (Gomoll and Wong, 1994), granted it sufficient clout, which contributed to its changing organizational culture.
- **Designing adaptive strategies for product development**. As suggested (Chapter 5), being a product is but one small phase in the life cycle of artifacts. Once in the public, an artifact enters the conversations of users, the promotional campaigns of competitors, the claims and counterclaims of critics, and the stakes claimed by powerful interest groups, causing its meanings to shift in unanticipated ways. Producers that want to stay in business are concerned with developing new or improved products. But they must be even more concerned with the meanings their products acquire and tend to spend much effort in ensuring that these meanings are favorable to them. But since competitors do this as well, any strategy for product development, launch, and retirement must not merely guide one's own technological development, it must also acknowledge the strategies pursued by competitors. It must account for the uncontrollable cultural or societal changes in the use of technology, the narratives that are mass reproduced by the media, life cycle and ecological issues, and adapt to changes in the economy and government regulations.

  One can conceptualize an adaptive product development strategy within a space that connects the current state of knowledge with several stakeholders' visions of what a particular technology should do, say, 20 years from now, and the proposal for a product, a system of artifacts, or

a series of actions that would amount to an advance on a path toward these visions. Cultural, economic, and political forces, which are outside designers' control, contribute to shifts in the meanings of these products, which need to be monitored. If the products in question are found to be no longer moving in the desired direction, say after a few years, the current strategy needs to be revisited and revised. This revision applies not only to the visions of desirable futures but also to the paths to get to them. Thus, a successively revisable product development strategy emerges that drives its visions ahead of the foreseeable future. Figure 7.9 sketches four revisions of a product development process over time.

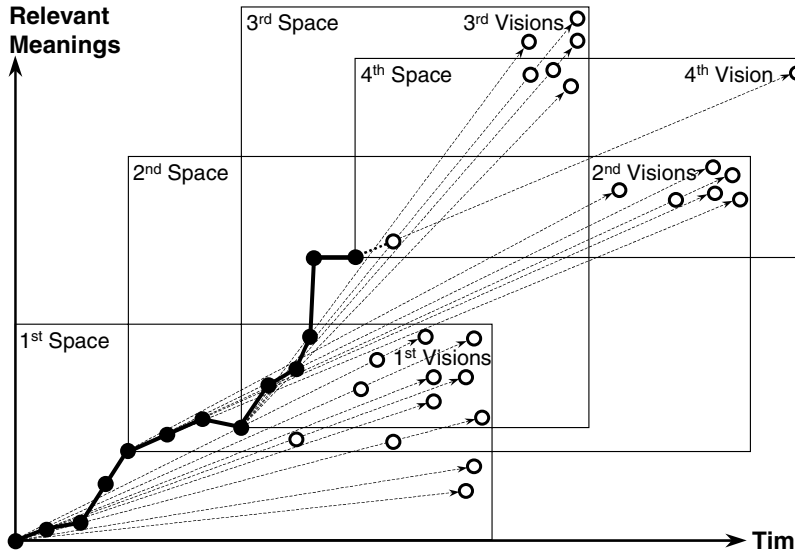


Figure 7.9 Re-visions of desirable futures and steps taken at different times.

To test whether visions of future products are meaningful to the public, automobile manufacturers develop and exhibit so-called concept cars at automobile shows and record public reactions. Some employ think tanks or skunk works to conceptualize future technologies, initially without constraints on their imagination, but subsequently to find steps that move in that direction. Others sponsor development projects at universities to obtain new ideas. The design of elevator interiors for Kohne, Figure 3.10, and the design of truck cabin interiors for Freightliner, Figure 7.3, had that aim, but we can also include a project for IBM to develop forms for a new generation of (massively parallel) super computers, still under development, but known for their computing power, compact size, modularity, special cooling needs, and probable place in office environments. Butter's students developed several proposals, three of which are shown in Figure 7.10. Any one of these could be adopted as the guiding visions for future technological developments.

These design strategies, while not easily describable in steps, share with the other design methods an understanding of the present situation; the search (not research for what exists, is true, or can be generalized) for what could be altered for effect; variables; the creation of a space of possibilities, which connect what is presently known with visions of desirable futures; and paths to realistic futures that come closer to those envisioned. For stakeholders to become enrolled at various stages of the design process, it is not necessary that they need to have the same visions as long as the path actually taken is considered desirable and leads to something meaningful to them.



(a) (b) (c)

Figure 7.10. Super computer designs for IBM expressing modularity and computing power.

7.4.5 Dialogical ways to design

The previous methods employed stakeholders at many of their steps but largely as respondents, informants, paid members of a focus group, or employees who rarely are allowed to pursue their own agendas. One of the more rewarding experiences I had at EDL was to work with designers and their clients in multidisciplinary development teams, using what one might call dialogical ways.

Dialogue (Bakhtin, 1988, Buber, 1958, Shotter, 1993) is inherently unpredictable, certainly for external observers of the process, but also for those involved, who, as participants, have at least the benefit of contributing to the direction that a dialogue is taking. Participants may have diverse backgrounds and expertise but dialogue grants each of them dialogical equality, the possibility of participation. In dialogue, nobody's voice may be suppressed — unless he or she violates the dialogical equality of the other participants. To engage in dialogue is not to debate, which implies winning or losing an argument; not to convince or persuade, which entails that one participant eventually yields to the opinions of another; not even to discuss

an issue in the sense of analyzing it into components with the expectation that participants eventually agree on the analysis. Being in dialogue means a suspension of judgment (Bohm, 1996:26) and of claims to be right or superior to other participants in the service of listening to what is said, acknowledging each other's contributions, and building on them. So conceived, dialogue is an inherently creative coprocess that is neither attributable to any one individual nor exclusive of any one participant. It could be regarded as a model of participatory design.

In one design project, already alluded to in an earlier section, EDL was asked to develop a new computer application, using the technology that the client had developed and relying on the client's engineers, but making marketable and user-friendliness the prime driver. The technology was sufficiently flexible to allow the applications to be taken in numerous directions. EDL organized a series of workshops, initially with representatives of the client, who were soon convinced to invite all those who had something to do with its development. In this case, participants included project leaders, hardware and software engineers, a potential user of the technology, and at some point marketing people and board members interested in the process. While the ball was clearly in the designers' court, after all, EDL was charged with the development of a benchmark technology projected to be used very much into the future. EDL developed new concepts, approaches, and metaphors and presented them in the form of reports and slide presentations that contained many questions for the workshop to address. A workshop agenda made the workshop more focused and not as unpredictable as dialogues typically are, but otherwise it preserved the main features of dialogue outlined above. EDL contributed to the project by:

- Introducing *a narrative* of the future technology in which the not-yet-existing design could emerge (without knowing exactly what that was). It used a vocabulary that was soon adopted because it allowed each expert to assume responsibility for the details they knew how best to handle, and granted them a voice in the emerging consensus on an overarching narrative. Each participant had the chance to make contributions and everyone else listened. Dialogue prevented the participants from withdrawing into familiar patterns of specialist reasoning that had kept the client stuck before EDL was invited. (Some asymmetrical relationships among the participating employees of the client may have existed prior to EDL involvement.)
- Generating *a wide range of optional paths* so as to enable the exploration of advantages and disadvantages within these meetings. These paths had to be sufficiently exciting to invite participants to go out of their usual ways of thinking and explore the positive and dark scenarios of these alternatives on their own. Because this was a new technology, between these workshops there was much concurrent development and communication across employment categories that the participants had not experienced before. Participants in the process pursued some of these paths

on their own, filling the narrative with the technical details needed to make it work.

- Finally the *decisions* of where to go were turned over *to the stakeholders* in this technology, who would then have to carry the project to its fruition beyond the designers' concepts. Ultimately, a viable project has to become owned by its stakeholders. The purpose of designers is to make themselves dispensable (see Section 3.6.12).

One of the more rewarding experiences of a dialogical way of designing is that nobody can really tell (nor cares to assess) exactly who made which contribution. This can present a problem for designers who are paid for doing work that is subject to their clients' specifications. A true dialogue proceeds by mutual respect, not by specifications, and to design by dialogical ways requires that the criteria for completion of the design be jointly acceptable as well. Although the client of the just-discussed design effort sent observers to learn from how EDL worked, thus fully acknowledging EDL's leadership role in the development process, after a sequence of workshops, one participant wondered what the role of EDL really was, feeling that the workshop participants had "done all the work by themselves!" This expression, while naively conceived, is a compliment for the dialogically working designer. Expecting credit and assuming to be rewarded for all ideas limits a development to the capabilities of the designers. Dialogue expands that capability to the designers' clients, which improves the chance that the development continues and becomes realized once the designers leave the scene.

A more difficult problem is to find the right moment at which other stakeholders need to become involved, how many and of which kind. One fascinating design project for a forward-looking office environment and furniture that EDL was asked to develop never went beyond the team that worked on it, and the client's contact person. EDL was not given the chance to engage in dialogue with those who mattered.

7.5 *Validating semantic claims*

While the ultimate aim of human-centered design is to propose artifacts that are self-evident, ideally requiring no explanations and no justifications, and to assure users of reliable interfaces without disruptions, a science for design cannot entirely rely on this "face validity" and has to offer ways to provide more explicit evidence for the claims that designers make to their stakeholders.

Because designs, by definition, are proposals for artifacts that do not yet exist in their intended forms, designers need to validate their claims in ways that are fundamentally different from how scientists argue for the validity of their findings. Scientific evidence is rooted in the past (and merely generalized to the future) and relies on systematically gathered observational data. By contrast, evidence for designs (a) concerns projections into a not yet

observable future, and (b) is conditional on stakeholders' actions, for example, to realize design specifications. In other words, scientists are concerned with explaining an observable world, designers with creating desirable worlds, and statements about either of these worlds call for vastly different methods of validation.

As suggested in the introduction to this chapter, designers' claims to artistic sensitivities or cultural foresights that their clients are presumed to lack, true or not, are neither empirically demonstrable (at present) nor relevant when the real issue is one of convincing skeptical stakeholders about the virtue of a design, about what needs to be done to realize future interfaces between future users and future artifacts. For this reason, Chapters 3 through 6 developed four concepts of meanings that provide designers with an indigenous subject matter, something to observe, something to work with, something to measure, and a new ground to stand on while presenting their proposals to others.

- *Meanings of artifacts* **in use** are manifest in the way users approach, interface, and learn to rely on them.
- *Meanings of artifacts* **in language** are manifest in how artifacts enter communication among stakeholders — in which categories they end up, which attributes they acquire, which user identities they support, which emotional expressions they evoke, and how their role in people's lives is described and rearticulated to other people, creating a stakeholder community — in other words, how artifacts appear in language, conversation, and discourse.
- *Meanings of artifacts* **in their life cycle** are manifest in how artifacts travel through networks of stakeholders and inform their successive transformations: how compelling proposals generate resources, sketches inform production drawings, advertisements create desires, appearances inform uses, and uses affect other technologies and the environment.
- *Meanings of artifacts* **in an ecology of other artifacts** are manifest in their evolution from one generation to another, in their interactions with other technological species, in the relations that develop between species of artifacts: cooperative, competitive, dominant, or submissive, supporting cultural complexes or subverting them, and affecting the availability of material resources and the ecology of living beings.

None of these meanings can be located exclusively in someone's mind, that is, they are not subjective or entirely cognitive in nature, nor can they be found on the surfaces of artifacts. While not observable directly, meanings account for what people do and therefore cannot be dismissed by those who celebrate objective accounts, chide human-centered accounts as subjective, or prefer hard quantitative evidence to soft impressions. The distinction between objective and subjective evidence is not the issue, however. One must be reminded that even in physics, the concept of energy, fundamental and well-established, is not observable as such. Energy is manifest in its

consequences. It has no reality without a conceptualizing observer, a physicist's interest in explaining certain measurable phenomena. Thus, meanings, as conceived here, are in good company. They are explanatory constructs that we live with but they become evident only in what we do with them. Thus, the semantic turn replaces the causal models of the natural sciences by social or linguistic models of meaningful interfaces with technology. This is a huge undertaking, but a necessary one in justifying human-centered design. Meanings add a new dimension to design, which was previously ignored or at least not sufficiently conceptualized. With various concepts of meanings, the science for design provides an empirical ground for designers to stand on where previous practitioners were floating in subjectivities.

Unlike the claims that natural scientists make and validate, the claims that designers make and need to validate always concern multiple descriptions of artifacts (see Figure 2.8 but also Krippendorff, 2005, in press): minimally the designers' understanding, various stakeholders' understanding, and the users' understanding included. Each employs different perspectives, at least as a beginning hypothesis. Each has good reason to see differently what they might all agree to be the same artifact. Some of the questions that designers seek to address and for which stakeholders need plausible answers involve a second-order understanding, here of whether artifacts can:

- Afford the actions encouraged by their meanings
- Acquire intended categories and characters
- Bring forth the future that their users are promised
- Obtain desirable cultural roles in particular communities
- Travel through a given network of stakeholders
- Find desirable ecological relations to other artifacts, etc.

Answers to such questions are "*semantic claims*," for they *assert what proposed artifacts are likely to mean and do in particular communities*. Semantic claims do not depend on other kinds of claims. They cannot be reduced to geometry, physics, psychology, or economics, and thus justify a science for design as independent of other sciences.

The claims that the natural sciences cannot be concerned with but that engineering shares with design concern *futures that could not come about by natural causes*. To be sure, claims in engineering and in design prove themselves only after a design has been realized and put to work, in the case of engineering, in the form of functional pieces of machinery, and in the case of design, in the form of reliable interfaces, social practices involving artifacts, or human communication through multiuser systems.

Engineering proposals tend to be justifiable by available theories or principles, for example, from mechanics, aerodynamics, hydraulics, chemistry, and material science. Engineers prefer mathematical theories and substantiate their claims by calculating the details of an engineering design as specific instances of established theories. Where details are missing, engineers run

tests. In design, calculations are rare, but the nature of the arguments is analogous: Mathematical theories of a technology are to engineering as communities of stakeholders of a technology are to design. Communities, even single individuals, are rarely theorizable, however. They participate, learn while in interactions, and use language to construct realities, which cannot be cast in mathematical–deterministic terms.

Herein lie the crucial differences in validation efforts between engineering and design. While mathematical arguments limit engineers to a first-order understanding, the arguments that designers must advance cannot ignore the perceptions, voices, and practices of their stakeholders, and are, hence, bound to be of a second-order kind. Moreover, when designing a unique bridge, for example, engineering calculations must convince other experts that it conforms to various rules and safety regulations. But when designing a consumer product, designers must convince a whole network of other stakeholders, including the targeted consumers, that their design will work as promised by its appearances. Engineers have the additional benefit of testing the functionality of the prototypes of products for conformity with performance specifications. Human-centered designers, however, must rely on far less clear stakeholder expectations the analogues of specifications that are not written down and may shift and move in all kinds of directions. In human-centered design, validity cannot be separated from the politics of the stakeholders involved.

The following list presents three customary and two new ways of validating semantic claims, ordered by their increasing strength.

- **Demonstrative validity**. To demonstrate is to show something with the intent to prove its value or workings, for example, what an artifact would do and how. Designers' presentations to clients typically refer to particular user groups, argue what a design would mean to them, and show how they could interface with it. Extending traditional slide shows and drawings, computer simulations and videos have been used to substantiate such semantic claims. Walking through a virtual building can give future inhabitants views that no drawing can adequately capture. And exploring the interior of an automobile engine, or of the human body for that matter, can provide designers or surgeons surprising views of a reality they've never seen before. Demonstrations, done well, can be compelling, perhaps because of their inconspicuous omissions: Computer aided design (CAD) of a cell phone, for example, provides no clue to how heavy it might be or how it would feel. Visual representations of automobile interiors cannot possibly convey how comfortable its seats are or how they smell, which are known to add to their characters. Demonstrative validity can speak only to the sensory modality on which the demonstration relies. Perhaps the more important limitation lies in casting those for whom a demonstration is intended into the role of reactive audiences, viewers, or critics, trying to imagine being users.

Within a network of diverse stakeholders that ultimately determine the fate of a design, users' meanings are only a small part of what matters.

- **Experimental validity**. Experiments with prototypes are familiar ways to gather evidence for whether a design affords the meanings, habits, and expectations that potential users bring to them. While the prototype of a mass product, by definition, stands for all copies to be produced, the subjects that participate in such experiments need to be sampled to represent all stakeholders conceivably involved. Answers to such questions as to whether people can recognize the prototype for what it affords, how many disruptions they experience while interfacing with it, the character traits they would attribute to it, and which user identities they are believed to support define statistical distributions. Experimental validity goes beyond demonstrative validity by allowing subjects to interact with a prototype, which can yield unexpected evidence in support of or against the claims advanced by its designer.

  However, experimental validity has its limitations as well. Although it is not inconceivable to experiment with stakeholder networks, experimental validity privileges easily manageable, usually tangible, artifacts and individual users, not social networks. Experimental validity for large multiuser systems would be difficult to obtain because they work by developing unique organizational cultures and breed the technological competencies of their participants, which cannot be tested with prototypes.

- **Interpretative validity**. Designers may not have the mathematical theories with which engineers can impress their clients, but they often justify their designs by propositions at home in other discourses, established theories and published findings about perception, cognition, ergonomics, sociolinguistics, politics, marketing, anthropology, economics, and of course engineering. Such propositions rarely come ready to use and must be interpreted, reframed, and rearticulated before they can be assembled in support of the claims put forth about a design. For example, UCMs come from a marriage of the idea of cognitive models in linguistic semantics with schema theory in psychology. The former provides propositions about conceptual relationships in linguistic expressions and the latter adds propositions about behavior. With the help of both, one can investigate the UCMs of potential users and anticipate how holding certain UCMs leads to certain interfaces. Neither discipline is interested in design. When designers use UCMs to validate semantic claims, they in fact interpret available theories and associated empirical findings.

  Interpretive validity is limited in at least four ways that designers need to be aware of. First, most of these research results are created under the auspices of disciplinary objectives, which may not coincide with the multifacetedness of a given design. Before using them, designers need to be cognizant of their biases. Second, scientific data are records of past observations. It requires a leap of faith to believe that scientific findings

will still be valid in the future in which a design must ultimately prove itself. Taking data from the past blindly renders design conservative of past conception. True inventions, original designs, by definition have no precedents to rely on. Third, scientific theories are generalizations that omit details in favor of theorized pattern. Design, however, is less interested in generalizations then in details. It needs to accommodate the diversity of stakeholders, not its most frequent members. A design must work in all details, not in theory. Fourth and finally, empirical research tends to address particular measurable variables. When designers argue for the interpretive validity of semantic claims, they must put them together into whole narratives much like stakeholders would. Generally, wholes work differently from the sum of their parts.

- **Methodological validity**. In the natural sciences, researchers have the freedom to state any hypothesis they please, and devise experiments and measuring instruments consistent with the hypotheses under consideration, but are obligated to account for what they did so that qualified scientific peers can critically reexamine or reproduce the steps leading to the reported findings. To safeguard that observational data, and only these, decide whether the stated hypotheses are true or need to be rejected, the scientific community insists on being able to evaluate whether the use of methods is conclusive. Thus, critical examination of the methodology used to process observations is an important source of the validity of scientific claims.

  A science for design offers a powerful analogue to this form of validation with three by now obvious differences. First, design is not concerned with what is or was, but with what is possible, what should be done, and how it could be accomplished. In design, the analogy to scientific data is the knowledge of the current practices of living with technology. Second, the validity of the semantic claims of a design cannot be established by observations of nature. It concerns how humans conceive of, interface with, and develop artifacts, which fundamentally concerns the ability and aspirations of the human use of technology. Third, unlike scientists who critically evaluate the research methods used by their peers, in design it is the communities that claim their stake in a design that must be convinced by the designers' use of methods. Design and science share the belief that the validity of their claims — semantic or concerning truths — can be supported by a critical examination of the methods used to derive them.

  The foregoing described design methods as involving a systematic expansion of the space of possibilities, followed by an informed reduction of that space to arguable proposals for artifacts that promise to bring forth desirable futures for particular communities or prevent undesirable ones from occurring. Accordingly, *methodological validity* can rely on three kinds of evidence of what designers did:

- Concerning **the number and kinds of alternative futures**, versions, or approaches that were considered and explored before coming to the proposed design. The sheer number of alternatives that designers examined seriously can serve as a first indication for whether a designer covered sufficient ground. Blatant omissions that a critical examination may reveal distract from the likelihood that a design is acceptable.
- Concerning **the number and feasibility of paths** from the present to many if not all of the alternative futures that had been considered. Examinations of this kind of evidence must take into account various kinds of resources that the designer considered in evaluating which designs could move through their life cycles and at whose cost and benefits.

  In human-centered design, feasible paths must not only be technically possible but also desirable by present and future stakeholders. Technologies always have their plusses and minuses. Methodological validity depends on their examination. There maybe trade-offs, for example, between classical and electronic guitars or in limiting telephone communication to voice while expanding its range to great distances. There are desirable and dark scenarios; for example, the very gun that can protect against intruders can also accidentally kill its owner. Technologies may benefit one community at the expense of another. Probably most important are the social costs of privileging stated ends at the expense of the processes of reaching them. For example, to install Communism in China, millions of entrepreneurs were executed; in the industrialized West, profit motives favored the outsourcing of production at the expense of maintaining a workforce that could purchase the products produced elsewhere. If those who will have to live through a proposed path are given a voice, such excesses of technological rationality are not likely to happen. Designs that are derived without having considered their plusses and minuses should be treated with suspicion.
- Concerning **the numbers and kinds of stakeholders consulted** in developing a design. Semantic claims that are not qualified or substantiated by any evidence involving the communities of stakeholders to which they apply can safely be rejected (except where they are demonstrably convincing to the very stakeholders that matter). Where stakeholders have been consulted, it would matter whether they are representative of the stakeholders that ultimately decide on the fate of a design, and how their diverse backgrounds, cognitive abilities, and aspirations entered the design process, how incompatible concepts were resolved and in whose favor. This would suggest which communities of stakeholders are likely to understand, use, and feel at ease with the design, and for which communities evidence does not exist. Whether relevant stakeholders have been shown to recognize their contributions in the proposed design or at least feel at home with it counts toward the validity of semantic claims.

- **Pragmatic validity**. If the stakeholders of a design are committed to support, promote, realize, or use it, this surely is a kind of evidence that no one could ignore. Pragmatic validity lies in the hands of its stakeholders, as it should. In some sense, pragmatic validity is the ideal of a self-evident proposal that requires no further explanations. The following example was told by Robert Blaich.* Philips marketing research had found that the roller radio, which came out of the 1984 product semantics workshop in Eindhoven, had no market. Subsequently, a friend of Blaich's saw the design and ordered a large number of them. On account of that order, the roller radio was first produced in 1985 and subsequently became a huge success, and is now copied by many competitors.

  However, stakeholder commitments may not be enough. It could be based on misinformed enthusiasm. Drawing on the scientific principle of always looking for counter evidence, especially in the larger context of a proposed design, to accept the validity of a design on pragmatic grounds calls on designers to turn the tables and inquire into the stakeholders' knowledge and motives to ensure that their enthusiasm is not based on unrealistic or unethical expectations. Making sure that a design can deliver what others expect is the easier half of pragmatic validity. The more difficult half is to ensure that a design does not aid one community of eager users at the expense of the well-being of others, or that it satisfies an immediate desire while constraining future developments. Enthusiasts may not recognize the dark scenarios that a designer is obligated to consider.

Post factum evidence is of course the ultimate proof of the validity of a design, being adopted by its stakeholders, working its way through its life cycle, and succeeding in the ecology of artifacts. Such evidence does not need to support designers' intentions. All that matters is that it is viable in terms of the stakeholders involved. If it has succeeded, designers might reap the benefit of having been part of a successful effort but should be addressing other issues. However, since design is always before the fact, *post factum* evidence cannot validate semantic claims.

The point of a science for design is not to encourage systematicity or to celebrate methodology for its own sake, but to provide designers with a solid subject matter, the rhetorical strength needed to render their proposals compelling, and the ability to collaborate with other disciplines on diverse aspects of a design.

7.6 Advancing design discourse

No profession, no academic discipline can survive without providing its members the ability to explain themselves to others, continuously improve their proficiency, and demonstrate the benefits they can bring to collaborations with

* Robert Blaich. Personal communication, 1989. See also Robert Blaich (1989, 1993).

others. Explanations of this kind are constituent parts of a discourse. For designers such explanations occur when they are offering their services to clients. They are implicit in how designers collaborate on a project and become explicit when consulting an outside expert rather than handling the task in the design office. They are manifest in the kind of arguments designers use to justify their designs to those who matter. Such explanations are also built into various institutional practices: meetings of professional designers, awarding educational degrees in design, publishing in a design journal, participating on a jury for design awards, and organizing a design office or a design department within a corporation. The discourse used among designers and in front of others is the most important albeit little recognized source of designers' identity.

Individual designers thrive or wane in personal influence with the public image of the design community, for instance, when negotiating contracts with clients. Speaking as designers and identifying themselves as members of the design community invokes the whole history of design — as far as known by those spoken to — its exemplars, its celebrities, and its landmark schools. Relying on the rhetorical strength of this discourse entails the obligation to contribute, advance, and celebrate this very design discourse for it is what holds the design community together. In this regard designers have been less successful.

Medicine could be a good model for design. Medicine and design are practical professions that fix things: In the case of medicine, restoring biological normalcy and in the case of design, proposing something better. Medicine derives its rhetorical strength not only from the successful treatment of patient ills but foremost from rigorously training its aspiring practitioners, engaging in research on treatment methods, and improving medical technology and practices. Medical professionals maintain a reasonably clear boundary between what is a medical concern and what is not, and what is inside conforms to certain models that the medical profession vigorously pursues. Medical discourse enjoys an enormous respect by nonpractitioners. Patients are in awe of its vocabulary, and know just enough to find a doctor and pay the bills for treatment often beyond their comprehension.

As suggested in the beginning (Section 1.4.3), design discourse can acquire a comparable standing if it adopts an easily recognizable and productive boundary. To this end the semantic turn advanced an axiom that recognizes that humans do not see and act on physical qualities but on what these qualities mean to them (Section 2.2). Much of the book elaborated on its theorems. The absence of counterexamples for the axiom, along with its theorems, suggests that it holds a basic epistemological truth. A vocabulary that is carefully developed under the umbrella of such a powerful axiom can lend designers a rhetorical strength that is unprecedented among other practical and problem-oriented professions, without the ability or intent of developing a science for their own practices.

Medical discourse does not quite recognize its social nature. For example, health is a largely socially or culturally constructed norm, and diseases

are constantly evolving. Medications are developed for profit by pharmaceutical companies, subject to governmental regulations. Illnesses are at least in part defined by how insurance companies pay for them, and research for which funding is not made available does not typically go too far. Nevertheless, medicine has drawn an effective boundary. Medical knowledge accumulates within this boundary, its vocabulary is continually refined and revised, and information about the kind of problems it claims to solve is communicated to the public outside its boundary to attract medical students, patients, and research funding. This is where the rhetorical strength of medical discourse comes from.

The semantic turn is introducing a new design discourse that lays the conceptual foundation of a science for design and identifies an area in which designers' competence can excel. Here are four activities that would advance the design discourse and strengthen the emerging science for design.

7.6.1 Postdesign research

It may be convenient for designers to stop any concern for their design on its delivery to their clients, but this would terminate the learning process. Figure 5.1 situates designers in a recursive process. It suggests that designers could make the additional effort to observe and learn from their own successes and failures. A science for design suggests doing so not merely anecdotally but systematically. Postdesign research inquires into what happened after a design was delivered, how it worked its way through the network of stakeholders or not. Postdesign research is re-search proper as it examines and seeks to generalize from what happened in the recorded past of design practices. The aim of postdesign research is less to establish what happened or to distinguish successful from unsuccessful designers, but to avoid our own and others' mistakes. The cases of designs that were considered outstanding, or even won prizes, without ever reaching the market, are as important as those of many products that were badly designed or not at all and became smashing successes. There seems to be a disconnect between what designers think good design is and what in fact succeeds.

7.6.2 Design literature

Written design discourse is the reexaminable repository of knowledge of design. Current design literature stems largely from art history, curated museum exhibitions, and designers' self-promotions. The creation of a literature that addresses the meanings of interfaces, how artifacts enter language, life cycles and ecologies; accounts of methods, their sources, successes, and failures; research reports; design theories; even data banks of visual forms and their meanings, are badly needed. Scholarship in design — writing design discourse, not just speaking it — is a political necessity for design. Documentation of design practices and research results could prevent reinventions and the repetition of mistakes. While major advances have been made in the

automatic retrieval of texts and images from the World Wide Web (Krippendorff, 2004a), they have not reached the level of synthesis of interest to a science for design. Textbooks and dissertations that can synthesize the design literature, extract recommendations, codify design methods, and identify important exemplars wait to be written.

7.6.3 Institutionalizations

Sciences require institutions to encourage consistency of their discourse and sustain their growth. Universities are the most obvious candidates for teaching, engaging in basic research, and intellectually expanding any science. In the United States, graduate education in design is undervalued and Ph.D.s in design are rare. For several years now, a lively conversation is in progress on the Internet concerned with design studies and introducing Ph.D. education in design.* Notwithstanding degrees, contributions to a science for design should be the primary focus of graduate education. This is where students learn to speak and act designerly. A science for design also needs its (re)search institutes, funds for basic investigations, support for conceptual development, literature searches that can bring diverse knowledges together, scholarly publications, and well-trained design researchers willing to channel their energies to advance design. Industry would greatly benefit from these exciting opportunities, especially in the design of information technology. Its support for the effort would be amply rewarded, speeding up human-centered design and reducing the risk of failures. Professional associations for designers do exist, but they tend to focus on celebrating traditional design accomplishments rather than making a science for design their home. There is research done in ergonomics and human factors design, which is addressed in Chapter 8. The U.S. government has recognized the need for human-centered design to contribute to technological development in the information age.**

7.6.4 Self-reflection

Any discourse needs some autonomy, without being closed to outside ideas. As suggested in several earlier sections, blind borrowing of research paradigms and epistemologies couched as empirical evidence from other disciplines entails the danger of subverting, colonizing, and in the end stifling design discourse. Taking the semantic turn seriously resists this surrender. It proposes instead a subject matter, and various meanings of technology, about which other disciplines have little to say; and it creates methods and ways of validating claims that are unique to design. But the continued

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** A workshop sponsored by the U.S. National Science Foundation examined the opportunities and need for human-centered design to contribute to contemporary technological developments. Its recommendations for a national policy on funding of design for the next century was edited by Krippendorff (1997).

viability and growth of design discourse calls for rigorous self-examination of its science and its practices.

In the natural sciences, the philosophy of science assumes the task of examining a science's practices. Owing to the representational notions dominating the natural sciences, there seems to be little connection between the philosophers of a science and the practitioners they examine. Productive scientists do not need to and often do not know anything about the philosophy of their science. In design, this separation is not necessary. Perhaps a science for design is more inclined to self-reflection than other sciences because second-order understanding lies at its core. From understanding stakeholders' understanding, it is but a small step to grant stakeholders the ability of second-order understanding as well. This brings the designer to an understanding of a stakeholder's understanding of the designer's understanding, which turns understanding onto itself — not as a solipsistic construction but as one that requires the presence if not cooperation with others. A philosopher of science who would target the science for design is condemned to remain outside it and therefore of only marginal importance to designers. Taken seriously, a science for design is inherently self-reflective, within its boundary, while always open to other stakeholders. It is destined to continually reflect on itself, on its strengths, and on its weaknesses. However, this is done always in collaboration with outsiders of the design discourse, ideally making all relevant stakeholder communities its validity check. This is the philosophical stance of a human-centered science for design. It is not driven by disembodied theory or monologue, but it emerges in the embodied linguistic practices or dialogue, involving all those who care or matter.

The dialogue that a science for design entails is probably the best assurance of its continued viability — but it must be practiced as such.

chapter eight

Distantiations

This chapter examines similarities and differences between the semantic turn in design and a variety of disciplines and approaches from which it needs to be set apart.

8.1 Semiotics

Semiotics probably is the one discipline that is most frequently confused with what the semantic turn seeks to accomplish. For those who have been exposed to semiotics, the adjective semantic — etymologically "relating to meaning" — tends to put an elaborate vocabulary in charge of their thinking, which defines semiotics as the study of signs and divides that effort into three branches: syntactics, semantics, and pragmatics (Morris, 1955). When applied to design, semiotic theories divert attention from the meaning of artifacts to the relations between signs and their referents, between signifiers and the signified, a move that unwittingly imports epistemological assumptions into discussions of meaning, which are alien to design and generally untenable. Consider five features of semiotics that are objectionable here.

1. **Semiotics is grounded in a two-world ontology,** as John Stewart (1995) aptly describes it: the idea that there exist a world of signs and a world of referents, a world of signifiers and a world of signified objects,* or a world of sign vehicles and a world of the contents they supposedly convey. This ontology was fundamental to Charles Sanders Peirce's (1931) Semeiotics, Ferdinand de Saussure's (1916) Semiology, and has continued in Charles Morris' (1955) failed attempt to integrate various theories of logic, scientific conduct, literature, and human behavior into

* This is already expressed in the word "semi-otics." Its prefix "semi" = "half" has roots in the Greek "sema," which suggests a sign to be the accessible half of a relationship between it and something else. It is speculated to come from the custom of breaking a coin in half when two friends part, each carrying one half in the hope that this would bring them back together. So conceived, a theory of signs should be built on metonymy, part–whole relationships, not reference.

a unified science. That these two worlds shall not overlap has been given logical force in Bertrand Russell's Theory of Logical Types (Whitehead and Russell, 1910/1958), according to which signs may not enter what they describe. The two-world ontology remains the backbone of contemporary semiotics as encoded in Thomas Sebeok's (1986) *Encyclopedic Dictionary of Semiotics* and continued in Umberto Eco's (1976, 1980, 1984) projects. The current French versions of semiotics, by Roland Barthes (1983) and Jean Baudrillard (1983), for example, have downplayed the material or objective part of this two-world ontology, privileging representations instead, and theorizing what representations do in society, including representations of representations — without overcoming the ontological schizophrenia of semiotics, however.

If designers of artifacts were to conceptualize the meanings of artifacts in semiotic terms, they would be led to artifacts as *representations* of something else. This is known to invite "pretentious semiotizations," deceptions of the users of artifacts regarding their nature: Creating forms that make a product appear more valuable than it is; using hard-to-distinguish but cheaper substitutes, for example, vinyl for leather, printed Formica for rare woods, plastic for chrome, or anodized aluminum for precious metals; or suggesting functionalities that do not exist, such as inoperative push buttons on telephones or impressive sound indicators on amplifiers without functionality. The practice of covering the interior mechanisms of artifacts with skins that suggest something else or using verbal instructions to override uninformative or misleading forms speak to the same phenomenon.

Already mentioned was Umberto Eco's (1976:7) definition of semiotics as the discipline that studies everything that can be used to lie. The examples of pretentious semiotizations demonstrate that possibility. By not separating artifacts from what they mean, the semantic turn pursues a nonrepresentational concept of meaning, one that encourages artifacts to be self-evident to their users and supportive of natural, obvious, and reliable human interfaces. Human-centered designers could fail in this regard, of course. Designers are humans too. But the concepts provided here do not encourage deception.

2. **Semiotics excludes human agency** in creating meanings. Peirce and virtually all semioticians after him recognize that the relation between signs and referents are found in nature, as for indices; based on existing similarities, as for icons; or established by conventions, as for symbols. Accordingly, what makes something a sign is attributed to conditions not controlled by the sign users. While this three-partite distinction recognizes symbols as human artifacts, it hides the fact that all signs are artifacts indeed. Even indices are of social origins, often with long histories preceding current use. For example, all medical symptoms of various bodily malfunctions have been proposed at one point in time, refined over long periods by systematic experiments and treatments,

and finally institutionalized in texts and medical practices. The presumption that symptoms have underlying biological causes, which would make them indices by definition, leads semioticians to believe that these symptoms were always there, thus denying their social constructedness. Medicine is a living discourse that continuously reexamines and reinterprets its diagnostic and treatment practices. A medical symptom is held stable by the institution of medicine and only as long as it serves that institution.

Consider the most basic semiotic distinction between signs and their referents. To begin, we must realize that all distinctions are drawn by someone, by people who have lived through their own history of conceptualizations. Distinctions create the very differences that those making them end up seeing. In the case of semiotics, one can point to a long history of conceptual struggles going back to ancient Greece. But by building semiotic theories on differences (the difference between signs and their referents) rather than on acts of distinguishing, semioticians effectively remove themselves from the responsibility for them, which leads to the delusion of studying objective or observer-independent facts. Semioticians seem totally unconcerned with or utterly unaware of how their own terminology constructs the world of signs they claim to study. In sharp contrast, the semantic turn thrives in the very awareness that the human world is created by human involvement if not design.

But semioticians go further. They not only deny their own agency but also the agency of the sign users. How humans occur in semiotic theories may be seen in the following painfully accurate rearticulation of semiotics' three primary subdisciplines*:

- **Syntactics** (in semiotic terms: the study of the *relations between signs* [signifiers or sign-vehicles]) constructs a reality in which humans do not exist or are not allowed to enter. Geometry, grammar, harmony, symmetry, proportions, character sequences, and natural laws, posit relations among signs that are either true or not and provide no place for those who recognize such syntactic relationships.
- **Semantics** (in semiotic terms: the study of the *relations between signs and the objects they refer to or signify*) constructs a reality that exists independent of sign users' conceptual participation. Semantic propositions of the form "X is a sign of Y" or "X stands for Y," sometimes qualified by adding "in the context of Z" (situations, time periods, user cultures), are either recognized for what they are or not. Witness the idea of having to learn signs, as if their signifiers and signified were there to begin with. Semioticians who address issues

* I am taking the standard definitions of these subdisciplines from Morris (1955). They are widely accepted.

of meaning expect sign users to realize the connection between signs and their referents correctly. It assumes that signs have their references regardless of whether people understand them or comply with their illocutions. In Peirce's semiotics, the three "interpretants," causality, similarity, and conventions, are factual and must be learned before use. Humans seem to be dispensable for establishing the validity of semantic claims.

- **Pragmatics** (in semiotic terms: the study of the *relations between signs and their uses*) constructs a reality in which signs appear correlated with psychological effects on their users, and with causing certain behaviors, including the reproduction of signs. So conceived, pragmatics establishes correlations between a stimulus X, conceived of as the sign of Y in the context of Z, and observed sign users' behaviors. By taking X as the sign of Y, semioticians seem to see no problem with imposing their conceptions on sign users, and using the latter merely to establish the degree of correlation between X and what they observe. In semiotic pragmatics, the sign users are at least acknowledged, but without conceptual participation and without creativity.

This is a grim picture. The vocabularies of the three branches of semiotics derive from ontological assumptions that reduce the image of human beings to a farce. They amount to a closed system of explanations, which provides no space for alternative conceptions. This kind of semiotics has little to contribute to what human-centered designers seek to accomplish: proposing artifacts that offer its stakeholders possibilities of interfacing with them in individually meaningful ways.

3. **Semiotics has perennial problems with polysemy**, the common experience of multiple interpretations, for example, of texts by different readers. If the truth value of such propositions as "X stands for Y" or "A has the function of B" would not be uniquely determinable, if the reference relations of signs change from one person to another or over time, then propositions of what something means are in serious trouble.

   Roland Barthes (1983) proposed a compelling solution to this (entirely semiotic) problem by supposing that signs can mean anything, and are inherently polysemious, except for the operation of social techniques that constrain the great variety of meanings, and enforce particular codes of conduct. He found what everyone who has received a traffic ticket has experienced, that powerful institutions are required to keep meanings (e.g., of traffic signs and regulations) from becoming polysemious and arbitrary. Accordingly, the validity of semiotic theories — presuming the existence of signs and relating them to specific references — derives from the effectiveness of social institutions — Michel Foucault (1980, 1990) calls them "regimes of power" — that keep meanings stable, in place, and hence describable propositionally. From the correlation between the absence of polysemy and the presence

of powerful social mechanisms of control follows that semiotic propositions of the kind "X means Y in the context of Z" are true only where strong institutions keep them stable. Thus, the project of semiotics unwittingly depends on the exercise of coercion and in turn supports regimes of power that demand sign users to comply with particular interpretations.

4. **Semiotics is content with static accounts, classifications, and taxonomies**. Although Peirce coined the word "semeiosis" to refer to processes of something becoming a sign, contemporary semiotics has become content with categorizing signs — as if the competence of sign users would be manifest in their ability to put signs into the same conceptual boxes that semioticians have created, and as if these categories would not be subject to cultural drifts and redesign. Already Peirce (1931) provided extensive logical taxonomies, which were taken to be analytical and not meant to be open to empirical investigations of or negotiation with sign users. Whereas the semantic turn recognizes that conversations with the users of artifacts, including about signs, are needed to understand what they mean to them, semioticians take their own accounts of signs as factual and construct explanations for when and why people fail to recognize signs for what they "really" are. Categories of signs result from monologue (one unified voice and one logic). This is anathema to the semantic turn, which is concerned with how meanings are created in dialogue (human interactions between two or more people and two or more logics) and emerge in interfaces with technological artifacts.
5. **Semiotics' belief in a rational consensus**. There is another semiotic twist that needs to be questioned, one that is more directly related to design, semiotics' belief in a consensual use of signs, in sharing the understanding of signs. Max Bense (1971) and Elisabeth Walther (1974) built instrumentality into their triad of "material," "form," and "function," which is thought to be common to all signs. Prietro (1973) articulated his conception of artifacts in the triadic relation between the "tool," its "utility," the set of operations executable by that tool, and its "operand," the operations that the utilities of a set of similar tools share. Martin Krampen (1979) reviewed these approaches before synthesizing his own instrumental notions of architecture. Gui Bonsiepe (1993) revived Tomas Maldonado's triad consisting of the material "tool," the "problem" to be solved, and the "agent" who executes the operations linking the two. Although it is not always clear what these triadic conceptions have in common and why they are associated with semiotics, commitments to an instrumental view of artifacts, that is, to the goal-directed use of technology, are far less common now than during the industrial era. There is much more to technology than its purposive use, and the pursuit of conscious purposes has been shown to have disastrous ecological consequences (Bateson, 1972). While the semantic turn

cannot possibly deny instrumentality, artifacts may be the constituents of ritual practices (without reason), invoke emotions (meaningful bodily involvements), be intrinsically motivated (in pursuit of processes rather than goals or results), and participate in ecologies (exceeding rational comprehension). If rationality were universal and shared, all humans would be machines that calculate the odds of going to places.

8.2 Cognitivism

The reasonable assumption that the human nervous system is implicated in how humans interface with their outside world cannot, however, turn a science for design into a cognitive science, which arose out of behaviorism and is now becoming an increasingly popular umbrella term for computational linguistics, psychology, ergonomics, and artificial intelligence. "Cognitivism," according to Lucy Suchman, "is the general doctrine that behavior can be explained by reference to cognitive or mental states. Cognitive science (theory) is the specific form of cognitivism that seeks to develop explanations of conduct in terms borrowed from computer science. Cognitive science thus deploys a model of human action that … is now being reified in the form of new computational artifacts" (Suchman, 1985).

Before considering the relevance of cognitive theory in design, it is important to realize what its proponents do and what they do not do.

- Cognitive scientists do not observe mental activity — although some have consulted magnetic resonance images (MRIs) with uncertain results.
- The concept of mental activity is constructed in the discourse (expert language) of cognitive theorists, and evidence of mental activity is obtained largely from linguistic accounts by individuals about how they think, see, understand, and decide, occasionally supplemented by observations of human behavior that is thought to result from mental activity.
- There is a commitment to computational explanations of mental processes, as Suchman points out above. Computations are assumed to be undertaken by the neurophysiologic mechanisms of the brain. Without actually studying these mechanisms, the mere belief that all humans are endowed with the same computational capabilities is extended to the belief in the universality of human cognition.*

* A good example of this reasoning is Noam Chomsky's construction of a universal language (UL) or universal grammar that all humans are assumed to be genetically endowed with and of which all natural languages are taken to be special cases. This formal abstraction posits that language and cognition are independent of practices of living, embodiment in individuals, communication or interaction, and cultural meanings, to which one could add processes of designing the physical realities that humans interact with. By contrast, the view accepted here is that the process of using language, languaging, much like an ecology of technological artifacts, is under continuous reconstruction by a multitude of interacting and more or less creative users and changes "as we speak" and act. Artifacts cannot be explained by individual cognition and by a theory of language that is independent of how language participates in the construction and reconstruction of the worlds of their speakers.

- The mental processes that cognitive theory ends up explaining are stripped of their social variables (Coulter, 1983), situational nature (Suchman, 1987), dialogical character, and second-order qualities.
- Cognitive theory is fundamentally individualistic, and mono-logical, that is, committed to a single consistent (computational) logic. It has no place for dialogic conceptions (Bakhtin, 1984; Bohm, 1996; Buber, 1958; Shotter, 1993), which inform the dynamics of interfaces with artifacts.

Users' conceptual models (UCMs) come close to what cognitive scientists aim to construct. But the semantic turn uses UCMs to explain how the circular process of sensing and acting adjusts to what particular artifacts afford (Section 3.4.1), not as explanations of how the mind works. Constructed in and from ordinary language use, UCMs naturally embrace human agency, plans, and situated actions, which the unilateral determinism of computational theories cannot address. Cognitive theories may well contribute to explanations of interfaces between individual users and artifacts, which Chapter 3 develops into a theory of meaning of artifacts in use, but it can make such contributions only where human–machine interaction is uncreative, repetitious, and unchanging.

The semantic turn in design attends to several kinds of meanings, none conceived of as mental representations. Its concepts of meaning are tied to alternative ways of seeing, to multiple descriptions, but always in conjunction with affording certain actions. Meanings arise in and in turn guide human interfaces with artifacts, and they enter and populate the communication with fellow human beings. Notwithstanding claims to universality, cognitivism has nothing to say about the meanings of artifacts in language, in their life cycles, and in their ecology of other artifacts. The semantic turn moved far away from mono-logical theories and avoids mentalizing or cognitivising linguistic, cultural, political, and technological phenomena. It suggests that meanings emerge when at least two "logics" come into contact with one another, regarding interfaces, the technologic of an artifact, and the conceptual world of a user. Mostly, two are not enough, for instance, when bystanders and stakeholders enter, and ecologies matter.

8.3 Ergonomics

Ergonomics emerged in the 1960s as the discipline aimed at improving the efficiency, economy, safety, and usability of designs: making chairs less tiring; finding the best position and shape of handles; determining the optimum lighting conditions for reading controls; suggesting principles for designing computer interfaces that are usable, that is, simple to learn, efficient in use, easy to remember, and causing a minimum of errors (Nielson, 1993). Ergonomics is committed to developing scientific, that is, objective and general standards for the design of technological artifacts, systems, and environments for human use. One of the earliest contributions of ergonomics was the publication of measurements of the sizes of humans in various positions,

compiled from military records for designers to comply with (Dreyfuss, 1960).

The aims of ergonomics and of human-centered design seem close. Both focus on human interaction with technology. Papers on product semantics have been presented with increasing frequency in conferences on ergonomics. The following looks into crucial differences between the two approaches, starting with their respective histories.

1. **Ergonomic research is at home largely within hierarchies of assigned responsibilities**. Ergonomics was funded initially by the military, and later by industry and large bureaucracies. It gathered its data, developed its methods, and made design recommendations in such situations as pilots flying fighter planes, tank crews operating within small spaces, workers on assembly lines, and operators of atomic power plants whose mistakes could have disastrous consequences. The human subjects that provided the data for ergonomic studies were soldiers, factory workers, or employees who were trained to do a job and could be removed if they failed. This social situation is virtually replicated in controlled psychophysiological experiments, which is the preferred source of ergonomic data. In them, experimenters are in charge of what human subjects are expected to do and observe the effects of their varying instructions on compliant subjects in the experimenters' terms.

   Human-centered design, by contrast, evolved in efforts to develop marketable products where stakeholders surface on their own accord; buyers and users exercise their choice of getting involved or not, applying their own criteria; advertisers interpret what they are interested in marketing, and consumer advocates speak freely about the nature and usability of products. Obviously, ergonomics and the semantic turn are concerned with very different worlds. Given the sponsors of ergonomic research, it made sense to adopt their clients' performance criteria and job descriptions, which were handed down to the operators of usually expensive equipment. Data generated under these conditions reduce human subjects to functional components of social hierarchies they are employed, paid for, and expected to support. For human-centered designers, by contrast, it makes no sense to rely on data that were generated under conditions in which an authority spells out performance criteria that users dutifully comply with as best as they can. Data generated under these conditions reflect the degree to which subjects can comply with what is demanded of them. In markets, in the public, and in everyday life, there are few authorities that can tell people what to do. Designers face stakeholders who cherish their choices, their ability to scan their world for clues to make their artifacts meaningful in their own terms, and use if not redesign what is available in ways that suit them. Ergonomic performance criteria are imposed

by an authority, scientists included, whereas meanings always are created by people as a way of managing themselves in their own world.

Moreover, the meanings of interest in the semantic turn are not restricted to individual users of single artifacts, to operators, workers, or consumers. While Chapter 3 addresses the issue of use, there are linguistic, life cycle, and ecological contexts in which artifacts find themselves. They are not addressable by cognitive theories and certainly not in ergonomic terms. While ergonomics thrives in hierarchically structured situations, the idea that meaning matters has already penetrated its walls, at least through the critical writing of two of its scholars (Flach and Bennet, 1996), and is undermining the comfortable stability that ergonomics heretofore enjoyed (Krippendorff, 2004b). Human-centered designers need to treat ergonomic data and the theories arising from them with considerable caution.

2. **Ergonomics constructs humans as the unreliable components of technical systems**. Ergonomics, also known by its earlier and less scientific-sounding name "human engineering," is still dominated by engineers with technology-centered views (Figure 2.1), which leads them to celebrate the orderliness and determinism of technical systems and treat user errors without contempt. Consequently, ergonomic research focuses on ways to immunize technical systems against the effects of the unfortunate flaws, imperfections, and biases of human operators for fear of ruining the systems or harming themselves. The paternalism that underlies this conception should be apparent. Virtually all human performance measures in ergonomics assess deviations from ideal performances, which invariably are mechanical ideals: unquestioning obedience, predictability, efficiency, and perfect reliability. Writing from within ergonomics, John Flach observed that human performance measures amount to measuring human responsibilities for systems failures, which is paramount to denigrating the human operator (Flach, 1994).

   True to its tradition in engineering, ergonomic research prefers objective, that is, observer-independent measures of the function that humans are expected to perform, for example by clocking the time for performing an operation and measuring energy expenditures or efforts, number of steps taken to complete a task, comfort levels, and especially error rates. There is the prevailing belief that without quantitative measures the phenomena of interest are not real. Such measures deliberately, preferably, or at least implicitly ignore users' conceptions, for example, by training operators to perform a task in prescribed ways, leaving users unaware of how their performance is being assessed, or, when asking questions and recording answers, using standardized question–answer pairs that reflect the ergonomists' conceptions and interests, not the interviewees'. Users' or operators' conceptions and concerns have little currency in this research tradition.

Product semantics, the precursor of the semantic turn, recognized early on that people acquired products for reasons that are not necessarily anticipated by their designers. Chairs, a favorite target of ergonomic inquiries, are hardly ever bought for ergonomic reasons alone. People surround themselves with furniture that means something to them, whether it reminds them of something, demonstrates social status, is in style, impresses others, or simply fits with other artifacts at home. Although chairs need to afford sitting, users are willing to sacrifice ergonomic efficiency for what they understand, and like, and are eager to talk about in their own words. Ergonomic considerations become important only when people consider them important or, as stated above, when they can be instructed or are paid to perform accordingly. No doubt, there are plenty of industrial and military applications where ergonomics matters, but in a market economy and information society, ergonomic data have become weak sales arguments. In human-centered design, ergonomic findings deserve no privilege.

3. **Semantic issues precede ergonomic issues**. In everyday settings people approach artifacts by first recognizing what they are and could do with them, then exploring how they might be handled, ideally leading to reliance or use (Section 3.2; Figure 3.4). Ulrich Burandt, who taught ergonomics to designers for years and published a widely used text (Burandt, 1978), observed that semantic considerations always precede ergonomic considerations. The meanings of technical artifacts must be understood *before* human performance measures make sense, if they do at all.* For something that is not self-evidently a handle, its ergonomic efficiency (as a handle) cannot become an issue, and if it is recognized to be a handle and used that way, ergonomic criteria may not become relevant either. If one needs to tell people what something is and teach them how to work with it before evaluating users' efficiency, the resulting measures are not generalizable to situations in which training is absent and artifacts are seen or used in ways other than anticipated by ergonomists. No doubt people can temporarily suspend their own judgments and blindly comply with given instructions, but humans work best when they succeed in figuring out what artifacts can do for them, follow their own motivations, and set their own criteria of satisfaction. If ergonomics is to contribute to human-centered design, its data have to be tied to how users conceptualize their world, what motivates them to interface with particular artifacts, and the criteria they apply to themselves.

* Ulrich Burandt, personal communication, April 1994, Essen.

8.4 Aesthetics

Design was born into an ambiguous relationship with aesthetics. Traditionally taught as an applied art, design always had one foot in the arts and the other in technology. Today, most designers are still consulted at least in part for their aesthetic sensibilities but end up solving larger problems. The semantic turn touches aesthetics at several points but then takes a distinctly different route (Section 4.3).

To begin, the classical project of aesthetics* was to find a theory that could explain the sense of beauty. But it soon sought such explanations in the objective qualities that make artifacts, but also objects of nature, pleasing, attractive, or charming, for example, proportions, geometries, colors, harmonies, movements, meters, melodies, and gestalts. Since Aristotle, aesthetics is considered a branch of philosophy, but it has always been of interest to artists, designers, and architects in search of guidelines where rational considerations seem to fail. Over the years, aesthetic theories have come and gone, casting doubt on the possibility of a universal theory.

1. **Theories of aesthetics have the effect of stabilizing the use of aesthetic vocabularies in society.** Some theories of aesthetics have endured for longer periods, and their vocabularies have inspired some artists, connoisseurs, designers, and critics more than others have. But, on closer examination, they endured not because they were right but because concurrent social institutions had a stake in them and the means to promote their normative use (Section 9.4). What was said about polysemy in semiotics (Section 8.1, item 3) applies to the stability of aesthetic theories as well. The nexus of the art world — artist communities, art museums, art dealers, collectors, educational institutions, including schools of philosophy — can mount considerable resistance to changes of a prevailing aesthetic consensus. Yet, there are always artists, designers, and cultural critics who challenge the aesthetic conventions of their time or genre by not only creating novel artifacts but also by talking about them in novel ways. Challenging and destabilizing the existing aesthetic consensus or proposing new conventions is a political process. Theories stay in circulation because they are convincing and of use to those involved. Thus theories of aesthetics, while claiming to describe and generalize certain qualities of artifacts, in effect fix, encode into propositions, and selectively stabilize the use of aesthetic vocabularies in society. Although aesthetic theories can inspire

* Actually, the term "aesthetics" was coined by the late seventeenth century German philosopher Alexander Baumgarten, who derived it from the Greek "aisthetikos," pertaining to sense perception. With this coinage, he wanted to designate an area of scientific exploration that does not follow strict logical reasoning. This project entails the paradox of searching for a logic (a syntax) to describe something that is defined as defying logic. Unfortunately, in the history of aesthetics this paradox came to be settled in favor of logic and of accounting for objective qualities presumed to cause senses common to all humans at the expense of the actual experiences to be explained. Contemporary aesthetics is still cursed by this turn.

for a while and grant artists and designers freedoms within their boundaries, their normative use can also stifle creative practices. In sharp contrast, human-centered designers see themselves embedded in stakeholder networks, which are inherently political, not governed by an overarching theory.

2. **Theorists of aesthetics blind themselves to the linguistic nature of their theories**. Stabilizing vocabularies, fixing aesthetic attributions, and not reproducing what could challenge the dominant networks of institutions ties theories of aesthetics to language, to prevailing linguistic conventions. The class of adjectives that aesthetic theorists theorize varies with genre, but invariably includes the synonyms and semantic relatives of the word "beauty" and their opposites. Aesthetic theories systematize this vocabulary and are bound in this effort by what their users — philosophers, artists, cultural critics; their discourse community — can understand and apply. Thus, theories of aesthetics cannot escape the language in which they are expressed. And yet, by attending to the qualities of *artifacts*, not to their own attention, theorists of aesthetics tend not to recognize the linguistic and political nature of their theories.

   The semantic turn, by contrast, highlights and refocuses the attention of designers on how people describe their senses, experiences, and active involvement with their artifacts. Human-centered designers will have succeeded in creating beautiful artifacts, should this be an aim, when their stakeholders *say* in words or in deeds that they are "beautiful" to them. Whatever the character of an artifact is intended to be, verbal accounts confirming that this character has been achieved is all one can hope for.

3. **Theories of aesthetics are limited, not only in what they explain, but also by merely theorizing**. Objectifying what it is that elicits aesthetic perceptions effectively eliminates the human element, which is central to human-centered concerns. Subjective accounts of inner processes, by contrast, have no meanings for those who have not had the experiences they describe. The latter can hardly serve as communications to designers who are likely to have experiences that are different from those who use their design. By addressing interactions between the two, the semantic turn provides not only accounts of embodied phenomena that arise in human interfaces with artifacts, acknowledging the role of senses, feelings, actions, and affordances, but also reliable design methods for converting aesthetic attributes into new designs (Section 7.4.1). Theorizing a phenomenon puts theorists and the readers of theory into the position of detached observers, intent to categorize objects, in this case, into aesthetic and non-aesthetic ones, into art or not art, or into whether the attribute of beauty applies or not. In the notion of language adopted here, making attributions does not separate the use of vocabularies from embodied experiences and actions. The semantic turn treats senses and

feelings not as abstractions (statistically removed from a body) or as something that could be designed into products (as Kansei engineers like to think they can), but as what can be expressed in words and in deeds. It also does not assign aesthetic attributes a privileged status among other vocabularies by which people organize their worlds. With a method in hand for converting attributes used and understood within a community into a design for this community (Section 7.4.1) theorizing the use of this vocabulary would stifle empirical inquiries.

8.5 Functionalism

Functionalism is part of enlightenment's rational agenda. Rationalism claims that human knowledge and action are obtained by reason from purposes that are ultimately innate properties, according to Descartes, of mind (in distinction to religious beliefs, of God). Rationalism is contrasted with empiricism, according to which knowledge is based on experiences, not entirely reducible to reason. This contrast resurfaces here in the difference between technology-centered design and human-centered design (Chapter 2). In design, the commitment to functionalism is enshrined in the common phrase "form follows function." As a proposition, it insists that the form of artifacts is logically derivable from, and hence subordinate to, a clear understanding of its function. Frequently mentioned exemplars for the relationship between form and function are the aerodynamic shapes of airplanes verified in wind-tunnel experiments; bridges whose girds follow mathematical stress lines; and the orderly arrangement of pipes at chemical plants. Functionalism's claim to universality is evident in the conviction that arbitrary relations between the form and function of artifacts are due to a lack of understanding of the function that a design is to serve. Adopting such beliefs entails moving in a direction designers may not want go. Following are three such entailments and suggestions for why human-centered designers may want to avoid them.

1. **Functions result from a logic of conceptual subordination**; they have no reality of their own. Propositions like "the *function* of A is B," "A serves the *function* B," or "A *function*s as B" presuppose a larger whole within which A *serves* in the capacity of B. In theater, we judge the performance of actors by how well it supports the performance as a whole. What matters is not who the actors really are but how well they can hide their identity behind the roles they are supposed to play. In theories of biology, a function is the particular contribution that any one part of an organism makes to the well-being of the organism, ultimately its living. For engineers, the intended function of an artifact is the very reason for engineering it. That intention silently implies taking the larger whole in which the artifact is to take its proper place for granted, as not in question. This is true whether one speaks of the

function of a screwdriver, which is part of and subordinate to a system of using fasteners; the function of a bridge, which is part of and subordinate to the system of cars and roads and the problem of crossing rivers and valleys; or the function of a communication satellite, which is part of and subordinate to a larger communication system within which it serves in its assigned capacity.

Part–whole relationships have been applied to many systems, to the hierarchy of biological entities: living organisms, organs, cells, macromolecules, and organic compounds (Simon, 1973) and social hierarchies, bureaucratic forms of organization, for example. Functional sociology (Parsons et al., 1961), which proposes a conceptual system based on the distinction between the functionality or dysfunctionality of individual acts, has been aptly criticized for preserving the status quo, the social standing of a society's governing elites, in effect demanding the subordination of all social activity to the preservation of the system.

Part–whole distinctions, it must be realized, are distinctions made by an observer. The systems of subordination that result from such distinctions are conceptual. Functions cannot be measured or photographed, and they do not exist without someone conceiving them as such and applying their conception to the larger whole they want to preserve. Functionalism is the mistaken belief in the reality of systems of conceptual subordination. It might facilitate the engineering of machinery. It may help managers to cope with the complexity of social organizations, but it can also have disastrous consequences when confused with reality. Functionalism is conservative of the conceptual hierarchies it takes for granted. Designing something to serve a particular function limits the space of available solutions to those satisfying the given function. Human-centered designers should be aware that talking of functions is talking within the limits of a chosen system of subordination and conceiving artifacts under the auspices of such a system can have major social implications.

2. **Functional hierarchies are giving way to political heterarchies.** In the transition from industrial era technological determinism to human-centered design, logical, functional, and organizational hierarchies are being replaced by networks, heterarchies, which are relatively flat forms of organization, informed more by politics than by authority or logic. This is not to say that functional hierarchies are out of fashion. Complex technological systems may well be built that way. There are economic advantages to combining elements into parts into subassemblies and into larger entities, etc., which eventually reach the taken-for-granted whole. Simon (1973) demonstrated the value of hierarchies of complexity, as he called them: In systems of component parts, replacing a defective part tends to be easier than rebuilding the whole.

In society, functionalism leads to various forms of oppression and the justification of many inhuman acts.* It is no surprise that functionalism in design emerged during the industrial era, which was governed by hierarchically organized authorities (Section 1.3.1). Karmasin and Karmasin (1997) provide a cultural analysis of different societies to show how concepts of design correlate with the kinds of societies in which they operate. In the market- and communication-driven postindustrial era, functional hierarchies increasingly are becoming an anachronism. New theories have emerged, largely in cybernetics, of complexity, emergent structures, self-organization, and autopoiesis, in which *wholes* have come to be explained as *constituted in the networks of interactions among parts*, not as superordinated to them. The Internet, for example, could not be conceptualized as a whole or designed for what it has become. It evolved thanks to millions of creative contributors. It is a heterarchical system and usable precisely because of its nonhierarchical nature. Similarly, interactive media are making major inroads into the previously dominant mass media, characterized by one-way communication. The logical hierarchies that functionalism instituted as real are subverted in many areas of everyday life, by engaging in cooperation and dialogue rather than looking for approval from higher authorities; by no longer accepting the excuse of having been ordered to do something unethical, even when a soldier; by becoming involved in networks, distributed organizations, ecologies, and self-organizing systems. The emphasis on stakeholders in design recognizes the fading functionalism.

3. **Functionalism entails a methodological monologism that leaves no space for human beings to enter.** Mathematical problem solving techniques (Simon, 1969/2001), for example, start from a functional objective and use algorithms that select the best "servant" of the objective. Once the algorithm is in place, no other logic can enter, and alternative conceptions of the problem or objections against unanticipated solutions have no place. Mathematical approaches, while basic to engineering and technology-centered design, cannot embrace diversities of meanings that heterarchically organized stakeholders may consider relevant. Imposing mathematically perfect solutions where meanings rule supreme often create pseudoscientific arguments. For example, automobile designers have invoked aerodynamics since the 1940s. At first, "aerodynamic" was the character attribute of forms that resembled the then considered striking photographs of water drops and the

* Herbert Brün (personal communication) characterized fascism as the belief that humans are not only subordinate to social systems but also dispensable in the system's favor. In the social domain, functionalism is indeed very close to if not an essential ingredient of oppressive regimes. It is a way of languaging that ends up privileging abstract purposes over practices of being. Inasmuch as artifacts are human accomplishments, serving the needs of many, not just of their designers, adopting a functionalist perspective regarding human conduct almost always invited oppression.

fins of airplanes, and it created streamlined products associated with speed and sports. When actual wind-tunnel experiments entered functionalist automobile design, the appearance of being aerodynamic proved quite different from the measured aerodynamic efficiency. Now, despite years of wind-tunnel tests and aerodynamic experiments that reduced the driving noise, the forms of automobiles are still variable. The argument of aerodynamic efficiency has given way to cultural meanings. Spoilers turn out to be fashion statements of uncertain aerodynamic value. Whereas human-centered designers acknowledge individual variations in perception, cultural differences, and ecological diversity, technological-centered designers consider this multiplicity as deficient, incoherent, technologically inefficient, or misguided subjectivism. The functionalist's monologism has lost much of its past relevance.

8.6 Marketing

Marketing is concerned with the behavior and management of markets, with monitoring the sales of goods and services, predicting trends, and developing strategies for launching new products, including creating advertising campaigns and evaluating their successes. Marketing has also served other causes: political elections, fundraising, and the promotion of sports. Marketing provides statistical data based on surveys, focus group discussions, psychological experiments, pretesting of products, and records of actual transactions, including profits. Because of its concern for the point of sales, at which the funds are generated that fuel industrial production, marketing was once considered an arm of industry. But it has lately emerged as a force of its own, driving industry rather than serving it. Because design is recognized to make a difference in sales, some marketers consider design as an adjunct to their own marketing efforts. Following are three ways in which the objectives of design, especially of human-centered design, and marketing differ.

1. **Marketing attends to only a brief moment in the life cycle of artifacts**: the point of sale. At that point, artifacts are merchandise. They change from corporate to individual ownership and much information about their possible futures come into play: trust in brands is an issue, the ability to identify with a product becomes important, the cost of acquisition is compared with their anticipated utility, safety, usability, repair records, social norms of use, and ecological impact. In justifying their designs, designers usually are concerned with these issues. Since, at the point of sale, these purchasing factors are anticipatory, it is important for designers to visualize them where possible. Market research can provide a service in finding out which criteria are statistically significant.

From the perspective of marketing, the point of sale is the only phase that designers need to address. According to Karmasin (2004), the purpose of design is to "add value" and "distinguishability" to products. However, from the broader perspective of design taken here, all meanings that advance artifacts through their life cycle are important (Chapter 5). For designers, the point of sale is as much a bottleneck as any other transition through a stakeholder network that an artifact may have to undergo to come to fruition. Information matters at any transition from one manifestation of an artifact to another. The semantic turn makes numerous devices available to carry a design beyond the point of sale: successive meanings, semantic layering, and polysemy, for example. The fact that buyers might not grasp or be interested in everything that could happen following their purchase is no reason for designers to limit their concerns to what buyers care about at the point of sale or what marketers want them to consider. Pretentious semiotizations, inflating the value through appearances, introducing insignificant differences, adding meaningless options, and making unrealizable promises may help sales, especially since buyers may not be able to verify sales promises, but they may ruin valued trust in similar products, the reputation of those that could be blamed for it, and come to be seen as the causes of long term decay of culture and the environment. Human-centered designers cannot afford to be so limited.

2. **Marketing marginalizes unprofitable populations**. In economics, a market is a metaphor for statistical supply and demand curves, which are abstracted from a multiplicity of observed or anticipated transactions. The market of marketing occurs in language, not on the street. To optimize profit, one task of market researchers is to locate potential buyers. In marketing, "THE buyer" of a particular product is not merely a convenient stereotype as "THE user" is for traditional designers (Section 2.4); it has the statistical reality of a category of people that could yield the largest sales at the lowest advertising cost. This category occupies the center of a normal curve. Focusing on this abstraction has the effect of ignoring people on the outliers of normal curves, people that have a lower probability of becoming buyers, are more difficult to convince, less well off perhaps, or handicapped. From a marketing perspective, this is the right thing to do. Human-centered designers cannot ignore marketing. After all, it has a legitimate stake in design as well. But, design is never about statistical abstractions, about number of buyers, but about the organization of affordances that move an artifact through the network of its stakeholders. Details matter, whether there are many users or few. For designers, therefore, frequency distributions of population characteristics do not matter as much as knowing of their diversity, the range of people that could come in contact with a design. The possibility of designing a product with multiple meanings and with affordances for different kinds of stakeholders enables human-centered

designers to address this diversity and counter the marketing logic of limiting attention to the profitable majority of a population. These semantic possibilities permit designers to make products available to a broader spectrum of people without necessarily incurring additional costs. This possibility probably is the reason why designers have a long history of considering marginal users in design considerations, enabling the unusual, attending to the poor, and overcoming the constraints experienced by the disabled. Designers need to look at marketing data, but differently.

3. **Statistical decision makers are bad designers**. Locating profitable markets for existing products is one thing. Inquiring into consumer expectations and motivations to write design specifications is quite another. Naturally, concerned with large numbers, the preferred method of gathering data about potential markets is large surveys. To be analyzable, survey questions and answers have to be comparable, that is, standardized. The questions that interviewees are asked must be understandable by everyone, and easy to analyze, preferably using multiple-choice answers. But above all, the questions that interviewees are asked are those of the marketers or their clients, not those of the interviewees. They cannot reveal anything that marketers did not anticipate, except for the frequencies within the marketer's categories. There is a history of failures when statistical majorities decide on the direction that a design is to take. The story of how market research failed to recognize the virtue of a revolutionary new design, Philips' roller radio, has already been told (Section 7.5; Blaich, 1989). The most celebrated example of a marketing failure, the 1957 Ford Edsel, was designed after the most thorough consumer research ever undertaken at that time. Design, by definition, always breaks new ground and introduces something unexpected, unconventional. The methodology of survey research is inherently blind to newness, whether it concerns changes in the consumer perceptions or thinking about new technologies. Untrained consumers rarely have the foresights that designers are expected to have and those that do easily fall through the filter of statistical analyses. Technological culture is always changing. Consumers learn all the time and abandon existing practices when presented with better options. Motivation researchers who faithfully record what people prefer today have difficulty anticipating what they will appreciate in the future. In the 1950s, a governmental survey is said to have determined that the world will need at most about five computers. It took a few designers to change that world against all prevailing odds. The art of design rests on knowing which data to distrust.

Large survey data usually come to rather different conclusions about the direction a product development should take than smaller collaborations between designers and stakeholders in a design, whether they are focus groups or participatory design processes. To be fair, market research has

used other methods to cope with newness. One method is to launch trial balloons, futuristic designs, for example, so-called concept cars at automobile shows, and listen to what potential users have to say about them. The management of fashion provides another way. New fashions tend to be introduced, not after conducting surveys for what is sellable, but by securing several designs and introducing them in stages, first to highly visible fashion leaders who act as both filters against fashions that are not likely to succeed and as models for a secondary target population to see. Fashions are passed on from smaller to larger communities of users until they are purchased in common department stores. In that stepwise process, information gathering is coordinated with design modifications, not independent of it, and data are gathered not by large surveys but by listening to articulate stakeholders. Participatory design (Section 7.4.5) is a third alternative to market research. It saves market researchers the uncertain effort of extrapolating from past surveys how a larger population of potential buyers may react in the future. It provides information about what more or less knowledgeable people can envision beyond the point of sales, and it introduces that information into the design process that is collaborative and timely.

8.7 Textualism

In search for a metaphor that avoids technological determinism, acknowledges the linguistic and social nature of artifacts, but maintains a distinction between what is uniquely technological and what is not, the "reading of text" has become an increasingly attractive candidate: Texts are composed by writers, who may speak for themselves or as parts of social institutions. Texts have some kind of materiality, they can be reproduced photographically, for example, but are meaningless unless read by someone — by readers but also by their authors who always are the first reader of a text. While writers compose their texts in anticipation of being read by others, there is no way to assure that texts are read as intended. Within culture-specific constraints, most texts afford numerous readings. In the social scientific literature, an increasing number of publications employ the metaphor of reading text.* The analogy: "a text is to an artifact as its reader is to its user" is attractive to designers, though limiting as follows.

1. **Textualism locates meaning in texts.** Like texts, artifacts are created by designers and producers, and their composition entails constraints on what they can afford. But the affordances of texts and of artifacts differ, and textualism and the semantic turn in design take different routes. The understanding of texts is demonstrable by rearticulation, by readers offering appropriate interpretations of their reading. Interpretations occur in language, as does the interpreted text. Judges of the

* For example, Brown (1987): *Society as Text;* Grint and Woolgar (1997): *The Machine at Work: Technology, Work and Organization*; Silverman and Rader (2002): *The World Is a Text;* Widdowson (2005): *Text, Context, Pretext.*

appropriateness of a reading find themselves in the position of a reader as well. Therefore, there is no way for text analysts to step outside the literary and cultural competencies they presume to share with readers.

The understanding of artifacts, by contrast, is demonstrable by interfacing with them: selecting one artifact among others, handling its controls, guiding its operation toward desirable ends, and especially redesigning, reconfiguring, reprogramming, or repairing it according to one's own conceptions. There is no way to observe *what* the users of an artifact understand without asking for their accounts of it, but it is possible to get an inclination of that understanding by observing *how* users unfold their understanding into interfacing with these artifacts, moving from recognition to exploration and to reliance (Figure 3.4). Thus, whereas textualism locates the meanings of texts in their interpretations, the semantic turn locates the meanings of artifacts within the dynamics of an emerging interface with them and through them with other human beings. The meanings of artifacts are not merely interpreted; they can also be tested by enacting them and monitoring the successes of these actions in one's own terms. Verifying the meaning (appropriate interpretation) of texts requires a literary authority, a dictionary, for example. Verifying the meaning of artifacts requires an additional involvement on a nonlinguistic level. Continued reliance on artifacts replaces textual authority.

2. **The theory of product language.** One approach to meanings in design has been developed at the Hochschule für Gestaltung in Offenbach. The approach taken probably stems from the same dissatisfaction that fueled early product semantics. The theory of product language (Steffen, 2000), as their principal authors, Richard Fisher and Jochen Gros, have called it, consists of a vocabulary that resembles that of the semantics of artifacts in use (Chapter 3), but departs from the more comprehensive semantic turn in several ways. Among the many points of agreement is the concern for the meanings of artifacts; the opposition to the formula of "form follows function," which produced geometrical shapes that are oblivious to meanings (Gros in Steffen, 2000:12); the attempt to adopt meanings as the defining focus of design (Gros in Steffen, 2000:13–14); and the acknowledgement of semiotics as not helping the understanding of the meanings of artifacts. The theory allies itself instead with hermeneutic and phenomenological conceptions of meanings (Steffen, 2000:23–26), two disciplines that evolved from analyzing texts or analyzing phenomena as texts, textuality for short.

   The chief objection to the theory of product language concerns its conception of language as a system of signs and symbols* (Section 4.1).

* Gros (1987:8) replicates the abstract-objectivism of Saussure's (1916) outline for linguistics, which deliberately omits the speaking of a language, its embodiment in speakers and listeners, and above all what poets and designers do, respectively, creating new ways of speaking and new interfaces with the artificial world.

Its premise is that products, their makeup and composition, can be designed to tell their stories, stories about their mode of production, purpose, history, use, and location in time and in society (Fisher in Steffen, 2000:17–22). Even textualism would not go so far. Technological artifacts, just like texts, say nothing and mean nothing on their own, except what their stakeholders grant them to mean. *Products cannot speak a language, humans do.* Adopting this theory of language diverts attention from the human use of language — in everyday life as well as in collaborations among designers — to abstract accounts of the properties of products. It *ignores how artifacts are languaged into being in favor of what a theory considers them to be*. Three points may illustrate this confusion.

- In the theory, **functions are considered as sensory in nature**. Superficially, this move makes the theory human-centered, at least in this respect, and bypasses some of the undesirable entailments of functionalism (Section 8.5). It starts by distinguishing between two kinds of sensory functions, formal aesthetic and sign functions. The former concerns perceptions of syntactical features, grammaticality, or textuality. Steffen pays much attention to the visual pattern and gestalt principles governing this function and offers examples for what this function is meant to describe (Steffen, 2000:34–62). Here, serious epistemological problems surface: Forms are never seen as such. They are the abstractions from the complexity of real images or objects by analysts who are trained in geometry and similar descriptive devices, which may not be shared by those being theorized. Since an individual cannot see what others see, the idea of a sensory function, of making the analyst-defined formal or syntactical features responsible for someone else's perception, elicits epistemological doubts. Taking textualism seriously, the only reasonable access to someone else's perceptions is through obtaining verbal accounts of them. But language not only reports, it also affects perception. To the extent aesthetic sensibilities are acquired, verbal accounts of aesthetic perceptions are social or cultural, not to be confused with what reaches the senses. The authors do not address such questions and do not say whose interpretations they discuss, presumably their own. The semantic turn resolves these quandaries by conceptualizing meaning in the relationship between designers' and users' accounts (Figure 2.8; Krippendorff, in press). This acknowledges the social role that language plays in perception (Chapter 4), including for designers.
- In the theory sign functions are distinguished into two kinds, **indicator functions** (Fischer, 1984) **and symbol functions** (Gros, 1987). Indicator functions make the "practical functions of a product visible and understandable." They are quite compatible with the informatives (Section 3.4.5) defined in conjunction with use to which

Steffen's (2000) account adds numerous instructive examples. Symbol functions pertain to the product's "cultural, social, technological, economic, and ecological meanings." They do not explain how the product might be handled or used and instead "point … to the super ordinate societal contexts" of the products in hand (Steffen, 2000:82). The authors of the theory suggest that symbol functions are responsible for likes or dislikes, and the attribution of such character traits as "bourgeois," "modern," and "avant-garde" to them, indicating memberships in product families, or identifying styles, whether of periods, nations, corporations, designers, or target groups (Steffen, 2000:82–95). Defining symbol functions by reference to the superordinate contexts of a product reveals the functionalist predilection for hierarchical conceptions (Section 8.5, item 2). Assuming that it is the products that speak to their users, the authors are led to focus attention to the sensory qualities of artifacts and be silent about the interpretations offered by different people. The ability to recognize the style of a product may be of academic interest, but it is not clear what difference its correct identification makes in ordinary users' lives. In contrast, by starting from the observation that stakeholders bring their own conceptions to their interfaces with artifacts, and create their own meanings in the course of their involvement with them, the semantic turn leads designers to look for the conceptual diversity of users and moreover provide designers at least four contexts with radically different meanings: the context of use, of language, of life cycles within stakeholder networks, and of the dynamics of artifacts in an ecology.

- All theories highlight some features in the world while blinding their theorists to others. One **blind spot** of the theory of product language is a core concept of human-centered design: **second-order understanding**, designers' willingness to listen and make an effort to understand diverse stakeholders' understanding — without attempting to impose a theory on them. It is inconceivable how products, when conceived of as compositions in the terms provided by a product language, could provide a space for diverse stakeholders to enter their own conceptions, which is what human language is capable of providing naturally.

3. **The container metaphor of meaning revisited.** A related approach to meanings in design, developed earlier by Siemens AG, München, Germany, deserves some attention. It consists of in-house guidelines (Käo and Lengert, 1984), written to improve communication between designers and their teammates in product development efforts, which is what the theorists of product language might want to acknowledge as an important aim. It is of interest here because it adopts an older but still widely used metaphor of understanding text, the container metaphor of meaning. Just as in much of literary scholarship, which

routinely analyzes texts with the help of the form–context distinction, the Siemens guidelines assume that products have forms and contents as well. Form, they explain, is what designers can describe without recourse to context and meaning, in terms of geometry, pattern, grammar, and arrangement, for example. Content is what these forms are capable of conveying to the users of a product. The guidelines further distinguish between two kinds of content: functional content (technical and use) and value content (addressing needs, expectations, style, and the product's image, which is to be conveyed). The similarity of this distinction with the distinctions between formal–aesthetic and sign functions, and between indicator and symbol functions in the theory of product language is striking, but the two approaches use dissimilar metaphors and therefore invite different problems to be dealt with.

All metaphors have entailments (Sections 3.3.2 and 4.5), mostly unnoticed, but acted out. The container metaphor renders forms as in-itself-meaningless vehicles capable of containing meanings or information, according to the guidelines about the makeup, use, and value of products. This metaphor renders contents as an entity that designers could enter into their container, to be shipped to a user, who removes it from its vehicle, and puts it to some use. Being considered a container, folk physics suggests that one could remove from a product form or message only what its sender, the designer of the product, had put into it. While ordinary users may well see artifacts as undivided entities, it is the talking about these artifacts in terms of the container metaphor that makes the form–content distinction quite real and shifts their user's attention from trying to figure out what it could afford them to what its designer or producer had intended (Section 8.5, item 1). It is this metaphor that puts the designer or producer of a product in the position of an authority on what their designs are to mean and encourages users to give up their ability to create their own meanings, uses, and unique ways of interfacing with them. The guideline's use of the container metaphor makes it difficult to see how stakeholders in a design conceptualize their worlds. Conversely, if this metaphor were to enter the language of ordinary people, it would encourage them to figure out exactly what the forms of products do contain, i.e., what designers had intended them to convey. The container metaphor, while a powerful organizer of perception, is not particularly conducive to understanding what designers need to do. One might as well say it is essentially misguided.

chapter nine

Roots in the Ulm School of Design?

Origins are hard to prove, and answers to questions concerning them offer at best arguable reconstructions. The following is no exception. Clearly, the semantic turn is not a mere extension of what was taught and deliberated at the Ulm School of Design,* where I studied industrial design from 1956 to 1961. For one reason, the culture of technology at that time did not require it as much as it does now. Ulm happened during Germany's reconstruction after WWII and during the transition from an industrial to a postindustrial society, a transition not recognizable at that time. For another, science, naively cherished in Ulm, has moved away from the then prevailing positivism. Finally and personally, I continued to study communication in the United States, became a professor at the University of Pennsylvania's Annenberg School for Communication, and am now weaving my understanding of human communication back into design, an understanding that developed largely outside and after Ulm. Although my ultimate involvement in communication theory and research can be traced to Ulm, product semantics was formulated only in the 1980s, and its growth into the philosophically grounded and social science informed semantic turn in design is a more recent phenomenon. Let me suggest where credit is due and where claims of origins might have to be qualified.

Surely, the Ulm School of Design was an intellectually unique and enormously creative place. It was also far less monolithic than it might appear from its outside and in retrospect. Its faculty was amazingly young and came from several countries. Numerous designers, architects, cultural critics, and scholars at the cutting edge of their fields passed through Ulm and were given opportunities to present their ideas and often worked with students for a while. Through these guests and a small but current library, new vocabularies entered Ulm's discourse. They were tested and contested, always in conjunction with particular design applications, and affected the curriculum unevenly.

* Hochschule für Gestaltung (HfG) Ulm, Germany, 1953–1968.

9.1 Bill's functionalism

Functionalism probably was the most deeply engrained vocabulary at the Ulm School of Design. There, much as in most design circles at the time, Louis Henry Sullivan's dictum "*form follows function*" served as a principle for rational justifications of designs. It asserted the conviction that once the function of an artifact was understood, its form would naturally emerge from that understanding, and conversely, if the form of an artifact had little to do with the function it was meant to serve, its function had not been understood well enough to start with.* We, the faculty and students at the Ulm School, relied heavily on this functionalist logic and tradition. We referred to THE function of cultural artifacts as if it was obvious that all artifacts had to have one or be worthless. There was no critical questioning of what a function was, where a function came from, and who or what defined the function in question. There was also no discussion or awareness of the conservatism that part–whole relationships entailed (Section 8.5). Accepting a function as the goal of a design meant not to question the whole of which it was to be a part, the larger context that the artifact was meant to "serve." For example, while designing tableware, eating practices are not questioned, much as while developing a radio, the whole system of radio transmission and programming is assumed given. Taking the larger whole for granted is, of course, a way to limit a design problem and get it done, but it is also a way to submit to prevailing conceptions that restricted spaces for design. More important, in Ulm, we reasoned with this functionalist dictum as if our vocabulary had nothing to do with merely seeing the world that way. Our discourse, and just that, made functions appear to be objective, amenable to scientific explorations, and the topic of seemingly rational discussions. Owing largely to the influence of Max Bill (1952) — founder, first Rector of the Ulm School, and on its faculty until 1957 — the vocabulary of functionalism became refined and ended up recognizing four functions: technical, material, production, and aesthetic.

1. **The technical function:** All designs were expected to satisfy their mechanical purposes (within a larger whole). Max Bill saw this as the primary function of products. The philosopher and long-time guest professor Max Bense called it the "technical function," one of three semiotic dimensions he believed all objects of design had to satisfy.** Reinhart Butter and I (Krippendorff and Butter, 1993) wrote critically about this functionalism, and the discussion of the logic of functionalism (Section 8.5) explored the implications of this part–whole relationship.

* As elaborated in Sections 1.2 and 1.3 and illustrated in Figure 1.3, miniaturization has made this design principle technologically obsolete or at least not generalizable to electronic artifacts. There simply is no natural form that could be derived from the miniscule circuitry. Often, the body of such artifacts reduces to a human interface or is absorbed in another device, in a computer, for example.

** Bense (1971:79, 81) considers design objects in terms of three semiotic dimensions: form, materiality and function and combines these with Peirce's (1931) triadic distinctions.

Thinking within established categories, obviously, a watch is expected to tell time, a chair is expected to carry the weight of a sitting person, and an automobile is expected to enable people to drive from one place to another.

2. **The material function:** This dimension entailed the obligation to use materials appropriately, in German, "Materialgerechtigkeit." This commitment tied Ulm to the design tradition of craftsmanship, starting with William Morris at the turn of the century. It encouraged designers to make use of materials in ways most fitting to their nature. For example, the Ulm stool in Figure 9.1 was used to sit on two levels, at a table or desk and more informally during discussions, parties, or lectures. It afforded carrying personal belongings, serving as bookshelves in the dormitory, and as a lectern on top of a table. It exhibited not only all the materials it was made of — the three flat pine boards, and the hardwood dowel and edges — but most important, it revealed and was meant to show the craftsmanship in their joints. Showing the (good) craftsmanship of their makers was considered a matter of "honesty" as opposed to "deviously" covering it up by other materials, paint, for example.



Figure 9.1 The Ulm stool — its form, true to its materials — by Max Bill and Hans Gugelot, 1954.

3. **The production function:** This function entailed the obligation to find forms especially suited to economic mass production, culminating in the demand that products should ideally express or at least not conceal their industrial origins, their means of production. This dimension went beyond the Bauhaus manifestos of the 30s, which sought to commit designers to make their products accessible to everyone, hence, mass producible. But in Ulm, this commitment to an egalitarian vision achieved the status of a function: the forms of artifacts had to accommodate the available means of industrial production. This function was

used to justify rectangular volumes, machined surfaces, repetitive patterns, and combinatorial systems over unique and artistically embellished products, for example, by celebrating the modularity of stereo equipment and prefabricated architecture. Figure 9.2 presents the design of a briefcase whose plastic shells were identical so that they could be produced from the same mold. When the briefcase was closed, the snaps complemented each other in perfect symmetry.



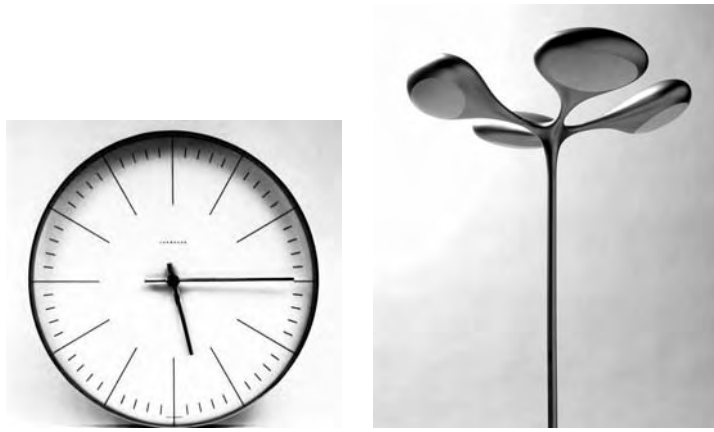
Figure 9.2 Plastic shell briefcase — identical shells in service of their mass production — by Peter Raake and Dieter Raffler for Hanning, Paderborn, 1966.

4. **The aesthetic function:** According to Bill (1952), this function embraced everything that could not be accounted for by technological, material, and production functions. He visualized the domain of aesthetic decisions as a space of all options that the other three functions did not rule out. The concept of an aesthetic function stayed within the functionalist discourse, obviously, but reserved artists a legitimate place in that discourse. In Bill's vision, design was the art to integrate the forms of everyday things into the industrial-cultural context in which they were to function. In Ulm, the aesthetic function came to embrace such virtues as consistency, simplicity, symmetry, clarity, cleanliness, and honesty, none of which were captured by the other three.

It is important to note that Bill defined the aesthetic function in the negative, as a leftover category, as what remained open to choices after the constraints of technical, material, and production functions had been satisfied. Logically, this was a remarkable move because now all qualities of form could become the target of functional explanations, which made his

functionalism "complete." By reference to these functions we could exhaustively examine and justify or dismiss any design, including arguing for the infamous gray boxes, even when they presented challenges to manufacturers. To achieve "good form" in Bill's terms meant that all four functions had to be satisfied.

In the absence of theoretical propositions in the domain of aesthetic functions, the prevailing contempt of arbitrariness and intuition, and of forms that failed rational, i.e., functional analysis and justification, Bill drew on mathematical justifications, originally for his nonrepresentational art, called "concrete art." In 1949, he wrote: "The mainspring of all visual arts is geometry, the correlation of elements on a surface or in space. Thus, even as mathematics is one of the essential forms or primary thought … it is also intrinsically a science of the relationship of object to object, group to group, and movement to movement" (Bill, 1949:86–90). In Ulm, mathematical forms came to be associated with nonarbitrariness, unquestionable purity, and the highest aesthetic quality. Figure 9.3 shows two examples, designed about 10 years apart but manifesting the same spirit. Remarkably, both designs, like most that were developed in Ulm, proposed forms whose aesthetics were hidden behind mathematical justifications, but envisioned no new uses — a manifest consequence of the conservatism inherent in functionalism.



(a) (b)

Figure 9.3 Mathematically justified forms — (a) wall clock by Max Bill for Junghans, 1957; (b) street lighting by Peter Hofmeister, Thomas Mentzel, and Werner Zemp; instructor Walter Zeischegg, 1966.

Bill's wall clock reduces the clock to its conceivably simplest graphical display: its face — today one would say to its (graphical) interface. All of its features are mathematically justified. For example, the length of the hour hand equals the radius of the face. Both hands reach the ends of the divisions of the circle into hours and minutes, exactly. The form of the street lighting

derives from minimizing the proportion of surface to volume while as a matter of obvious functional necessity connecting the lights to a pole. To the designer's credit, after nearly 50 years, Bill's wall clock is still on the market and today, the street lighting could easily win design competitions.

Logically, Bill's aesthetic functions, defined by negation, would include product semantics in its domain. However, in celebrating mathematical forms — geometry, symmetry, systematic repetitions, and formal aesthetic principles (Section 8.6, item 2) — which are inherently a-contextual, supposedly determinate, and objective, nonabstract meanings are ruled out. There was no way that the semantic turn could have found a place and grow in Ulm's functionalism.

To be fair, technical functions, to the extent they are commonly understood, could be regarded as common meanings of some kind. For very simple tools, like tableware, bicycles, or umbrellas, there probably is no difference between the functions they are to serve and what they mean to most users. For more complex artifacts, the technical functions that engineers have to work with typically deviate from the meanings that users have of these artifacts. Computers, transportation systems, and the Internet, for example, could not be realized entirely with the understanding that their users have of them. The semantic turn acknowledges diverse stakeholders to have different meanings. The technical functions that engineers are concerned with are important only for a short period in an artifact's life. In use, in language, and in an ecology, artifacts have different meanings, and designers may have to consider all relevant meanings. In Ulm, we acknowledged that products could and indeed should serve several functions, like the Ulm stool shown in Figure 9.1. But all of these functions were thought to be universal, not cultural or social. Universality of meaning is utopian and utopians we were.

Another anecdote might shed light on the difference between how these matters were attended to at Ulm and now. To fulfill an assignment aimed at clarifying how various semantic character traits (Section 4.3) can be made manifest, a student of mine at Philadelphia's University of the Arts, where I teach occasionally, presented a Slinky for its expressing the character traits "flexibility," "lightweightedness," and being "intrinsically motivating" (as opposed to extrinsically or instrumentally motivating — Section 3.5.2). Her photographic image of that Slinky resembled one that was published prominently in Bill's seminal book, *Form,* which defined his notion of "good form" by a collection of aesthetically exemplary designs. Bill's caption of the Slinky celebrated it as an "ingenious" toy whose movements and the sounds it makes are "indescribable" (Bill, 1952:108).* Evidently, this was true then. Now, a vocabulary, measuring scales, and practical procedures have been developed for mapping semantic attributes into the characters of human

* The caption of this photograph reads: "Diese Stahlfeder ist eines der genialsten existierenden Spielzeuge. Seine Bewegung in der Hand und sein Klang können nicht beschrieben werden. Diese Feder wandert, unter anderem, selbständig die Treppe hinunter." — "This spring of steel is one of the most ingenious existing toys. Its movement in one's hand and the sound it makes are indescribable. This spring walks, among others, down a flight of stairs, on its own."

interfaces, the very phenomena that could then be described only in the geometrical terms (Slinky = a perfect spiral) of an "aesthetic function."

Bill deserves credit for recognizing that the human qualities of forms could not be reduced to technical, material, and production functions, which, in the 1960s were increasingly recognized as the cause of failures of strictly functional designs. Functionalist mass products caused not too many problems when they turned out socially unusable or lacking meaning. They could simply disappear from the market. But functionalist buildings often became unlivable horror stories, and functionalist city plans just did not work as intended, often eliciting high human costs. Without human-centered conceptions, without openness to second-order understanding in Ulm, we could not see that talking, working, and justifying designs within these functions boxed us into our own limitations.

9.2 Bense's information philosophy

An idea that shaped my interest in semantics is Claude E. Shannon's *Mathematical Theory of Communication* (Shannon and Weaver, 1949), called information theory for short. Max Bense presented a version of it in lectures between 1954 and 1958, and his books on information aesthetics (Bense, 1954, 1956, 1958, 1960), followed by Horst Rittel and finally Abraham Moles. Bense subscribed to the Cartesian distinction between a material-energetic reality and an intelligent "co-reality." From the perspective of information theory, reality, determined by thermodynamic contingencies, was seen as converging toward a state of entropy, a measure of disorder. And coreality, shaped by human intelligence, was seen as increasing the amount of neg-entropy, a measure of order. For Bense, both worlds were conceived of as dynamic and digitized. Reality was subject to natural laws, hence probable; coreality was governed by intelligent choices, creativity, newness, and hence improbable. Aesthetic production therefore was inherently neg-entropic and order creating. With these conceptions, Bense sought to transform aesthetics from a philosophy of appreciating beauty to a philosophy of the artificial, of the cultural production of technological artifacts, works of art, and texts.

In information theory, neg-entropy increases with the improbability of the produced work, with the number of possible works from which it was chosen. Entropy increases with duplications, with the mass production of the same kinds, resulting in an increase of the likelihood of any one of them. By equating the measure of the neg-entropy of a cultural artifact with its aesthetic value, Bense's aesthetics entailed a quantum-dynamic theory of culture: According to this theory, the products of the avant-garde, being radically new, highly improbable, and maximally surprising, turn also out to be most informative aesthetically, but in their extreme also incomprehensible. And as these products become more common, by mass reproduction or increased familiarity, their information value declines to the point of losing any aesthetic appeal. Naturally, students and faculty of the most innovative and advanced school of design at that time wholeheartedly embraced

Bense's aesthetics and intellectual justification as a theory of their own mission in an industrial culture.

Shannon's information theory did not address issues of meaning. This went well with Bill's project of concrete art, an antirepresentational program that can be traced to the Bauhaus, Bill's alma mater. I am convinced that his project is what drew Bense to Ulm. A consequence of Bense's information aesthetics was that students started to count, and experienced how problematic it was. Entropy measures were relatively easy to obtain for the typography of texts, but they could become quite arbitrary when applied to paintings, even of a nonrepresentational kind, and to industrial products.* Counting and computing everything by hand severely limited comparisons of information quantities to very small sample sizes. An avant-garde poem instead of poetry, a printed page instead of newsprint. However, the idea of information as a relative quantity associated with the probability of one artifact, text, or character within a large repertoire of them unwittingly diverted our attention from the aesthetics of single artifacts (works of art or design) to an aesthetics of the distributions of artifacts, to an aesthetics of the dynamics of mass culture. The absence of representational meanings in this theory conveniently bypassed the two-world ontology of semiotics. It led us to talk of information as a measure of textuality — as one would call it now — and nudged us to think of communication in nonrepresentational terms — in my case, of how artifacts could inform potential users of their own use, without standing for something else.

Subsequent to my time in Ulm, I had the fortune of studying with the cybernetician W. Ross Ashby at the University of Illinois. Shannon's theory was part of the tool kit of cyberneticians, and I became more thoroughly acquainted with it. I contributed several papers, including a book (Krippendorff, 1986) on the subject. Looking back to how information theory was handled in Ulm: we, faculty and students at Ulm, had a vastly overextended and mainly discursive understanding of it. The lack of mathematical training to look through its formalism, the absence of computers, and perhaps naively equating counting with being scientific, prevented us from experiencing its limitations and contributing to it. However, Bense managed to stir our attention to our own role in the dynamics of cultural production, in what I would now call a quantum theory of designed artifacts or, more broadly, of technology.

Finally, in his heart, Bense was a taxonomist. In lectures on the philosophy of science, he presented a systematic classification of all the sciences, filling holes with scientific disciplines to be realized, but being quite oblivious to the political gerrymandering of the boundaries among the sciences. After he left Ulm, Bense (1971) elaborated Charles S. Peirce's theory of signs, iterating its distinctions and generating thereby a whole system of ever-finer differentiations. Although the title of this elaboration, *Sign and Design*, promised to

* Garnich (1979) is an example of these efforts. His 1968 dissertation was supervised by Max Bense for a doctoral degree in philosophy at the Technische Hochschule Stuttgart.

be about design, it did little to clarify its practice. Bense was a creative philosopher, a prolific writer, an enormously effective teacher, and probably the most influential intellectual force in the early part of Ulm. However, I am not sure if he would have been willing to make the semantic turn here described. I do not recall him mentioning Giovanni B. Vico, Ludwig Wittgenstein, Mikhail Bakhtin, J.J. Gibson, or Jacob von Uexküll who prepared me for this turn. Accepting the notion that artifacts are social constructions would have meant for him to give up his Cartesian *Philosophy as a Mirror of Nature* (Rorty, 1979). He was too impatient to acknowledge divergent and especially "intellectually inferior" understandings, which is a presupposition for second-order understanding to arise. When I met him at the 1989 International Design Forum, held on the site of the former Ulm School of Design, I had hoped to tell him about the exciting developments of the semantic turn and to ask him whether he could relate to it. Before I could, early in that conference, he stormed out in protest over a speaker's irresponsible philosophizing. He promised to visit me in Philadelphia but passed away before this could happen.

9.3 Maldonado's semiotics

In 1956, Tomás Maldonado replaced Max Bill as the rector of the Ulm School. A year later he introduced semiotics into the curriculum. Bense (1956) had already paved the way for its reception in lectures and publications on a theory of signs, largely in the service of his information aesthetics, later focusing on Peirce's distinctions (Bense, 1972). Semiotics was Maldonado's effort to put his "scientific operationalism" into the driver's seat of design, partly in opposition to Bill's vision of a designer as a kind of aesthetic coordinator of cultural artifacts. To the best of my knowledge, Ulm was the first design school that acknowledged the subject. Maldonado was particularly fascinated with the politics of semioticians. In his seminar, he traced the influences that various schools had on each other, who adopted whose definitions, and so forth. During these explorations, he became committed to the work of neo-positivists, particularly Charles W. Morris (1955) and more generally to the Unity of Science movement. Count Alfred Korzybski's (1933) somewhat obscure General Semantics entered Ulm through guest lectures by Warren Robbins, a cultural attaché at the U.S. Embassy in Stuttgart. As his own contribution to semiotics, Maldonado (1961) published the definitions of 94 terms. These definitions were intended as analytical tools and as a way to canonize the semiotic discourse for design. Subsequently, Maldonado's students added rhetorical concepts and applied them descriptively, largely to visual phenomena (Bonsiepe, 1996).*

* Pages 91–103 of this book contain a rewritten article, originally published in the journal of the HfG Ulm: *Ulm*, 14/15/16, 1965.

The absence of empirical investigations *with* these terms — using human subjects to verify the semiotic distinctions — and *of* these terms — critically evaluating how the vocabulary of semiotics aided design — attests to Maldonado's belief in the universalism of semiotics, following the hopes of the already mentioned Unity of Science movement. But universalism leaves no space for alternative constructions of reality. Its propositions are not taken as mere hypotheses but as objective and hence culture-free generalizations of facts (Section 8.1). In condensing the diverse definitions published in the semiotic literature to a single consistent set — without theoretical propositions — Maldonado demonstrated a taxonomic or classificatory conception of science, which is essentially closed to empirical validation.

The neo-positivism of that time prevented semioticians from recognizing that semiotic distinctions — actually all terminological distinctions — are drawn in language and negotiated among people who care about them. Terminological distinctions create differences, constitute the very reality they claim to describe, and call on authorities to keep them in place. Maldonado's semiotics relied on the authority of philosophers such as Rudolph Carnap, I. A. Richards, Charles S. Peirce, John Dewey, mostly, however, on Charles W. Morris, whom he cited most often, but also on his own. After all, he was the rector of the most progressive design school in the world.

Naturally, any taxonomic conception of science invites struggles for being right, and there were many among Ulm's faculty. For example from 1960 to 1961, I was working at Ulm's Institute for Visual Perception, led by Mervyn W. Perrine, an empirical social psychologist from Princeton University. One funded research project concerned color perception. Arguments arose, to the point of threatening the very existence of the Institute, between Otl Aicher, Maldonado, and Perrine. Aicher insisted that empirical findings with subjects should not contradict his own color theory — no longer taught in Ulm after 1958 — because subjects could not possibly know as much about color as he did. As a successful graphic designer, Aicher was confident he knew better. He could not care less about how others perceived colors, which made empirical investigations meaningless for him. At another point, Maldonado was befuddled when subjects ranked visual stimuli in ways inconsistent with each other and quite different from what his ideas led him to expect. To him, this called scientific methods into question.

True, statistical generalizations do not make good design decisions, but done well, empirical studies certainly can shed light on the reality of other people, the very people that would come in contact with designed artifacts. In Ulm's antiempirical climate, vocabularies took the place of unchallengeable truths, and the realities they brought forth were lived intensely, defended against alternative versions, and generalized to if not authoritatively imposed on students, leaving little room for the uncertainties that nourish the testing of empirical hypotheses. Not surprisingly, the Institute for Visual Perception was short-lived. Empirical studies of meaning, on which a semantics for design could have been built, were pioneered elsewhere, by Charles Osgood et al. (1957), for an early example. Osgood's work

stimulated studies of meaning outside Ulm and influenced the work of some Ulm graduates but only after they had graduated.*

The inability of semiotics to grant human agency (Section 8.1, item 2) is evident in Maldonado's definitions as well. Their carefully chosen definienda did not envision people as having the choice of seeing something as a sign, sensing something "as it is," or experiencing multiple versions of the same reality (Section 2.3.2). Peirce's (1931) notion of "semiosis," the process of something becoming a sign for someone, which could have opened a space to enter human agency, came to be defined with reference to Morris (1938) as the "process in which something is a sign to an organism. The object of semiotic investigations" (Maldonado, 1961). Morris was a behaviorist and Peirce's pragmatist conception did not translate well into Morris' and Maldonado's definitions. They offered no possibility of conceiving of semiotic theory as a byproduct of the language used to state it, as the result of the dialogues, social interactions, and the interfaces in which it is realized and could differ by contexts of use, culture, or intention. The idea that acts of defining, publishing, and using the semiotic vocabulary could create the very phenomena that semioticians claimed to study, that semiotics relied on the semiosis of its own reality, did not occur to anyone at that time.

The positivist prohibition against self-reference (Whitehead and Russell, 1910/1958), the absence of recursive accounts and the universalism in Maldonado's semiotics precluded second-order understanding — designers' understanding of stakeholders' understanding — which is a key to human-centered design (Section 2.5). It also prevented conceptualizing the kind of interactivity, now built into human–computer interfaces, which are guided meanings and in turn create new meanings without prescription by a grand semiotic theory. The hermeneutics of reading text could have introduced such notions even before the advent of computers. But it had no currency in Ulm.

Fundamental to Maldonado's semiotics was its two-world ontology of a world of signs and symbols and a world of material objects (Section 8.1, item 1). An anecdote of how this played out in practice may make the point. When I wanted to write my graduate thesis, later titled *On the Sign- and Symbol-Character of Objects* (Krippendorff, 1961c), it was only natural to think of working with Tomás Maldonado. Predictably — but only in retrospect — he told me that signs refer to objects, but objects cannot be signs of something else; and that the whole idea of objects having meanings therefore was a categorical mistake. For him, meaning was a semiotic reference relationship and this settled what one could legitimately say about the objects of design. With this conception of meaning, graphics, photography, and texts became natural candidates for semiotic analysis — but the objects of industrial design were not. It was this positivist ontology of semiotics that created a blind spot

* Zillmann (1964), for example, wrote as the director of the research component of a Zurich advertising agency and now teaches in the United States.

for its proponents and put a theory in charge of explorations at the expense of everyday experiences. I had to write my thesis with Horst Rittel instead.

9.4 Chernyshevsky's political economy of aesthetics

As I was writing this chapter I recalled a perhaps more personal influence. It was in Ulm that I first read the work on aesthetics by the nineteenth century Russian materialist philosopher Nikolai Chernyshevsky (1855/1953). Radically departing from universalist claims, Chernyshevsky saw aesthetics as a ruling class' self-serving theory of beauty. He asked, for example, why fair skinned women with rose-colored cheeks are considered more beautiful than women with tanned and wrinkled faces. His answer was plain. Women of the ruling class did not have to work in fields and could instead attend to their attractiveness in appearance, dress, and conversation. He deconstructed — as one would say now — the prevailing aesthetics, especially its claims to universality, showing that an aesthetics was an elite's instrument of domination, setting its own values above those of others. In his thesis, beauty was — in today's terms — a socially constructed phenomenon and hence a politically variable concept. Aesthetics is not true or false, as theories in the natural sciences are supposed to be, but it is constituted in particular social practices. It is kept stable through the tacit enforcement of powerful institutions, including the network of artists, critics, collectors, and connoisseurs who benefit from working within that dominant theory of beauty. Chernyshevsky's ideas foreshadowed those of Roland Barthes, Jean Baudrillard, Michel Foucault, and especially Antonio Gramsci (1971), whose theory of hegemony covers similar grounds, now shaping the understanding of postindustrial culture.

Bringing Chernyshevsky into the twentieth century, it became obvious to me that art required name recognition (brands) and arguments (advertisements) at least as much as providing sensory experiences (works of art). Although creating art was banned from the curriculum at the Ulm School, several students and some faculty painted on the side. There were places in the city of Ulm where art could be exhibited and discussed. But the determining factor for whether something was worth showing largely seemed to depend on how it was justified and by whom. Those able to wrap their art into compelling narratives succeeded more readily than those who merely exhibited their work. Max Bill, for one, was a well-known Swiss painter, sculptor, and architect before he cofounded Ulm and became the first rector of the school. He, and an international group of artists and poets, in manifestos, exhibitions, and readings, had developed a vocabulary, including a theory of what art "really" is, to distinguish their "concrete art" from everything else. Using this vocabulary with the implicit backing of the loose community of concrete artists generated discussion and granted recognition. Those who did not have compelling narratives were less fortunate. For example, Friedrich Vordemberge-Gildewart, an early faculty member at the Ulm School with an impressive record in painting and typography, did not

go for the dominant discourse, and felt marginalized.* Bill and Bense were close as long as Bense used Bill's works in his writings. When he had an argument with Bill and used Vordemberge's paintings, there were tensions. Unlike Bill's interest in justifying individual products of culture, Bense was interested in distributions of artifacts in society. Neither realized that aesthetic perception is acquired in language and, hence, is social or political in nature.** Explaining aesthetic perception as physiological correlates of the objective properties of art objects — as attempted by Bill's reliance on ideally mathematical forms (geometry, proportions, rhythm, or grammar), and much later Gros' (1984) concept of a formal-aesthetic function (in terms of the classical principles of gestalt perception) (Section 8.7, item 2) — merely supported the aesthetics of the dominant cultural elite. Chernyshevsky might have taken Bense's information aesthetic to task as well by asking who is being served by the cultural production of improbable objects.

The sociology of Ulm's aesthetics and functionalist rhetoric cannot be understood without situating it in the period of its use. When the idea of a design school surfaced, a few years after WWII, Germany was in ruins. Inge Aicher-Scholl, whose siblings were killed in 1943 for protesting Nazism, wanted to create a memorial in their name. Some hoped to revive the Bauhaus, whose teaching was tragically interrupted by its forced closing in 1933. Bill, who had been at the Bauhaus, saw unprecedented opportunities for a new kind of design school that would contribute to the industrial culture of a new democratic Germany. From its beginning in 1953, the Ulm School of Design presented to the public an impressive array of consumer products with radically new appearances and proposals for larger projects. The clarity of its designs, its astonishing productivity in many spheres — industrial consumer products, visual communication, industrialized architecture, journalism, and film — fueled the emerging consumer industry in Germany, Braun AG, for example, and gained international acclaim.

Ulm's success was due largely to its highly effective rhetoric, a vocabulary for explaining its designs. We, students and faculty, were easily drawn into the functionalism just described by what it accomplished rhetorically. We challenged each other's designs, learned to justify every detail, and what succeeded was defensible indeed and typically also outstanding. We spiked our arguments with concepts from novel disciplines, ergonomics, semiotics, cybernetics, scientific methodology, and information aesthetics, to name but a few. We chose easily defeatable adversaries — advocates of American styling and commercialism, traditional arts and crafts–oriented designers, and the mindless producers of kitsch — and easily "won" debates by "proving" their products "indefensible," their methods "unscientific," their ideas "obsolete," or overlooking essential functions. We played a successful

* Friedrich Vordemberge-Gildewart, personal communications on several occasions. Even Lindinger's (1987/1990) Ulm retrospective had not that much to say about him.

** The social psychology of perception revealed the social background of subjects as a major factor in their perception of things. To explore this was one of my motivations for joining the Institute for Visual Perception at the HfG Ulm.

language game on top of visible design innovations. We saw ourselves as challenging an old aesthetics but — entering Chernyshevsky — we did not recognize that what Ulm provided served the needs of a generation of younger and progressive consumers to set itself apart from the generation that lived through WWII and subscribed to values considered out of date. With the marketing of successful designs from Ulm, industries naturally adopted and unwittingly promoted Ulm's justifications and thus entered Ulm's vocabulary of product forms into that intergenerational struggle.

Moreover, Ulm's alleged culture-neutral celebration of technology played right into the needs of postwar industry — not just in Germany — to expand its markets internationally. The explicit denial of pursuing a particular aesthetics, celebrating arguments in the name of science and technology had the effect of rendering cultures that held on to their own traditions or resisted the supposedly culture-neutral functionalism of the industrialized West as "backward" or "underdeveloped." We also did not see any contradiction between battling for a new aesthetics and the universalism we claimed for what we were arguing. Chernyshevsky's notion of what aesthetics does could have easily undermined these claims, but only a few of us read his work.*

Regrettably, the successes of Ulm's aesthetics also deafened us to different voices. We knew the users of industrial products only through texts on ergonomics, gestalt psychological principles of perception, typographical rules, and color theories, some developed in the Bauhaus — not by listening to or surveying the concerns of users, social classes, or other cultures. The Ulm School contributed many original investigations. But they were compelling demonstrations, like the Ames demonstrations of visual perception, not empirical inquiries into how others conceptualized their own worlds. We took for granted that consumers had to be educated in the new industrial culture and assumed their acceptance of our ideas to be merely a question of time. We hardly doubted our own convictions. This contrasts sharply with the semantic turn in design, which starts with the recognition that people (designers, consumers, and other stakeholders) may not think alike and create their own meanings for the innumerable ways of interfacing with technology. The disinterest in potentially different but equally valid conceptions prevented us from embracing Chernyshevsky's social relativism and from considering that artifacts are also the medium of social or cultural struggles.

9.5 *Rittel's methodology*

In Ulm, all design procedures and product forms were questioned and had to be justified or, failing this, be dismissed as arbitrary and inferior. We practiced quite unconsciously what Chapter 4 poses as a theorem of the semantic turn — that "the fate of artifacts is decided in language." The

* Dolf Zillmann, for one.

vocabulary for such deliberations was enriched by a curriculum that, by comparison to other design schools, was academically demanding and included several design related disciplines, notably physiology of perception, ergonomics, social psychology, sociology, economics, political science, cultural anthropology, semiotics, information and communication theory, but also traditional topics from physics to the cultural history of art. They were all made relevant to design, and the intellectual framework intended to assure their integration was the philosophy of science, originally taught by Bense. Bense was an abstract thinker and his presence enhanced the intellectual quality of the deliberations and added to Ulm's academic aspirations, but it did not challenge design as practiced in Ulm.

In 1958, Horst Rittel, a mathematician by training, replaced Bense. His first assignment was to lecture on information and communication theory, which everyone at that time thought to be the key to the future. I recall his first lecture in which he presented the technical details of Shannon's theory that Bense had glossed over. Being far younger, more practical, and quick to adapt to the needs of Ulm's curriculum, in the course of Rittel's tenure at the school, the philosophy of science he was asked to teach became increasingly relevant to design. It turned into a systematic exploration of the heuristics that designers were using. Rittel's mathematical training gave him access to models, theories, and conceptual frameworks that ordinary designers had barely heard of. He immensely enjoyed making abstract ideas relevant, and the echo he found among students brought the design discourse to a different level. He introduced methods of operations research, mathematical decision theory, game theory, systems analysis, and planning techniques, and thus enriched the repertoire of design supporting methods. Again, the Ulm School of Design was probably the first and may still be the only design school in the world in whose curriculum such topics flourished.

Whereas Bill's designer was to find a new aesthetic unity between contemporary culture and forms of mass produced artifacts without catering to commercialism or relying on sentimental kitsch, Maldonado's designer was to be equipped with the instruments of science in order to coordinate decision making from the centers of industry. Under the influence of Rittel, the conception shifted to a designer who would be able to handle heuristic planning and design methods, and work as equals in product or strategy development teams.

Bruce Archer, a guest professor and pioneer of design methods in the United Kingdom, also played a major role in introducing systematic, that is, step-by-step, design methods into the curriculum of industrial designers, later adopted by Butter (1989) into product semantical considerations. Other faculty, notably the macro-sociologist Hanno Kesting, also supported Rittel's teaching. Some of the more senior faculty, Otl Aicher, for example, felt threatened by talk about these design methods. And although his opposition was born largely out of ignorance — he abstained from lectures of his colleagues and did not participate in applying and testing these new ideas — the reason for this rejection may have been due to his interest in graphics,

which while innovative, concerned simple and obvious problems for the solution of which good intuition sufficed. Initially, Maldonado may have seen in Rittel someone who contributed to his scientific vision for the school. But Rittel was not a taxonomist who categorized the world or a positivist in search of evidence in support of a theory. He was a pragmatist, full of compelling ideas. Rittel's design methods projected and in turn prepared us for a more complex world, opened the possibilities of empirical research, encouraged examinations of the assumptions made when deciding with uncertain information, raised the awareness that our designs would have to resist possible counterdesigns (game theory), and realized that in complex systems in which almost everything is connected to everything else, designs can have rather unanticipated consequences. Most students soon recognized that the future of design was in being able to solve complex design issues with defensible methods. Designing photogenic products was no longer seen as an aim of design education. Some students shifted attention from architecture to urban planning, from feature writing to organizing advertising campaigns, and from conceptualizing single products to systems. Many current writers on design methodology (e.g., Cross, 1984) acknowledge Ulm's contribution to design methodology. A science for design, as proposed in Chapter 7, bears the influence of Rittel and Archer as well.

As Rittel's heuristic methods started to enter and guide the design decisions of students, it was inevitable that the language of students shifted from justifying the functionality of photogenic products to justifying the design methods that led to them. The arguments that were correlated with this shift proved far more compelling than functionalist and semiotic ones had been. This gave rise to tensions between the visible and invisible Ulm, as some now see it. Faculty who ran increasingly profitable design studios, working for industrial clients inside the school, left teaching largely to those who did not have such studios. They could not see virtue in the shift in design thinking, created a false schism between science and design, and used it as a smokescreen to hide their economic interest. Some historians (e.g., Betts, 1998) have fallen victim to this rhetoric, blaming scientification as the major factor of Ulm's eventual demise. This was not so, however, as Rene Spitz (2002) revealed in his fascinating political history of Ulm.

Rittel's planning methods, developed further after he left Ulm for the architecture department at the University of California at Berkeley, must be sketched here because they took off from what he learned and taught in Ulm. In Berkley, Rittel moved beyond design methods to formulating a theory of strategic argumentation under condition of conflicting objectives (Cross, 1984:135–144, 317–328). Starting from the obvious that planning always concerns future conditions and requires commitments (by stakeholders) from which facts (art, artifacts, and social facts) arise, Rittel's approach came to embrace a dialogical notion of language (Section 4.1). He conceived of designs as plans (I prefer proposals to indicate that they must always address stakeholders), as networks of defensible claims, together with their appropriate warrants, against counterclaims and diverging interests in the

presence of which they would have to succeed. In terms of the trajectory of artificiality (Section 1.2), Rittel started to develop a multilogic for projects. He anticipated stakeholder theories but did not concern himself with meanings, I suppose largely because he became involved in large public projects and complex architectural designs that are often driven by a handful of economic players and political interest groups. Already mentioned was his formulation of so-called wicked problems (Section 1.4.2), which defy rational considerations by being definable only from their possible solutions. This is quite consistent with the semantic turn's emphasis on visions of desirable futures as opposed to traditional problem solving à la Simon (1969/2001). I am convinced that had Rittel become concerned with artifacts of mass culture and information systems, which are driven by voluntary use, market considerations, and multiuser and multicultural participation, and are of political significance, some of his planning methods, posthumously published in Rittel (1992), would have led him to a semantic turn as well.

While in Ulm, Rittel supervised my graduate thesis on the procedural meanings of designed objects (Krippendorff, 1961), the precursor to product semantics. It mapped planning theory into everyday interactions between users and their artifacts.

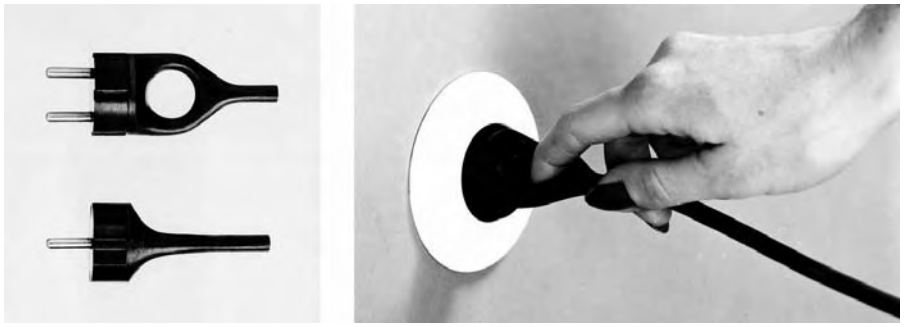
9.6 Barriers to considerations of meaning and some exceptions

Clearly, meaning had no currency in Ulm. The preference to reveal a product's technical function, its materiality, and its method of production — using the kind of forms, that we thought were easy to manufacture but actually were only easy to draw: straight lines, right angles, geometrical forms, unobtrusive gray colors — and the equation of aesthetics with culture-free mathematical forms left few opportunities for considerations of different users' conceptions and meanings. Under the aegis of functionalism, artifacts could not be conceived to mean different things in different contexts and for different users without being considered wrong. The only contexts we considered were far removed from ordinary people's discourses about artifacts: modern society, technological trends, and visual culture, on the one hand, and physiological or ergonomic and perhaps social textbook generalizations of users, on the other.

If it is true that meanings are central to any design intended for human understanding and use, axiomatically (Section 2.2), then it stands to reason that designers always had to cope with them in some form or another. I maintain that this was so, indeed, even in Ulm. Functionalism in effect provided one kind of meaning. Without adequate vocabularies, other kinds were either left to habits or intuition, or hidden behind the façade of a dominant rational consensus. Bill's notion of an aesthetic function, followed by Maldonado's semiotic scientism, exemplified efforts to capture territories lost to formal reasoning but not quite getting there for lack of second-order

understanding. In Ulm, meanings got lost between the cracks of its dominant justifications. Following are six examples whose semantics were not recognized at that time nor entered Ulm's retrospective (Lindinger, 1987/1990).

1. The first example is Aribert Vahlenbreder's electric plug, shown in Figure 9.4. This is one design that is semantically obvious to everyone who has handled electric plugs before. There is no need to ponder and figure out what it is for. The hole of this plug is just the right size to invite putting a finger through and pulling it out.* It is also flat and can be handled conventionally as well, by holding it between the thumb and the index finger, for plugging it into a socket. The absence of screws and its plastic material gives a sense of being safe from electric currents. Its use is self-evident despite its unfamiliar form, which is a semantic ideal. When Vahlenbreder presented his design, it was unfavorably compared with the design efforts of his costudents who had produced more elaborate (but now mostly forgotten) models. His was considered too simple. He had used available parts, and it was completed the night before presentation. His design was recognized, published, never produced, but recently reinvented in Japan. Interestingly, it is only one of two out of almost 60 products that the Ulm retrospective presents as being handled, "in use." All other products are shown in peopleless photographs. Even the model of a car, photographed for contrast on a street with several contemporary models, is shown on a city street without a soul present. From these photographs alone, readers cannot escape the impression that Ulm celebrated the forms of artifacts over their use, an impression that has been independently examined and



(a) (b)

Figure 9.4 Electric plug in soft PVC — self-evident handling — by Aribert Vahlenbreder, 1959.

* Note that European safety plugs are far harder to insert and pull out than those conforming to U.S. standards.

verified (Wachsmann, 1991). Vahlenbreder's use-determined design was an exception. In the retrospective, the caption for this design mentions the material it was to be made of but nothing about accomplishing obviousness in meaning.

2. The second example is Reinhold Weiss's laundry iron, Figure 9.5. Ulm's retrospective praises the uncommon and ergonomically justified position of the handle. But it fails to note Weiss' considerable effort to (semantically) distinguish the part that users could handle safely and the part he wanted to discourage users from touching. He accomplished this by introducing a discontinuity (Section 3.4.5) in the form of several seemingly heat-insulating ribs, visually highlighting the difference between the two parts. As a witness to the process of this graduate project, I can attest to his unacknowledged concerns. There was no accepted language to describe the meanings of these lamellar discontinuities.



Figure 9.5 Clothes iron — distinguishing hot and handleable areas — by Reinhold Weiss, 1959.

3. Design students in Ulm selected their own projects. Instructors did not pose particular themes and all criteria were open to discussion. In designing a caliper, Figure 9.6, my intention was less aesthetic than to render this unfamiliar instrument easy to control. I examined the handling of numerous familiar tools that required a squeezing action: pliers, scissors, can openers, pincers, for example, even sponges, and ended with how people held stopwatches. Stopwatches and calipers shared a dial for numbers to be read with the expectation of being accurate, the more the better. I used two colors to distinguish the part that one was supposed to grasp and the arms with the exchangeable precision tips that should ideally remain untouched to obtain reliable measurements. I also explored ways to express the precision of this

instrument through the fine divisions on its dial. The design received attention in publications and exhibitions, but nobody saw or cared to mention what its form was to accomplish and whether it did.

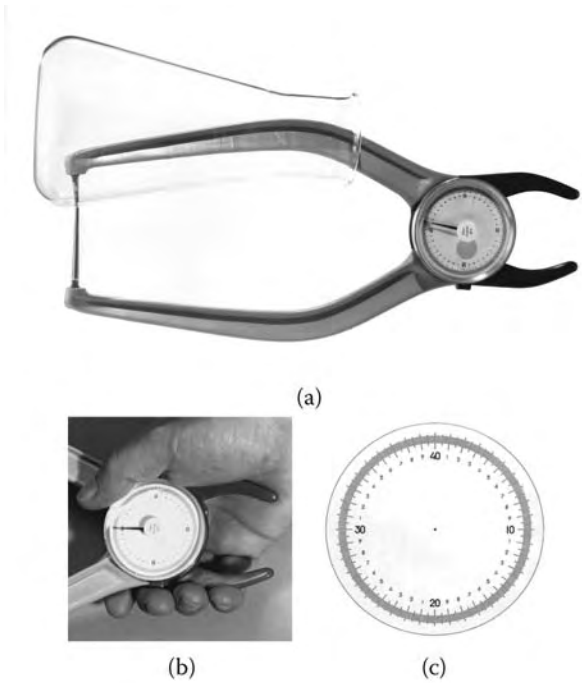


Figure 9.6 Precision caliper — making its handling familiar and recognizable — by Klaus Krippendorff, 1958. (a) Caliper in use; (b) its handleability; (c) its precision.

I also recall an example that was disastrous semantically: my design of the head of a drafting machine. It had a bright yellow squeezable handle, or knob, whose mechanism eliminated the need for several push buttons and levers. Its technology resulted in a patent. But, against the background of its white circular scales, the yellow knob led many to describe the instrument as a fried egg. Clearly, something went wrong here. Holding on to a squishy egg yolk was certainly not appealing. Many had such associations. How could one avoid these meanings? Color theory was too abstract to embrace meanings and was, in any case, no longer taught in Ulm. Semantic criteria for color choices were simply unheard of.

4. The fourth example is Reinhart Butter's precision balance, Figure 9.7. Butter followed the general preference in Ulm for straight and simple appearances. He remembers deciding on the color of the two release knobs for the scale. Common sense would suggest a dark color to camouflage accumulating dirt from handling. However, he chose white to motivate cleanliness, instead, which was more in line with the fragile

character of a precision balance. He adds that there was no vocabulary to discuss meanings, no discussion of semantic issues, and certainly no method to systematically consider them. What made his balance a precision balance was left to intuition or subjective impressions. Only in retrospect can one see how the meaning of precision, now described as a character trait (Section 4.3), was achieved.



Figure 9.7 Precision balance — implicit semantic concerns — by Reinhart Butter, 1959.

5. As a fifth example consider Hans Gugelot's 1956 Radio-Phono-Kombination SK4 for Braun AG, Figure 9.8. It shows a longish white metal box, topped by a transparent Plexiglas cover through which one can see its turntable and radio controls. Unlike traditional radios, SK4 was also attractive from all sides, did not have to be placed against a wall, and commanded attention in a living room. SK4 was an instant conversation piece, a moderate market success, but became something like a public icon of Ulm design. Interestingly, Gugelot's explanations for this revolutionary design are not mentioned in Ulm's retrospective (Lindinger, 1989/1990:75–77), and I do not recall ever hearing any, except for the insistence on its being functional, clean, and simple. Interestingly, its technical sounding name "SK4" did not appeal to buyers, and so it acquired an unintended but with hindsight meaningful nickname. The see-through cover reminded the public of the glass coffin of Snow White in the Grimm Brothers' fairy tale by that name, and so it came to be known as "Snow White's coffin." Its unexpected naming demonstrates that industrial products are not merely used as intended by their designers or producers, they also enter language (Chapter 4) where they come to live a life of their own and may become

infused with all kinds of meanings, stories, and myths that in turn shape its perception. Luckily, "Snow White's coffin" was affectionate and positive. Names can also ruin a product.



Figure 9.8 Radio–record player combination SK4 for Braun AG — "Snow White's coffin" — by Hans Gugelot, 1956.

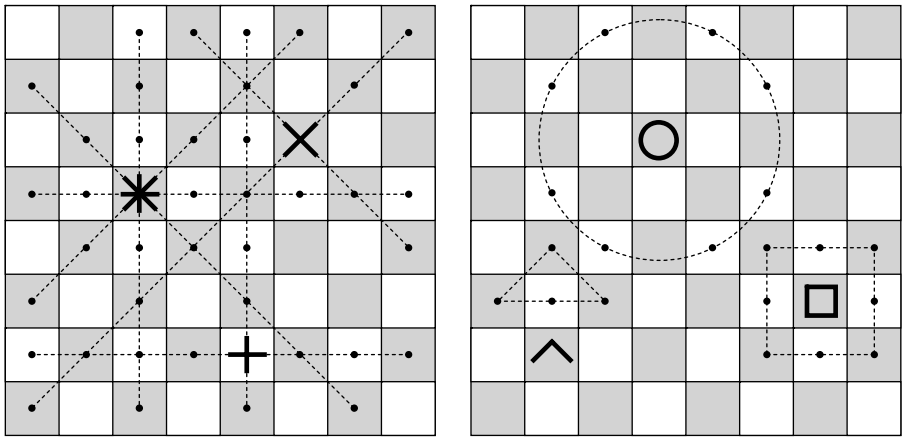
Most important, where SK4 entered the ecology of hi-fi equipment, it replaced the ideal type of wooden or plastic radio boxes with their distinct front: cloth-covered loudspeaker, station and volume controls, and a hidden turntable, if any. It had the ecological meaning of weakening the then popular radio stereotype and opened the possibility of subsequent designs.

The sound quality of SK4 was clearly below the performance expectations, even of its times. But buyers willingly sacrificed technical quality for meaning. The exciting and bold universalism that Ulm stood for became a convenient means for younger users to distinguish themselves from the "provincialism" of the declining prewar culture of their parents. "Snow White's coffin" became a conversation piece with totally unexpected sociopolitical meanings. It still is a conversation piece but now in different contexts: design museums, connoisseurs and collectors of design, and published histories of design, this book included. It acquired its meanings in the latter contexts as technological culture moves on. Today, we can see the semantics of this radio–turntable combination more clearly. "Snow White's coffin" had made no functional contribution by which Ulm wanted to be judged. In the absence of an adequate language to address semantic considerations, meanings were left to fortuitous circumstances.

6. The final example is a set of chess pieces that I developed in 1960. To appreciate the visual nature of the game, I started to play chess daily while writing my thesis, which, not coincidentally, concerned itself with how artifacts could inform their use (Krippendorff, 1961a, c). Chess is a game that requires players to recognize strategic patterns of possibilities and constraints. It consists of moves and countermoves that threaten or protect pieces from being lost; each aims at improving one's strategic position relative to the other player, ultimately cornering the prize: the opponent's king.

Chess is alleged to have originated in seventh century India as a war game. It took 300 years to travel via Persia and the Arab world to Europe. As it traveled, its nomenclature and the forms of its pieces adapted to the cultural contexts of their new users. In seventh century India, war was fought with elephants, horses, chariots, and foot soldiers, and the earliest known accounts of chess use these terms. In Persia, the game revolved around a shah and his faithful counselor and messenger. In the Arab world, the name shah stayed but the counselor became a vizier. When chess entered Europe, the names of its pieces came to represent the then prevailing feudal hierarchy. In English, the shah was translated into the king, literally, but the vizier became the queen, elephants became bishops, representing the church, horsemen became knights, chariots became rooks (castles or towers), and foot soldiers became pawns. In Europe, the abstract shapes of Arabic chess pieces became more or less elaborate representations that followed their names, not the roles they played in the game. This says much about how language acculturates artifacts in use.

I did not like war, and found the feudal structure encoded in the traditional European chess pieces unfortunate and unnecessary to playing the game. I knew the 1924 Bauhaus chess set, which preserved its feudal heritage, but expressed its hierarchy by cubes and spheres, except for the bishop, who was represented by a cross. I was also familiar with other designs, for example Man Ray's 1927 set whose pieces had nearly nothing to do with the nature of the game. Consistent with my simultaneously evolving graduate thesis, I wanted my chess pieces to inform players about what each was capable of doing, and reveal the threats they could pose to each other in the course of a game. I hoped that making their operational roles "self-evident," as one would say now, would allow the game to be learned more easily, render the strategic options that a particular configuration of pieces entail more readily transparent, and improve strategic thinking. In my analysis, chess has three directional pieces that can move any number of squares in any one of several directions, threatening everything in their possible paths. It also has three pieces whose movements are circumscribed. Two of them control fields that surround them and one can move only in one direction, forward. It made sense to adopt two simple and obvious rules: the directional pieces should visualize the direction of their mobility and control, and the circumscribed pieces their ranges, the limits of their mobility and control.

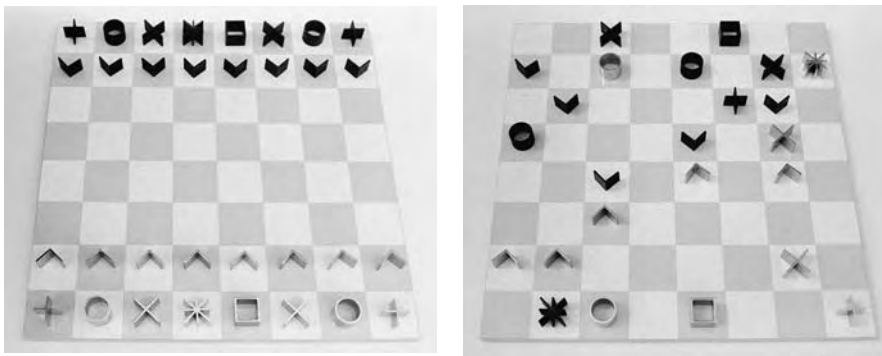


Directional pieces: Queen = Rook + Bishop

Circumscribed pieces: Pawn, Knight, and King

Figure 9.9 Correlations of chess pieces with what they control.

This correlation may be seen in Figure 9.9. All pieces were of equal height, naturally. The Bauhaus' bishop resembled mine, suggesting that its albeit differently motivated semantics could make sense here as well. In the beginning of a game, the row of pawns visualizes the protective shield they collectively provide for the pieces behind them. During a game, when pawns typically appear in "birds' flight formation," they continue to visualize their defensive roles, as may be seen in Figure 9.10. The queen, combining the capabilities of the bishop and the rook, combines them visually as well.



(a) (b)

Figure 9.10 Chess pieces — visualizing possible movements — by Klaus Krippendorff, 1960. (a) Starting configuration; (b) intermediate configuration.

The pieces were to be cut from profiles of extruded nonmagnetic metal, chromed and blackened respectively. To center the figures on the squares of the chessboard and prevent literally turning the more mobile bishops into clumsy rooks or vice versa, a system of four positive and four negative

magnets were to be inserted into the board, which would ensure the correct directionality of the pieces during a game. Carving these shapes into wooden cubes or using inlays on one face of squares was envisioned but not realized.* To use the set immediately, a compromise was made, replacing the + of the rook with a ⊥, which depicts the fields it controls at the beginning of the game.

Much after I had finished the set, I realized that designing these chesspieces had been my way of working out at least part of the thesis that I was writing at the same time.

The six examples just discussed are not representative of Ulm's many contributions to design. They were selected as semantically interesting exceptions, pertaining only to product design and representing but a brief period of Ulm's short but productive history. They are also relatively small artifacts, hand tools, appliances, and furniture, which are hardly representative of the kind of design problems the semantic turn is facing now. Computer interfaces did not exist. Multiuser systems did not play the role they do today. Large projects that required work in multidisciplinary teams were talked about but not really tackled. Ecology had not been recognized as something to worry about. And the lack of self-reflective use of language made it difficult to consider our own discourse as a possible way of redesigning design.

In concluding this chapter, I am suggesting that the approach to design that Ulm had taken, while reaching far into the future, was essentially blind to systematic considerations of meaning. These examples of designs slipped through the cracks of its functionalism. Maldonado's semiotics saw meanings only through the lens of an unproductive theory of signs, as representations. Nevertheless, Ulm's discourse certainly informed my own path, even where I disagreed. It is not surprising that from Ulm, many former students were led to considerations of meaning, each for their own reasons. Richard Fischer (1984), one coauthor of the theory of product language (Section 8.7), expressed his dissatisfaction with functionalism already as a student in Ulm and subsequently found his way out of its box. Jochen Gros (1987), the other coauthor of this theory, studied with Martin Krampen who had studied with Maldonado. Krampen (1979, 1989) wrote on the meaning of urban architecture and did much to develop behavioral methods to experiment with representational meanings. During his short interest in semantics, Dolf Zillmann (1984) developed novel measuring scales. Gui Bonsiepe (1996), keeping the legacy of Maldonado alive, recently acknowledged human–computer interfaces as a new ontology of design. Shutaro Mukai (1979, 1986, 1991, 2003) explored the nature of language, Japanese characters in particular, advocating a concrete poetics, and teaching industrial design with this in mind. Reinhart Butter encouraged me for years to continue working on the meaning of meaning in design, which he recalled from our overlapping tenure in

* These shapes may gain currency in contemporary computer chess, which is played on a flat screen. The contemporary two-dimensional depiction of the classical chess pieces has no correlation with what they are able to control.

Ulm. He practiced product semantics long before it had a name, and this book is full of examples of his Ohio State University student's work. Hans-Jürgen Lannoch (Lannoch and Lannoch, 1989) became an early contributor to product semantics. Sudhakar Badkarni coorganized the first product semantics conference, in 1987, in Bombay, India, titled Arthaya, which is an ancient Hindu word for meaning. Bernhard Bürdek (1991/2004, 1997) contributed to HfG Offenbach's theory of product language and began to address issues of computer interface design. For all of these diverse contributors, and probably others as well, experiences in Ulm were the common thread. Ulm enabled its former students to boldly pursue paths not envisioned at that time.

Despite Ulm's blind spots for meanings, the semantic turn does not oppose Ulm's values. It could be considered a radical reformulation of the Ulm School's moral mission, now moving design into what was then an unarticulated domain of human-centeredness: The design of meaningful human interfaces with technology. Some of Ulm's esteemed virtues reappear in the well-known qualities of human communication: clarity, economy, expressivity, and informativeness, but now conceptualized in interactive and culture sensitive terms. While the semantic turn creates new options for design, it does not endorse arbitrariness, meaninglessness, or dishonesty either. By playing on the now dated design principle "form follows function," the semantic turn offers designers a new covenant:

Interfaces follow recognizable meanings.

The semantic turn acknowledges design as a fundamental human right, the right to construct one's own world, interact with fellow beings in theirs, and make contributions to the ecology of humanly accessible artifacts. It is a matter of ethics to acknowledge multiple stakeholders in design, but it is a matter of survival for the design profession to make design discourse — the language that creates possibilities and proposes collaboratively realizable futures — methodologically sound, easy to use, and accessible to everyone who cares to exercise their right to design. With this in mind, this book starts a new foundation for design.

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